

What is Reflective Practice?

Reflective practice is an ongoing and dynamic process for the teacher. It involves thinking critically and deeply about what is happening in the class room. It occurs spontaneously, immediately but, also goes in a very planned and structured way. Most importantly, reflection leads to action. Teachers need to value reflection and evaluation of their own practice and their own professional role in the class room. In addition, teachers' professional knowledge and understanding includes ways to reflect and ways to evaluate what is happening. This is part of the professional role of the teacher.

Teaching is a profession and as such in a professional setting reflection is very **deliberate**. It is also **purposeful**. There is a reason for it. It is **structured** and it **links our understanding to theory to practice** and also includes deep learning and not superficial learning. Therefore, reflective practices aimed at development becoming a reflective practitioner is becoming an excellent teacher.

What are the key words that we can take from this message?

- **Mental processing** (reflection is structured thoughtful, it is not superficial)
- **Better understanding** (we gain a clearer, thoughtful understanding of what is happening in the classroom and reflection includes the teacher in the process of teaching)

Reflective practice encourages the teachers to understand their:

- learners,
- their needs and
- their abilities.

Why Reflect?

Reflective practice leads to

- Higher quality teaching and therefore better outcomes for learners.
- Awareness for the teachers of the values of teaching and their own beliefs about teaching.
- Challenge and to change
- More inclusive environments for learners.
- Improvement
- promotes collaboration between teachers as professionals.

Why is reflective practice so important?

Research shows that reflective practice is crucial for teacher's learning and for extending their skills as professionals. It is a highly personal process for any teacher and the teacher requires the professional to take their daily class room experiences to internalize, understand them to pull them over in their mind. In this way, it is possible for the teacher to filter through what is happened to focus on what is most important, to understand their personal values and their biases as teachers before deciding how to better move forward as a practitioner.

Research also shows that teachers, who regularly reflect on their teaching, reflect on why they do things and reflect on new knowledge which is being created, use this to achieve the best outcomes for their students.

Reflective practice is a process by which you stop and think and you consciously analyze the decisions you make.

Critical analysis and evaluation for the teachers refocuses the thinking of the teachers and helps to generate the knowledge needed for good quality teaching. We can represent this visually:

- subject knowledge,
- teaching skills,
- interpersonal relationships in the school as well as
- their own personality with a focus on research and teaching.

What Is to Be Gained from Reflective Practice?

Clearly, reflective practice means best practice. It provides a real shift in focus for the teacher away from deterministic views of teaching, a shift in the power of teaching. It creates a general dialogue about what is teaching and learning.

The benefits of reflective practice

Most importantly, reflection helps us as teachers to develop our own learning and provides a teaching framework. Stephen Brookfield (1995) explained this and proposed the following framework. According to him, reflective practice

- Allows us to consciously develop our own repertoire of strategies and techniques in the classroom. We can draw upon our teaching and use this in the context and discipline of teaching.
- Helps us to take informed decisions in the classroom to take the actions which are meaningful and justified and be able to explain these to others.
- Helps us to adjust our teaching, respond to problems and issues which we have faced as teachers in the classroom. For example, rather than just reacting to a poor evaluation, we investigate and try to understand the underlying reasons for a poor evaluation and use this to take appropriate action.
- Helps us to become aware of the underlying beliefs and assumptions that we have as teachers about teaching and learning and as a consequence we can try to understand what we do and why we do.

- Helps to promote a positive learning environment. Through reflection, our teaching becomes much more responsive to the students and their feedback and through this we get good quality learning.
- Helps us to locate our teaching in the broader institutional and even social context in which teaching and learning takes place.

Advantages and Drawbacks of Reflective Practice

The key principles in relation to reflective practice and the drawbacks associated with it are:

- Lack of conceptual clarity
- Too much focus on the individual
- Failure to acknowledge the discussion, dialogue (discourse) around reflective practice in teaching.
- Failure to understand the complexities involved in teaching in the classroom.

Consequently, the concept of RP is abstract.

- It is ill-defined.
- It is open to multiple interpretations.
- It is applied in a wide variety of ways in different practice environment.

Why Is Reflective Practice Important?

Reflecting critically allows us to synthesize different perspectives to help, explain or to justify and challenge what we have encountered in our teaching. RP may provide evidence to support our

views or our practices or it may even challenge them. Critical reflection also allows us to analyze what we have learned and how we have learned it. Most importantly, it enables us to take control of our own development. It is in light of these two functions of reflective practice that a great deal of importance is now placed on critical reflection in the process of professional development for teachers.

Why it is important?

The passage discusses the concept of reflective teaching, which involves more than just casually thinking or talking about teaching experiences. While many teachers already reflect on their lessons and student behavior, this reflection can often be superficial, focusing only on immediate reactions. Reflective teaching requires a more systematic approach, involving the collection, recording, and analysis of thoughts, observations, and student feedback. This process helps teachers understand the reasons behind their successes and challenges in the classroom, allowing them to make informed changes to improve their teaching practices. Reflective teaching is crucial for continuous professional development and competence..

How RP can develop your work further

- Address student's needs
- Working with colleagues
- Your role with your students
- Planning students' learning
- Working with external agencies/parents

Principles of RP

1. **RP is problem-oriented** or in other words problem-posing or problem-solving.

2. **RP is based on action research.** Action research is a very powerful tool for change and improvement based on a cyclical process of thinking, planning and doing.
3. **RP ensures that teaching and learning are research-led**
4. **RP is data driven.** This means that there is ownership by the teacher of the data which is central to the teacher professional development.

3 specific types of reflection

1. **Technical rationality:** (Behavior and skills) This is essentially the practical skills that the teacher brings to teaching the students. It is the behaviors and the attitudes, the beliefs that the teacher has about learning that impacts on their teaching in the classroom.
2. **Reflection-on-action:** This involves description and justification. It is dialogic, in other words a discussion. It is explorative and also engages multiple perspectives and factors. Reflection-on-action happens after teaching.
3. **Reflection-in-action:** This is the teacher's thinking 'in the moment'. Thinking which occurs in the second by second, the minute by minute. Teaching that occurs between the interactions, between the teacher and the student in the classroom.

So, there are three types of critical thinking, three types of reflection, technical rationality, reflection-on-action and reflection-in-action.

- **Reflection-in-action:** It refers to the thinking the reflection that the teacher does moment to moment while teaching is happening there in the classroom. It is a reflection on what the students are doing or saying and also what the teacher is doing. Reflection-in-action allows the teacher to see clearly what is happening to consider why is happening and then to respond to do things differently in relation to that reflection.
- **Reflection-on-action:** It occurs after teaching has taken place, after learning has happened. You may think more deeply at this point about what the pupils didn't understand. What caused something to happen in the classroom and what options were valuable to you, while you were

teaching. Your responses will depend therefore, on the level of knowledge and experience and your understanding of theories and the values, the beliefs you have as a teacher in relation to this type of reflection.

Hatton and Smith (1995) identified two types of reflective practice.

1. Dialogic reflection
2. Critical reflection

Dialogic reflection: it refers to a less intensive process than critical reflection. It involves a self-discourse with the teacher. It's an attempt to explore an event internally. It involves considering the judgments and decision that the teacher has made during their teaching.

Example

Dialogic reflection is an example of a basic form of reflection proposed initially by Brockbank and McGill (2000). They said that in dialogic reflection, the individual thinks about what they are going to do in their teaching, about the information they'd like to convey to the students and how they would do this, the methods they would use to convey that information. They also consider the level of engagement of the student in the class. Afterwards they then reflect on learning. They think about how far the individual goes for the students, the objectives for the lesson of the teacher have actually been made. This can be represented in the following model.

Critical reflection: This refers to efforts from a much broader perspective, thinking about historical even political perspectives on the problems which are being encountered. This is been described by Boyd and Fales (1983) as "the core difference between whether a person repeats the same experience several times becoming highly proficient in one behavior, or learns from experience in such a way to be cognitively and affectively changed".

Therefore, critical reflection facilitates transformational learning in a way that dialogic reflection does not and transformational learning can happen gradually, slowly over a period of time or it can happen quite suddenly based on some critical situation in the classroom.

What do these considerations mean for RP?

It suggests that RP is

- Non-judgmental
- Collaborative, and
- Should be flexible and based on open-mindedness.

Critiques of Reflective Practice

Ethical concerns

The teaching and application of RP have generated these concerns over the period of time and they continue to be discussed. These ethical concerns relate to issues of confidentiality, privacy, informed consent and also in relation to the professional relationships that teachers have. Therefore, practitioners who are engaging in RP need to be aware of these concerns, and the risks involved in the process of RP. A further problem can be with the way reflection takes place. It can lead to feelings of disapproval and self-rejection because the process is so critical. Further, if an individual understands the word 'critical' to mean 'negative', that can have very serious consequences long term for the teacher.

Professional concerns

If RP is done badly or ineffectively, this has a consequence in terms of what can be missed for the teacher and the value of it goes diminished. If RP is applied in an uncritically way, then reflection

can actually reinforce prejudices and pre-conceived ideas about teaching and learning and about practice. This can actually lead to continuation of the way sudden practices are used in the classroom. RP at the level of the individual practitioner means that organizations can actually remove themselves from the responsibility of professional development, moving it to the individual rather than the organization.

Pedagogic concerns

Reflective teaching raises two main pedagogic concerns;

Developmental readiness

Are the teachers ready for this sort of developmental activities?

Do they have developmental readiness? And also, the extent to which forcing teachers to reflect may be counter-productive. Developmental readiness identifies the difference between a novice teacher and an experienced teacher. The novice teacher may not be ready to reflect critically. Therefore, developmental readiness defines a distinction between a novice teacher and the expert who has practical mastery.

Forced RP

If RP is forced on the teacher for example, as part of the course, this can actually affect the quality of the reflection and especially where it may be linked to an appraisal process for teachers, any sort of genuine and honest critical self-evaluation could actually be diminished.

Conceptual concerns

There are conceptual concerns about how ideas can be transplanted from one place to another.

Is there clarity of understanding of what teaching and learning is and what reflective practice is as a process? It can be understood in multiple different ways and actually therefore, the exercise of RP can engage in different traditions and therefore, is no uniform between teachers and organizations (Finlay, 2003). Therefore, some may see it as a primarily personal or even confessional process that deconstructs socially situated action and therefore does not have broader implications for teachers.

Informal reflection: This relies on experiential learning. Experiential learning is an approach to informal reflection is often quite dysfunctional. It is always incomplete. It is a cyclical process, one thing leading on to another and therefore, is never ending. This is based on reflection on present experience and this can therefore, lead to misinterpretation. Such an approach can be manipulated to fit our own beliefs and therefore, can be quite biased because we view the world quite selectively. What we see and what we hear, we view through our own internal lens. Informal reflection can be both planned and unplanned but will always involve a conscious awareness that learning is actually taking place. A by-product of some other activity, usually unintentional or unexamined and can actually test here our beliefs.

Formal reflection: learning from formal reflection is a little bit different. It requires becoming aware of conscious learning in non-routine situation. Learning from formal reflection also requires an awareness of learning to surface, to not be tacit (understood), to move from being tacit to being more explicit. In literature this referred to as ‘mindfulness’. Therefore, formal reflection requires models and frameworks for the teacher to use. These provide a very specific and structured approach to reflection. All formal models rely on following parameter:

Learning

Behavior	Visible behavior Function: teaching, maintaining professional relationships
Reasoning/Judgment	Student-oriented understanding Function: problem-solving

Professional thinking	Critical appraisal of literature and own practice Function: optimizing teaching based on experience
Personal reflection	Mindful awareness; exploring Function: balance

1. Behavior: There is a visible behavior and the function associated with that is teaching and maintaining professional relationships in the school.
2. Reasoning/Judgment: This relates to student-oriented understanding and also about function of problem solving.
3. Professional thinking: The teacher engaging in a critical appraisal of literature in relation to their own practice and the function is to optimize teaching based on experience.
4. Personal experience: This is a mindful awareness of what is happening in the classroom. The teacher exploring what is being going on and the function of this is to balance, what is happening to eke out (use) all the good things and to identify the problems which need to be resolved.

Teachers reflect on the

- philosophy that provides the foundation for their teaching,
- their values and beliefs about teaching and learning,
- their practice

Reflection on your philosophy

Reflecting on your philosophy is whether you gave children the opportunity to express their thoughts and their feelings and the support you give to that level of independence. It is also about

investigating your own prejudices and your own biases and understanding and appreciating your own beliefs and values about teaching and learning.

Reflecting on beliefs

You can reflect from your own perspective about yourself. For example, he might believe “I am student-oriented” ... or... “I see myself as providing a service” that would be another perfectly good belief to have.... or.... even, “I have professional expertise” if you have been teaching for many years. These are beliefs about yourself. There can also be beliefs about families. For example, families are the most important part of children’s life, or you may actually see families as part of the problem that interfere your work in the classroom.

You also have beliefs about children. You could see for example, that children are very capable as learners. Children are interested as learners from birth or you may just see children as very vulnerable and needing a lot of support and guidance.

Reflection on values

Reflecting on values is also very significant for the reflective practitioner because it informs the decisions that you make in the classroom.

These come from our life experiences, especially the early ones from young childhood, from the family and the community. They come from study and from experience. More interestingly, these can be shared in every setting in terms of how children learn and develop.

Professionals reflecting

It is very significant part of reflective practice. It is important for you to become aware of the assumptions that you hold and the values and beliefs that you have about teaching and learning, and how all these things come together to create your professional identity.

There are some assumptions that we need to consider about reflective practice.

- **Assumption 1:** The reflection takes place in the classroom and is conducted **by** the teacher and **for** the teacher. Therefore, such reflection is contextualized and very private to the individual teacher.
- **Assumption 2:** Teacher reflects and acts on what they observe. This means that reflective practice is both a process of research and action.
- **Assumption 3:** Reflective practice requires dialogue for true understanding (Kemmis & Mc Taggart, 1992; Wells, 1999, Vygotsky, 1999). This places reflective practice within the social constructivist model of learning.

Consequently, a **social constructivist model of learning** suggests that reflective practice is a process of thinking which is mediated by language or in other words potentially influenced by the language that we use to describe our understanding.

Assumptions and reflective practice

Reflective practice is therefore about becoming aware of our own assumptions and how these assumptions govern our practice (what we do in our classroom) and how these assumptions must shift in order to change and change is necessarily in teaching. There is also an understanding that there can be a resistance to an assumption change or to a shift in assumption and finally a change to assumption to support a better state of affairs improved teaching and learning. The reflective practitioner is therefore someone who lives reflection on a day-to-day basis. Therefore, becoming reflective practitioner takes a lot of time, lot of commitment and requires discipline and an understanding of the responsibility about being a reflective practitioner.

Key points about our assumptions

Reflective teaching involves discovering and researching our own assumptions. Our own assumptions can be explored through a number of different lenses. We can explore our assumptions through

1. autobiographical writing usually things like learning journals
2. student feedback whether written or verbal
3. discussions with colleagues regularly
4. reviewing the literature on teaching and learning

Stephen Brookfield puts this very well. He says “Teaching in a critically reflective way involves teachers trying to discover the assumptions that frame how they teach. Teachers have four complementary lenses through which they can view their practice. The lens of their own autobiographies as learners,

the lens of students’ eyes, the lens of colleagues’ perceptions, and the lens of educational literature”.

Why are these lenses important?

Reviewing practice through these lenses helps us to admit and surface the assumptions we have about teaching, methods which we use or techniques that we feel appropriate. In so doing therefore, we can ensure the best for student learning.

Why best practice?

The best practice is important because the 21st century is a different century in which we were educated and therefore the needs of children in the 21st century are changing. We need to constantly think and rethink what we are doing in the classroom. What children need in the 21st

century is something **that is** going to help them survive better and thrive more in this new global technological environment and the reflective practitioner has a means through achieving best **practices** to help their students help children to achieve their dreams. **Moreover, children will need creativity, problem-solving abilities, a passion for learning, a dedicated work ethic and lifelong learning attributes.** The opportunity is therefore that can be given to children are dependent on the reflective practice inner achieving the best practice the best approaches that they can in the classroom.

What Does Best Practice Look like for the Reflective Practitioner?

Students are engaged and focused on their work. Well one thing to consider is that a best practice classroom is where students are on task for very much of the time. They are engage in the process of learning.

We can also think of a best practice environment as: one where **teachers are collaborating and working together** on very realistic tasks, which place students at the center of learning or in other words **students-centered learning.**

And perhaps most importantly is that the best practice environment is an **activity-based environment** and this is a phrase **that** teachers use a great deal these things.

And a very important aspect of this new **21st-century** learning of the best practice environment in our classrooms which are environmentally important learning spaces or in other words **conducive to learning.** **Also, there is a joyful feeling of purposeful movement industrious thinking and a vital and vibrant atmosphere and environment.**

Core Best Practices:

- One best practice is that the teacher will **educate the whole child.** A math teacher for example: what does teach math but is aware of the child as a whole learning individual and their needs for learning in their interests for learning.

- And so, therefore **teaching is not only based on the knowledge of the subject but on the knowledge of how children develop and learn.**
- Another core best practice is to think of **preparing children for success not only in school in terms of academic achievement but also in later life.**
- And we should also consider that best practice is about **providing an environment that is enriched** in other words one where children can thrive **from** a mixture of approaches and resources.
- Another core best practice is to think of **students as active participants in the learning** process rather than as passive observers.
- And a learning environment that **provides a challenge** but also is very rigorous in terms of what students are learning.

Core Worst Practices:

Some bad practices for example:

- Worse practices are teachers who **plan and teach in isolation** without collaboration.
- **Teaches to the test “one size fits all”.** Teachers teach simply so students pass the test and teach one subject in the same way to all students.
- Teachers who assess who **teach without assessing students’ needs** so that some students may get left behind.
- Teachers who do not challenge themselves and **stay within their comfort zone**. All is also considered to be not the best practice.
- **Applies student-centered/activity-based approaches in name only.** Teachers who say in label only that they are using student-centered and activity-based practices. This is another aspect of what we were consider not best practice.

Operationalizing Best Practice:

- We can talk about operationalizing, these best practices simply by thinking of the title and status of teacher as a skillful reflective practitioner.
- The teacher therefore needs to be concerned really about the consequences of teaching, as well as what might be called technical proficiency. The long-term consequences on the child not just about passing the test or getting through the curriculum.
- They need to be prepared to experiment with the unfamiliar things in the classroom and learn from their experiences. This is another aspect of best practice.
- And the reflective practitioner who we can see is using best practice has an approach which is very open-mindedness and honest and is wholeheartedly about the student.
- The best practice practitioner is committed to professional dialogue with others, talking, collaborating with peers and sharing, understanding and acquired knowledge.
- And also the best practice practitioner is someone who focuses on action and evaluating and revising what they are doing regularly on an ongoing basis.
- And perhaps most importantly the reflective practitioner who can be seen to be one who is using best practice is themselves a lifelong learner and responsible for their ongoing professional development.

Guiding Reflection:

RP develops an awareness of how professionals come to hold their assumptions and informs reflection on how this informs and shapes practice. So reflective practice develops all of these skills potentially and so in the classroom, we can see best practices.

Implications for Achieving Best Practice

- **Reflecting in practice** is very much about thinking on your feet the moment to moment reflection that happens while you're teaching.
- **Reflection on practice** is very much about thinking back after the event after the classroom teaching has happened.

- What is important in terms of this reflecting in practice and reflecting on practice? It is about the teacher reflecting also on their own values and their own attitude about what has been happening in terms of the teacher-child interaction and children's overall learning.
- And in this way very often in the literature you'll see that the reflective practitioner needs a lot of guidance to do this in a very constructive way and therefore make a positive change.
- And to make changes teachers need to not only deepen their understanding of the subjects they're teaching. But, also an understanding that their own values and beliefs and assumptions about teaching and learning. And this is what requires a lot of formal practice and some very good quality guidance.
- Studies show that the without effective tools without a structure for reflection then what can happen is that reflection tends to be quite superficial. And so what you can use for reflective practice. Our questionnaire, or another structured techniques such as recording formats. Guided questions to guide the reflection.
- Most importantly reflection needs to be ongoing. It needs to be a normal part of the day-to-day routine for the teacher not something which is done just once in a while.
- And therefore we can break the reflective practice down the structure into certain components. And one component is about collaborating with other teachers
- This is very significant because, it indicates that teachers are part of a learning community. If there is collaboration then there is a long-term commitment to reflective practice. While, guidance and structures are important factors in engaging a professional's long-term commitment to reflective practice
- And more than anything else therefore, reflection is not only encouraged but mutually supported and that is a very significant process to occur. A professional can bring a different perspective to the reflection, and encourage a teacher to think about what values they hold, why they hold them, and what works best for children and their learning.

Regular Learning Networks:

Another structure we can give to reflective practice is regular learning networks. Reflecting regularly on other professionals' comfortable and predictable worlds has been proven to create reflective teaching "hubs". So, in other words, opportunities for teachers to get together. This does require an organizational perspective on the reflective practice process and how the school or the educational organization can support the teacher's reflection. The interesting thing about such learning networks or hubs is that they don't necessarily need to focus on the events in the classroom. But, they can include a wider dialogues-around policy in curriculum and around other issues, in teaching, pedagogy, and learning.

Time and Space for Reflection:

Research does show that a significant impact on the ability of the reflective practice to achieve best practice is that the organizations give time and space for reflection to occur in the busy school schedule. This can be difficult but whereas cool allocates five or ten minutes at the end of the day this become extremely valuable. Networks are significant in this way because networks give chance for groups to get together and for reflection to occur.

Informed by Contemporary Research:

Another structure that we can give to reflective practice is that it does need to be informed by contemporary research. Research tells us that regular reflection is not only about what happens in the classroom but linking theory to practice. Also an effective practice is regular, and it includes both reflection in and on practice. The one-off workshop has been shown to be quite ineffective in terms of teacher development. So, any sort of reflection which is ongoing and they are therefore, support best practice through development. Simply, reflective practice in this way can become a much more strategic and collaborative suite of the long term rather than just a passing on of information.

Comparison with Traditional Models of Teacher Learning:

We can contrast with the traditional model of professional development which is very much about that **transitioning** from one person to another of accepted knowledge, and accepted experience **and that is not personal or based on individual experience.** And the primary assumption there is that knowledge about teaching needs to be passed on is acquired by one person. And then, shared with other people around them rather than something which is mutually believed and understood by everyone. Therefore, theory needs to be thoroughly understood and **carefully applied** in this regard. Because, theory then means that the reflective practitioner is achieving good practice. So, the Traditional approach to professional development reflects a basic assumption that information is passed on and reflective practice is not about that. Reflective practice is about learning from acquired experience. **The traditional approach to professional development reflects on this assumption that information is a stimulus for behavioral change, and that individuals receiving knowledge will use it to improve performance.**

The Reflection Process:

The teacher needs to review,

- What happened to think about?
- Why certain things happen in a particular way?
- **How** they happen?
- **And what** were the consequences?

And most importantly how could the insights from this sort of experience be used in future lessons.

What have they learned? What is the teacher **learn? What could they do now in relation to?**

What happened before?

Collect Information: **Then** there is a part of the cycle which is about collecting information that could be about asking the students for feedback. It could be about asking colleagues to come and watch you teaching to give you feedback. It could even be about you watching other teachers in their classrooms. So that you can consider how they are approaching their teaching. This leads into **the** questioning and analyzing phase and where you are posing questions for yourself.

- What happened?
- Why did that happen?
- How did it happen?
- Who was left out in the class?
- Who succeeded most of all in the class?
- What could be improved?

This whole process then leads back into another planning phrase ready for teaching, so you can see the reflective cycle. It is an iterative and it leads to a best practice approach.

- So collecting information can be talking to children, colleagues, and parents.
- It can be about observing class students in the classic could be observing other teachers.
- It can be having teachers observing you and giving your feedback.
- It could even be about video in your own teaching or taking an audiotape and listening or watching those things over and over again.
- Collecting information could include looking over the lesson plans for the lesson and looking over other people's lesson plans to see how yours relates to theirs. And certainly looking at your lesson evaluations. If the students evaluate what they say if you evaluate what were you thinking at the time when you filled out the evaluation.

Question/Analysis:

- Question and analysis is about looking for a solution to in order to correct a problem which is very often considered a **single-loop learning**.
- If it takes a wider perspective looking at bigger questions and looking at how theories are being used or applied in the classroom this is considered **double-loop learning**.
- It could be about asking questions about changing and **different perspectives**
- And **asking why** things happened.

Plan:

Planning is about developing the hypotheses and using the **reflective practice strategies** to apply in the classroom and test out the results. There are some Reflective practice strategies as follows: Journals or diaries, Meeting with Mentor or critical friend, reflective practice blog, Action research, Video, and Observation.

It's (planning) very much about asking well, So, what if I do this what's the result? If I don't do this what effect will that have?

Act/Do:

Acting and doing in the classroom is testing your assumptions do they work, don't they work. Things can go wrong and failure is not a failure. Failure is a learning opportunity and it is an opportunity for the teacher to say okay. Now, what now what is next now what can I do.

Review:

- Reviewing is very much about reflecting on practice reflecting on **what has happened** after the event. Asking the question well what did happen.
- Van Manen proposes is in terms of **hierarchical levels**.
- The first level he talks about his **technical reflection**. This is the actual practice the teaching, the skills for teaching, and the pedagogical activity that takes place in the class. At this level, our teachers are concerned with purely the technical aspect of being in the classroom How they're talking, how they are presenting, and how they're interacting with the students. Teachers are also concerned with applying knowledge to achieve instrumental outcomes, and actions taken are evaluated based on of their success or failure in the classroom.
- **Practical reflection** is the next level. It focuses on the learning experience of the student. It goes beyond that technical rationality aspect and starts to investigate into questioning a "What is going on in the classroom? What learning is taking place? And how is teaching affecting that learning?". And clarifying the end objectives and the assumptions behind teaching activities designed to achieve those objectives.

- And finally there is **critical reflection**. This is the highest level of reflection that you can achieve. It focuses on what knowledge is of value and to whom that knowledge is valuable. At this level, teachers **are** not simply concerned about the goals and the activities and the assumptions that underlie those goals and activities. But the larger context of teaching and learning **is** the bigger picture of education. And it is very much connected with the teacher's beliefs values and assumptions. Their ethical and moral issues **are** associated with teaching children in schools.

Reflexivity:

There is a term used very often in texts about reflective practice and that term is reflexivity. Reflexivity for teachers means not only engaging both in a personal reflection but also engaging with wider discussions about teaching and learning. So what's very good for us to consider what skills were talking about when we use **the** word reflexivity?

Skills Underpinning Reflexivity:

Self-awareness:

The first skill we can talk about is self-awareness. Self-awareness is very much about your thinking and your feeling about how you sense the world **you are** how you into it. What is going on and very much helps to develop your understanding of what goes on in the classroom? And how teaching and learning affects the students.

Reflection:

Another skill underpinning reflexivity is of **course reflection**. This is very much a process of learning and evaluating what is going on around us. This can be day to day can be weak too weak. It can even be career-long.

Critical Thinking:

And of course, the third component underpinning reflexivity is **critical thinking**. Critical thinking has its roots in critical theory. And is very much about the teacher identifying and challenging their assumptions about learning. Challenging the importance of the context in which teaching takes place and thinking through how **they are** teaching impacts on learning and the alternatives which are open to them in terms of how they approach their work in the classroom.

So, we can put these **three** things together and see reflexivity as being made up of self-awareness. This understanding of self-reflection is **an** ongoing activity career-long potentially. And critical thinking is very much based in an understanding that critical practice is not superficial at all.

Types of Reflexivity:

So, **where** reflexivity becomes more of introspection. We can look at reflexivity from some very specific perspectives.

- One perspective is in terms of **hyper reflexivity**. This is about the teacher deconstructing what they're doing deconstructing their teaching in very fine detail.
- **Hypo reflexivity** we can also think of system **systematic reflexivity** **which** is looking at the assumptions and beliefs that underpin the teaching practice.
- And we can also think of **epistemic reflexivity**. **This** is exploring the deeper beliefs about teaching that the teacher holds and where these three aspects of reflexivity come together then we get very deep thinking about what is going on. Therefore good practice from extensive thorough critical reflection. **Consequently**, we have to see that reflexivity is not a one-time thing, **And it is** very much based **on** day-to-day experience and practice.

Topics for Critical Reflection:

Pedagogy and how you know the teacher connects with practices in the classroom needs to be think thought about in terms of the topics which directly being considered. This leads to a number of crest questions, **How are practices and the physical environment acknowledge**. And what is

important that **we are** thinking about the autonomy that **we are** giving children the sense of empowerment the children get from being in the classroom.

We have to consider also the wider picture because there's also the empowerment that we're giving whole families **about** to what we do as teachers. **So**, we go beyond questions like well.

- Who is not listening or paying attention in the classroom?
- And why and **we are** thinking about what strategies might change that and perhaps even the physical environment.

So there are **several** questions that we need to consider and I've listed some I'd like to share with you. The questions relate to culture and communities, and the lives of the families, as well as the individual practitioner and so, they are **embedded** into certain sections. **There** are some questions I put down.

Modelling reflection with students:

Let's just go through them the first one,

- 1- **Was this activity successful while why not?** That's a very basic question. It's a good starting point for the teacher.
- 2- **The second question if we do this again what can I do differently to help you learn more?** This praises the teacher's reflection from the student's perspective rather than their perspective.
- 3- **Did this activity help you learn more than others we have done and Why?** **This** gets you to understand the student's perspective on their experience of the methodologies that you use.
- 4- **Are the relationships that I have with my students helping or hindering their ability to learn?** This will get you to think about your day-to-day interactions with students. How you're talking to them and how they're talking to you.
- 5- **Could the problems I have in my classroom be solved by pre-teaching my expectations or developing rules and procedures to deal with these issues?** This will get you to think

about the problems the issues that you may be experiencing from a very pragmatic point of view.

6- **Was my demeanor and attitude towards the class today effective for student learning?**

This gets you to think about how students may perceive your emotional state.

7- **Am I excited to go to work today?** Students will know if you're not happy in the classroom. And therefore what you need to do is be sure that you are enjoying what you're doing.

8- **Are my students excited to come to my class today?** This is connected to your attitude and whether **you are** excited about being in the classroom.

9- **What choices have I given my students lately?** Remember students are there to learn but **they also have the** potential to be empowered by what is taking place.

10- **Can I explain at least SOMETHING about each of my students' personal lives?** This gives you a focus on the individual student or maybe their interests and needs as learners.

11- **Does my assessment accurately reflect student learning?** It's all too easy to use checklists and to put ticks in boxes without really thinking **about** how much the student has **learned**.

12- **Do my assessments reflect learning, or merely task completion or even memorization skills?** This sort of question will get you thinking about the higher-order learning of the students.

13- **Why did I REALLY choose this particular lesson to cover a certain objective?** There could have been many others this will get you to think about the different approaches that you can take.

14- **What evidence do I have my students are learning?** There can be more evidence than just assessments there could be feedback or your observation of what teachers that students are doing.

15- **What new strategies have I tried likely that might benefit a student I am struggling with?** This is a good question because it will get you thinking about expanding your skills and trying out new techniques.

- 16- **In what ways am I challenging students who are clearly being successful in my classroom?** Very often teachers tend to teach to the middle and the people at the extreme end the lower end and the top end get missed. Those at the top end get missed most of all what are you doing to challenge them.
- 17- **What do I do when students are not learning in my classroom?** How do you react what do you do?
- 18- **Which students benefited from a particular activity?** It's worth knowing. What worked well and why it work well?
- 19- **Which students did not benefit from an activity?** Of course, if something goes horribly wrong this may be an opportunity to change your approach or it may be just understanding that activity didn't match with those particular students.
- 20- **In what areas can I still improve?** This is a good question to think about in terms of your reflection. Because it will take you into that cycle of constant renewal.
- 21- **What's stopping me from improving in these areas?** What are the barriers because there will always be things which will get in the way?
- 22- **In what ways can I support my colleagues in my student's learning?** This places you as a teacher in a community of practice where there is lots of support.
- 23- **Do my actions of the teacher show my belief that all students can learn at a high level?** A part of the student's learning is dependent on your approach to teaching and your expectations that they can achieve much higher than they think they can.
- 24- **Do my actions as a teachers show that I take pride in my work?** Remember teaching is a profession and as a teacher, you are a professional. Are you a professional?
- 25- **Are the relationships I have with my colleagues conducive to creating a collaborative culture focused on learning?** with collaboration comes a best practice.
- 26- **Are the relationships I have with students' parents conducive to learning?** Remember teaching is not just about kids in the classroom but it's also about whole families.

- 27- **What new ideas have I tried in my classroom lately to keep myself energized about teaching?** Don't stick to old patterns. If you're using this question then it is challenging for you to extend your repertoire.
- 28- **What have I done lately to relieve stress and to focus on my health, to ensure I remain an effective teacher?** Teaching is a very stressful profession. And what you must do is always check that you are in tip-top condition to be able to teach in the classroom.
- 29- **And finally what things am I currently doing that you could you know reasonably.** Do with more priority.
- 30- **How much time have spent with friends and family in the last two weeks?** Health is important and you must look after your health as a reflective practitioner. This is how you will stay in the profession for a long time.

Learning from Reflection:

- So reflection is a learning tool.
- Reflection is the process of consciously thinking through what is happening. This helps us again an understanding and to make meaning from what is going on in our lives.
- Therefore it's associated with learning that's occurred through experience and activity. How do we internalize this in terms of our roles as professionals.
- Reflection is a means to assist the teacher to think and to explore, to go beyond what is gone in the classroom. To understand thoughts and feelings and to work through the experience to get a new understanding as well as self-awareness of how things are happening and why things are happening in the teaching-learning situation.
- As a tool, it's there for active consideration and learning from all the actions that are taking place. These thoughts and actions come together to help us develop meaning through reflective thinking.
- Perhaps the most important aspect therefore of engaging in reflection is the on-going personal and professional learning that needs to happen. And that you can demonstrate that

to others this is all about achieving a good standard of proficiency as a teacher and a good standard of professionalism.

- This process of reflection is very often undertaken through reflective writing. You'll undoubtedly have heard of reflective journals. Reflective writing can lead to a greater understanding and a better knowledge about why things are happening. And it should also challenge your preconceptions about what is going on and the theories as they are applied in the classroom. When you reflect on a situation you not only see more of it but you start to see it differently.
- This different way of viewing things is about situating reflecting in a process where the action is the final stage of reflection.
- So what do you know about and how confident are you about certain aspects of the reflective practice process and learning from reflection. Well for example,
 - ✓ What do you know about the **roles and responsibilities of the teacher** this is worth reflecting on?
 - ✓ What do you know about **learning styles**?
 - ✓ What do you know about **planning a course** or planning a lesson?
 - ✓ What do you know about **and how people learn**, how children learn?
 - ✓ What do you know about **lifelong learning**? This is something which is talked about a lot now.
 - ✓ What do you know about **communication skills**? This is the practical aspect of the teacher in the classroom.
 - ✓ What **do you** know about **presentation skills and demonstration skills**? What's the difference between these things?
 - ✓ How confident are you with **questioning and explaining** things to students?
 - ✓ How confident are you using **different teaching methods** to teach subjects?
 - ✓ **How** comfortable are you **in** using **ICT** information and communications technology in the classroom?

- ✓ What do you know and how confident are you are with **assessing learning** and also the process of **reflection** itself and course your own **subject knowledge and the skills** used to teach that subject knowledge?

So, learning from reflection as we can see from those questions is very holistic. It covers a whole range, a whole spectrum of professional aspects. Beginning to develop a reflective behavior is about becoming a professional at least now you should start with your own cultural and historic background. As a beginning for being a best practice reflective practitioner.

The concept of **cognition** is actually quite narrow gives a narrow focus to what we're talking about here. But reflective practice enables the individual to think beyond that and to create more knowledge based on their experience of actually what's going on, **and** what they're experiencing day-to-day in the classroom.

This means that reflective practice cannot be an isolated activity it is something that has to be cooperatively generated. And therefore words like collaboration become interesting for us. As reflective practitioners collaboration helps us extend what we're doing beyond the individual learner. Facilitate a contract the relationship into something which is more interdependent between those involved the actors the students and the teacher in the classroom.

Reflection-on-Action:

- Teachers consciously review, **describe, analyze, and evaluate** in reflection on practice. It's a process of higher-order thinking, analysis, and evaluation as; well as being ongoing. It provides insight for the teacher in terms of future practice.
- Teachers in reflecting on practice, will think about their experiences **and responses as they occur**. But most importantly will consider their responses to those experiences as well.

Reflecting “in” and “on” action:

- In both types of reflection. Reflection in practice and reflection on practice. The teacher focuses on the connection between their feelings. How **they are** responding to a situation

and the theory that they understand about what is happening, what **they have** been taught about what is happening.

- Teachers are seeking new understanding in this light of; what their experience means for that theory and what that theory means for their experience.

The Core of Reflection:

- For Schon, reflection in practice was the core component of what he called professional artistry. He saw teaching as very much art, which was different from positivistic views of teaching, **where** teaching was very much a science, very rational and technical.
- **So**, in Schon's view, this technical rationality really sorts of failed to pick up the idea for teachers in terms of teaching as a very rigorous activity as opposed to what it was doing in terms of children's learning. Schon explains this by saying that teachers have to do more **than** just follow set procedures that **they are** told is the best way to approach. But to think about drawing on the experience, and drawing on practice and making experiences practical. **So**, they start to grow their theories in terms of; what **they are** doing in the classroom.
- And in this way, the core of reflection is very much intuitive. And is quite creative for teachers. Both reflections on action and reflection in action means for teachers to revise, modify, and ultimately refine themselves as professionals **or experts**.

Levels of Reflecting

There are different levels of reflecting. Grushka, Hinde-McLeod, **and Reynolds (2005)** identify **reflection on action** and **reflection in action** but, they link this to **reflection for action**. What this does is offer both technical and practical considerations of critical questions for the teacher to engage with. Under reflection for action, they say that teachers need to consider their resources as well. For example, how long will the lesson take? **These** are sort of very practical technical issues. They also talked about how to make the **resources** relevant in different learning situations **or styles**. Again, a very practical perspective on reflection. In this sense, teachers are always questioning,

why are they teaching in a particular way. Why are they using certain resources? Why do they believe, and what do they believe?

- As; Zeichner and Liston (1996) go on to differentiate between five different levels of reflection which take place during the process of teaching. For example,
 - ✓ They talk about rapid reflection.
 - ✓ This is very immediate.
 - ✓ It is ongoing
 - ✓ It is an automatic reaction to what they're experiencing with the children in the classroom

Repair:

They also talk about a level of reflection as repair. This is where the teacher is making decisions to alter what they are currently experiencing to try and change. Maybe the way children are behaving or to change activity. So that students can learn more or to even slow students down so that they are thinking more about learning.

Review: Another level of reflection; they talk about is review. This is where the teacher is thinking about and discussing or writing about some element or some key components of their teaching.

Research:

Another level of reflection is research. This is where the teacher is engaging with theory and with the dialogue in texts thinking over time about collecting data and understanding whole pictures of teaching and learning.

Re-theorising and Reformulating:

Another level of reflection that they talked about is re-theorising and reformulating. This is where the teacher is critically examining their understanding and their approaches and trying to understand how these links back to what they have been taught to link back to the theories which have been explored for years and years in the text.

Guided Reflection

Let's look at guided reflection which is a very specific form of reflective practice. John developed this concept in 1994.

- Maughan and Webb (2014) consider several key questions that can help guide reflection for example,
- They say that teachers can ask, what is most important or interesting, useful, and relevant about what they did or something that they used a resource that they used?
- They sort of asked the question, how can it be explained with theory or just through experience?
- They offer the question, how is it similar to and different from other issues, events, or experiences that the teacher had? To make connections between their teachings.
- The teacher can ask, what have I learned from this? And what does this mean for my future teaching? This is where the teacher is engaging as a learner in the teaching process.

Biggs and Tang (2007) also offer some questions for reflection they ask questions like,

- What was the problem again, a very practical technical type of question to ask what went wrong? What's the evidence of that problem?
- What were the causes of that particular problem?
- How did the teacher deal with the problem? It is a very practical consideration.
- They asked a question how did the solution to the problem relate to the theory that the teacher understands? This is helping the teacher make connections between theory and practice.

Guided Reflection and Bloom's Taxonomy:

Perhaps, the most significant form of guided reflection would be to use Bloom's taxonomy. And Bloom's taxonomy provides a framework where the teacher can ask questions of themselves and their teaching from lower-order through to higher-order thinking.

First, let's try to reflect on Bloom's taxonomy about a taxonomy of reflection. In terms of **lower ordering order thinking**, we can think of the teacher just remembering to describe **what I do**. And this goes through to understanding. What was important about what I did? Beyond this, we move into **middle-order thinking** applying and analyzing. **Where** could I use this again this particular technique? **How** can it be applied elsewhere and analyzed? **Did** I see any patterns; which I need to be aware of or any considerations? And then through to **higher-order thinking** evaluating, how well did I do **and** then also creating, how can I use all this information next time? **Or, what** can I do in the future?

A Taxonomy of Reflection:

Blooms Remembering:

Bloom's level of remembering, **Or in** other words, what did I do?

Teacher Reflection:

The teacher can reflect by asking such questions as,

- What was the lesson?
- Did it address all the content?
- Was it completed on time?
- How did students "score" on the assessment?

These are again very practical, **and** easy questions to ask, but they are the important starting point for the reflective practitioner.

Blooms understanding level:

- What was important about what I did? **Did I meet my goals?** Can be reflected on by the teacher in terms of, Can I explain the major components of the lesson? **And** Do I understand how they connect **with the previous and next lessons?**

- Where does this fit into the curriculum? What instructional strategies were used? Did I follow best practices? These questions are linked to the understanding level of Bloom's taxonomy.

Bloom's Application Level:

Bloom's application level which is in relation to questions like,

- When did I do this before?
- Where could I use this again?

Teacher Reflection: can be reflected on by the teacher in terms of questions like

- Did I build on content and processes which I have taught in previous lessons? This would underpin students' learning.
- How does this lesson scaffold the learning for the next lesson? How does it provide a foundation for what is to come?
- How could I adapt the instructional approach to another lesson?
- How could this lesson be modified for different learners? Again, this is a level of reflection which is in the middle of Bloom's taxonomy.

Blooms Analysis Level: This is now getting into higher-order thinking. Do I see any patterns or relationships in what I did? This in terms of the reflective practitioner is very much focused on questions like:

- What background knowledge and skills do I need to teach a particular lesson effectively?
- Were the instructional strategies right for that lesson? Were they effective?

Teacher Reflection:

- Do I see any patterns recurring in my teaching? Which I need to consider are goods or other destructive patterns?

- And do I see patterns in the style of the approach that I have? Do students react to me when I teach in a particular way?
- Do I need to consider how that is affecting their learning? What were the results of the approach I used? Was it effective? Or could I have eliminated or recognized steps?

Bloom's Evaluation Level:

- How well did I do?
- Do what work?
- What do I need to improve?

Teacher Reflection:

These get to be translated in terms of teacher reflection to questions like,

- Did some aspects of what I did need improvement? Could they be better?
- Where are the needs of the individual learners met in the lesson?
- What levels of mastery of good quality learning took place for the students?
- What have I learned about my strengths, weaknesses, and my areas in need of improvement? That again is a good level of analysis. And Am I progressing as a teacher? , Am I developing myself?

Teacher reflection: can further take on aspects around the incorporation of lessons for the future.

- What can you do in the future to?
- What you have learned from this?
- What changes would you make?
- And what would you keep the same?
- How can you best use your strengths in the future?
- And what steps should you take in terms of using resources or applying new sources to challenge students' learning?

Teacher reflection is the training that would help me to meet my future intentions. The teacher needs to look beyond what is happening in the classroom. In terms of, how to develop themselves as professionals, they need to think about their role development over time.

Topic 022: Action Research

Action Research

When you are thinking about reflective practice, it very much resembles action research. Action research involves systematically changing your teaching about problems or issues you are experiencing in the classroom. Therefore, it is about taking responsibility for change. It is important to understand that action research is targeted at the teacher and it is not about changing anything else.

Educational Action Research as a Critical Social Science – Carr and Kemmis:

This is put very succinctly by Carr and Kemmis who say that Action research is a form of research carried out by practitioners in their practices. In this way action research is participatory. It is a very democratic form of educational research for educational improvement. This is because action research must involve the learners in the classroom, they are part of the change action that the teacher is taking.

Action research emphasizes teaching. It emphasizes practice is committed to acting action and committed to changing practice. It is also a way to express a commitment to improvement for the teacher. It is about not standing still, and it's about trying to look at the theory and make connections with practice. In this way, action research is seen as a very practical form of inquiry and it's characterized by self-reflecting through cycles that spiral continually for the teacher to plan do review (planning, acting, observing, and reflecting).

In this way, it involves a number of critiques

- One is **reflexive critique**. This is about questioning things, questioning arguments, and trying to find new possibilities for **new** actions.
- There is also the idea of **dialectic critique**. This is very much about discussing actions and discussing theory from different perspectives that the teacher is aware of.
- And what we have to consider also that action research is very much dependent on a collaborative resource. It is very much about people working together whether these are teachers or whether these are teachers and students. The collaboration is there and it is also true to say that there is an awareness of risk in terms of action things may go wrong and that's not necessarily a bad thing as long as the teacher can learn from what goes wrong.
- Action research is very much a pluralistic approach to critical thinking. It's very much about discussing the questions and identifying the **problems**. **But** it is very much about reporting them from very different perspectives.

Donald Schon states **that** action research is very much a relationship between understanding and change action.

- He believes that the practitioner has an interest in changing the situation, in changing the status quo. And this is better than leaving things away they were. It also there develops a better understanding of the situation.
- He sees action research as his idea of reflecting **on the** action. It's very much about learning while doing and learning from doing. **(Action research is something that any practitioner can do as a part of his/her everyday practice, and it does not just imply research).**

Some Suggestions for Thinking about Practice:

Bryant's researcher states that there are many themes and aspects to practice that can be questioned. There is certainly a way of looking at theories what he talks about is that,

- The teacher can look **at** action research from the point view **of** a setting, where the acting activities are taking place. What is the classroom environment actually like?

- He also talks about the play you know there was looking at ideas in terms of a game. Where are the rules of the game, the regulations which are a play in the classroom, and the rules that govern the teacher?
- He also talked about the reflection as thinking in terms of script. So in other words, texts that identified but don't you know completely explain performance. This is an approach where the teacher can really monitor using a script whether that is an internal dialogue or something written down in order to uncover a problem and resolve it.

Finally, in communication, there can be different interpretations of meanings and so, the way the teacher communicates these interpretations becomes a very unique perspective.

The Critical Friend

The Critical Friend:

- The critical friend is a very useful technique to think of as a reflective practitioner. The critical friend is a process that focuses on a professional relationship and encourages reflective practice with the colleague. It's very much built on cooperation and collaboration.
- It also addresses a number of situations in which teachers find themselves. For example, working in isolation, working as independent units within the school. A critical friend can actually help address this.
- It can also address the idea that the teacher is once they're in the classroom considered to be the person that is certified to know everything. Therefore, the critical friend is a way of continually professionally learning and not just sticking with what is been taught previously.

Background:

The background to this comes from the Annenberg Institute for school reform at Brown University in the USA. It was the first to develop the critical friend model for collegial dialogue. Interestingly

it is currently used by an estimated 35,000 principals and professors across the world in about 1,500 schools and universities. So, it is a very popular approach.

Key Contributions from the Critical Friend:

- The critical friend is about providing a balance between a friendship something which is very informal and supportive with something which is very critical and analytical.
- The critical friend is somebody who asks why things have been done in a particular way. They are helping the teacher seek to gain a sense of the context in which things are working. And the way things are working and also to challenge assumptions and perceptions.
- The critical friend, therefore, is helping the teacher think creatively and to encourage others in the environment to do so.
- As a friend, a colleague in the school becomes very significant. Therefore, the critical friend is very friendly and honest but also critical and analytical. As a colleague, they can share professional reflections. And help teachers look beyond the classroom and look more to the educational horizon.
- The critical friend is using their skills in a way to ask very superficially even “naive” questions of the teacher. This is a way to promote reflection and to get the teacher to self-evaluate. What is going on simple questions like, why did you do that? And What happened? The critical friend is also there to provide advice and some guidance and hopefully some very practical suggestions for the teacher.

Critical Friend Ethos:

- So, in a way, it is like keeping or maintaining a close distance.
- The critical friend is supportive is very friendly just like friends are. But they hold on to the role of an evaluator.
- As critical friends, they are very knowledgeable about teaching and they understand the subjects that teachers are working with. But they are often very supportive as well.

- And what the critical friend does is identify the challenges that the teacher has. And they have no responsibility for overcoming those challenges instead, they support objective and reflective debate to find solutions.
- Critical friends of course within this role need to make sure that they maintain confidentiality with the teacher whilst being frank. At the same time sensitive and independent the teachers experience the problems that they are facing.

What the Critical Friends actually DOES NOT DO?

So, what do friends do and what do they not do? Well, the critical friend does not make decisions for the teacher. And reporting formally and informally to others is something that they do not tend to do unless it is with the agreement of the teacher that they are talking to.

Limitations of the Critical Friend Role:

The limitations of the Critical friend role are as follows:

- The primary limitation is in terms of their role of advisor and supporter, and provider. Because they have to provide a constructive critique it is not just about friendship.
- Critical friends are not responsible for and they of course can never contribute to the quality of teaching. That's why the teacher does it themselves.
- In particular, critical friends have to take more roles in terms of becoming a negotiator on behalf of the teacher with the school administration. If there are very significant problems to resolve. Critical friends of course in this way may add value through engagement with a number of teachers, and so sharing knowledge around the community.

Participatory Methods for Reflection

I am putting aside time in school for focused discussion maybe with a critical friend or with a group is essential in order to ensure that reflection becomes a regular practice for the teacher.

It is important to have a facilitator in this regard somebody who uses critical questions to help guide thinking and stimulate discussion. Also, to ensure that there is a deep reflection or thinking about the issues. Maybe, even to record the discussions for the teacher to reflect on at a later date.

What Can Teachers Reflect with?

Such conversations and debates can occur with:

- **Colleagues**
- They can occur with **other professionals in the teaching environment**. For example School counselors
- It could be with **other professionals working in different educational environments**.
- It could even be of course with **families, parents, and children**.

Method:

Essentially what the teacher has to consider in this regard is the method there needs to be:

- **Careful description**
- **Thoughtful listening**
- **Questioning feedback**

And for me, the two most important components here are **thoughtful listening and questioning feedback**. It is very difficult sometimes to listen to critique from others and that does take a lot of courage from the teacher.

Types of Questioning Feedback:

There are types of questioning feedback and these can be considered in terms of hot feedback warm feedback and cool feedback.

- **Warm feedback** consists of feedback that is very supportive and appreciative of what the teacher is saying.
- **Cool feedback** offers very different perspectives and questions about teaching and raises questions about, Why the teacher took particular actions?
- **Hard feedback** challenges and extends thinking. It raises the level of concern to a very critical and analytical point of view.

The Participatory Process:

- So, as a participatory process reflective practice allows colleagues and others to share ideas, and issues and to seek suggestions and practical solutions between the group members in order to positively overcome the issues and problems. This is best of all done in small groups not so well in large groups. Because there is too much interaction with smaller groups. Also, it is easier for individual teachers to feel more comfortable sharing issues and problems.
- Hard feedback challenges and extends thinking and raises concerns in this sort of small group environment.
- As a participatory process then the teacher in a small group can:
 - Share the issue
 - Also, provide the context in which the problem or the issue arose.
 - And themselves frame key questions which the group can discuss and explore in order to try and find a resolution to that particular issue or problem.
- So, what really relies on our questions which are very probing questions? Questions that help to clarify understanding. The group can ask more questions to learn about the issue that the teacher is raising.
- And this leads to then hopefully an analytical evaluative discussion. The group discusses the issue using all types of feedback both warm and cool feedback. And the group finds a way to address possible suggestions and solutions to the issue using the hard feedback

approach. So, using cool, warm, and hard feedback makes truly analytical and evaluative thinking about the issues in question.

Foundation Models 1: John Dewey

So, let's look at some models of reflective practice that will introduce in this lecture. We will look at the models in more detail. First of all, we can consider Dewey model of reflective practice.

What is Reflective Practice?

- He saw **the** reflective practice **as** something which is very dynamic, quite spontaneous and requires a lot of honesty and **deep and critical thinking** from the teacher.
- He said that reflective practice actually should **lead to action**. It is not something just to think about it is an important thing to lead for the teacher to do.

Routine Action vs Reflective Action:

Dewey model is based on a fundamental difference between the sort of traditional approach to teaching which is about instruction and imitation **(quoted in Furlong and Maynard:31)**. And really said that the teacher in terms of reflective action needs to confront perhaps some **pre-assumptions, prejudices, and not judgments proper that rest upon a survey of evidence** about teaching. **(Dewey 1910: 4-5), and Reflective action**

- Reflective action, therefore, is very spontaneous or should be persistent, very active, and needs to be carefully engaged with by the teacher. **(1910:6)**
- For Dewey, it is in problem-solving that we find. "the steadying and guiding factor in the entire process of reflection" and without which "the course of suggestions flows on at random". **(1910:11)** Therefore, becomes a focus for the teacher in terms of how they go about teaching it affects teaching fundamentally.
- He saw reflective action as more than just thought it that's got a wider dimension to it.

- And indeed, he actually said that reflective practice can have emotional aspects for the teacher as well. While we cannot learn or be taught to think, we do have to learn to think well, especially acquire the general habit of reflection. (Dewey 1933)
- And what this actually means is that the process of reflective practice is a rational, purposeful action, and a very complex thing to do. Not straightforward and should not be entered into lightly or superficially by the teacher.
- It includes a good level of rationality or in other words a lot of clarity of thought by the teacher about the reflection process as well as the teaching that they are doing. (Dewey 1933)
- And interestingly, from a modern educational perspective, this does actually draw links not just with the cognitive aspect of teaching. But the way the teacher is responding to what is happening in the classroom. The emotions which are being created in their interactions with students.

Dewey's 5 Stage Model:

So, let's look at the model the five-stage model.

- First of all, he says that what the teacher does is to **identify a problem** something which is vexing, perplexing to them, and needs to be fixed.
- From then the next step is to observe and **refine the identified problem** to clarify the understanding of the model.
- Beyond that the third step of the 5-stage model is for the teacher to develop a hypothesis to explain the problem, or to try and understand it from a technical point of view. Also, as its origins and possible solutions.
- And this must lead to deeper scrutiny of that problem.
- **Test the hypothesis or** a deeper understanding before leading into some sort of action. Dewey, J. (1938) **Logic: The Theory of Inquiry**. Troy, MN: Rinehart & Winston

So, we see that placed a great deal of emphasis on this reflective thought process which is why he was saying that it is not natural to do it is something **that** has to be learned and has to be based **on** experience. He believed that reflective thought began with our needs and **having an experience that raised some difficulties or dilemmas (the “felt difficulty”).**

From this experience, we actually start to ask questions. For example

- In terms of **the problem**, the problem stages the question is well, what is going on here? What is the problem, what is happening?
- And in terms of **conceptualizing the problem** we might say well, how **do** I fix this? What might I do to put it right? And that leads to experimentation about solving the problem **or acting by trying out a possible solution.**
- Finally, in terms of solving the problem, we are **exploring the avenues** that we can take to do that. **And how it might be further adapted in the future.**
- So, within professional development, **our interest is not so much in learning from a single experience, or learning in the short-term, but more in the long-term.** Because it fits into the cycle of teaching and learning.
- It is **a** very developmental process and **enables well-grounded professional knowledge and skills** in the experience of the teacher, not focused so much on theoretical applications.
- And interestingly, what Dewey said is that although it is a cycle it should really be viewed as a spiral one cycle leading into another cycle being **correctly** connected by experience.
- This spiral actually leads to the reflective aspect of the process. We can see this in a model. We can see here, that the cycle of reflection and thinking about action and then experimenting **with** the action itself through experience goes into another cycle thus creating a spiral.

Foundation Models 2: Donald Schon

Foundation Models 2: Donald Schon (1983)

Reflecting “in” and “on” Practice: Schon’s model of reflective practice is based on reflecting in practice and reflecting on practice. These are terms we have come across before they are very popular terms and approaches. We can represent these ideas of reflecting in practice and reflecting on practice in a very simple diagram.

- We can see here; that the starting point is the actual teaching and the learning that is going on in the classroom. And it is the thinking at the moment and the reflection in practice that occurs at this point.
- After the session is when the teacher reflects on practice. This is when the teacher has more time to think about what happened and to think back on what happened.

And this whole process of reflecting in practice and reflecting on practice then leads to preparing for the next teaching session with the students planning to get the resources ready. Then, this leads into the next teaching which again will have a reflection in practice. So, again you can see this repeated cycle being engaged by the reflective practitioner.

Development of Schon’s Model:

- Schon made the distinction in this model between **technical rationality**. Or in other words, the application of learned theory and what he called **tacit knowledge**. Or in other words, experiential learning is gathered from the classroom. This can be more easily understood as the **“theory-practice gap”**.
- He believed that this was in a sense of theory experience gap. Because the theories that the teacher is trying to apply in the classroom may not generate the sort of outcomes that the teacher is expecting. In other words, a different experience. Therefore, the teachers may have theoretical knowledge (technical rationality).
- And what this can actually mean is that. It changes the way they are thinking about teaching. Actually, may not even define the way they actually think about their own practice in the classroom.

- So, in terms of the Schon model what we are actually thinking about is the experiences in the classroom. And how these are combined, and how these add together to generate learned and acquired theory for the teacher.
- Schon saw that the real teaching environment, therefore, was the place where the teacher applied, tested, and evaluated the theory for themselves in terms of what they are actually doing.
- Theory is, therefore, not a rigid thing here. It is something that can change and grow with the teacher's experience. Although, a new teacher would be applying theory quite rigidly as their experience develops. And as they learn more about their own approach to teaching and their own beliefs and values about learning. Then, the theory starts to become a little bit less rigid and they start to apply things differently.
- So, reflection in action and reflection on action allows teachers to begin to continually improve their own practice to change their teaching adapt learn and evolve.
- And what we have to remember is that this does not make theory invalid at all. It simply begins to change the theory that the teacher is acting and its theory is, therefore, a valuable starting point for the teacher.
- And this is what Schon calls **theory in use** rather than just theory.

Schon's Model:

So, in this model, we can see what Schon is talking about and the teacher is attempting to make sense of, and use theories in their teaching putting into action what the books are saying, what the literature is talking about. Therefore, this leads to some sort of real-world experience. So, in other words, what they are learning from what happens in the classroom. So, the teacher moves from knowing that or in other words, knowing about the theory. To know how. Or in other words, knowing how to put the theory into use.

Foundation Models 3: Kolb's Reflective Cycle (1984)

Another very important model of reflective practice is Kolb's reflective cycle where active experimentation leads to a transfer of learning from the current cycle to a new cycle. It is a very popular model you probably already know about it. What Kolb is talking about is a cycle that is based on experience entirely. In terms of this model, there are four main components. And there is a diagram to illustrate this.

The four components are,

- concrete experience
- reflective observation
- abstract conceptualization
- and active experimentation

So, the experiential learning cycle is made up of these four components. And we can look at them in a little bit more detail to try and understand what each of these four components are.

Components of Kolb's Model Applied to RP.

Concrete Experience: So, concrete experience is very often seen as, the starting point when reviewing the cycle. Although because it is a cycle the teacher can enter at any particular point. For example, the reflective practitioner could start with abstract conceptualization which would lead to active experimentation and then concrete experience. But concrete experience gives a very hands-on starting point for the teacher very practical.

Although this model can be entered at any stage, it tends to be the main point of entry. The concrete experience for the teacher is a very convenient and immediate place to start the reflective practice process.

Reflective Observation:

Reflective observation is the first level of reflection. Initially, thinking why did that happen? trying to understand their concrete experience, and trying to rationalize it. It also entails stepping back

and viewing the event or incident from an objective perspective. And It also gives the teacher that point to stop and think, about what you did and why you did it. With this model there is no reflection in action and no reflection on action, there is just observation of the process that took place.

Abstract Conceptualization:

Then this leads to abstract conceptualization. A little bit of a difficult word but in principle very simple to understand. It is trying to understand what has happened in relation to theory or to what the literature is telling us should be taking place. Therefore, it is making a cognitive connection with practical experience. It is looking for connections between interpreting, theories, events, actions, feelings, and all the experiences that happen in the classroom.

Active Experimentation:

And of course, with all of these cycles we see, there has to be some action involved. Otherwise, reflective practice has no point. So, in this cycle, we can see that after abstract conceptualization the teachers move into a process of experimentation. Experimentation suggests more than action. It is about trying something to see if it works it might fail which is why the cycle leads then back into concrete experience. So, the whole process is again more of a spiral rather than just one cycle. The spiral means that the cycle is continuous and never-ending.

Foundation Models 4: Greenway's 3-Stage Model (1995)

Greenaway's 3-Stage Model (1995)

Let's look at Greenway's three-stage model. It is a very popular model and is simply called the "plan-do-review" model.

The starting point I suppose is to talk about DO having an experience in the classroom. This leads to a review of that experience and then planning. The next steps in teaching planning the approach that you will take. We can look at this in terms of a model again very simple.

Benefits of Greenaway's 3-Stage Model:

- One benefit is it is a very simple easy-to-apply model.
- Because it is focused entirely on the action. It is grounded in the teacher's everyday experience and that gives it a lot of value and gives it a lot of meaning for the teacher.
- It is a very simple model it's also very quick to do. It's not at all time-consuming and especially for the beginning teacher. It has a lot of benefits over more complex models. Because it is much easier to understand and apply.
- And what we are looking at with this model. Therefore, is something which can be practiced in schools very quickly and easily.

When to Use

'Plan –Do- Review'

- What you actually need to do with this model is to think of it in very practical terms. It is a model for continuous development of the course.
- It is especially useful for new teachers.
- Or if you are starting with a new class of students. And beginning a new term or improved approach in school.
- It is especially useful also with activities in the classroom which can be quite repetitive; where you are doing things over and over again. Because the plan-do review cycle will help you refine what you are actually doing in those lessons.
- It can be a very useful approach. If you are collecting any data for further analysis in order to verify and prioritize problems or root causes. If you are doing some action research or if you just want to gather more data about your students.
- Most importantly, it is very useful if you are trying to implement some sort of change to your teaching. It is very quick to apply you can monitor the change process very easily.

The "Plan-Do-Review" Procedure

Plan:

So, in the procedure plan what you need to do is recognize an opportunity, and actually make a change or do something in your teaching.

Review:

The reviewing is actually analyzing the results of the action that you are undertaking and identifying what you have learned. And what you need to do then is to plan new improvements, which you will then action in the DO phase of the cycle.

Questions to Ask for “Planning”

So, Some useful questions about what we can look at in terms of each of these three stages in terms of the planning stage. Some questions might be well,

- What data indicates a change is required? Rather than just initiating some sort of change for any reason. There is got to be some good foundation for the change to take place.
- What change is to be made?
- How do you know the planned change is appropriate? What other alternatives are there? And what are you actually going to do? So, there are some fundamental questions that we can look at.
- What sequence of steps is needed to implement this change?
- Who will be responsible for carrying out each step? It might not just be the teacher alone; it could be others involved in the school.
- Who will need to be consulted?
- Who will the change affect? How long will the change take? It is another consideration for the teacher. It might be actually just a space of half a lesson, or one lesson. Or it could be over a period of a week or even a month.
- How long will each step take? Is it one action in the classroom? Or is it a whole approach to a whole lesson?

- How will you know you have completed each step?
- How will you monitor the change's effectiveness and the benefits of the change?
- How will you monitor and track the progress? That's another question is worth considering in the planning phase. Because you have to be prepared for all aspects of the reflective process.
- How will you collect, review and act on the information?
- What will you do about unexpected problems?

Questions to Ask for "Acting":

- In terms of acting, how will you monitor the progress and the effectiveness of what you are doing? How you will collect the data? How will you know? and the results of what you are doing.
- And of course, How will you save the recording of observations and results? (Planned and unexpected) in comparison with the original data or the goals, measures, and objectives. Will you use video? Would you just make notes for yourself in a journal? Could you get another teacher to watch you observe and give you some feedback?
- And part of acting is, what did you achieve? And what did you learn? from the whole process of going through this cycle. So, checking goes on continually throughout the planning phase the doing phase, and the review stage. You are continuously checking, you are not just engaging in either doing or planning.

Questions to Ask for "Reviewing":

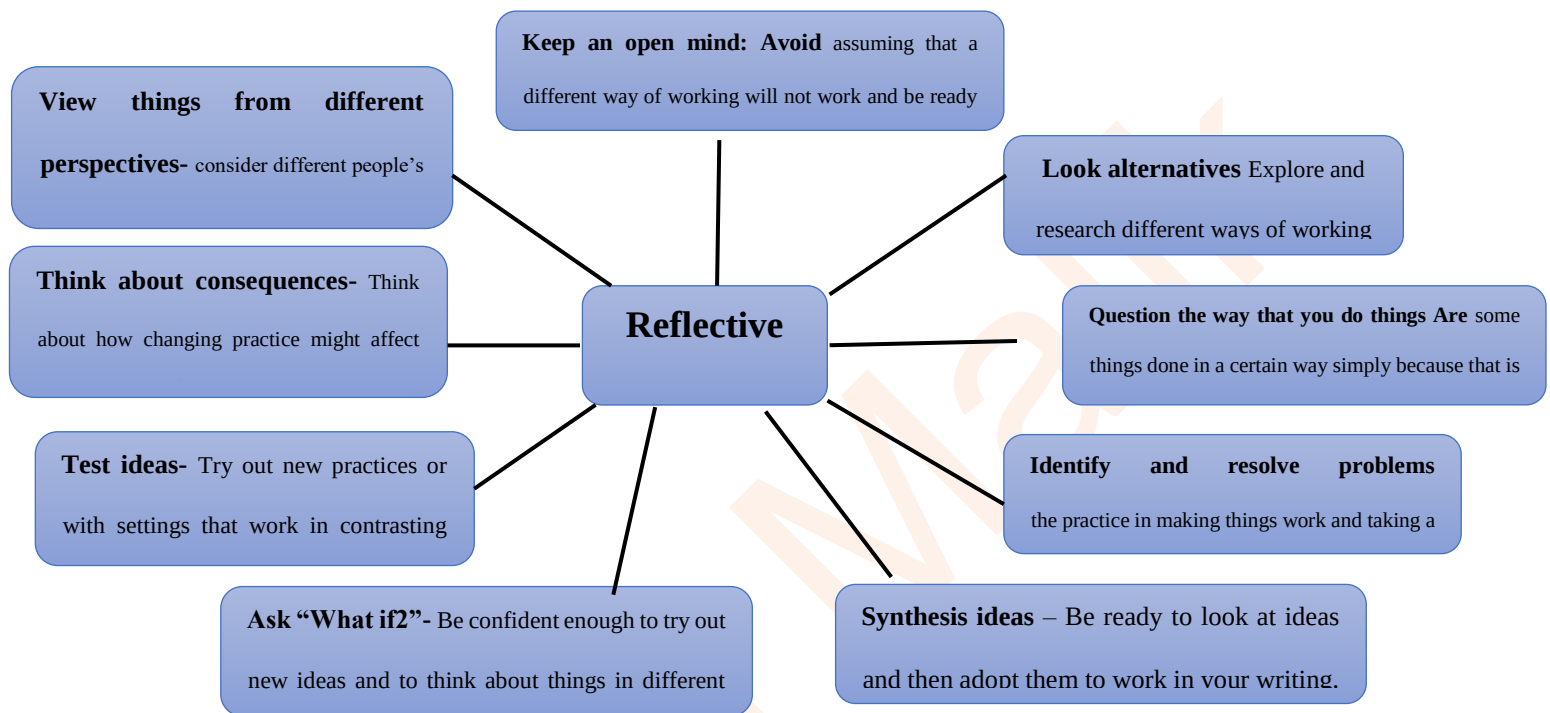
- So, what will you do with that learning is a significant question to ask during the review stage? How are you going to use the information? Or what will you do with the learning-adopt it abandon it, run it through another cycle again to test?
- What did the information that you collected? Impact in terms of change, what was the effectiveness of what you actually did?
- What can be done to further improve what you are doing in the classroom?

- How can you refine your own teaching in terms of what subject you are teaching?
- What lessons have you learned that can be applied elsewhere?
- How can these lessons be communicated to others? These are all aspects of this very simple process. There are probably many more questions that you can consider for yourself in relation to plan do and review.

Topic 029: Foundation Models 5: Roth (1989)

Roth's (1989) summaries of Reflective Practice processes as follows:

- In Roth's model of reflective practice. He very much sees it as a process of questioning asking what, why, and how as a teacher you are doing something. And also asking what, why, and how other teachers are doing what they do.
- **Seeking alternatives**
- And in this light is very much about the teacher keeping an open mind about the teaching.
- Also comparing themselves with other teachers, comparing and contrasting the approaches that different teachers have.
- In this way, it is very much about seeking a framework rather than using a framework. Although, this model is a theoretical basis. It is very much a process of action and reaction that's taking place.
- As a reflective process, it is very much then about asking questions like well, what if I did this thing in my teaching? What if that happened? And then seeking ideas looking for ideas, seeking opinions, and viewpoints from others in order to find answers to those questions.
- Therefore, it is very much using a prescriptive model of thinking, considering the consequences of actions, hypothesizing, and also synthesizing or testing the information with the theory. Also seeking, identifying, analyzing, and resolving problems

Diagrammatic Representation of Roth's Reflective Process:

We can see these questions or ideas are put into a very simple model. The **reflective practitioner at the center**. If you like conducting reflective analysis. And all these aspects surrounding the teacher everything from keeping an open mind, looking for alternatives, and comparing and contrasting. And the different questions that all these aspects of reflection bring to light.

So, Roth's foundation model describes 24 procedures for the reflective practitioner. The procedure ranges from those that correspond to cognitive models of reflection and teaching techniques to an embracing like "adapt and adjust to instability and change." And what this really means is the model is quite complex but very thorough. Because it is based on questions from the actual teaching is quite simple for the teacher to use.

- It is very much about hypothesis and synthesis, **test**, and quite interestingly.
- Griffith and Tans (1992) provide a very comprehensive way of categorizing reflective processes that consider both time frames and purpose. This means that it is always about higher-order thinking rather than lower-order thinking. It is not about identifying and recalling. It is very much about deep analysis or critical thinking.
- Then, ideally a process not only of reflection but, also of research and thinking about teaching, review, **research**, and even repair putting things right which have gone wrong.
- These labels imply how reflection occurs in the individual teacher's mind, but do not occur in isolation. They occur within a context with other teachers.
- So, you can see that this list is reminiscent it reminds us of something. It very much reminds us of the upper levels of Bloom's taxonomy. The levels of evaluating, analyzing, and also creating. And while it does seem to take into account the notion of different time frames, it does not locate the processes in the day-to-day work. So, the model fits into Bloom's taxonomy very neatly.
- And this constitutes professional practice. According to Roth, the labels that he applies within the reflective practice process, all of these questions related to these labels are about the teacher's thinking, acting, and believing themselves to be a professional.

Topic 030: Foundation Models 7: Peters (1991)

Peter (1991) described the reflective practice as a four-step model which he called DATA. DATA stands for:

- describe
- Analyze
- Theorise
- and act

Describe:

It is the first step of reflective practice. It needs to be detailed, needs to be thorough. As we have seen before the word honest comes up again it needs to be honest about what is actually happening.

The description phase, involves paying very close attention to what happened? And what is happening in the classroom. Who was involved? And does not remove that from the feelings, thoughts, or emotions but the teacher is having the way they are reacting and thinking about what has happened.

Analyse:

This leads to the analysis step of the model. This includes the assumptions and the beliefs that the teacher has about their teaching, learning, and what happened. It aims to link at this point or begins to link the theory with the practice. It is about the approaches and the underlying beliefs, rules, and assumptions that the teacher has tried to understand not just what happened but the thinking behind it.

So, here the teacher is very much exploring their own practice, and why this particular practice operates in a particular way. This is why it is quite a deep critical approach. In the analysis phase analyzing the beliefs, values, and assumptions, the source of the information that the teacher draws on what really is required in order for the teacher to make sense of what's going on in the classroom makes sense of the situation.

Theorise:

And this leads to the theorizing step of the model. This is where their practitioner is trying to understand from a cognitive, a conceptual point of view not only what has happened in the classroom and what is happening in the classroom, but what they understand from a theoretical point of view. And in the third step, practitioners theorise about alternative ways to approach learning. They take a theory developed from the previous step and create a new theory.

This requires teachers to think about new or additional sources of knowledge that they can draw on from the literature that they can go to. And even colleagues in the school that they can talk to

about what is happening. Remember sources of theory and understanding are not just what you learn on a course. But they can be much wider reaching than that.

Here, the teacher is thinking about the practice from different perspectives in order to formulate a new or different understanding and consider a new or different approach. Therefore, not just one perspective in the book but a range of books, and also from other teachers' experiences. So, the theory is very wide, very broad and the teacher is looking for as many different viewpoints as possible.

Act:

And the final step in the process is Act. In this step, the practitioner tests the new theory. As in the Greenway model, this is done, but the teacher is actually going to do it in terms of implementing actions, and theories in the classroom. The success of the process is, therefore, not only reflection but the thought is a very clear part of this model.

In the final stage, the teacher put their new thinking into practice by trying out a new or a different way of doing things in order to enhance and improve their practice. And what this would do as with all the models is lead to another cycle of reflection, acting, thinking, and doing. Teachers can use this model as part of a cycle of ongoing learning and continuous improvement. The goal is to integrate scientific and practical theory, and one must reflect on one's practical theory. Therefore, it is not although it looks quite linear DATA and the teacher can always go back and go through another cycle of the process.

Topic 031: Critical Models 1: Brookfield's Critical Lenses

We look at some foundation models for reflective practice. Now, we need to look at some critical models. Again we will be reviewing this briefly and looking at more detail in a later lecture. Perhaps, the most significant critical model is Stephen Brook's field critical lenses (1995). He said that teachers look through different classes, and different eyeglasses in order to reflect. And he identified these four different lenses as,

1. Our **own view of** autobiography.
2. He also said that we can look at what happens in the classroom from the **student's point of view**
3. And also from our **colleague's point of view**.
4. We can also look at what happens in our teaching from a **theoretical standpoint**. And he very conveniently labeled these. As the
 - **Self-lens** (our own viewpoint)
 - **The student lens** (the viewpoint of the students)
 - **The peer's lens** (in other words, the viewpoint of our colleagues)
 - **The theory lens** (which is why it's called the four lens approach.)

Despite the fact, that teacher's professional experience can often be considered as a little bit invaluable from a theoretical point of view. Stephen Brookfield makes a significant point in saying, "The experience and the autobiographies of the teachers is really the foundation point, and the most important aspect of being a teacher." (1995 p.31)

He says that as teachers we must consider this to be more important sometimes than the actual theories. Teachers also should consider whether the learning experience was inclusive and motivational. And he goes on to say that these four lenses are very important because, The most important thing that the teacher can do in terms of teaching, is trying to get inside the student's heads, what are they thinking about learning, what are they thinking about being in the classroom trying to understand how they are learning. (Brookfield, 1995 p.35)

Reflective Practice:

And therefore he made a number of significant points in terms of these four lenses.

- One point is that he says that, reflective practice and teaching actually become about identifying power and how power can support, and also hinder the educational process.

- He also goes on to say that what the teacher needs to do, therefore, is to challenge certain **untested** assumptions in order to address that power imbalance that may be in the classroom.

Applying the Four Lenses:

And in this light, he came up with a number of very useful frameworks for the reflective practitioner.

- a) One is the **critical incident questionnaire (CIQ)**. It is a series of questions that the teacher can ask themselves in order to, reflect on the teaching in the classroom.
- b) He also came up with the **Critical practice or Audit (CPA)**.
- c) Perhaps, most significantly he also identified how the critical incident questionnaire can be used as a framework for a teacher's journal writing, if they are keeping a reflective journal.

Critical Incident Questions:

So, the critical incident questionnaire has a number of significant questions. For example:

- At what moment in class did you feel most engaged with what was happening? This is a significant question for the teacher to consider.
- At what moment in class were you most **distant or** most removed from what was happening?
- What action that anyone **took** whether as a teacher **or a** student was most helpful or most productive?
- **What action that anyone took did you find most** puzzling or problematizing and **confusing?**
- And also what surprised you most? As a teacher, in terms of what you did in the classroom? What were you not expecting? **(This could be about your own reactions to what went on, something that someone did, or anything else that occurs).**

Critical Practice Audit:

- So, the critical practice audit takes a number of critical incidences say two or three in a week and uses the questionnaire to reflect on those incidences.
- The idea is that as a teacher you write a brief description about what happened in a particular incident, including details of what happened, who was involved, where and when it took place, and what made the incident critical to you. And use those questions to reflect on the incident in detail.
- Part of the audits also to reflect on your own assumptions as a teacher about what happened to think about what were the interactions that generated a particular incident. And how you as a teacher challenged what was happening both with yourself and with the students.
- The critical practice audit is also to think about the information the evidence that you have got as a teacher and to check its accuracy. Are you really seeing what really happened? Or are you looking through your own lens the self-lens that autobiographical lens which is perhaps skewing the reality? If you were not able to check them out at the time, how could you check them in the future? What sources of evidence could you consult? And what sources of evidence could you consult?
- What different perspectives could be taken on the incident? Therefore, as you think about it through the eyes of other people (the student lens, and the colleagues lens, the peer's lens, and also the theory lens) involved. Are there different ways the situation could be seen, Or that your behavior could be interpreted?
- Therefore, what we see through these four lenses is a very critical approach self-critical by the teacher and critical of the processes which are taking place looking not only within but also without the school.

Topic 032: Critical Models 2: John Model for Structured Reflection (2000)

John's model for structured reflection is another critical model that we can consider for the reflective practitioner. It is quite a simple model. It has two phases to it.

The **first phase** is “**looking in on the situation**”. This involves the teacher focusing on particular aspects of themselves, and paying attention to **their thoughts, emotions, and** what was happening in the classroom. So, looking **into** the situation is the first part of this process.

The **second phase** is quite simply “**looking out of the situation**” and trying to describe it so that others might actually understand it from your perspective and even from a theoretical perspective. And as such, it is situated then in **the** five sources of knowledge **that** Johns describes and each of these sources of knowledge have very particular cues that we can use as teachers to guide our thinking and reflection. For example:

Look In-

Looking in he said very important that you need to find a place where you can actually focus on yourself. He saw this as a very solitary activity in the initial stages because you need to concentrate and have space and time for reflection. Pay attention to not only your thoughts but the feelings, the emotions which have been generated by the experience. And then of course you need to record the experiences that you **have** writing them down or maybe even using an audio tape.

Looking – Out:

Looking out then requires **writing that description** within the sort of situational surrounding of the thoughts, and the emotions and then trying to find what is significant in that. This is called looking out because in a way, it is like trying to describe it for somebody else to understand, and in this way you get a clearer understanding **of** yourself.

This particular **description of the looking out process can be focused in the following ways.**

Aesthetic:

And we can actually talk about it from an **aesthetic point of view**. Or in other words, the teacher can reflect on what they were trying to do? What they were trying to achieve in their teaching?

And why did they react the way they did? to say some poor behavior or some extremely good behavior in the classroom.

Personal:

There is also a **personal perspective** to this. Or in other words, the teacher's reaction to the experience in the classroom. A good reaction, a bad reaction. How do they feel? what are the emotions associated with that?

Ethics:

We can also look at it from an **ethical perspective**. Or in other words, the teacher should think well. If I did something was it in the best interests of the students or the best interest of the school? And what were the influencing factors on the teacher in terms of their own reaction?

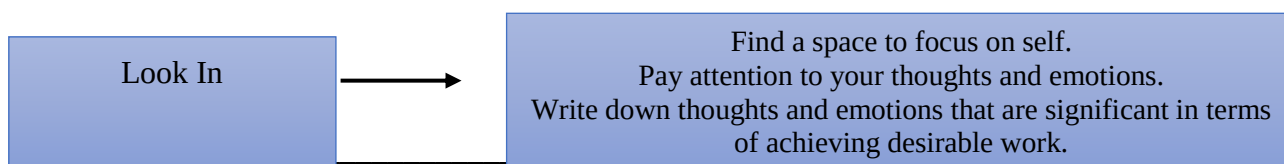
Empirics:

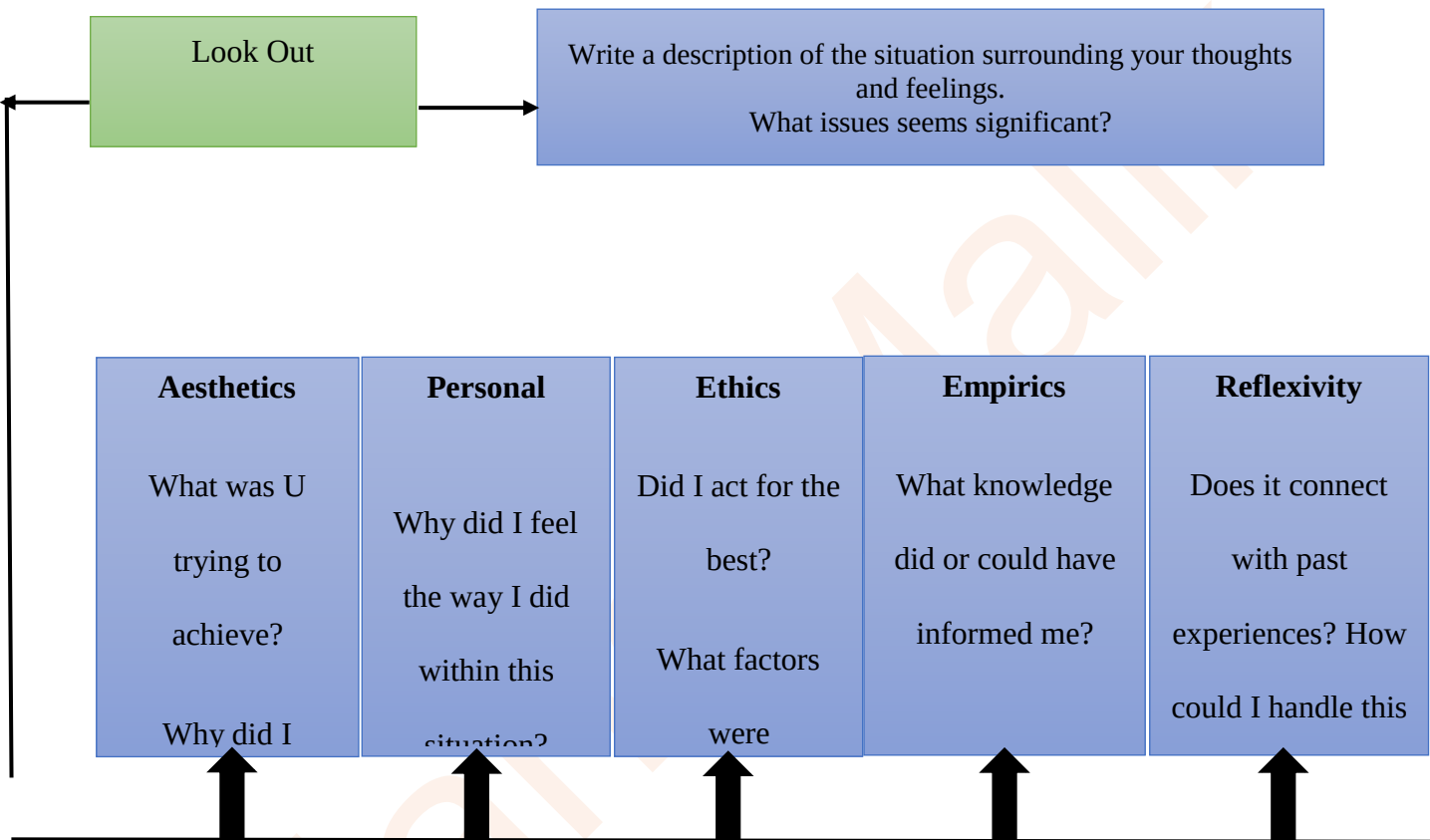
Another perspective identified in this model is the **empiric perspective**. This is very much starting to relate the experience with what the literature says. Or in other words, Is there information or is there some knowledge that could have actually helped me deal with this situation in the classroom?

Reflexivity:

A very significant part of this model, and the particular perspective is about **reflexivity**. This is a term that will come across in future lectures. Reflexivity is about making connections with our understanding. Does it reflect previous experiences that we have in the classroom? Therefore, Can we handle the situation better? And what's very good about these perspectives is that we can represent them in a model which is very easy to understand.

John's Model of Structured Reflection





In one component, we have **looked in** finding the space: describing what happened. That very solitary component but, then **looking out**: describing it for somebody else for our own purposes, and within the looking out we get those different perspectives. From the **aesthetic perspective**, What was the teacher trying to achieve? From a **personal perspective**, how do they feel? The ethics, what factors were influencing them? The empirics the knowledge that they are drawing on, And then **reflecting back** on previous experiences and making cognitive connections.

Topic 033: Critical Model 3: Rolfe, Freshwater and Jasper (2001) – The “What Model”

Another critical model to look at is simply called the “What” model. It is called the “what” model because it uses three very simple questions.

- What?
- So what?
- Now what?

Levels:

Let's see what these questions actually referred to.

- In the **first level**, the individual reflects on the situation in order to describe it. It is very much about **reflecting on the situation**. What happened? What was going on describing? As we have seen in other models.
- The next level, **So What** is reflecting on what that really means in terms of the literature that we have read and our theoretical understanding of the concept that we understand?
- And finally that leads into **Now What**. In other words, their action step of the process of the reflective cycle. At this level, they reflect on the action, what can be done to improve the incident, and the consequences of such action. So, we have, What, So What, and Now What.

Level 1: The Descriptive Level of Reflection:

What

We can very easily think of these in terms of very specific questions and for example, in the first level though what level is? What is the problem? What reason is there for us being stuck at this particular point? And what was my role in the situation? What action did I take?

So, it is always using that keyword what to describe? What actually happened? And what were the consequences for me? What were the consequences for the students? These are just sample

questions. The idea is that the teacher him or herself will use the key **what** in order to frame the questions, and as part of the reflective process.

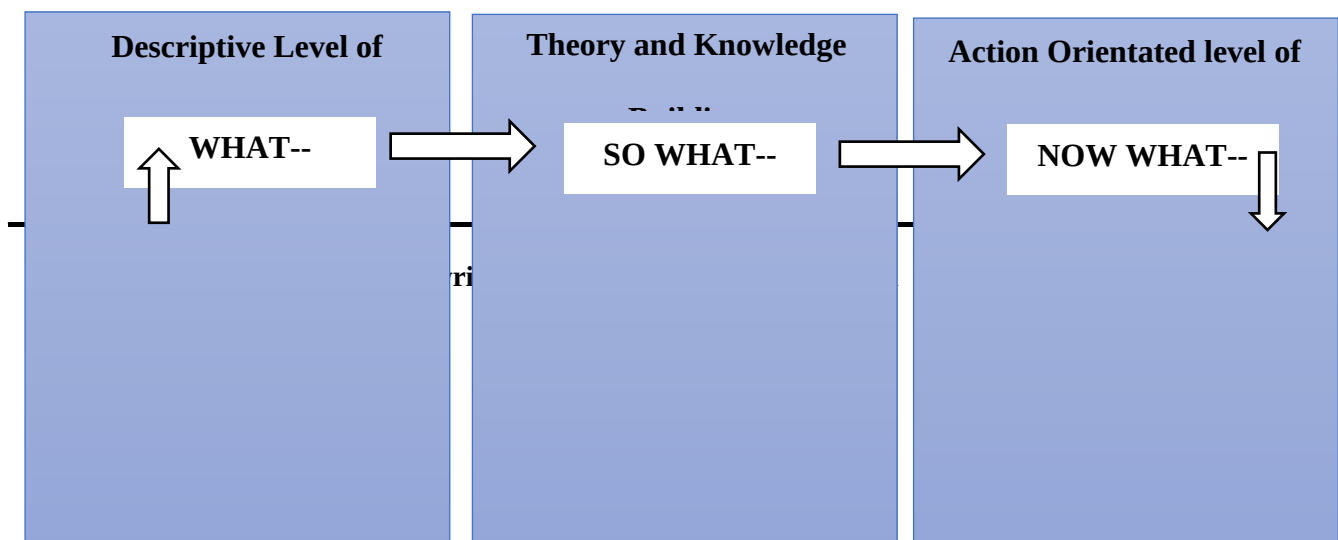
So –what:

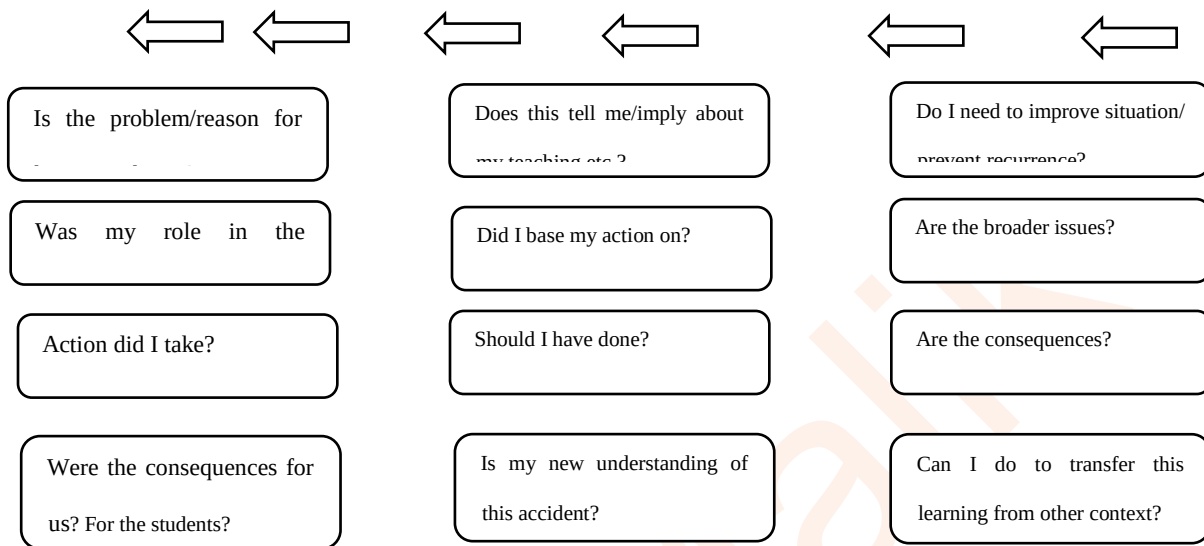
The next level the **so what** theorizing, the knowledge-building component. So what, does this tell me is a good question to ask? So what did I base my actions on? In other words, why I did **do** that? So what should I have done in that particular situation? You can start to see how you can frame particular questions using that cue so what? Another questions is my understanding, so what is my understanding of the incident? So, these questions **are** the first, second, and third level questions you can phrase them in any way you choose, and you can ask any questions to help you think the way you want to think about your teaching.

Now-what:

The action phase now what question. Now-what do I need to do **to improve the situation/prevent recurrence?** Now what are the broader issues at stake here? And now what are the consequences for me or for the students? And now what can I do to transfer this learning to my next teaching session? Again the questions that you can ask here a person you can identify **as** your own they are not fixed. **As** in the other models, we have seen this gives a lot of freedom to the teacher. These **are** very useful frameworks **so what** to know **is** what gives you the starting point for the questions you can ask. And what seems important with all reflective practice is that whatever process you take, whatever model you choose to use that it is actually part of a cycle. So, these questions these what questions, **what, so what, now what** needs to be iterated a number of times not just once but again and again.

Overview of Rolfe's Reflective Model





We can look at these in terms of again a nice straightforward model of the **what, so what, and now what** components going through repeatedly a number of iterations. Whatever questions you ask at those particular levels is up to you as a practitioner.

- But at the **What level**, it is very much about description, whatever viewpoint you take is entirely up to you. But the **What** question will help you describe the things happening in the classroom.
- The **So What** is where you will take that description and start to link it to your own understandings and theories.
- And this then will lead into the **Now What**. In other words, the action-oriented component of the model.

Critical Models 4: Rodgers (2002)

Rodgers's model of critical reflection very much encompasses Dewey's idea of reflective thought. In this model, Rodgers illustrates reflection as a cyclic process. He has broken down the reflective process into **four phases**.

- The first phase is described as **Presence in experience**. This is different from any other models we have seen presence in experience is about the teacher learning to see really, what is happening, seeing clearly and honestly.
- The second phase is **describing that experience**, describing what has been seen and trying to understand it to differentiate the different components.
- The third aspect is **an analysis of that experience** trying to link to deconstruct, link it to theory and to literature
- And then of course the final stage is **experimentation**. Or in other words, learning to take action from the analysis of the experience.

According to **Rodgers**, interestingly in this model, the idea is that the teacher although going through a cycle can actually move backward, and forwards in the cycle at any particular point. So, it is not necessarily a smooth process but something which can go backward and forwards.

Presence in Experience:

This is because the first element is the teacher being present in the experience and how they perceive it. Because in the model, the idea is that the more the teacher can actually see, the more the teacher can perceive which will affect the rest of the reflective practice process. Therefore, the teacher may need to move backward through the model in order to keep perceiving clearly what is really happening to question their version of reality. So, it is very much about learning to see, differentiating the different parts, giving them meaning, and responding intelligently to what happens in the classroom.

Description of Experience:

It is the differentiation and naming of an experience's diverse, and quite complex elements. So, that it can be looked at seen, and told from as many different perspectives as possible. Because you won't just describe the experience once you would describe it potentially a number of times. Because you'd be moving backward and forwards through the model. There are a number of description that you could make, and you would be constantly trying to refine the description, and

refine your perception of what is happening. And what this is very much about is the teacher being very clear about their own perception and their own apprehension of the way they see. This is the cornerstone of this model. It is quite different from other models, and is present in the description.

Analysis of Experience:

The analysis involves generating a number of different explanations. Therefore, there could be explanations from previous descriptions, as well as later descriptions in order to get that very critical in-depth, and honest perspective of what's been going on in the classroom. It is the phase where meaning-making happens.

Although the analysis comes from the description, there is often a dialectical relationship between the two. The model says that the teacher can move backward. Therefore, the analysis and the description can become very closely linked each depending on the other. The description can become a reaction to the analysis as much as the analysis can be a reaction to the description.

Experimentation:

Experimentation is the final as well as the initial phase of the reflective practice because it doubles as the next experience. This final phase of reflective practice is about action. In this model action is not the word that is used but clearly experimentation. It is very much focused on the teacher trying things out, and testing things in the classroom. And not being afraid of the consequences but learning from the consequences in order ultimately to improve their teaching.

Therefore, what this clearly says about this model is that knowledge from experience acquired knowledge is not something that is constructed in isolation, but it is very much linked to the students that the teacher is working with, and colleagues that the teacher interacts within the school.

Topic 035: Critical Models 5: Gore and Zeichner (1991)

The Gore and Zeichner model of reflective practice is another critical model to just briefly consider. What they suggest is that there are four major aspects of the work of the teacher, professional

work, and what they say. Therefore, the teacher needs to ask and reflect on particular questions in relation to these four aspects.

The purpose of this is to put it for the teacher to gain a thorough whole and holistic understanding of their teaching and their role in teaching, and their role in the classroom. The four components are very clear.

Academic Reflection:

The first component is an **academic reflection**. There are certain questions that the teacher can ask themselves in relation to academic reflection. Do I know my content really well? For example; Am I using the right strategies for teaching and my students' needs? Am I well-organized for my lessons? These are all focusing on the professional role from a very technical and practical point of view. Other questions that can be asked about the sequencing of the content. Was the lesson at the right sort of pace for the students? Was it too fast or was it too slow? Have I completed the planning cycle with suitable, relevant assessment strategies to evaluate student learning?

Also, it is useful to think of this aspect of the model in terms of a creative or innovative you are being in the classroom. And are you trying out new approaches not only to excite your students but to excite yourself or are you just going through a routine. So, it is good to reflect on the academic aspect.

Developmental Reflection: They also talked about developmental reflection. Am I providing teaching and learning contexts that give students a chance to develop their own perspectives on their learning? I like to think of this part of the model in terms of; how well the teacher is helping the students learn, and how to learn. And for me, that is a very significant part of being a teacher in the classroom.

Some more questions to ask in terms of developmental reflection. Have I evaluated the students' skills and thinking to determine the stages at which each of them is able to engage I different learning contexts? So, that you can actually determine how much they are learning.

And what about changes to the instructional approaches you can reflect on that try things out can you extend the use of a particular approach? Or Have I planned suitable instructional and task modifications to accommodate the differences in the students' thinking, emotional and physical capacities? Have I designed teaching and learning activities for individuals as well as groups? Have I taken into account and effectively utilized students' various interests to design lessons and curriculum? For example, if you are using group work, how are the students learning in a group? How are they supporting each other? Or how are they contradicting each other?

Social Reconstructionist (Critical) Reflection:

The most complex aspect of this model is the critical reflection or the social Reconstructionist approach. This takes a very much more holistic view of reflective practice. Actually, looking at what the teacher believes about teaching? What the purpose of education is? And, what the value of education is in society? And so, questions you can ask are things like, Do I have particular philosophical beliefs, and values about the values, purposes, and function of teaching which will affect the technical approach that I take in the class?

Have I critically evaluated certain aspects of teaching from the school's perspective? How do you articulate those and who helps you articulate those? Have I considered who determines the curriculum? And what does that mean in terms of; how you teach and how the students are learning and indeed what they are learning? Have I considered in what way a curriculum supports or neglects the individual learning needs of students from different social, cultural, and individual groups? This is taking you beyond just pure description and is being very critical about the imposing factors that can affect you as a teacher and affect the students as learners.

Are there ways in which I can implement components of the curriculum or different curricula in the school in a more unique and innovative way? So, you are actually meeting the needs of both worlds both the school education system but also the individual learning needs.

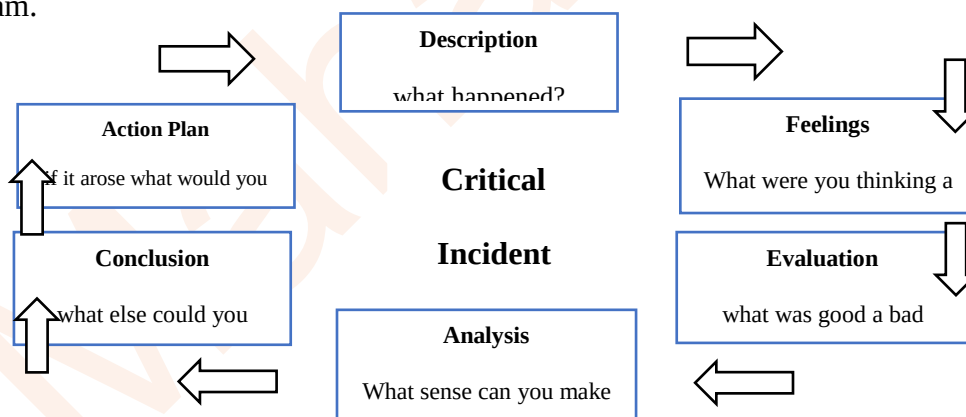
How can I overcome short come shortcomings or problems with particular systems in schools or particular issues in curricula? Therefore, make teaching and learning a much more interesting and creative process.

Have I analyzed the assessment processes may be no compulsory tests are they really testing learning or is it just about testing memory? There are some fundamental questions for the teacher to ask here from this broader critical perspective about their role as a teacher.

Critical Model 6: Gibbs Reflective Cycle (1988)

A very popular critical model is Gibb's reflective cycle. This is a way for the teacher to think about the different aspects of something which has happened in the classroom. Again, it is in the form of a cycle. It is very much about understanding something an event, or a situation describing it, evaluating, and establishing a course of action in relation to that reflection.

This model is very much about helping the teacher understand a particular situation from a very detailed point of view. Because there are quite a few steps in this cycle and we can see this in the diagram.



The starting point here is the **description** of what actually happened in the model. This leads to **feelings and evaluation**. It is important to consider the aspects of the teachers' feelings who knows what were you thinking and feeling about that particular incident? And then **evaluate** what was good about it and what was bad about it. After the evaluation comes, the **analysis** what sense

can you make from this particular model? What sense can you make from what happened? again linking it to the real concept. And from this, the teacher can make some conclusions and come to some sort of decision that will lead to action. Of course, those from the action there will be another chance for a description. So again, we get that spiral of thinking.

So, a great Gibbs reflective cycle has a lot of components to it but again it's all focused on asking particular questions, a particular point in the cycle so. For example:

Description:

In the description phase "What happened?". It is very much about the teacher describing in detail the event that took place and including particular focus points. In other words, where were you; who else was there; what the students were doing; why were you there; why were other people there; what was the context of the event; what happened? Asking very simple questions but quite probing questions.

In terms of what happened, what was your part in any particular incident that happened in the classroom; what parts did the other people play; and what was the result? Again, with previous models the questions here are not fixed they are very much for the teacher to think about and design for themselves. It's the individual components that add up to make the reflective process.

Feelings:

The feeling component is very much about the teacher trying to recall the reaction to incidents in the classroom. How were they feeling about it? What were the emotions that were involved? How does this feeling make you feel about and the event; when it started; and during; what you were thinking about at the time; how did it make you feel? This is very much about linking the teacher to the activity in the classroom, the actions which are taking place. How did you feel about the outcome of the event? So, in terms of feelings, you are thinking before, during, and after what happens in the classroom?

Evaluation:

The next phase of the cycle is evaluation. What was good and what was bad about the experience? This is for the teacher to make a judgment for themselves about the teaching in the classroom and the actual incident. This leads to further questions about what was good about the experience and what was bad about the experience or what went well; what didn't go well and perhaps to start to consider why.

Analysis: What sense can you make of the situation?

Because from this phase the teacher moves into analysis after the evaluation. What went well; what didn't go well the good experience the bad experience; How does this link to what we know about education about teaching and about learning from all the theories which were exposed to? So, it is very much about breaking down the analysis into component parts and could include questions like well what did you do; what did others do that was good; how does this actually relate to; what we know about theory in what way; did others contribute to what happened in the classroom?

Conclusion: What else could you have done?

This ultimately leads to the point where some judgment is made by the teacher. Some conclusion what else could you have done? For example this is quite different from the evaluation and the analysis stage. Because it is very much about putting a stop to the reflection that's taking place. It has to end at some point and there have to be some decisions. So, it forms part of the reflective cycle. Even though it does actually stop thinking. Because after the conclusion phase, when you are thinking about; how things could be done differently; what were the results; and how this will impact the future.

Action Plan: If it arose again, what would you do?

In the Gibbs cycle, the action plan that the teacher comes up with is a result of all that previous thinking including the conclusion, and the judgments that were made about what was happening in the classroom. With the action plan, you have to consider how you would do things; how you

would do things differently based on the reflection. Perhaps most important to remember is that again from action comes another cycle of reflection. So again, it is continuous even though there is that point of stopping conclusion built into the cycle. Here the cycle is tentatively completed and suggests that should the event occur again it will be the focus of another reflective cycle.

From Theory to Practice: Boud's Triangular Representation

From Theory to Practice

So, we've been introduced to some models of reflective practice. And the value of these models is in helping the reflective practitioner. The teacher begins to deconstruct their experiences and ensuring develop a deeper understanding of the questions and the issues involved in the work that they do.

Boud's Triangular Representation

David Boud (1987) has defined reflection as:

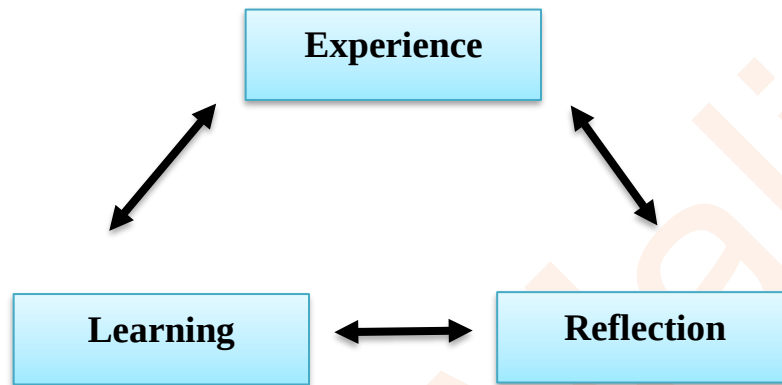
"A conscious activity in which we engage to explore our experiences and develop new understandings and conceptualisation."

Boud, 1987

The thing about the models of reflection is that they're all cyclical in nature, representing the idea that reflection leads to learning. Cyclic learning is never complicated nor transferable without reflection further and what Boud's adds to this discussion is something quite significant and that's the idea that reflection needs to happen in different contexts without any reflection on application of learning in one context in relation to another means that actually reflection is a waste of time.

And so, this model is an attempt to actually start to address that imperfection and we concede in the diagram here it's a very simple representation.

Boud's Triangular Representation



Boud's idea is that the reflective cycle is there and it can be clockwise or counter clockwise as a reflective process. But clearly, we can see that expression experience and reflection are part of the cycle and what is added here is learning by the teacher learning which can be taken to another context. So, this model attempts to capture the essentials (that experience and reflection lead to learning), the essence of what reflection is really about and what it is aimed at doing is making a clear connection a proper translation in terms of experience and learning.

So let's think a little bit more about this triangular representation.

1. What the reflective practitioner to do is return to the incident the event upon, which they're reflecting and to not only try to describe it for themselves.
2. But also, to understand their emotional response to it the feelings that they have in relation to their teaching.

3. And from this very personal reflection comes an evaluation a revaluation of the teaching in light of experience, knowledge and experimentation, trying to seek a deeper understanding for it.

4. And from this reflection comes some sort of thinking about action in relation to change so there is a planning phase to the cycle.

Boud's recognizes that this is a very individualistic process of learning and this can be criticized because reflective practice is very often seen as a collaborative activity.

So Boud's poses three alternatives:

1) should we reject models a reflection

Altogether this is just adding to the discussion about reflection and reflective practice.

2) Should we rehabilitate them to capture their previous potential

we should try to think differently about what reflection really is for the teacher and what reflective practice really is.

3) Should we move to new ways of regarding reflection that are in keeping with what we know about the context of practice?

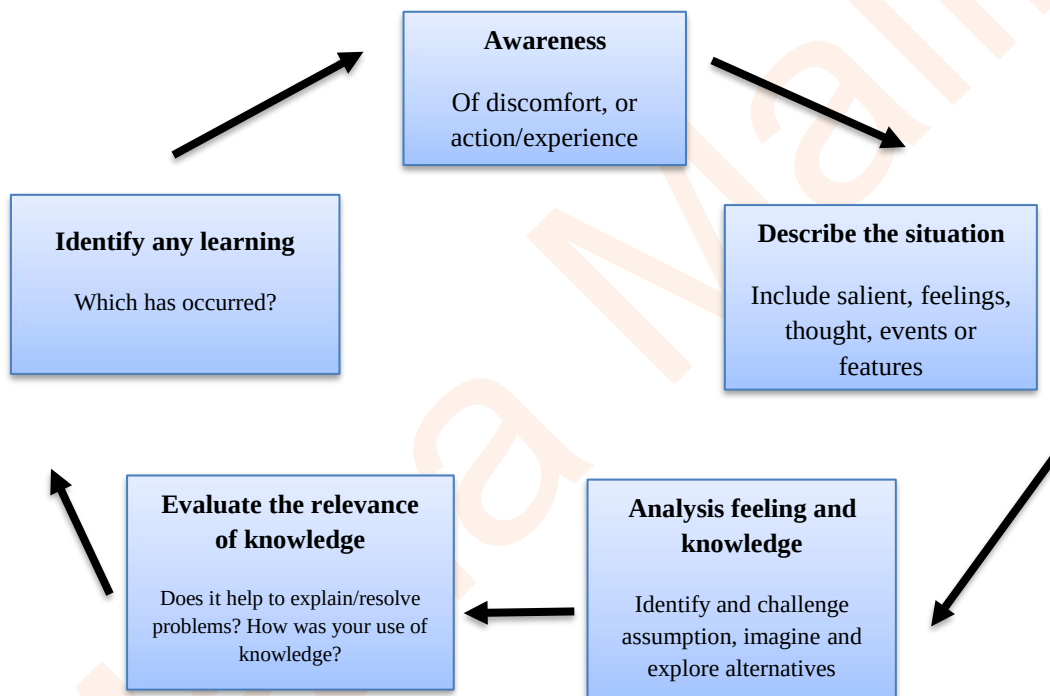
He starts talking about whether the reflection can actually be applied in new contexts and what a context adds to the process of reflection.

From Theory to Practice: The Atkins and Murphy Model

Another model of reflective practice which is aiming to move reflective practice from theory into action is the Atkins and Murphy model. Atkins and Murphy (1993) address many of the criticisms identified by Boud with their own cyclical model:

We can see the model here it has five components.

- Awareness
- Describing the situation
- Analyzing the feelings and the knowledge associated with that situation
- Evaluating the relevance of the knowledge within that situation
- Identifying the learning which comes from that



And we'll look at these in some detail Atkins and Murphy model has five stages to it and the stage is linked together in a cycle.

Stage 1: Awareness of uncomfortable feelings(usually due to new, unfamiliar or negative situations)

Stage one is very much about the teacher becoming aware of some need for change. Because of some uncomfortable feelings due to the situation that they're confronted by.

Stage 2: Examination of components of the situation

Stage two then comes from stage 1 and stage 2 is about the teacher examining those feelings and the components in the situation which generate those feelings.

Stage 3: Analysis of the components of the situation

Stage three leads then into analysis of the situation based on the evaluation

Stage 4: Summary of outcomes of reflection

Then into stage four there is some sort of conclusion a summary of what all those feelings and all that thinking generates.

Stage 5: Recognise learning from the reflection

And then finally into a stage five which is the learning stage and what the teacher has actually gathered from that experience.

So, this model attempts to provide a deep level of support for the reflective practitioner more so than other models tried to do.

And we must remember that this is not to say that the other models that exist about reflective practices are redundant or a waste of time, **but** this is just another way for the teacher to think about their role in the classroom. **It is important to remain alert to the potential to provide superficial responses as the critical, questioning and challenging...** What is important is to realize that as a reflective practitioner part of your role, part of your duty is to critique not only yourself and what is happening in the classroom but also, to add to the critique on the models which can be used for reflective practice.

Atkins **and** Murphy suggest that **for reflection to make a real difference to practice, we need to follow reflection with a commitment to action.** Any sort of reflection is not useful unless the teacher has a real commitment to action. Action is what reflective practice is all about and they

make an interesting connection in relation to action here which is action can happen after the event but it can also happen before the event in terms of planning and preparation, the problem with such models of reflection is that they do not take account of the importance of reflection before action. So, we must remember in this model that it is actually trying to make a connection not only with the reflective process but the actions that the teacher needs to take in the classroom.

Self-Awareness (mindfulness)

- Insight into personal sensations, emotions, thoughts and actions.
- Acceptance/ownership of response
- Recognition/acceptance of emotions, sensations thoughts in others involved.

This involves honest examination of how situation has affected the individual and how the individual has affected the situation.

Description

- Comprehensive account of the situation either verbally or in writing
- Involves the ability to recognise and recollect accurately salient events and key features of an experience.

Critical Analysis

- Involves examining the components of a situation
- Identify personal cognitive and emotional schema

- Identify existing knowledge
- Identify biases and assumptions
- Imagine/explore alternatives
- Integrating new understanding
- Formulate generalisations
- Predict likely consequences of actions

Evaluation

- Validation and appropriation of knowledge
- Along with synthesis this is crucial to the development of a new perspective.

From Theory to Practice: John Smyth's Framework for Reflection on Action (1993)

Smyth's Framework takes the teacher through a number of phases:

- A Descriptive Phase
- Reflective Phase
- An Emancipator Phase

\

1) The Descriptive Phase

- A Description of Practice events
- Examination of Descriptions for genuineness and comprehensiveness

2) **The Reflective Phase.**

Reflective analysis against adopted theories:

- Reflective analysis of the situation
- Reflective analysis of intentions

The Reflection phase is again the analysis of what's taking place, analysis not only of events but also the **situation of the context in which they occur** and trying to understanding the intentions behind them

The Reflective Phase Includes:

- Knowledge about practice processes and applications
- Self-awareness

And this model does not remove the teacher from this there is a sense that is the part of the process. Is the teacher developing a much stronger Self-awareness of themselves as practitioners?

3) **The Emancipator Phase**

- Critique of Practice regarding conflicts, distortions and inconsistencies

Now the interesting part of this model is the Emancipator phase this is where the teacher is very much critiquing their own practice and looking at inconsistencies in terms of what they're doing in relation to what they know they should be doing ought to be doing.

The Emancipator Phase Involves:

- Engagement in Emancipatory and change processes learning and change in practice
- Self-critique and emancipation

This is not to say that therefore the teacher is behaving incorrectly, it is very much about liberation. It's about the teacher beginning to realize that things can get change and those things are not static and that the self-critique is an opportunity for growth and development.

Smyth operationalizes these phases of Reflection for teachers as a set of activities developed from question cues:

Activity: Describe

Cue: what did I do?

That is very simple question, of course you could come up with any number of questions as part of the description

Smyth proposes Reflection for teacher as a set of activities developed from question cues:

Activity: inform (analysis)

Cue: what does this mean?

Another question could be well that what does this actually mean in terms of me as a teacher or

What I'm trying to do within a school?

Smyth propose Reflection for teachers as a set of activities developed from question cues:

Activity: confront (Self-awareness)

Cue: How did I come to be like this?

For example in term of developing a self-awareness that needs some sort of confrontation personal confrontation by the practitioner and cue could be how did I come to be like this in the classroom? , how did I come to feel these things?

So, you can see that the teacher is not extracted from this process at all and very much is within the action and the question help the teacher to focus on this aspect of their practice

Activity: Reconstruct (evaluation and synthesis)

So, in terms of the teachers starting to synthesize and learn from what is taking place there is a sort of reconstruction process and there are particular cues which the reflective practitioner can use.

Cues:

- What do my practices say about my assumptions, values and beliefs?
- Where did these ideas come from?
- What is it that causes me to maintain my theories?

From Theory to Practice: Agyris and Schon (1978)

Perhaps the most well-known model of reflective practice which attempts to link theory to practice is the model by a Garrison. Agyris and Schon pioneered the idea about two types of learning from reflection.

Those reflections are:

1. Single Loop Learning
2. Double Loop Learning

The theory is built around the recognition and amendment of a perceived fault or error.

Single Loop Learning

Single loop learning is when a practitioner or organization, even after an error has occurred and a correction is made, continues to rely on current strategies, techniques or policies when a situation again comes to light.

Single loop learning is about the practitioner reflecting and then acting in relation to something which they have experience in the classroom. It relies on standardized techniques practices in order to solve a problem and does not change those techniques or practices at any particular point.

Double Loop Learning

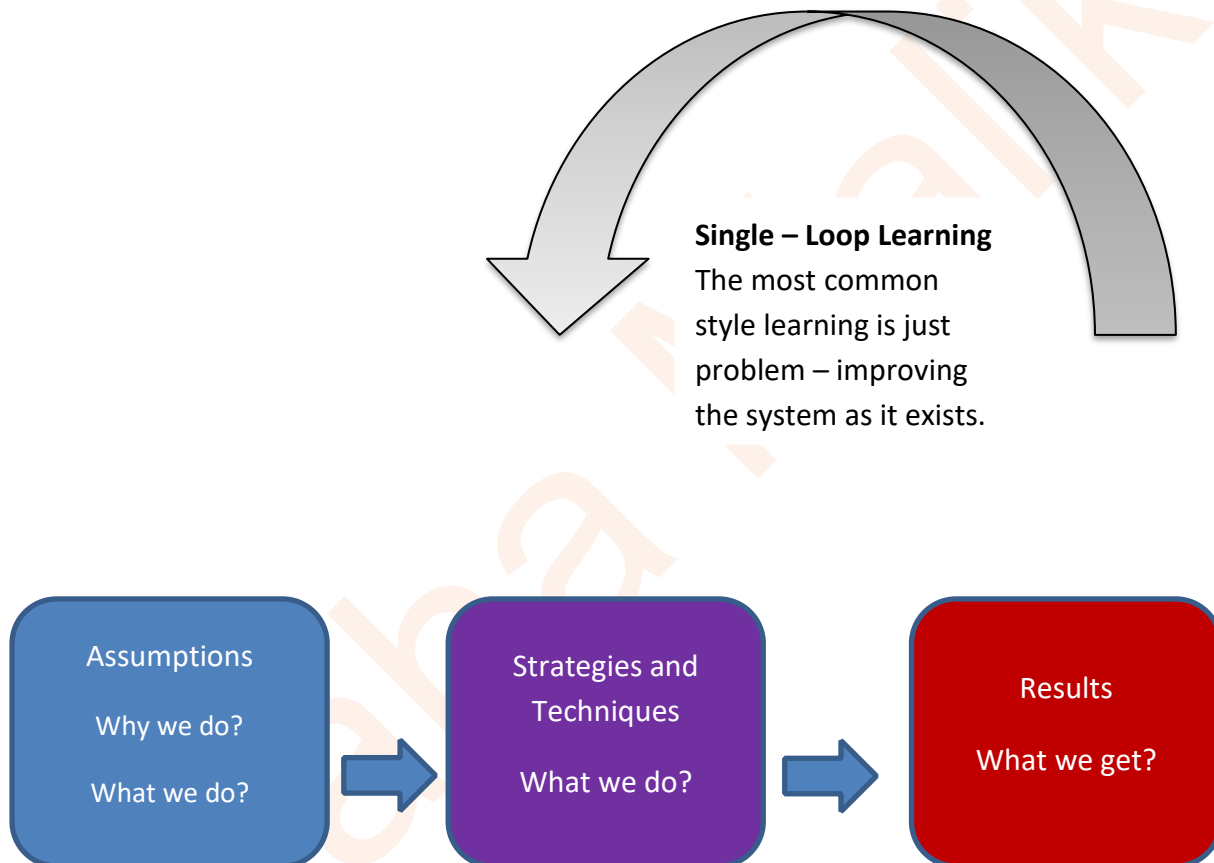
Double loop learning involves the modification of personal objectives, strategies or policies so that when a similar situation arises a new framing system is employed (i.e. new approaches are used to solve it).

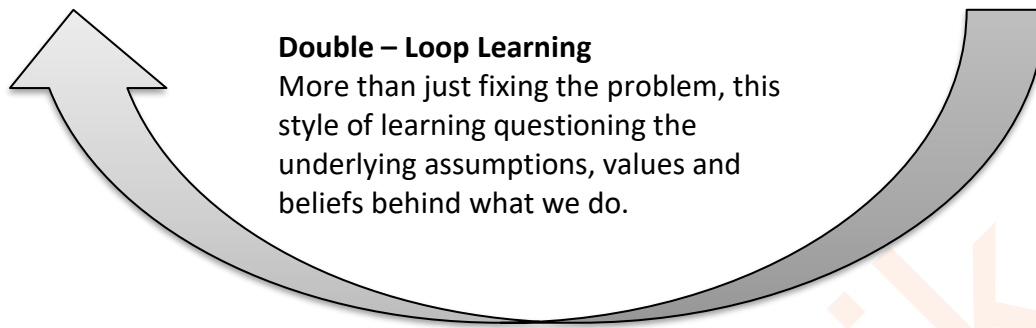
Single loop learning is in contrast to double loop learning where the learning can impact on the practices or the approaches which the teacher uses in order to change them.

So therefore, a reaction to an event can actually change with time with single loop learning their reaction would not change.

Explanation Through a Diagram

It is possible for us to look at this in very simple diagram (i.e., given below) and we can see here that single loop learning just takes one step back from the results, the event and the experience that the teacher has and through to strategizing approaches to deal with that event in the future. They can be very standardized approaches.





With double loop learning, it is a little bit deeper in the sense that the teacher is thinking about their assumptions about the event why things happen, their beliefs around why things happened which they can inform the strategies and therefore can possibly change the strategies and then these strategies are used to deal with the event in future.

So, single loop learning and double loop learning are two approaches within this model to change practice but one is quite fixed and other is very flexible.

So, single loop learning then does rely on fixed practices single ideas and repetition of approaches rather than trying to extend practice.

Single Loop Learning involves connecting a strategy for action with a result. For example, if an action we take yields results that are different to what we expected is through single-loop learning. Therefore, there is a certain amount of **automaticity** about how the teacher approaches their work in the classroom. There is less thinking and trying out new ideas rather than just relying on old practices.

Double – Loop Learning is the cyclical process of applying a new strategy to achieve an outcome may occur several times and we may never succeed. Running out of strategies may push us to re-evaluate the deeper governing variables that make us behave the way we do.

Double loop learning is much more creative as a process in a way somewhat experimental for the teacher changing assumptions means that the approaches will change, the approaches may work

or the approaches may not work and therefore this leads to a new experience and therefore a new double loop.

Complex Ways of Processing Information are:

- Re-evaluating our goals
- Reframing our goals, values and beliefs

The above ways are more complex way of processing information and involves a more sophisticated way of engaging with an experience.

The key point of **double loop** learning is about the teacher re-evaluating their own perceptions of what they are doing by looking at their goals and trying to reframe their assumptions and beliefs about teaching. This is called double-loop learning and looks at consequences from a wider perspective. Whereas single loop learning focuses very much on a much smaller, a much more mechanistic perspective.

In situations where teacher employ double loop learning, **not only will we evaluate our own goals and beliefs, but also those of the people we interact with.**

Double loop learning is very much about the teacher looking much more deeply at their own role where single loop learning actually is becoming quite instrumental in terms of how the teacher is approaching teaching and both can have a constructive outcome.

To have a constructive outcome, we may need to modify and adapt some of these variables to create an attitude that is open to many cultural values and application methods. Double loop learning is considered to be much more constructive potentially because of the opportunity for change and the opportunity for growth.

Mezirow's Transformational Learning

Mezirow (2000) argues that reflection only leads to learning if it leads to transformation. Mezirow suggests that transformation occurs where the original starting point causes a dilemma that needs to be addressed. and then suggests that there are a series of possible phases which may be gone through:

Mezirow suggests that there are a series of possible phases which may be gone through:

- A disorienting dilemma includes loss of job, divorce, marriage, back to school or moving to a new culture.
- Self-examination with feelings of fear, anger, guilt or shame.

In this way there are a number of phases for the teacher to go through. Mezirow talks about a disorienting phase. Something which causes some issues, some problems for the teacher and because of that issue or problem, a teacher needs to go through a process of self-examination. For instance, what are their feelings in relation to the experiences? how they are thinking about it? And how they are trying to rationalize what happened?

- A critical assessment of assumptions
- Recognition that one's discontent and the process of transformation are shared.

This then leads into an exploration of the assumptions that the teacher has and the value of **those assumptions about their own role as a practitioner. And from this, then there can only be a process of transformation taking place.**

- Exploration of options for new roles, relationships and actions
- Planning a course of action
- Acquiring knowledge and skills for **implementing** one's plans

The exploration **of self** that Mezirow is talking about here is very much a process of transformation. In the sense that it's about exploring, planning and **developing** new knowledge and understanding about the role that's what transformation is about in this model. It's about the teacher developing a much clearer perception, much deeper understanding and more clarity in the knowledge about what they do.

- Provisional testing of new roles

- Building competence and self-confidence in new roles and relationships

So, part of transformational learning needs to be about testing out the different roles that the teacher can take on. And within those roles, building up a level of confidence in relation to the role in the classroom. With time and with transformation, the teacher can become more confident as a practitioner.

- A reintegration into one 's life on the basis of conditions dictated by one 's new perspective

In order for that confidence to occur, that self-confidence to develop, feelings and emotions along with assumptions and beliefs have to be integrated into the teacher's perception of themselves. And this starts to develop different perspectives for the teacher on their role in the classroom.

What Is Particularly Striking about the Mezirow's Ideas?

- Particularly Striking About the Mezirow's Ideas is the emphasis on conscious, self-managed learning.
- Learning is not a direct result of the experience; it happens because the individual takes charge of their critical reflection and explicitly plans and carries out steps to learn from it.

Therefore, learning is not a direct experience. It is learned experience. It is acquired experience in the classroom. And it is about linking knowledge that is acquired from training, literature or research with the experience that is had in the classroom.

The following is a sequence for reflectivity draw from Mezirow 's model:

- A disorientating dilemma
 - Picture the event
- Critical Assessment of Assumptions:
 - What does it mean to you to feel this?
 - What advice are you giving yourself in the picture?
 - How do you interpret what is happening?
 - What is your intention?

And, in terms of that critical reflection, it's very important here for the teacher to think about what they were actually trying to do rather than externalizing the process of teaching. It is about internalizing teaching and understanding what their purposes were.

- **Exploration of new roles**

- How would you prefer this to be different?
- When this begins to occur for you, what will be different about you?
- How do you interpret what is happening?
- What is your intention?

The exploration of roles becomes part of this transformative learning and really is the underpinning concern in relation to this model. How does the teacher interpret? What is a teacher interpreting? How critical is the teacher being? It is only with questions like these that the level of reflection developed transformation. If the reflection is quite superficial then learning is actually not going to be transformational, change is not going to happen. So, it does require a certain level of criticality.

- **Planning a course of action**

- What are you aware of that keeps this from happening?
- Dangers to change?
- Benefits to staying the same?
- Does it happen a little bit sometimes already? What's different at those times?

The transformational model has to lead to some sort of action. It's not just about thinking and it's also about doing and really then the teacher needs to think about the action in terms of change as a change process. And what are the consequences of that change? Could there be some dangers involved in it? How will it affect students' learning? What are the negative effects as well as the positive? And, this is very much then as a transformational process something which through a cycle is repeated.

- **Acquiring knowledge and skills for implementation**

- What will you need to know/accomplish/overcome for this to occur?

The interesting thing about this because it's focused on learning. It's very much about acquiring knowledge from experience rather than knowledge from texts. And, a lot of people say that this is perhaps the most valuable knowledge for the reflective practitioner to acquire.

- **Trying out new roles**

- **How will you know when you are more on track?**

So, it is about trying out new roles, experimenting and acquiring knowledge through that experimentation to become a better teacher.

Modelling Reflective Practice: Criticisms of Reflective Practice

- A number of writers discuss key concerns of RP in terms of its problematic side: Brookfield (1994, 1995) explains the cultural and personal risks involved.

You can see by now that there are many models of reflective practice and now, we should also look at some criticisms of reflective practice. And, perhaps the most significant **figure** to look at in terms of critique of reflective practice is Stephen Brookfield. He actually identifies that for the reflective practitioner, there can be a significant number of personal risks involved both in terms of their confidence and also in terms of reflecting critically on their role.

- Ash and Moore (2002) say that new teachers can choose not to reflect constructively and critically, falling back on pre-conceived understandings.

Ash and Moore actually say that there are certain preconceived ideas and preconceived understandings about reflective practice which can interfere and, in a way, deconstruct the actual process.

There are four areas of concern about reflective practice:

- Ethical
- Professional
- Pedagogic
- Conceptual

Whatever criticisms we look at, they tend to fall into four categories. And we can see the categories related to the ethical considerations around reflective practice as well as professional

considerations. There are of course the considerations around teaching (that pedagogic considerations) and of course, there is the conceptualization of reflective practice itself.

Ethical Criticism of RP

- Concerns related to confidentiality, rights to privacy, informed consent and professional relationships.
- Practitioners engaging in RP need to be aware of the risks and the potential for conflicts of interest

Professional Concerns

- Relevant when RP is done badly or inappropriately. In this case the value of RP is missed and it simply rationalizes existing practice
- If applied uncritically, it can reinforce prejudices and bad practice
- The inappropriate use of RP may devalue teachers' professional work instead of promoting it.
- Where teachers follow RP models in mechanical, routinized or instrumental ways, they fall into the trap of engaging neither critical analysis or their emotions.
- At the level of the individual teacher, it can be means for organizations to ignore their responsibilities so that fostering good practice is a matter for the teacher rather than the organization.
- Two key concerns relate to 'developmental readiness' and the extent to which forcing student teachers to reflect may prove counter-productive.

Pedagogic Concerns: Developmental Readiness

- Teachers need to be developmentally ready to engage in critical reflection and some individual may not be capable (ready) for reflection.
- The respective abilities of 'novice' and 'expert' are relevant here. Novices tend to follow models mechanically and experts tend not to rely on models.
- As compulsory element of teacher education courses or of organizational requirements, RP ends up being superficial, strategic and guarded.

- As part of an assessment process, self-examination may will be discouraged.
 - Reflection and assessment are incompatible.

Conceptual Concerns

- Problems are posed with the way teachers embrace RP in uncritical and reductionist ways
- Often ideas are transplanted without sufficient care across philosophical, disciplinary and cultural boundaries
- Reflexivity is a concept which is easily miscommunicated; it can be understood in many different ways according to the aims and function of the RP activity.

Criticism of RP

- Regarding these issues therefore, care needs to be taken to recognize complexity and problematize what is involved in RP rather than accepting it as ‘fact’ or as ‘self-evident’.
- The ethical, professional, pedagogic and conceptual concerns presented here neither condemn RP nor suggest that it should not be used.
- What these concerns indicate is that RP should be applied selectively, taught sensitivity and generally used with care.
- Practitioners need to be critical and reflexive about the tools they are being given and not use them blindly.

A Critical Thinking Framework: What Is Critical Reflection?

Critical thinking involves:

- Analyzing
- Conceptualizing
- Defining
- Examining
- Inferring
- Listening
- Questioning
- Reasoning
- Synthesizing

What is Critical reflection?

- With a purpose to evaluate information in a disciplined way to help us evaluate and refine our thinking.
- As reflective practitioners we must want to be better at thinking, to seek out and be guided by knowledge and evidence that fits with reality:
 1. Belief
 2. If shown to be unfounded
 3. Leads to a change of position and an appropriate response

We can look at this from a very simple point of view that the teacher, the reflective practitioner has a set of core beliefs which guide the actions in the classroom. And through reflection, they can decide whether these beliefs improve or support what is taking place or indeed can actually detract can cancel out the effectiveness of what is happening. And this process leads into some sort of change in understanding, some sort of change in position for the teacher in relation to the way they are teaching.

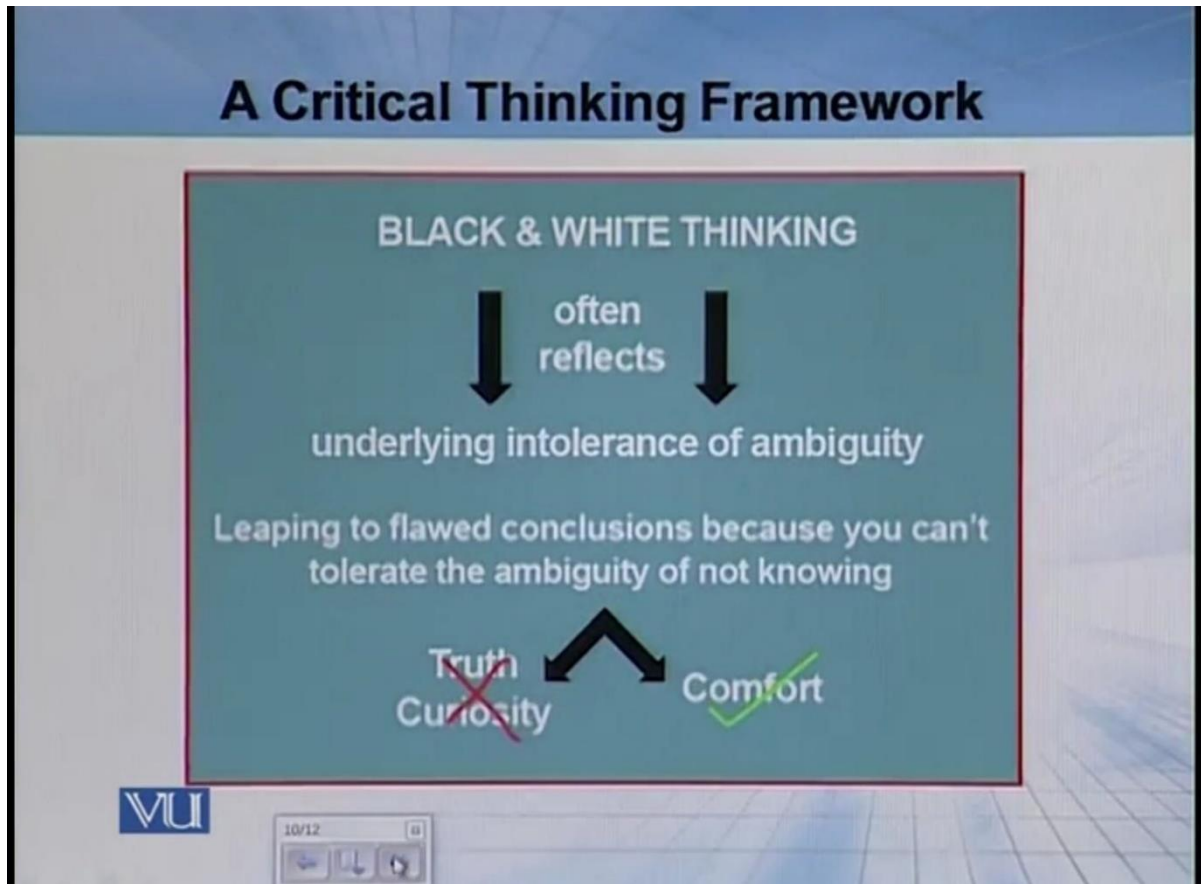
Critical Thinking Involves Reasoning

- It based on sound and consistent logic
- Not on emotions or social pressure
- Truth is not determined by the emotions that accompany them nor on the beliefs of social groups.

This can actually get in the way of critical thought; it can actually prevent critical thought from happening. And so, the teacher's role here is not about dismissing reason but really about looking honestly at themselves and perhaps having to be honest about the flaws that they have as teachers.

Critical Thinking Involves Understanding

So, it is very clearly based in black and white thinking, right and wrong thinking about what is taking place and where the flaws are not accepted. This can actually move the teacher to a point of comfort which is not good necessarily if bad practices are being continued rather than accepting truth and then becoming curious about what is actually happening.



The difference between truth, curiosity and comfort here is quite significant. Because with comfort, there can be complacency and with complacency, people can get into bad habits in the classroom. Where there is an honest reflection and critical thinking approach to reflective practice then there is truth and with truth comes the opportunity for change.

Critical thinking moves us away from:

- **Rash conclusions**
- **Mystification**
- **Reluctance to question**

So, critical thinking then is moving away from the emotionality of the role of teaching and becoming much more rational about it. And it's moving away from making quick decisions and it's moving away from superficial understandings and beliefs and it's moving away from being scared of criticism.

Critical thinking moves us towards:

- Intellectual discipline
- Clear expression of ideas
- Acceptance of personal responsibility for own thinking

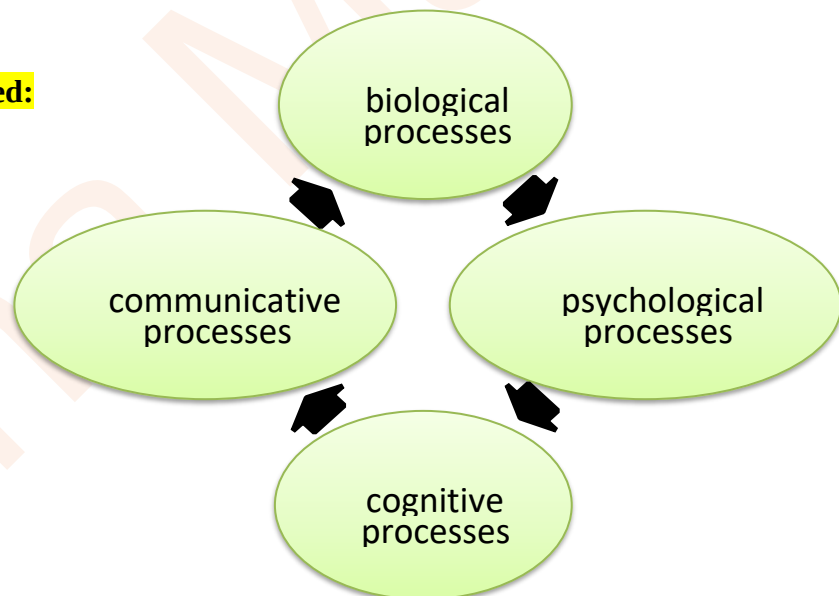
It is actually moving towards a much more intellectual perspective in terms of the discipline of teaching and an acceptance therefore that the teacher has a particular responsibility in terms of what is happening in the classroom.

A Critical Thinking Framework: What Is Critical Reflection?

Critical thinking is part of the reflective practice process. We need to consider some of the mechanisms involved in critical thinking.

The thinking process is as followed:

- Biological process
- Psychological process
- Cognitive process
- Communication process



Other types of thinking or “non- critical” thinking

- Habitual Thinking
- Brainstorming
- Creative Thinking
- Prejudicial Thinking
- Emotive Thinking

There is habitual thinking, the ways that we regularly think. Brainstorming, creative thinking, there's also prejudicial thinking and emotive thinking.

Let's look at these each in turn:

Habitual thinking

- Thinking based on past practices without consideration of current data.

Habitual thinking is what we will do regularly, in a way sort of unthinking and we will rely very much on that in everyday practice.

Brainstorming

- Saying what comes to minds without edit, judgement or evaluation.

Brainstorming is very much about making judgments and evaluating thinking creatively and innovatively about a situation.

Creative Thinking

- Putting facts, concepts, ideas and principles together in new and original ways.

Creative thinking is making links between facts and concepts, looking at the principles of what we know and questioning those principles.

Emotive Thinking

- Responding to the emotions of a message rather than the actual information presented.

With any experience that we have, there is always an emotional response and part of critical thinking is to understand that aspect of the process.

We are thinking critically when we are:

- Thinking rationally so that we rely on reason rather than emotions
- Requires evidence, ignore no known evidence, and follow evidence where it leads and are concerned more with finding, the best explanation than being right.
- Analyzing apparent confusion and asking questions.

We are thinking critically when we are self-aware so that we:

- Weigh the influences of motives and bias, and
- Recognize our own assumptions, prejudices, biases, and point of view.

We think critically when we are honest so that we:

- Recognize emotional impulses, selfish motives, immoral purposes or other modes of self-deception.

We also of course need to be honest because the word critical in critical thinking assumes for us that there will be no a cover-up of reality.

We are thinking critically when we are:

- Being open-minded so that we evaluate all reasonable inferences.
We also as critical thinkers need to be open minded because without that mindfulness, it would
- Consider a variety of possible viewpoints or perspectives, remain open to alternative interpretations.
- We're thinking critically when we're being open-minded so that we accept a new explanation, model, or paradigm because it explains the evidence better, is simpler, or has fewer inconsistencies or covers more data.
- By being open-minded so that we can accept new priorities in response to a re-evaluation of the evidence or re-assessment of our real interests and do not reject unpopular views out of hand.

By being open minded:

- we're accepting new visions
- new versions
- new priorities about teaching and learning

What we are clearly doing is not necessarily rejecting particular views but trying to understand all the different views which we're being confronted with.

We are thinking critically when we are disciplined so that we:

- Are precise, meticulous, comprehensive and exhaustive.
- Resist manipulation and irrational appeals, and
- Avoid snap judgements.

We need to be very disciplined. Critical thinking is not something which can happen superficially. It is very meticulous, very comprehensive, very thorough and therefore, what the

teacher cannot do is just jump to conclusions very quickly. Conclusions need to be arrived at overtime very carefully.

We are thinking critically when we are able to make good judgments so that we:

- Recognize the relevance and/or merit of alternative assumptions and perspectives.
- Recognize the extent and weight of evidence.

Critical thinker needs to be able to make good Judgments in relation to their role as teachers. And what this requires then is a look at the evidence, the value of the evidence, the weight of that evidence. If we react too strongly to something which is perhaps inconsequential not so important then that's actually not a good judgement, we need to be able to understand the weight of the evidence.

A Non- Critical Thinker

- Sees things in Black and white
- Sees things as “yes” or “no” with no subtitles.
- Fails to see linkages and complexities in information and situations.
- Fails to recognize related elements.

We can look at the opposite side of this or what is a non-critical thinker? A non-critical thinker is somebody who sees something very plainly. For example, they see questions as yes or no not as a maybe or it could be. And of course, somebody who is a non-critical thinker, is not able to see the connections and the linkages between things which they are experiencing. A non-critical thinker fails to recognize all the many elements involved in teaching.

Nine Attributes for Teachers to Become Critical Thinkers

- Be open-minded to new ideas
- Know that people have different ideas about the meaning of words.
- Separate emotional and logical thinking.
- Question things that don't make sense.
- Avoid common mistakes in your own reasoning.
- Don't argue about something that you know nothing about.
- Build a string vocabulary to better share and understand ideas.

- Know when you need more information.
- Know the difference between conclusions that could/must be true.

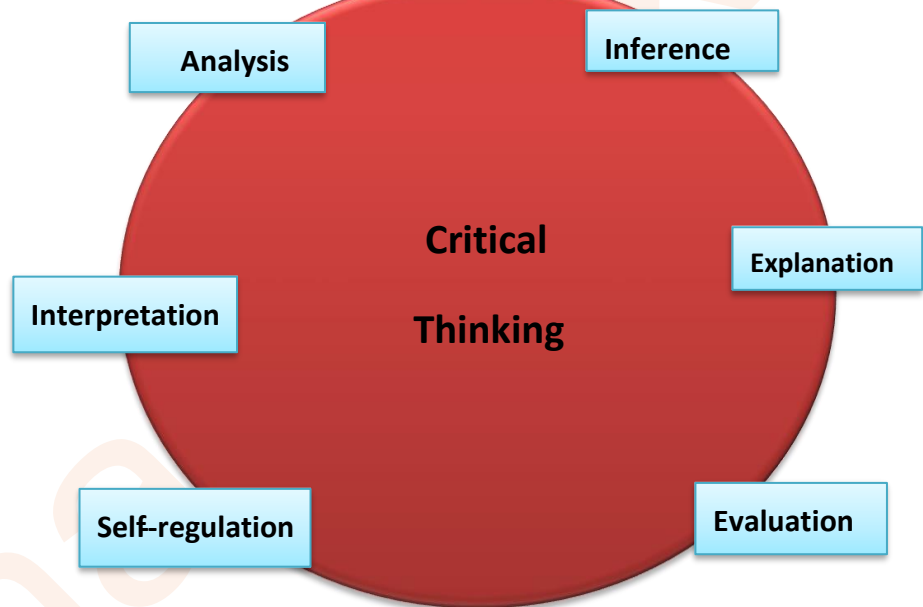
A Critical Thinking Framework: Core Ideas About Thinking Skills

- Critical thinking skills are extremely important in developing as a successful reflective practitioner.

Core Critical Thinking

Skills

1. Analysis
2. Inference
3. Explanation
4. Interpretation
5. Self-regulation
6. Evaluation
7. Interpretation



Interpretation

What it means: Having the ability to understand the information you are being presented with and being able to communicate the meaning of that information to others. As a teacher you will be presented with a variety of information in many different types of situations. Interpretation skills will enable you to better decode information and add clarity to what you have discovered which in turn will help you better understand your students.

Analysis

What it means: Having the ability to connect pieces of information together in order to determine what the intended meaning of the information was meant to represent. Having this skill will better

provide you with the ability to ‘read between the lines’ and help you understand how things you do will impact on your students and your organization.

Inference

- What it means: Having the ability to understand and recognize what elements you will need in order to determine an accurate conclusion or hypothesis from the information you have at your disposal.
- It is important for you to be able to understand what additional information you may need in order to determine what happens. The ability to understand the information you already have and determine what you may still need to find is an important skill.

Evaluation

- What it means: Being able to evaluate the credibility of statements or description of a person’s experience, judgment or opinion in order to measure the validity of the information being presented.
- You will need this skill to evaluate your students’ responses, so you can determine if the information you receive is valid and whether or not it needs to be further looked into.

Explanation

- What it means: Having the ability to not only restate information, but add clarity and perspective to the information, so it can be fully understood by anyone you are sharing it with.
- Your ability to clearly explain your ideas while keeping in mind who you interact with is important for making sure the information is understood and well received.

As a teacher, you try to look at the information and **phrase** it in a different way, so you get clarity about what it actually means. It’s a restating of the problem.

Self - Regulation

- What it means: Having the awareness of your own thinking abilities and the elements that you are using to find results.
- Your ability to understand the extent of your knowledge and that referring to a colleague with more knowledge on the subject is the best way to develop your skills and knowledge.

The final core **component** of critical thinking is the Self –regulation. This is the component about self-awareness of the teacher. The teachers’ own ability to think through a problem, to come to conclusions and to make those conclusions lead to actions

So, your ability to be able to think through, to rationalize and to come to conclusions is the core process of critical thinking and these ideas are significant in terms of that.

Critical Thinking Development: A Stage Theory

For critical thinking Development we can look at a staged approach and there are a number of stages that are at an individual can go through as a critical thinker.

Stage 1- The Unreflective Thinker

Stage 2 – The Challenged Thinker

Stage 3 - Is the Beginning Thinker

Stage 4 – The Practicing Thinker

Stage 5 - The Advanced Thinker

Stage 6- The Accomplished Thinker

Stage 1

- Unreflective thinkers are largely unaware of the determining role that thinking is playing in their lives and of the many ways that problems in thinking are causing problems in their lives.
- Unreflective thinkers lack the knowledge that high quality tinkering requires regular practice in taking thinking apart, accurately assessing it, and actively improve it. In fact, unreflective thinkers are largely unaware of thinking as such, hence fail to recognize thinking as involving concepts, assumptions, inferences, implications, points of view, etc.

Stage 2

- Thinkers move to the “challenged” stage when they become initially aware of the determining role that thinking is playing in their lives. And of the fact that problem in their thinking are causing them serious and significant problems.

From being an unreflective practitioner the unknown reflective thinker, the reflective practitioner can move to being challenged thinker. This is where they’re starting to think a little bit more and understand the role that thinking plays in their lives. At this point the thinker the teacher is starting to identify problems and trying to understand the causes of problems.

Again, it's involving standards for the assessment of thinking, clarity, accuracy, precision, relevance, logicity etc., a level of clarity and which the unreflective thinker wouldn't have.

Through the process of thinking that the reflective practitioner is starting to get a feel for certain standards of teaching and to understand that they are not meeting those standards.

Challenge thinker is also developing some understanding of the role of self-perception and self-concept for themselves as a teacher and the impact that this has on teaching.

At this stage the thinker develops some reflective awareness of how thinking operates for good or ill. At stage two also the reflective practitioner is not really thinking about the impacts of the teaching process on the students.

Stage 3

Those who moves to the beginning thinker stage are actively taking up the challenge to begin to take explicit command of their thinking across multiple domains of their lives. Thinkers at this stage recognise that they have basic problems in their thinking and make initial attempts to better understand how they can take charge of and improve it.

- At stage three we have got the beginning thinker, this is where the reflective practitioner is starting to develop a level of competence as a thinker.

Stage 4

- Stage 4's aspect of thinking is much more developed where the reflective practitioner is taking more charge more responsibility of the thinking process itself
- They not only recognize that problems exist, but they're beginning to think about those problems within a bigger context not just from a personal point of view but from a whole school point of view.
- Based on their need to practice regularly, they are actively analysing their thinking. However, since practicing thinkers are only beginning to approach the improvement of their thinking in a systematic way. They still have limited insight into deeper levels of thought and thus into deeper level of

Stage 5

Thinkers at this stage have now established good habits of thought which are "paying off."

Based on these habits, advanced thinkers not only actively analyse their thinking in all the significant domains of their lives, but also have significant insight into problems at deeper levels of thought.

Stage 6

- The accomplished thinker, this is where thinking is very technical it is actually very natural for this level of thinking.
- They are able to monitor and revise and adapt very quickly in relation to experiences in the classroom.

- This leads the reflective practitioner to be able to strategize very quickly in order for continual improvement to occur.
- It is conscious, but it is also very intuitive, it feels very natural for the thinker at this point.
- This level can only be achieved with experience and you would not expect a new teacher to be able to reach this level of critical criticality until they've had some number of years of experience.
- Therefore, it's linked very much to their lives as teachers, the experiences of teachers and the range of experiences that they have the problems, that they are confronted by.

Critical Thinking

- Is not a matter of accumulating knowledge- a person with a great memory and a good memory and who knows a lot of facts is not necessary good at critical thinking.
- Is being able to deduce consequences from what is known, to solve problems.
- It's not a matter of just building up knowledge of acquiring knowledge over time it's very much about the teacher starting to make connections and to develop an understanding of consequences from actions in order to solve problems.
- It's not about having a good memory it's about having a good ability to reason and to understand.

Someone with critical thinking skills is able to do the following:

- Understand the logical connection between ideas.
- Identify, construct and evaluate arguments
- Detect inconsistencies and common mistakes in reasoning.

- Solve problems systematically.
- Identify the importance and relevance of ideas
- Reflect on the justification of one's own beliefs and values
- Is very important in the new global knowledge economy

We can see then that critical thinking is actually a very important thing, the 21st century needs people who can think much more globally and with a greater understanding and application of knowledge, and the teacher is somebody who has a significant impact in this new global thinking.

- The teacher has to be able to deal with changes quickly and effectively
- The modern era places demands on flexible intellectual skills and integrate diverse sources of knowledge

This new 21st century global perspective about critical thinking and therefore about reflective practice, does suggest that what we need as teachers are very flexible thinking strategies, flexible approaches to understanding things and intellectual skills which can adapt and change, integrate and also be very diverse in terms of using sources.

- Enhances language and presentation skills
- Thinking clearly and systematically can improve the way we express our ideas

Critical thinking also enhances language skills, presentation skills, being able to think clearly and systematically does involve the need to express ideas to ourselves and to others.

Critical thinking:

- Promote creativity
- Critical thinking plays a crucial role in evaluating new ideas, selecting the best ones and modifying them if necessary
- Critical thinking is crucial to self-reflection.
 - Teachers need to justify and reflect on their values and decisions.

➤ Critical thinking provides the tools for this process of self-evaluation.

- Critical thinking is a rational and intentional cognitive skill

We use critical thinking to take a decision and solve a problem

The key characteristics therefore about rationality, about integrity, about reflection and about being able to evaluate.

- Critical thinking follows rules of logical and scientific reasoning
- It is about scientific reasoning to an extent, it's also very logical and part of being intentional then is about seeing how things can combine to make sense for us as thinkers how do all the connections come together for us as a critical thinker.

A Scheme for Critical Thinking and RP

Venn Diagrams and The Reflective Practitioner

- It is possible for the reflective practitioner to frame Venn Diagrams as a tool for thinking about practice.
- A simple Venn diagram for RP only has 2 circles. This represents the structure of professional learning.
- The diagram presents a concrete experience and the teacher conceptualizing the meaning of that experience.
- We can see in the diagram above, on the left-hand side the concrete experience and there can be a number of elements that the teacher could include in this part of the Venn diagram.
- The circle on the left represents “self-structure”. We can more easily understand this as the learning already held by the teacher from previous experience and even perhaps from formal professional learning. This area represents the entire perceptual experience of the teacher.

- It also of course represents that their values and their beliefs and the assumptions that the teacher has about self, others and society and about teaching and their role as a teacher.
- in this field, we can see that, for example, there are three elements of internal understanding- labeled here as '1', '2', and '3'. Of course. in reality there would be more than 3. And for any specific aspect of critical thinking and reflective activity, these would be very focused on only certain aspects of the individual teacher's self-concept.

For example:

- Element 1 may be the teacher's own competence with algebra, element 2 could be the teacher's own feelings of low self-confidence doing algebra. And element 3 may be the teacher's ability to communicate simply the ideas and mechanics of algebra.

The teacher, if they were teaching elementary level mathematics could be reflecting on elements of their own competence in terms of the subject that they are teaching. I was considering algebra at this level and maybe the teacher needs to think about themselves and their own knowledge about algebra.

Element two could be the teachers own feelings of confidence in relation to that subject area. Element 3 would refer to the teachers' feelings and ability about communicating that knowledge to the students so very much based in the concrete experience.

Next consider the right field

- The circle on the right represents 'experience' which is something the reflective teacher draws from action in the classroom.

On the right-hand side, we can actually see a number of elements and these refer to aspects of how the teacher is conceptualizing the process drawing on action in the classroom.

- It could be about the teacher trying out new things with students, or teaching something for the first time.
- It represents the immediate field of sensory experience-direct observations of occurrences as well as the feelings generated by those occurrences as such it represents all that is expected by the teacher and it will be very fluid in content because experience is ever-changing.

It represents a lot of immediacy and therefore, generates a lot of feelings associated with that and as such it can be very fluid, it can of course contain many more elements.

- In this field there are three elements of experience for this example- of course, there could be more. For example element 4 might relate to pupil's misbehaviors during the algebra lesson, element 5 could be the effectiveness of the use of hands-on manipulatives in the lesson, and element 6 could be the pupil's inability to understand the technical language used in the lesson.

In this field there were there could be i.e., elements **about** the student's behavior or about how the teacher is using materials, resources to teach the concepts in algebra or it could be about the students' inability to understand the technical language which is included in this subject.

Now consider the overlap field:

- where the left circle, relating to self structure, and the right circle, relating to experience, come together and interact
- It is important to be aware that experience will impose on us and affect us, but also our own internal frame of reference, our values, beliefs and assumptions will impact on our perceptions and processing of the experience.

Where the two circles intersect, where the concrete experience and the conceptualization come together is another field for the teacher to explore. In this area, this is where the conceptualization the teachers thinking about themselves how they working, comes together with the experience that they have in the classroom. So, it's where the teachers values and beliefs about what they are doing and about their own role as a teacher links to the concrete experience of the students.

- We had a number of elements 7 in this part of the diagram and these can relate to an increased lack of confidence with algebra from the experience due to their own feelings, because of the reactions of the pupils element 8 could be about an increased understanding of how to use resources to teach subjects as a positive attribute.
- Understanding of how manipulatives could aid the teaching of algebra and make the subject easier to access and more fun for pupils, and element 9 could be improved ability to communicate algebraic understanding based on the experience of which mathematical language pupil's understood most easily.
- Venn diagram represents the process of reflection and provides a scematic format for critical thinking.
- It is important to note here that this diagram is only an example. There could be situations where you might have more than 2 intersecting circles.
- So, Venn diagrams give a very individual approach for the teacher to think about their practice of teaching and to deconstruct it in a very clinical and scientific way.
- Venn diagrams can be a very useful graphic organizer which orders complex relationships visually; they allow what is abstract to be more visible.

Bloom's Taxonomy and Critical Thinking

How does Critical thinking fit into Bloom's Taxonomy?

The cognitive Domain comprises

- Knowledge
- Comprehension
- Application
- Analysis
- Synthesis
- Evaluation

Knowledge

- Knowledge of specifics
- Knowledge of terminology
- Knowledge of specific facts
- Knowledge of ways and means of dealing with specifics
- Knowledge of conventions
- Knowledge of trends and sequences
- Knowledge of classifications and categories
- Knowledge of criteria
- Knowledge of methodology
- Knowledge of principles and generalizations
- Knowledge of theories and structures

Comprehension

- Translation
 - Interpretation
-

- Extrapolation

Application

- The use of abstractions in particular and concrete situations.
- The abstractions may be in the form of general ideas, rule or methods.

Analysis

- Analysis of elements
- Analysis of relationships
- Analysis of organizational principals

Synthesis

- Production of unique communication
- Production of plan or proposed set of objectives
- Derivation of a set of abstract relations

Using Bloom's Taxonomy for Critical Thinking

- The Cognitive Process Characterized as Essential to Higher-Order Thinking (that is, analysis, synthesis and evaluation in the unrevised taxonomy) are integrative of the basic critical thinking concepts.

A critical thinker can think for following aspects

- Assumptions
 - Facts
 - Concept
 - Value
 - Conclusion
-

- Relevance
- Evidence
- Premise
- Irrelevance
- Inconsistency
- Fallacy
- Inference
- Prejudice
- Bias
- Prejudice
- Consistency
- Implications
- hypothesis

Techniques for Critical Thinking

Here are 16 basics techniques of critical thinking:

1. Clarity

- State one point at a time. Elaborate, give examples, ask others to clarify or give examples.
- If you are not sure what you are talking about you cannot address it.

For example:

Unclear: “How can we fix education?”

Clear: “How can teachers better prepare students for the future?”

2. Be Accurate

It's important for the critical thinker to be accurate.

For example:

Inaccurate: “Most students don’t know how to learn”.

Accurate: “**Most** Students in Pakistan need to develop thinking skill”.

3. Be Relevant**4. Know Your Purpose**

It's important for the critical thinker to know the purpose of what you are trying to accomplish?

What is the most important thing here? Distinguish your purpose from related purposes?

5. Identify Assumptions**6. Check Your Emotions****7. Empathise**

- Try to see things from others’ perspective, imagine how they feel. Imagine how you sound to them.

8. Know Your Ignorance

- You still might be wrong. Educate yourself as much as possible, but still be humble.

9. Be Independent

- Think critically about important issues for yourself. Don’t believe everything you read.
- Don’t confirm to the priorities, values and perspectives of others.

10. Think Through Implications

Not thinking through Implications: “Not planning my lessons in a particularly good fashion”.

Thinking Through Implications: “If the lessons are unplanned then there are implications to that the implication is that the lesson won't go very well and perhaps the students won't learn in that lesson and therefore not achieve the outcomes in the curriculum”.

11. Know your own Biases

Another thing for the critical thinker to consider is a particular policy. It is that they have about education and about teaching and learning.

For Example:

Biased: “I believe all students should play sport”

Unbiased: “The evidence linking physical activity to academic achievement is compelling. I will have to think it through before I can say whether it has an impact on the way I teach.”

Bias might be that you believe that students should play sport whereas it would be much better from an unbiased point of view to consider how sports activities may in fact have some influence over academic achievement

12. Suspend Judgment

- Do not make a decision and then use critical thinking to back it up.

Immediate judgement: “We are here to promote the school's plan for curriculum reform. What arguments can we construct in its favor?”

Suspended judgement: “What do we want from our curriculum? Let us use critical thinking to find the best ways to do this.”

13. Consider Other Point of View

14. Recognize Cultural Assumptions

15. Be fair, not selfish

The 5-Step Model of Critical Thinking

There are five step model of critical thinking. Let's look at each of these steps in turn.

Step 1

- Determine goals/objectives
- Define behaviours/skills you need to exhibit.
- Target those behaviour

Step 2

- Develop appropriate questions
- Engage in discussion with colleagues

Step 3

- Practice before you assess
- Choose activities that promote active learning.
- Utilize all components of active learning.

Step 4

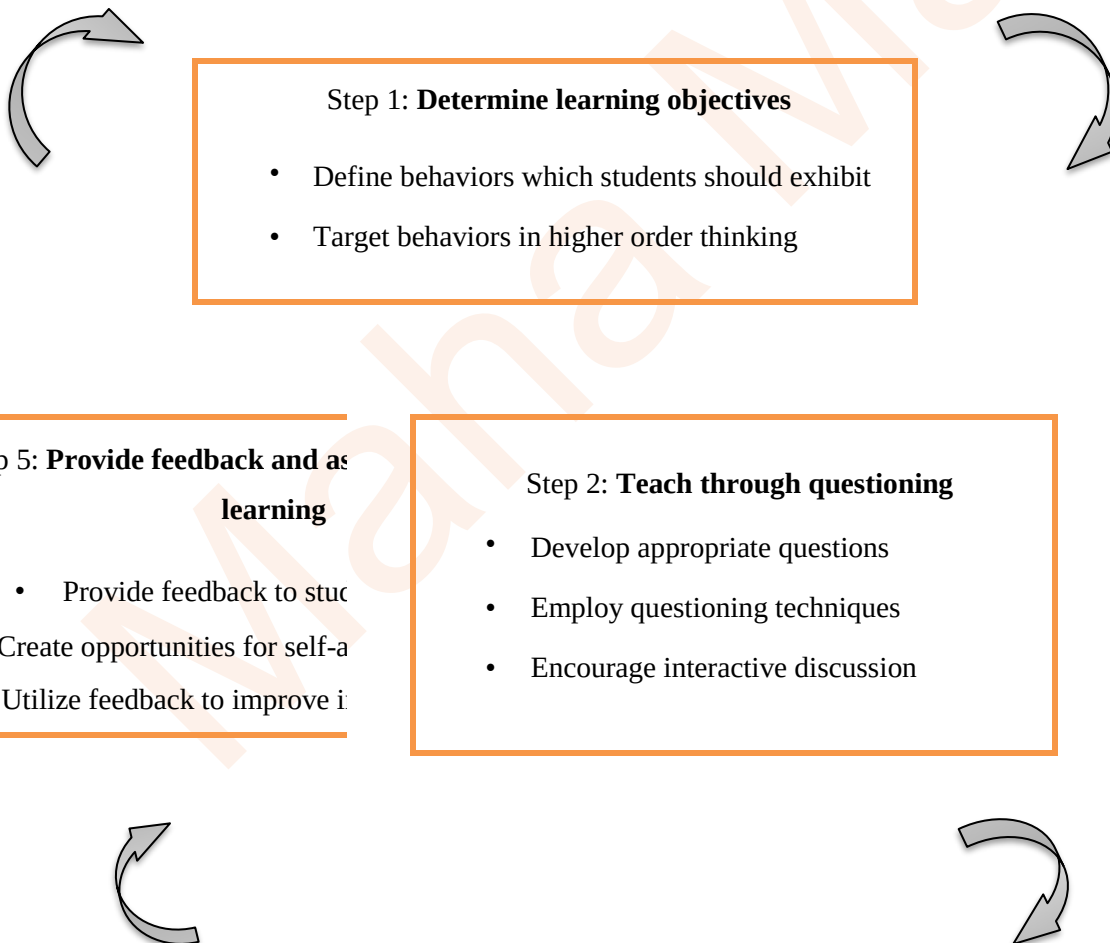
- Review, refine and improve
 - Monitor activities in the classroom
-

- Collect feedback from students and other

Step 5

Gather and review feedback and assessment of learning.

- Review feedback
- Create opportunities for self-assessment
- Utilize feedback to improve instruction



Step 4: Review. Refine and improve

- Monitor class activities
- Collect feedback from students

Step 3: Practice before you assess

- Choose activities that promote active learning
- Utilize all components of active learning

Halpern's Framework For CT

Halpern proposes a framework based on different kinds of critical thinking:

- Thinking and language
- Argument analysis
- Hypothesis testing

Looking at particular theories around why things happen and finding ways to test them out and see whether they are true.

- Dealing with probability and uncertainty
- Decision-making
- Problem-solving
- Creative thinking
- Memory skills

Thinking and Language

- **Recognising persuasive language**
-

- Mis-use of Definitions
- Leading Questions
- Paraphrasing
- Using multiple representations

Argument Analysis

- Identifying premises, counter-arguments and conclusions
- Judging credibility of information
- Difference between opinion, judgment and fact
- Avoiding common fallacies

Creative Thinking

- Re-defining a problem
- Generating possibilities
- Seeing multiple perspectives
- Taking risks

Hypothesis Testing

- Predicting and confirming hypothesis
- Operational definitions
- Controlling variables
- Sampling and generalizations
- Limits of correlational reasoning

Dealing with Probability and Uncertainty

- Estimating probabilities
- Risk assessments
- Using baseline data

Decision-Making :

- Seeking disproving evidence
- Effects of memory on decision-making

Problem Solving

- Defining and redefining problem situations
-

- **Setting goals**
- **Generating solutions**
- **Using alternatives representations**
- **Recognizing constrains**
- **Evaluating solutions**

Critical Incident Analysis

CIA helps teachers focus on the extent to which critical analysis is evident in their practice. The term Critical incident refers to any work teachers do that involves analyzing situations, reflecting on past experience making judgement and decisions, and taking actions without the benefit of a standard protocol or uniform response that takes care of each and every problem they encounter.

So, let's take a look at some of this particular step:

- As a teacher you should ideally do a weekly audit to understand more about your practice.
- In particular, it will help you to understand the assumptions that underpin how you analyse situations, make decision and take actions.

Think back over the last week:

Identify the critical incidents that have happened during that time.

A critical incident is an event that can be called to mind easily and quickly because you remember it vividly.

We usually consider events to be critical when they are unexpected a surprise .

They can be positive highs or demoralizing Lows,

Examining Your Personal Filtering System

The meaning we attribute to our experience is influenced by various factors that effectively screen out some responses while letting others through.

- The screening process leads to differing perceptions of circumstances and events ,resulting in different interpretations,and subsequently in different responses.

SITUATION

The way we respond to circumstances and situations is determined by this personal filtering.

Conceptual Learning: Learning through Conceptual Innovation

It involves:

- Seeing farther
- Understanding deeper
- Making the right decisions

It's not about:

- Working harder
-

- Getting more resources
- Increasing our collection of methods and tools

Its more about the reflective practitioner

- Re-thinking what is already known
- Renewing the concepts we use for understanding
- Getting wiser

For the reflective practitioner, conceptual learning is a process of rethinking or in other words looking beyond what is already known **and then renewing** concepts in relation to that thinking and we use concepts in order to understand situations with greater clarity. For the rich reflective practitioner this is very much then about becoming wiser in relation to the job role.

Benefits of a Concept-Based Model

So, what we can do is think about the benefits of conceptual learning or learning based on a concept-based model.

- **Facilities “synergistic thinking” – the interplay between factual and conceptual levels of thinking.** Synergistic thinking is about linking facts with concepts rather than keeping them isolated)
 - **Requires deeper intellectual processing as students relate the facts to key concepts and principles.** It requires intellectualizing situations, thinking much more deeply about processes that students are going through in terms of learning facts and relating facts and key concepts and principles.
 - **Develops conceptual structures in the brain to relate new knowledge to prior knowledge.** It's very much about developing a structure for thinking conceptual frameworks in order to explore and understand what we're actually seeing.
-

- **Illuminate the patterns and connections of knowledge.** It is based on an approach which attempts to illuminate or in other words give us much more clarity and to understand the patterns and the Connections between pieces of knowledge.
- **Facilitates the transfer of knowledge at the conceptual level.** A concept-based approach, concept learning helps the reflective practitioner to facilitate the transfer of their knowledge that they are acquiring into other contexts and other situations in which they work.
- **Provides opportunities for personal meaning- making through processes of thinking, creating and reflecting.** This provides opportunities for the reflective practitioner to create meaning to develop a personal understanding of our experiences in relation to the theories that we understand and the practices that we employ

Technical and Conceptual Learning

We are more trained in learning new methods and techniques than for learning new concepts. In terms of conceptual learning as a practitioner, we tend to be more focused on the technical aspects, the techniques and the methods that we use rather than acquiring and understanding new concepts. One reason for this is that new methods impact on productivity almost immediately. This is primarily because of teachers, if we're focused on the technical aspect of our role as teachers then we can see the immediate impact of what we're doing and there's a sort of productivity in relation to the actions that we take.

But by limiting our learning to technical issues:

- **We are not flexible enough when we are confronted by complex situations.** From the critical thinking point of view, we're actually limiting ability to act in very flexible, adaptable and highly productive ways because we're focused on techniques and methods rather than whether those techniques and methods are actually productive.
 - **We and our techniques, get out of date in a short period of time.** Therefore in terms of the techniques we can, as teachers become quite outdated or stuck in the ways of teaching year after year without change.
-

- **We get 'over-learned' - in a few years, our curiosity decreases.** What this actually means is that the quality of the teaching actually begins to reduce in a way the techniques, the approaches over learned and become innocence meaningless.

According to Dewey, Schon and others' concepts are **cognitive tools for coping with the world, and for solving problems.**

What Is Conceptual Learning?

Conceptual learning is a process by which the reflective practitioner needs to learn how to organize **information in logical mental structures** and organize knowledge rather than just acquire and store knowledge.

- Conceptual learning therefore focuses on organizing that knowledge and organizing principles into meaningful whole
- **Conceptual learning is a catalyst for challenging ourselves to think at more advance levels** and it is very much a way that the reflective practitioner can challenge their assumptions and find ways to change and innovate.

Inductive Vs Deductive

Conceptual learning then can move in two directions it can be inductive or it can be deductive.

Inductive Learning

Inductive learning is the process of learning and reasoning from detailed facts to general principles.

Deductive Learning

Deductive learning is a process of learning and reasoning general principles (concepts) towards detailed facts.

Conceptual Learning: Concepts and Frames

Conceptual learning relies on concepts and frames. We are more trained in learning new methods and techniques rather than learning new concepts and this is essentially about productivity rather than immediacy in learning.

Concepts determine the way we **perceive** something and the way situation is **framed** rather than just the facts of what we're actually seeing.

Donald Schon

This is exemplified by Schon, who says “while the given situation can be conceived in a variety of ways it is always **concept-structured** situation”

What is saying is that although we can see facts, we can observe the world. What we actually perceive is the based in the concepts that we hold. So, concepts enable **perception and understanding**. It moves beyond observations and facts.

The process of concept formation has three significant phases:

Perception

Experiences or learning in any form is the starting point of the process of concept formation. The experience of learning that we have and in terms of the concepts that we hold what are we actually seeing? how do we see the world? and how is that information going to be viewed by us?

Abstraction

The mind analyses the perceived images and synthesis what is common to all, neglecting what is not needed. Beyond perception is abstractions. Our minds attempt to analyze the message that is perceived and tries to

make sense in relation to the experience and the knowledge that we already have. It attempts to synthesize the new information coming in with the old information that we already have.

Generalization

After making such observation in the form of abstraction for a number of times we are able to form a general idea about common. what we have to do as reflective practitioners is make sense of that information in terms of what we already know and what we have to do is to make generalizations of information we have because what that will do is to develop new concepts. The concept that we already hold.

Conceptual Learning: New Realities

Every time, we're confronted by new situations, we have to deal with new experiences. This is an opportunity for us to make sense of them and we make sense of them based on old concepts that we hold.

- Once we detect something new, some new information what we tried to do is make sense of it in relation to the concepts - concepts that we already have. What we do is, we make analogies with other things that we already know and we do this by thinking back through the experiences we have and finding points of comparison and points of contrast.
 - Gradually as a reflective practitioner, we go through a process of extending and changing the concepts that we have. They start to evolve with the experiences that we have.
 - This means is that new concepts begin to emerge and new concepts can actually trigger new learning processes. For example: the assimilation or the incorporation of new concepts will begin to transform the way we think and the way we behave in any situation and what this means is that this will begin to have an impact on us and our world, the people around us and the whole world around us. With these new learning process and what emerges are new concepts. Concepts which have changed and therefore, we begin to perceive and understand things very differently.
 - New concepts emerge as new questions are made as we start to question differently the experiences.
-

- Conceptual innovation is like reframing, but deeper in other words, we start to look at a situation from a very different perspective.
- This re-framing is part of the new reality what we do then:
 - we improve our understanding of a situation.
 - We see things more deeply more clearly and it is also then an opportunity to solve the conflicts that we had previously
 - begin to find new strategies in order to dealing with situations that we face.
- Conceptual innovations and these changes give a chance to re-think the whole approach that we have and also our values beliefs and assumptions as teachers
- This actually does for us - is lead us to a point where there are many new opportunities many new possibilities that we can begin to engage with as reflective practitioners because
- This new reality will provide for us is an opportunity for a **Paradigm Shift**.

What Is Paradigm Shift?

Paradigm shift is a complete change in perspective on how the world is viewed on how situations are perceived and our roles and actions within those situations as they occur and after they occur.

A Critical Thinking Framework

Conceptual Learning: Paradigm Shift

Paradigm Shift

A paradigm shift brings a new set of concepts:

- From things to people
- From product to process

The paradigm shift brings about new set of concepts for us; concepts from things to people and concepts from products to process. A paradigm shift is a way to view things differently; from one point of view to a

different point of view, from one perspective to another perspective and therefore, leads to a very different frame of perception.

A paradigm shift could be

- From teacher to student
- From standardizing to managing diversity

Paradigm shift should be from teacher to student or from standardized approaches to managing diversity in the classroom rather than looking at a situation from your own point of view as a teacher. A paradigm shift would be looking at a situation from the students' perspective. Or it could be instead of looking from a school point of view at what's happening in the classroom to a very unique individualist perspective from yourself and your students in the classroom in terms of what's learning.

A paradigm shift could be

- From replication to inspiration
- From controlling to managing change and uncertainty

It could be moving from replication. In other words, just repeating strategies which you have been taught in which everybody else is using to some sort of different approach, perhaps inspirational, testing different ways of doing things in the classroom.

It could be about moving from controlling the students in the classroom to actually managing change and managing uncertainty which is very different way of working with things.

We can represent the paradigm shift for the reflective practitioner in the following diagram and it's a very simple diagram to review. It has three components:

- New understanding
 - Paradigm Shift
-

- Emergent knowing

There's not necessarily a starting point but we can think of the paradigm shift as the change in perspective, the different point of view as leading to new understanding for the reflective practitioner. A new way of doing things, a new way of looking at things. And, this new understanding, this new way of seeing and doing can actually lead to a different way of knowing what things are about or in other words, emergent knowing. Because this is conceptual learning what that can do as it builds up overtime and lead to another paradigm shift which can of course lead to new perspective, new ways of seeing and doing in the classroom. So, it becomes a cycle.



Different kinds of learning

The paradigm shift leads to different kinds of learning, and we can think of three different kinds of learning.

Single-Loop Learning (following the rules)

- Rigid strategies
- Spend time detecting and correcting deviations from the 'rules'

Double-Loop Learning (changing the rules)

- Reflect on whether the rules themselves need to change
- Requires 'thinking outside the box'

Triple-Loop Learning (learning about learning)

- Learning how to learn
- Searching for meaning and understanding

A Critical Thinking Framework: Conceptual Learning: Theories-in-Use or Espoused Theories

Theory –in-Use

- Theory-in-use is embedded in the logic of the action, it is the theory that commands the thinking of the action.

Espoused Theory

- Espoused theory is that theory used for arguing about the actions, it is the theory used for explaining the action to others but not necessarily for conducting the action.
 - Espoused theories are explicit but most theories in use are tacit.
 - How to combine the espoused theories and experience for creating solutions?
 - Reflective practice is about ownership of knowledge
 - Reflective practice is about awareness of the knowledge we use, how we use it and how we can improve our action in real time.
-

Reflective practice helps us do this. Reflective practice is very much about ownership of the knowledge that we generate from **theory-in-use and an** espoused theory. And, it's about awareness of that knowledge and how we use it in order to improve what we're doing in the classroom and to improve on the actions which we are using in order to teach.

- Reflective practice is about how our minds work and how we use and create theories in practical situations.
- It is about invisible and visible, tacit and explicit, blindness and sight.
- Reflective practice is about flexibility, adaptation and effectiveness.

Reflective Practice = Evolving Brains

- If our theories-in-use are stagnant so are our brains
- If we reduce our learning to the methods and techniques, our knowledge goes out of date in a short period of time.
- If we can renew our theories (espoused and in-use) and the conceptual frameworks that are behind them, we will be able to dialogue with the disruptive changes that the future will bring to our roles in the classroom.

Frameworks for Reflection: Reflective Writing

What Is Reflective Writing?

- Reflective writing is evidence of reflective thinking:
 - Looking back at something (something that happened, or an idea or a project)
 - Analyzing what happened (thinking in depth and from different perspectives)
 - Thinking carefully about what it means for you as a professional

It is therefore an aid for the teacher for the reflective practitioner to think. It's a thinking tool and this is very much about helping the reflective practitioner **develop a** professional stance as a teacher in the classroom.

Reflective writing is much more personal

- Reflecting writing is more personal than other kinds of academic writing.
- We all think reflectively in everyday life but not to the same depth as that expected in good reflective writing for a reflective practitioner.

We already know that reflective writing:

- Describes
- Links theory to practice
- Questions experience
- Has a focus on improvement

Example of Basic Reflective Writing

Specific tasks were shared out amongst members of my team. Initially, however the tasks were not seen as equally difficult by all team members. Cooperation between group members was at risk because of this perception of unfairness. Social interdependence theory recognizes a type of group interaction called ‘positive interdependence’ meaning cooperation (Johnson and Johnson 1993, cited by Maughan & Webb, 2001). And many studies have demonstrated that ‘Cooperative learning experiences encourage higher achievement’ (Maughan & Webb, 2001). Ultimately, our group achieved a successful outcome, but to improve the process, we perhaps needed a chairperson to help encourage cooperation when tasks were being shared out. In future group work, on the course and at work, I would probably suggest this.

For example, social interdependence theory recognizes a type of group interaction called positive interdependence. And at this point, you can see the reflective practitioner is beginning to analyze and to think deeply about the occurrence. This is the interpretation aspect of **reflective** writing and the writing goes on at the end, ultimately our group achieved a successful outcome but to improve the process, we perhaps needed a chairperson to help encourage **cooperation**. And so, at this point in the reflective writing you can see where the teacher is actually looking at outcomes from the reflection. So, in this example, you can see the three components of the reflective writing process and these components (Description, interpretation, outcome) are very important to consider when you're actually using writing as a tool for reflective practice.

- So, you can see that reflective writing is a far greater focused activity than simply writing in a journal.
- Description
- Interpretation
- Outcome

Frameworks for Reflection: A Structure for Reflective Writing

- Reflective thinking especially if done in discussion with others can be very free and unstructured and still very useful.
- Even reflective writing can be unstructured, for example, when it is done in a personal diary.
- Reflective writing, however, is normally carefully structured.

Stages of Reflective Writing

Description

- What happened?
- What is being examined?
- Keep this part SHORT!

Interpretation

- What is most important/interesting/useful/relevant about the object/event/idea?
- How can it be explained? (e.g., with theory? from literature on the subject?)
- How is it similar to and different from others?

Outcome

- What I have learned from this?
 - What does this mean for my future?
-

Remember the following 4 key points

- Reflection is an exploration and an explanation of events not just a description of them.
- Genuinely reflective writing often involves revealing anxieties, errors and weaknesses as well as strengths and successes. This is fine (in fact, it is often essential!), as long as you show some understanding of possible causes and explain how you plan to improve.
- It is necessary to select only the most significant parts of the event or idea on which you are reflecting.
- Do not try to 'tell the whole story', this would be description rather than interpretation.
- It is often useful to 'reflect forward' to the future as well as 'reflecting back' on the past.

Frameworks for Reflection: Vocabulary for Reflective Writing

- The following are a few suggestions for words and phrases that might be useful in reflective writing.

Description

- Because the range of events, ideas and objects can be so great it is not possible to suggest specific vocabulary to help you reflect.

Remember:

- We tend to talk of events using the past tense
- We tend to discuss theories or models using the present tense

WARNING!

- Using only these words and phrases will not make you a good reflective writer.

So, reflective writing can be quite straight forward, if you give yourself some nice language than simple language structures to use and not to get bogged down too much in the spelling of the grammar. What's important is the fact that writing is reflective and by using particular vocabulary, particular words, that can help your thinking **processes** much more.

- What I would say is that these are just examples and do not just rely on these only. As I said earlier, use them as a starting point and develop your own writing style as a reflective practitioner. The
-

important thing is to remember that what you will be doing is supporting your thinking and not just writing a diary.

Vocabulary for Reflective Writing

The following are a few suggestions for words and phrases that might be useful in reflective writing.

WARNING!

Using only these words and phrases will not make you a good reflective writer!

This is just a guide.

Whatever language you use, make sure that you (1) describe; (2) interpret and (3) reflect on outcomes.

There are three stages to reflective writing:

- Description
- Interpret
- Outcome

Reflective Writing: Limitations of Self-Reflection

Limitations of self-Reflection

- Hebermas (1974) suggested that reflection requires a level of detachment and objectivity which can be confused by self-deception.

Let us look at an example of this to put it into perspective.

Dr. Brown is worried about his class. He knows that if he doesn't stick to the time allocated for each section he will run out of time and the material at the end would not be covered.

- What can happen in terms of reflection is that as a there's a difference between the teacher's perception of what happened and what really happened in the classroom.

	Perception	Reality
Problem	Poor time management	Can't relinquish control
Reflection	Question reduce time to cover material	Time management becomes the problem & the disguise
Solution	Better lesson plans: don't allow question	Not discovered via reflection

- The concern is, then, that there may be much that you keep from yourself (consciously or otherwise),...
- A further concern relates to 'memory bias'

Reflective Writing: Strategies for Overcoming Barriers

- There are always barriers before, during and after any reflective practice.
- Some of these barriers are self-imposed. Others originate within an institution.

Lack of Time and Space to reflect

- Use the journey to and from your workplace is a chance to reflect.
 - Using the first 20 minutes after your learners have left for the day while events are still fresh in your mind.
 - Jotting down notes in a journal to reflect on when you have time later.
 - Talking with a trusted colleague over lunch.
-

- A group discussion with several colleagues at the beginning or end of week – perhaps this could be scheduled into the timetable?

Negative Ideas about Reflection

- It is common to find negative preconceptions about reflective practice.
- It can be a difficult and time-consuming process.

Organizational Culture

- Some organizations are supportive of reflective practice and your management might model reflective behavior. Others may not.
- At the level of the individual practitioner the aim of RP is not about changing the culture of your organization but about changing your own practice and skills. Do not be put off by a culture which does not embrace this.
- Be resilient!

Fear

- RP can be an intimidating idea, as it requires a critical, honest and open view of yourself. This can be difficult.

Lack of Knowledge and Experience

- This can be a problem for new and experienced practitioners alike.
- You need practice to be able to do it well.
- Re-read about some of the theories discussed about RP and use some of the techniques to practice.
- See what works for you.

Seeing it a success or failure

- This can be a problem for new and
- You should not see the work you do with learners as either a success or a failure.

Anxiety about introspection

- Reflective practice is a critical process, and looking for an area to develop and improve on can be disturbing.

RP as an uncritical routine

- RP should never be done as a simple 'tick box' activity.
- Real change comes from hard work and determination to improve.
- It would be better for you to reflect less and do it right than go through the motions every day without thinking about it.

Frameworks for Reflection: Developing Competences: Professional Competences

Professional competence can be categorized into three broad headings;

- Professional values and practice
- Professional knowledge and understanding
- Professional skills and application

1. Professional Values and Practice

- The reflective practitioner should demonstrate a commitment to a personal code of values.
 - What would you consider to be core values for the teaching profession?
 - Trust
 - Respect
 - Integrity
 - Honesty
 - Fairness
-

- Tolerance
- Commitment
- Equality
- Service

- A commitment to serve lies at the heart of professional behavior.
- Teachers need to exemplify these values in their relationships with their students.

In applying their values in a commitment to learners, reflective practitioners must:

- Exemplify these values in their relationships with their students.
- Motivate and inspire students with a view to helping each realize his or her potential.

And therefore, there is a particular commitment that we need to consider here. There is a **commitment to the learners** by the reflective practitioner. Because the reflective practitioner must be somebody who can motivate and inspire students and these competencies which we've just listed; integrity, honesty will help us with this need to motivate and inspire other people.

In applying their values in a commitment to colleagues, reflective practitioners must:

- Work with colleagues to create a professional community.
- Ensure that relationships with parents are characterized by trust and respect.

There's also a **commitment to colleagues** as well as learners as well as students. And in relation to this commitment to our colleagues, what we need to think of in terms of professional competence is our commitment to the educational community and the relationships that we have with people within that community. This is where the most significant competencies of things like trust and honesty, I think become more important than any others.

In applying their values in a commitment to profession, reflective practitioners must:

- Contribute to the continual review of policies and practices.
 - Assume responsibility for on-going professional development
-

Frameworks for Reflection Developing Competences: Professional Competences (Part 2)**Professional Knowledge and Understanding**

- The reflective practitioner must enhance knowledge and understanding of the subjects they teach.

So, we've looked at developing professional values as a competence. Another competence, we need to look at is that of knowledge and that of understanding. And in this regard, the reflective practitioner needs to always be enhancing, building on their knowledge and understanding of the subjects they teach. There's a particular commitment here for the teacher.

- The reflective practitioner must develop an understanding and knowledge of social and policy contexts for education.
- The reflective practitioner must develop knowledge and understanding of a range of strategies to promote and maintain positive behavior.
- The reflective practitioner must develop knowledge and understanding of factors that promote and hinder effective learning.
- The reflective practitioner must enhance skills for communicating effectively with students and colleagues.
- The reflective practitioner must develop knowledge and understanding of how to use technology to support student learning.
- The reflective practitioner must understand the interrelationship between schools and the communities they serve.
- The reflective practitioner must develop a knowledge and understanding of students social and community contexts and address the implications for learning arising from these.

Developing Competences: Professional Competences**Professional Skills and Application**

In planning and leading the reflective practitioner must set appropriate learning objectives/outcomes intentions, taking account of what students know, understand and can do.

- **Manage workloads effectively and efficiently and maintain a work/life balance.**

In terms of planning and leading we can also think in terms of how the reflective practitioner is achieving the curriculum aims. Any teaching has a particular workload associated with it. And therefore, the reflective practitioner needs to think about how they are managing to get through the curriculum and manage all the work of the school the administrative duties and also have a personal life. The reflective practitioner is somebody who manages to get a good work-life balance.

- **In terms of this professional competence relating to skills and applications we can also look specifically at teaching and learning.**
- **The reflective practitioner also needs to use a broad range of teaching strategies in their teaching.**
- **In terms of teaching and learning the reflective practitioner also needs to consider the strategies that they using throughout the course within every single lesson to motivate the students.**
- **The reflective practitioner needs to set very clear standards of behavior and needs to promote these in the classroom in order to make sure that the psychological the emotional space is conducive to learning.**
- **In terms of professional skills and application**
- **The reflective practitioner's role at holistically in terms of assessment**
- **A reflective practitioner is consider which assessment procedures you will use .**
- **A reflective practitioner in terms of this competence area of skills and application**

Developing Competences: Professional Competences

Dimensions of Development

- **The reflective practitioner needs to consider are different dimensions of development. And we need to think about the progress of careers as professionals.**

The challenge is that can entail the opportunities which can go along with that and our own **expectations** associated with all of this. This means that as we progress through this professional life but **our confidence** as reflective practitioners will grow and we will continue to learn from experience and use that **experience** to help us develop as quality practitioner and say what we needed to do is think about these dimensions or what are these particular dimensions of development that we can consider?

- **The reflective practitioner becomes much more sophisticated at reflective practice.**
 - **Reflective practice and with using a wider range of teaching strategies these will help us build up our experience and actually help us as reflective practitioners.**
 - **The** reflective practitioner the developed experienced practitioner somebody being very sophisticated is somebody who can expand. They're learning and expand their experiences and therefore expand their impact beyond the immediate classroom.
 - **This** ultimately will lead to much fuller and complex participation within the whole school. And of course, we've growing competence then there is increased autonomy and as a reflective practitioner autonomy means that we're much more self-assured and we have a much greater self-esteem as teachers.
 - **This** then gives us a much greater capacity for self-criticism. And also, self-improvement simultaneously.
 - **Then** if we're looking from a whole school protect perspective will help us consider the impact that this will have on our colleagues especially where we as reflective practitioners can enter into a mentoring situation or a coaching partnership with others.
 - **This is** where good practice gets shared in the school environment. And of course, there is also the opportunity for the reflective practitioner to model good practice. And then this helps to enhance the
-

quality of teaching across the whole school. So, at this point what we need to do is think about well what are the core values?

- **Which** underpin professional practice and encourage attitudes and conduct also in terms of what we're doing in the school. So, the competencies that we've been talking about the competencies which have been described. About values and about understanding and knowledge or will help to build a professional persona professional identity.

Developing Competences: Provisional Model

The Provisional Model 1

This model attempts to combine particular strengths as well as approaches for the reflective practitioner and at the heart of the provisional model are four key components.

Knowledge/ Cognitive Competence one component is knowledge and cognitive competence or in other words the work-based knowledge and experience that we built up the acquired knowledge that we build from experience.

Functional Competence

There's also functional competence, this is in relation to the reflective practitioner performing specific work-related tasks.

Personal Behavioral Competence

We can and also look at personal behavioral competence within the provisional model. This is about how the teacher is adapting to a work-related situation and the behaviors that they bring as a professional into the school environment.

Values/Ethical Competence

The model with values and ethical competence that we need to consider. this is about the teacher making appropriate correct judgments about their role and responsibilities within the school.

Metta competence

- **Communication**
- **Self-development**
- **Creativity**
- **Analysis**
- **Problem solving**

Meta Competencies			
Communication, Self-development Creativity, Analysis, Problem solving			
Knowledge/ Cognitive Competence	Functional Competence	Personal Behavioral Competence	Values/ Competence Ethical

Developing Competences: The Provisional Model

The Provisional Model 2

Developing competencies, we've looked at the provisional model. The provisional model made up of Metta competencies and core competencies. What we need to do is consider the components the individual components of these core competencies. And what we can do is look at each of these in turn.

The Provisional Model 1

Knowledge/ Cognitive Competence			
Tacit/practical- this knowledge linked closely to functional personal competences; linked closely to Schon's 'reflection- INaction'	Technical/theoretical- this relates to underlying knowledge bases including their application, transfer synthesis etc.	Procedural – this consists of the how, what, when etc. Of the more routine tasks of teacher	Contextual – this is general back ground knowledge about education / teaching

Reflecting Effectively

The reflective practitioner to be effective. Must have the right mental attitude for reflection. And if we look closely at Dewey's paradigm of reflective practice, what we can do is draw out three key attitudes, which make reflective thinking effective and these three attitudes are:

- **Open-Mindedness:** This is the ability of the teacher, the reflective practitioner to have an attitude which is questioning and not closed.

- **Responsibility:** It drives the process of reflection for the reflective practice
- **Whole-Heartedness:** The desire to improve and to change and to manage ongoing professional learning.

Perhaps the most important attitude that we can draw out from these three is open-mindedness. It refers to the ability to remain open to multiple alternatives possibilities, with an idea that we can always learn and improve and develop our skills as practitioners that the reflective process becomes truly effective. This means that the open-minded teacher continuously questions routines and practices their validity and value.

In order for these three attitudes to come together and become very powerful to improve effectiveness. What teacher need to do is actually look at what he/she know, what are their values and beliefs as reflective practitioners. And once we begin to think along these lines, it actually starts to underpin particular ways of knowing for the reflective practitioner, our values and beliefs are what, direct us as teachers and actually give us a focus for our approach and the ways of knowing actually then how purse bring together all these elements, these attitudes, and values and beliefs to be effective. Therefore the reflective practitioner does not believe in one single truth or in one right way to teach and develops 'ways of knowing' and 'patterns of reasoning'.

And we can look at different ways of knowing. For example, we can talk about **absolute knowing**. This is perhaps the most simplistic level where knowing is either about right or wrong, correct or incorrect. We can also talk about **transitional knowing** this is where we know that some knowledge is correct and other types of knowledge we need to explore or discovered.

Frameworks for Reflection		
Reflecting Effectively		
Ways of Knowing	Patterns of Reasoning	
Absolute Knowing Knowledge is either correct or incorrect. Correct knowledge obtained from authority.	Mastery Peers debate and quiz each other for mastery.	Receiving Peers talk to create a relaxed atmosphere.
Transitional Knowing Some knowledge is correct; other types of knowledge can never be discovered. Learner should understand rather than memorize facts. Not all knowledge comes from authority.	Impersonal Learner exchanges views with peers and express their opinions.	Interpersonal Learners collect ideas from others who provide exposure to new ideas.
Independent Knowing Most knowledge is uncertain so learner thinks independently. Values knowledge that does not come from authority.	Individual Learners focus primarily in thinking independently; exchange with others is of secondary importance, though appreciated.	Interindividual Learners place equal value on hearing others' perspectives and thinking for themselves.
Contextual Knowing	Peers enhance learning via quality contributions. Learners exchange and compare perspectives, then integrate and apply knowledge.	

And, this actually comes from the reflective practitioner, developing in confidence, becoming more authoritative as a teacher. Beyond this, we can talk about independent knowing this is where the reflective practitioner and the authoritative, they have as a teacher start to really take control and become very powerful.

All of these ways of knowing is something that we call **contextual knowing**. This is where the reflective practitioner is bringing knowledge together from theory, from experience into a particular context, into a particular setting, into their own classroom and therefore, synthesizing that knowledge. This is where reflection becomes extremely effective within ways of knowing there are different patterns that we can apply.

In terms of independent knowing we can talk about individual patterns of knowing and also inter individual patterns of knowing. This really underpins the idea that the reflective practitioner is not only somebody who reflects independently, but also reflects with others. As a community of practice with colleagues to create a

very powerful understandings and it's through these perspectives that we can begin to integrate and also create knowledge as reflective practitioners.

Reflecting Effectively: Professional Development Journal (PDJ)

Professional Development Journal (PDJ)

For effective reflection we can consider a number of tools perhaps the most prominent, the most used method of reflecting is the professional development journal. This is a written record of experiences in the classroom and their response that the teacher has to those experiences in the classroom, feelings about planning, preparing and delivering your lessons.

Because the reflective practitioner is writing in a journal:

- The professional development journal is quite subjective - it is written by you and for you, so it provides a dialogue with yourself. It is the individual's response, the individual's perception of what has actually taken place.
- It is very important in the professional development journal to take time to relate theory, which has been explored in the literature with the experience of practice.
- One of the most valuable functions of the professional development journal is to help you identify development points for action planning. Once the critical incidents, once the experience has been noted down, once the feelings have been identified, is for there to be some indication of action in relation to the experience.
- A professional development journal is not a diary of what happened. It does follow a structure in terms of, reflection and action.

So, what you can be doing is to be looking for themes longitudinally through the journal, to see if things recur, and then starting to develop some ideas about actions that you can take in terms of your experience.

- It's very important to get into a very good habit of writing in the journal, try to do it as soon as possible after teaching and as regularly as possible as you go through your teaching.
- The most difficult thing to consider as a reflective practitioner is writing in a journal. When do you get the time? the important thing to realize is that, it is a very critical process, but try not to think too hard about it that can actually stop you from writing in the journal. What you want to do is to let the writing flow and to get the experience, capture the experience and some critical incidents on the paper.
- A very good and simple rule to follow is little and often in the professional development journal. You don't need to write pages and pages, you simply need to note down the key incidences, the key approaches.
- A very good thing to do with the journal is to share your reflections with your colleagues in school - it can be very helpful to find that colleague are experiencing the same uncertainties or difficulties. So, you can get their perspectives on your reflections. And of course, it doesn't have to be just a word. You can actually put diagrams or drawings in there.
- Therefore writing in the professional development journal is a very structured process. It's not free form. And what you need to make sure is that you include **description** under the words (what happened?) you need to add **analysis** (how, why?) and **evaluation** (how effective was it?) and also **conclusions** (suggestions for future practice) that you can draw. The conclusions are important because these will help you identify action points.

Reflecting Effectively: Professional Development Journal (PDJ)

Reflecting effectively using the professional development journal. What is useful to consider here is, a framework that can support the process of writing. Driscoll and Teh (2001) provide a very simple but useful framework for reflection based on three questions:

- **What?** Start off writing in the journal by talking about what happened, or in other words, a description of the event, what was it that, occurred in the classroom that warranted the reflection.
-

- **So what?** In other words, what's your reaction to what happened? What's the actual analysis of the event that occurred? What's the analysis and evaluation of the experience of teaching on that particular occasion.
- **Now what?** or in other words, what's the next step? What will you do? What are their actions, which will come from your reflection?

What?

So, for example, in describing the occasion, describing the event in the classroom, the reflective practitioner can use what questions or in other words, what happened? What did I see? What did I do? What did my students do? These are very simple trigger questions to the first part of the frame. They also provide trigger questions for the other components in the framework as well.

So What?

These are part of the analysis and evaluation aspect of the reflection. And so, the trigger questions need to reflect analysis and evaluation, or in other words, how did I feel at the time? what were the effects of this particular thing happening? why did I react, to this particular student in this particular way? this is where the critical reflection, has most power.

Now What?

And then the trigger questions for now what, or in other words, the action planning component of the framework: what are the implications of the analysis? And then what are the next steps? For example, it could be, how can I modify this aspect of my teaching?

You can design your own template for your PDJ using the following categories (headings) can help:

- Description
-

- Analysis and evaluation
- Conclusion for future practice

Reflecting Effectively: Critical Incidents

Critical Incidents

What we need to do as reflective practitioners is be continual in terms of reflection, and we can use tools like professional development journals to do that. And we know that a professional development journal can have, a useful structure that can help reflection.

However, if we just telling a story in the classroom, the effectiveness of reflection can be reduced. What many reflective practitioners do is. Actually just focus on critical incidents that occur in their teaching and only reflect on these in their professional development journal.

Critical incidents are those specific occurrences within teaching and learning which you consider significant or important.

Reflective Action Planning

- **Select a specific target** (e.g. effective rules and routines or widening your range of strategies)
- It certainly involves **planning a program**, not only of reflection, but the actions in relation to that reflection. This could include discussing with colleagues, notes in the reflective journal and your actions which you're planning. It could also be a process of observing other teachers or having other teachers observe you to give you feedback and for you to acquire more knowledge about your approach as a teacher.
- It's very much about not only action planning, but focusing on the **effectiveness of quality** of what you're doing as a teacher.

The active action planning process is about:

- Identifying the opportunities that you have in discussing these opportunities. And also using this process to reinforce what you're doing well and to identify where you need to make improvements, where you need to make changes.
- Deciding what evidence of planning and practice goes into professional development.

An effective action plan then is something that we need to think about. It's not just about putting action points on paper. The effective action plan is a very useful document for the reflective practitioner.

- It helps you build up a profile of your work and your development over time.
- It's certainly something that is a professional document and therefore should not be just for your consumption, but also it should be shared with your line manager, the principal in the school and others.
- The action plan needs to be very precise in terms of how it's written, not vague.

And so, an action plan answers certain questions for the reflective practitioner.

And really the success criteria, what we'll **demonstrate** through the evidence that you collect, that you've been successful with the action planning process.

Indicators of a Reflective Practitioner

There are some key teachings and learning practices which contribute to what we do as reflective practitioners and to effective pedagogies. And what we can do is identify particular attributes, which will indicate when a teacher is being a reflective practitioner.

Intellectual Quality

Key attribute really is intellectual quality.

- The reflective practitioner is somebody who is engaged in **higher order thinking** rather than common sense thinking this is very critical thinking.
-

- In terms of intellectual quality, the reflective practitioner is somebody who has a **deep knowledge** and a **deep understanding** of the subject they're teaching and also the pedagogy of the subject that they're teaching.
- The intellectual quality of the reflective practitioner is exhibited through the way they **communicate** with other professionals, or how they talk about teaching, how they talk about language or in other words, the **meta-language** that they use to talk about the profession.

Supportive Classroom Environments

Another indicator of an effective reflective practitioner is that the classroom environment is supportive of teaching and learning.

- There is very clear **student direction** in terms of activities and learning, which is taking place.
- There is **social support** for our students within the entire room.
- Their classroom environment is very **active** rather than passive and therefore students engage much more with the teaching and learning which is taking place.
- And what is key to identify here where you have effective reflective practice is that the students are great degree **self-regulated**, or in other words, learning happens almost without direction from the teacher.

Framework for Reflection: Five Key Features of Reflection

Five Key Features of Reflection

- Teaching is a complex activity, in which decision are made in complex contexts.
- In addition, there are theoretical perspectives to consider, and the process of reflection brings all these aspects together.
- There are some key features of reflection that are widely considered:

1. Reflection Results in Learning

- Reflection results in learning- through changing ideas and your understanding of the situation.

2. Reflection is an Active Process

- Reflection is an active process of learning and is more than thinking or thoughtful action.

3. Reflection Involves Problematizing

- Reflection involves problematizing teaching by recognizing that practice is not without dilemmas and issues.

4. Reflection is not a Linear Process

- Reflection is not a linear process, but a cyclical one where reflection leads to the development of new ideas which are then used to plan the next stages of learning.

5. Reflection Encourages Looking at Issues from Different Perspectives

- Reflection encourages looking at issues from different perspectives, which helps you to understand the issue and scrutinize your own values, assumptions and perspectives.
- Therefore, when the term ‘critical reflection’ is used, it refers to a combination of the analytical, questioning (or critical thinking) and reflective approaches.

Framework for Reflection: Ensuring Reflection Leads to Learning

Ensuring Reflection Leads to Learning

- There is a distinction between surface level reflection (routinized, bland and unthinking) and a deeper level (conscious control, self-analysis, metacognition).
 - Many theorists make a distinction between “**common sense thinkers**” and those who are “**pedagogical thinkers**”. This distinction gets to the heart of how to ensure reflection is an effective learning tool.
-

Common Sense Thinkers

- Laboskey defines “**common sense thinkers**” as those who reflect in an unconscious way, suggesting they are happy to use a “trial and error” approach to learning to address short-term issues that are context specific.

Pedagogical Thinkers

- “**Pedagogical thinkers**” are more conscious of their actions, asking the questions ‘what is my intuitive response to this, and why am I feeling or acting this way? Laboskey suggests that it is by taking conscious control that pedagogical thinkers ‘take a long-term view of how to solve problems ... remaining open to learning ... recognize there are no simple answers and the conclusions they reach are likely to be tentative’. (Laboskey in Cartwright, 2011).

Four Key Principles for Learning for Pedagogical Thinker

And what we can do then is look at four key principles of learning for the pedagogical thinker and these define the process of pedagogical thinking. **We have continuous reflection which is ongoing, connected reflection, challenging reflection, and also contextualized or situated reflection.**

1. **Continuous Reflection**
2. **Connected Reflection**
3. **Challenging Reflection**
4. **Contextualized Reflection**

Continuous Reflection

- This ensure that reflection is an ongoing component in learning. Often short-term experiences can lead to more complex long-term involvement which gives the teacher extensive material for observation, reflection and experimentation.

Connected Reflection

- This is essentially the component that links experimental learning to formal training.

Challenging Reflection

- This is the teacher being prepared to pose questions and propose unfamiliar or even uncomfortable ideas for making changes.

Contextualized Reflection

- This ensures that the reflection activities or topic are appropriate and meaningful in relation to the experiences of the teacher. For example, should the reflection be more formal or informal?
- Contextualized reflection takes into consideration all these components, which makes for more meaningful and educational reflection.

Framework for Reflection: Tools for Reflection

Tools for Reflection

Reflective Journal

- Most frequently used.
 - Involves teachers in self-assessment, collaborative critique, self-reflection and goal setting.
 - May become a shared dialogue with a colleague.
-

- May be an individual journey of professional self-reflection.
- May be supported by a framework (list of questions/checklist etc.).

Types of Reflective Journals

- **Reading response Journal**

- Teachers reads recommended content and then record responses in the journal.

And there are many different types of journals that we can use. The **reading response journal** is where the teacher reads, it could be some recommended text or has access to texts in the school or at home and uses the journal to reflect on the content of what they've read.

- **Double entry Journal**

- Involve a statement of context on the left-hand side and reflections and proposed actions on the right-hand side.

We also can get something called a **double entry journal**. This is a very popular approach for the professional development journal where one side of the page, the left-hand side describes the context and the situation and on the right-hand side is where the reflection, the analysis, the evaluation takes place.

- Based on Smyth's (1989) 4 phase model of "describe, inform, content, confront and reconstruct".

Based on Smyth's **four phase model** of describe, inform, confront and reconstruct, this is where journals become quite powerful.

- **Interactive Journal**

- The teacher shares the journal writings with an 'audience'.
- Provides encouragement and deepening reflection.

In **interactive journal**, this is where the teacher writes regularly in a professional development journal. But these writings are regularly as a matter of process shared with an audience that could be one individual or it could be a group of individuals. This is a means for encouragement and support of the teacher.

- **Dialogue Journal**

- Supports the process of deconstructing and reconstructing experience.
- Helps teachers make sense of the conceptual and theoretical understandings about teaching.

And you can also get a **dialogue journal** or sometimes called a **dialogic journal**. This supports the process for the teacher of deconstructing and reconstructing the experience very methodically. It helps the teacher makes sense of the conceptual and theoretical understandings that they have in the classroom by critically taking them apart, looking at the components and then reconstructing them in a meaningful way.

- **Narrative Journal**

- The teacher reflects on the values and beliefs which form the essence of teaching.
- Teachers can draw inferences from experience.

There's also something called the **narrative journal**. This is where the teacher uses the writing process to reflect on values and beliefs about their role as a teacher, what is the essence of teaching? And it is very much here where the teacher can draw on inferences, make inferences about their teaching.

- **Practicum Journal**

- Provides guiding questions about teaching.
- Enables teachers to critique and modify their teaching.

There's also a **practicum journal**. This is perhaps something that teachers will experience during training, where there would be a sequence of guiding questions which need to be answered by the teacher. This gives a very strong structure and helps the reflective practitioner critique and modify their teaching in a very processed and sequential way.

Self-Assessment

- Self- assessment framework help to develop reflective skills.
-

- There are many different self- models which teachers can use.

Portfolios

- A collection of structured professional ‘artifacts’ that demonstrate accumulated knowledge, skills and practice.
- Paper-based or electronic.
- Teachers can portray higher level cognitive thinking and self-reflective growth.
- Examples of artifacts include: lesson plans, students work, feedback, recordings, photos, formal appraisals, etc.

Group Reflection

- Collective reflection with colleagues in a small group.
- Provides a richer experience.
- Generates safe reflection and greater feelings of confidence.

Mentoring

- Can play a crucial rule.
- Mentors can model reflection on own practice and by directly challenging and affirming the critical thinking process.

And mentoring is another significant process for the reflective practitioner. This is a dual process where the teacher is coached and supported as part of reflection.

Framework for Reflection: Social Criticality

Social Criticality

- Being socially critical means starting with reality, with seeing injustices and contradictions, and beginning to overturn reality by reasserting the importance of learning.
- Only when teachers take an active reflective stance that they are able to challenge the dominant “factory” metaphor of the way many schools are conceived, organized and enacted.
- In theorizing practice and practicing theory, teachers are able to see reflection as embedded deeply in our teaching, rather than something that we do to it afterwards

Self-Development is based on:

(a) **Received knowledge**- the ‘intellectual content’ of the profession

(b) **Experiential knowledge**- classroom experience PLUS reflection.

- Dialogue and discussion are central to development.
- Articulation of ideas is crucial to shaping pedagogical thinking.

And what is also involved in that is a process of dialogue and discussion. Social criticality recognizes that as reflective practitioners, we're working within the context, with other teachers and therefore development is very much a process of shared understanding and dialogue and not an isolated individual activity as such, and therefore, it's very much about articulating our understandings and using that language to shape our pedagogical thinking.

- Cooperative development involves another.
- Self-development is based on social constructivism:

(a) The mind is mediated by language.

(b) Understanding is jointly constructed through dialogue.

(c) Teachers are guided to higher planes of understanding through dialogues they have with other professionals

(d) Knowledge is socially derived.

(e) Learning takes place in ZPD (zone of proximal development).

And what we have to consider then is that there is a **zone of proximal development** as a reflective practitioner. There are those that who are closer to us, can have much more impact and can support us. Those who are not so close, they can offer less support and less criticality in terms of our reflection.

- The **process of reflection allows reflective practitioners to develop an “inquiry stance” on their practice** and construct their own knowledge base and their own professional skills.
- Teachers advance their effectiveness through reflective practice by challenging uncritically accepted assumptions.
- Critical reflection within a group/social setting gives teachers multiple opportunities for reflection.

Framework for Reflection: Hegarty's Reflective Framework

Hegarty's Reflective Framework

- The framework consists of a template that you can follow to structure your reflective writing in three steps. This encourages you to move beyond basic description about your experience to analyze your actions, learning and emotional reactions and thus examine your practice more critically and from different perspectives
- **Take notice and describe the experience-** pay attention and record what happened; What did you do, know, feel, think, need? What decisions were made?

So, the first step is about ‘**taking notice and describing the experience**’; pay attention and record it. The journal here is the place where you can use this framework to go through each of the three steps. The first step is where you're describing. What did you do? How did you feel about it? **How** do you think the students experienced that particular lesson?

Analyze the experience- what were the actions and decisions, and why these actions and decision? What was your reaction?

The second step is ‘**analysis of the experience**’. What were the actions and decisions, and why were these actions and decisions taken? what was your reaction at this particular point to the teaching or the experience that you had? Analyzing and evaluating is the second step where the reflective practitioner is beginning to understand what needs to be learned.

- **Reflect, take action-** what did you learn? How will you use this learning? plan how you will apply this learning

And then the final step is ‘**reflect and take action**’. What did you learn? How will you use this learning and, how do you plan to apply this learning?

Framework for Reflection: Personal Practice Theory (PPT)

Personal Practice Theory (PPT)

- Researchers exploring the relationship between teacher beliefs and their classroom practices found that teachers’ experiences impact what they believe teaching should be like and that teachers form their own theories in teaching.
- PPT is the systematic set of beliefs (**theories**) guiding teachers’ practices (**practical**) that are based on their prior life experiences (**personal**).

Personal practice theory then is very much about a personal, a systematic set of beliefs that a teacher has, which describe theories for the individual teacher, and how the teacher operationalizes these theories, or in other words the theories which guide their practice. So, there is a practical component to personal practice theory. And then this is all brought together through the experiences, the work experiences as well as the life experiences that the teacher has. Personal practice theory is something that all teachers will develop in time through the experiences that they have and their evolving values and beliefs.

- What does this mean for the reflective practitioner?
 - Surfacing their PPTs enables teachers to be more aware of their rationale for on-going decision-making and empowers them to become reflective practitioners.
-

- **Developing Your Own PPT**

- You can use the following chart.
- There are no right or wrong answers.

Understanding Self and Others: Professional Identity

Professional Identity

- What is it?
- A more or less conscious apprehension of what constitutes good and professional teaching and learning (Huggen 2005)
- Develop through participation in professional communities in cooperation with others on concrete tasks (Wenger 1998)
- Knowledge and skills are transmitted through language and communication (Vygotsky 1978)
- These definitions imply that
Theoretical perspectives and the habit to reflect must be experienced by teachers as normal teaching activities.

Reflection in Teacher Development

Practice Oriented

- **Practice oriented reflection (practice as training)**
- **Based on experiences and what works**
- **Theory is not important**

One premise is that, professional identity is **practice oriented** or based in practice-oriented reflection. This means that the teacher uses their experience, finds out what works and what doesn't work and uses this in

order to create that their own identity as a teacher in this sense theory is not important. It is the experience which has more value.

Practice as Education

- **Professional reflection (practice as education)**
- **Critical reflection, self-reflection**
- **Theory is an important tool.**
- **Professional reflection is necessary to educate authoritative independent teachers**

But there is another premise about professional identity, which is that it is based in **professional reflection** or in other words **practice as education** in this sense theory is an important component of professional identity and actually professional identity is developed through critical reflection and self-reflection. So professional reflection is necessary to the educator in terms of being authoritative as a professional and in terms of being independent and confident as a competent teacher.

Reflection in Advance

- **Reflection in advance**
- **Central to intellectual and professional growth**
- **Reflection is mostly instrumental/practice oriented, focusing on “technical” aspects and what the students could have done better.**

Understanding Self and Others: Developing Professional Identity

Developing Professional Identity

- **Meaning is constructed through communication and cooperation in concrete situations with others.**
-

- By participating in normal teacher activities together with competent others as legitimate peripheral members.
- It is by participating in this communication and cooperation that teachers start to develop a professional identity.

Finding A Mentor

- Mentors are competent teachers serving as models for the reflective practitioner.
- Professional growth, thus implies that reflective practitioners meet mentors who focus on professional aspects of teachers' work.
- Experience critical reflection/thinking and pedagogical argumentation as part of professional teaching practice.
- Assumption are verbalized and challenged.
- Experiences with theory based practice will inform the reflective practitioner that critical reflection and theoretical perspectives are normal parts of professional competence.
- Such experiences will, thus be expressions of competence and good teaching (professional identity)

Constructing your Identity

- The construction of a professional identity and its development are individual maturation processes.
- It begin when one enters training for the professional as a novice and these processes evolve through many development stages, Until finally one perceives one's professional identity.

Understanding Self and Others: Multiple Intelligence Theory and the Reflective Practitioner

Multiple Intelligence Theory and the Reflective Practitioner

The benefits of utilizing (MI) theory in educator professional are multifold

- It emphasizes the process of learning in delivering desired outcomes.
- It promotes thoughtful consideration of learning experienced in a variety of diverse modalities.
- It generates a broad survey of experiences across levels and disciplines.
- It encourages development of strategies beyond the strengths and interests of the teacher
- It provides a varied pacing of activities throughout the development experience.
- It fosters authentic, real-world experiences that promote deeper understanding.

Understanding Self and Others: Multiple Intelligence Theory and the Reflective Practitioner

Multiple Intelligence Theory and the Reflective Practitioner

- There are now nine MIs Multiple Intelligence Theory and the reflective practitioner
- How does the reflective practitioner's approach their professional identity growth when applying MI theory.

And we can also diagram this for ourselves as well. What we can see is that for the reflective practitioner. (MI) theory can be broken into three areas, **the analytic** component of reflection. **The interactive** component of reflection and **the introspective** component of reflection.

In terms of analysis therefore, the reflective practitioner something the reflective practitioner engages with all the time then the intelligences which are based in a **logical and rhythmic and naturalistic, learning**, become very useful.

Where the reflective practitioner is being more introspective or in other words, thinking about the experience, reflecting on the experience and the interpersonal existential and visual components of (MI) theory become very useful. And what the reflective practitioner can apply from (MI) theory in terms of how they action their reflection, are based in **linguistic interpersonal and the kinesthetic** modes of learning.

- Let's Consider each path to Growing professional identity.

Visual/Spatial

- Learning visually and organizing ideas spatially. Seeing concepts in action in order to understand them.
- The ability to "see" things in one's mind in planning to create a product or solve a problem.

The visual spatial component is about learning visually and organizing ideas, **visually spatially**, seeing what concepts, come together in action and how we understand them. This might be translated into diagrams. For example, in a learning journal. It's the ability to, to see things very clearly to get a picture of things in the mind. how are things working out?

Verbal/Linguistic

- Learning through the spoken and written word. This intelligence was always valued in the traditional classroom and in traditional assessment of intelligence and achievement.

Mathematical/Logical

- Learning through reasoning and problem solving. Also highly valued in the traditional classroom where students were asked to adapt to logically sequenced delivery of instruction.

Bodily/Kinesthetic

- Bodily/kinaesthetic learning through interaction with one's environment. This intelligence is not the domain of overly active learners. It promotes understanding through concrete
-

experience.

Musical/Rhythmic

- Musical/rhythmic learning through patterns, rhythms and music. This includes not only auditory learning but also the identification of patterns through all the senses.

Intra-Personal

- Intrapersonal learning through feeling, values and attitudes. This is a decidedly affective component of learning through which we place value on what we learn and take ownership for our learning.

Inter-Personal

- Interpersonal learning through interaction with others. Not being simply talkative or overly social. This intelligence promotes collaboration and working cooperative with others.

Naturalistic

- Naturalistic learning through classification categories. The naturalist intelligence picks up on subtle differences in meaning. It is not simply the study of nature. It can be used in all areas of study.

Existential

- Existential learning by seeing the big picture. Why are we here? What is my place in my family, school and community? This intelligence seeks connections to real world understanding and application of new learning.
-

Understanding Self and Others: Multiple Intelligence Theory and the Reflective Practitioner

Verbal Intelligence

Verbal

Lecture, reading, writing, reporting, presenting, reciting, discussing, online discussion

Logical Intelligence

Logical

Problem-solving, brainstorming, hypothesizing, investigating, experimenting, Socratic method, online searching.

Visual Intelligence

Visual

Observing, symbolizing, drawing, outlining, conceiving, imagining, visualizing, video.

Kinesthetic Intelligence

Kinesthetic

Building, imitating, performing.

Musical Intelligence

Musical

Listening, patterning, mirroring, repeating

The intrapersonal Intelligence

Intrapersonal

Supporting, advising, advocating, characterizing, defending, evaluating, judging, challenging.

The interpersonal Intelligence

Interpersonal

Sharing, leading, helping, managing, collaborating, influencing, team-building

Naturalistic

Naturalistic

Categorizing, contrasting, classifying, organizing, semantic/concept mapping, graphing.

- For any sort of identity-building activity, you should focus on at least three intelligence but no more than five intelligence.

Understanding Self and Others: Making Professional Identities through a Mentor

- Finding a mentor in your work place can have a significant effect on both the mentor and mentee regarding professional identity and development.
- Under a formal mentoring process the mentee not only reaches the competency level of the mentor but grows beyond the established baseline along with the mentor.
- The mentor may share materials but also goes beyond the sharing and moves into the development of materials within a collaborative relationship. This requires reflective practice and collaborative planning coupled with a joint action plan by the mentor and mentee.
- Through this process both mentor and mentee (the reflective practitioner) can soar to new heights of professional growth and competence.

Functions of Mentoring

What do you do when you are seeking a mentor? Well. What you want to do is actually understand what the functions of a mentor are and try to identify or find somebody who can meet those functions and who can provide that facility for you.

Seeking a Mentor? Know the functions of Mentoring

Teaching

- Teaching the skills that teachers practice on a daily

Well, one function of course is **teaching**. The mentor is somebody who you would regard as highly experienced and highly proficient at teaching. Somebody who you could watch, learn from, and experience with.

Sponsoring

- Sponsoring someone who can identify your strengths and advise as to what activities would be most successful. Sponsoring then requires the mentor to support you when you attempt a new practice.
-

Another function of the mentor is that they are somebody who can **sponsor** another person. Sponsoring means that they support and guide the growth of that individual. They can identify the strengths of the individual and help the individual build on those strengths, identify weaknesses and build their weaknesses. There is somebody who may be will even advocate in the school environment for somebody else.

Encouraging

- Encouraging is a key mentor function. By helping you see the positive side of your teaching practice and building on those reflections, the mentor is supporting and encouraging your growth.

Another function of a mentor is to **encourage**. Encouraging is very important because teaching is a very complex and very difficult profession. And therefore, encouragement especially for the reflective practitioner is a vital component of building confidence and developing professional identity.

Counseling

- Counseling is the fourth function of mentoring. You need sound advice regarding teaching practice, professional conduct and the culture of the school and community.

And also, the mentor needs to be a good **counselor**. Somebody with good listening skills, somebody with good communication skills, somebody who is non-judgmental perhaps unbiased, and somebody who and we can confide in knowing that anything we say is confidential and will not be judged.

Briefing

- The fifth function of mentoring is befriending, you need someone to whom you can speak freely.

And finally, we need to think about in terms of looking for a mentor is somebody who in a sense can become a friend. Most importantly a critical friend. We are looking for somebody who can **befriend** us as reflective practitioner and, in a while, look after and guide us as teachers as we build our identity.

The Importance of Self-Awareness: How does RP Support Professional Practice?

How does RP support professional practice?

The importance of self-awareness to teacher identity and what we have to ask is how reflective practice in this sense supports professional practice. Well, what we can consider is a model called the **conscious competence model**.

The Conscious Competence Model

2. Conscious Incompetence (Awareness of the existence and relevance of the skill)	3. Conscious Competence (Can perform a skill at will)
1. Unconscious Incompetence (not aware of the skill (and lack of skills) and its relevance)	4. Unconscious Competence (the skill is so practiced that it becomes second nature)

We can see that the **conscious competence model has four quadrants**. The first quadrant is **unconscious in competence**. This is where the reflective practitioner is unconscious of what they do not know.

The second quadrant is **conscious in competence**. This is the reflective practitioner growing in awareness of what they do not know.

The third quadrant is **conscious competence**. This is where the reflective practitioner becomes more aware of the skills that they do have.

And finally, the fourth quadrant is **unconscious competence**. This is where reflective practice seems very easy where the teacher (the reflective practitioner) is unaware of what they do know. So, we can see that this model demonstrates how a teacher how a reflective practitioner moves through different levels of knowing.

The model is very useful for the reflective practitioner to consider because it is important and useful also to know where we are at any particular point in terms of our learning. And what we have to remember is that we can be at any stage in the model at any time indeed. For example: if you become a student on a new course, you could be in the unconscious incompetence phase of learning rather than any other phase.

So, the model highlights two key factors here. These factors are about the teachers' awareness of what they know and what they do not know. **The consciousness aspect (awareness) and also the competence aspect (skill).** The skill level that the reflective practitioner is at any particular point.

Understanding the Conscious-Competence Model

According to the model, we move through these different levels as we learn and we will always start at level one, which is **unconsciously unskilled**, or in other words, **we don't know what we don't have this skill.** This clearly relates to entering into a new situation where learning is new for us and of course for teachers that could be just starting out in teaching.

The next level is **consciously unskilled- we know that we don't have this skill.** This is where a teacher becomes aware of the skills, the knowledge, and the abilities that we don't have. Therefore, what we need to develop.

The next level is **consciously skilled- we know that we have this skill.** This is where we realize that we have learned the skills. We are aware that we have those skills. We are actively thinking about using them and actively regarding the knowledge that we have.

The fourth level is the level that we aspire to this is **unconsciously skilled- we don't know that we have this skill.** This is where teaching seems very easy. This is where we have learned sufficiently to develop a professional identity to point that everything seems very easy. It is an application of skills in a way that we do not even think about. It seems very natural. And instead of the matrix which we have just looked at. We can also consider it, therefore, in terms of a ladder from level one to level four and here we can see the learning ladder. The conscious competence model is applied to a ladder of learning. Level one is unconsciously unskilled. Three to level four which is unconsciously skilled. We can refer to this model as a reflective practitioner in many different ways.

The Importance of Self-Awareness: Applying the Conscious Competence Model Level 1- Unconsciously Unskilled

Let's look at how we might apply their conscious competence model as reflective practitioners. First of all, we can start at level 1: unconsciously unskilled. For the reflective practitioner, this is where we are ignorant of what we do not know, what we do not know is invisible to us, we have not yet uncovered what is missing from our practice and what we need to develop. Your confidence therefor far exceeds your abilities.

To move out of level one. to move away from being unconsciously unskilled, it can be very useful for the reflective practitioner to do a personal SWOT analysis. Looking at strengths, weaknesses, opportunities, and threats in other words to conduct a personal training needs assessment to identify those areas where you are unskilled and where you need to develop.

As part of this, it is very useful to ask other people, to ask colleagues for their input to your training needs assessment. This can help you uncover weaknesses that you are not aware of because you are unconsciously unskilled. There will be areas that you do not know are lacking and therefore input from outside is incredibly valuable. Also, it is important to make sure that you are very clear about their learning objectives that you have as a teacher because this will give direction to the skills that you need to develop. There's no point learning skills in areas that don't align with your personal or work goals.

Applying the Conscious Competence Model Level 2- Consciously Unskilled

At level 2, you are consciously unskilled, in other words by this stage, you discovered those areas which are weak, those areas which need to be developed and you are aware that you need to learn new skills and acquire new practical ways of doing things. You also realize that you are perhaps much less competent in the role than others that you see around you. It is very important at this point to remember that you are in a development process and not to lose confidence because it is very easy when you are at level 2 when you are consciously unskilled and when you are aware that you have many weaknesses and you have a lot of growth to go through. It is very easy to become unconfident. This level can be demoralising, causing you to lose confidence or even give up on your learning efforts. Therefore, it's important to stay positive at this stage

Applying the Conscious Competence Model Level 3- Consciously Skilled

At level 3, you become consciously skilled. This is where you realize that you have learned and you **are able to** apply skills in practice. It is where you are starting to put your learning into practice in the classroom and you can see the benefits of it and you are also reaping the reward of that. It is at this point, where you are starting to build your confidence. You are becoming more confident with the skills even though you are **actually** applying them in a very measured and thought-out process. It is not unconscious, it is still conscious at this point. Therefore, as a practitioner, you are concentrating very hard on applying skills and making sure that you are applying them in the right sort of way. It is very much about being more aware of what you are doing and what is important to move from level 3 to level 4 is to look for opportunities where you can practice your skills, develop your skills in new ways in order to move away from the consciously skilled component to the unconsciously skilled aspect of the model.

The Importance of Self-Awareness: Learning to Learn

Learning to learn

- The ability to be able to pursue learning and to persist in learning throughout one's career, even when learning can be very difficult. In this regard, it is very much about the individual knowing what works for them as a learner, their different approach to learning.
- To organize one's own learning and also being able to organize the way they learn and what they learn.
- It is very much also about effective management of time and also of things like motivation and approaches both individually and in groups that you bring as a learner to a learning situation.

Learning Perspectives

In this regard, therefore, we can consider several perspectives about learning. We can think about their **orientation** that we bring to learning or in other words, is learning purely an **individual approach** or is it about learning within a **group** or learning with it within a team and it can also be much wider than that it could be learning within and for the **organization** meeting the needs of the organization as much as meeting the needs of the individual.

Another perspective is the actual **approach to learning** that we take as reflective practitioners. Are we focusing on the **causes of situations**? or are we actually focusing on **subjects**? Or are the subjects that we are interested in not so important? so, we reject those and just focus on the situations so the perspectives that we have to guide the learning process. We can look at those perspectives diagrammatically.

The Importance of Self-Awareness: The Jo-Hari Window (Self-Evaluation in Practice)

The Jo-Hari Window is a communication model that is used by people to develop a better understanding not only of themselves but also of other people that they work with. “Johari” comes from the two people who originated the model **in 1955**: Joseph Luft and Harry Ingham. The model has developed a long time ago but it is a very popular model for communication and therefore for learning,

The Jo-Hari Window Model

In **the Jo-Hari Window** there are four quadrants. The top left quadrant is called the **Open Area (Quadrant 1)**. This area represents for the reflective practitioner the things that you know about yourself and the things the other people know about you as well. This includes things like your behaviour, your knowledge, the skills that you have, and the attitudes that you have as a professional in a way it is almost like your public history.

The next quadrant area 2 is known as **Blind Area**. This area represents the things that you are not aware of but are known by others in a way this is like being unconsciously incompetent. And this can involve simple information that you do not know or it could be even deeper issues that you do not know about yourself but other people do know.

The third quadrant is known as the **Hidden Area**. It is hidden because it represents the things that you know about yourself but the others do not know. This is the private, closed-off area that you keep secret to yourself and the others do not know.

The fourth quadrant is known as the Unknown Area. This area represents all those things that are unknown to you and which also unknown to other people.

So, we can see we have these four quadrants in this model quadrants which relate to how much we know about ourselves and how much other people know about us. And what we are always trying to do in terms of

developing our teacher identity is to raise our own awareness about our skills and knowledge about ourselves and also helping others to become more aware of those things in us too.

The Importance of Self-Awareness: Using the Jo-Hari Window in the Team Context

Feedback:

The important thing in terms of a new teacher joining a school where they are open-air is smaller and will be the importance of feedback because this will be vital in terms of reducing the size of the blind area and increasing the size of the open area. And where feedback becomes significant is in terms of helping an individual understand where they fit in an organization, the spoken and unspoken rules, and the consequences of working there.

Feedback for group members is very much than about striving to help the whole group increase the size of the open area within this model. And therefore, feedback needs to be constructive.

Any non-constructive feedback will actually reduce the size of the open area because the hidden area will start to increase. People will stop disclosing and because of the non-constructive feedback that they might get. So, feedback should always be constructive and therefore the size of the open area can expand vertically downwards reducing the hidden area and increasing the size of the open area for reflective practitioners. This is due to increased disclosure telling people about feelings and experiences the problems that you have the worries and concerns. These are all things which can help increase the open area and therefore get more support and reduce the hidden area.

From Theory to Practice: Theory, Pedagogy and Reflection

What we need to look at as reflective practitioner is how we translate theory into practice and how this is professionally and personally empowering for teachers. **Research shows us that:**

- Reflective practitioners are people who are regularly evaluating their approaches to teaching and the impact that this has on learning.
-

- Reflective practitioner is somebody who understands more about the positive impacts that good quality teaching has on learners and the quality of education that occurs in schools.
- Reflective practitioners are teachers who can draw upon a range of strategies and approaches to helping students learn.
- Reflective practitioners are teachers who can co-construct learning with students, with the people that they are working with in a way that is mutually empowering.

Therefore, what we need to do is think about what this means for the reflective practitioner. What does critical reflection actually involve? Well, it really means that it involves understanding our own learning and understanding ourselves as learners. So that we can bring this to the pedagogy that we use the teaching approaches that we apply in the classroom. **These are relevant to several areas.**

- One particular area is that the reflective practitioner understands children and their needs as learners.
- The reflective practitioner also someone who can build effective and supportive relationships within the context that they work.
- The reflective practitioner is also somebody who can establish learning environments that are flexible and adaptable.
- The reflective practitioner is somebody who can explore what children are learning, what they are interested in learning, and also how they are learning.

This means that teachers hold multiple roles and certainly the teacher as a reflective practitioner is somebody who takes on these many roles within the classroom. And therefore, the reflective practitioner is someone whom we need to look at through various lenses because of the different roles that they take on.

This raises several questions for us as reflective practitioners. Questions such as:

- How do we build and manage relationships with people, relationships with colleagues, relationships with management, and also relationships with students?
-

- How we support and guide students' learning? This is a significant question for the reflective practitioner because what we are always doing is questioning our approaches in order to reinforce and develop them.
- How well do we plan for learning and how good is our preparation to maximize the effectiveness of what we do.
- How would we describe ourselves as a learner part of being a good teacher is understanding what makes us good learners?

From Theory to Practice: Application

"It is because professionals face complex and unpredictable situations that they need a specialized body of knowledge; if they are to apply that knowledge, they need autonomy to make their own judgments. Given that autonomy, it is essential that they act with responsibility; collectively they need to develop appropriate professional values"

Furlong et.al (2000)

From Theory to Practice: RP and Professionalism

Autonomy

Well, one thing we can identify as an attribute of being a professional is our autonomy as professionals; as learners within the complex environment of teaching. There is a conflict between the unpredictable nature of the context we teach in & centralisation of control in education. This illustrates the fact that there is a conflict for us as reflective practitioners between the work we are doing and the nature of working in an environment that tends to be quite rigid and quite strict. By developing autonomy, the reflective practitioner develops becomes more empowered as a person and as a professional.

Professional Knowledge

Another aspect of professionalism is the knowledge that we bring to the role of teaching. Their professional knowledge that we have and perhaps more than in any other profession. This is a complex issue because it

has two strands. Of course, a teacher has to have **subject knowledge**. For example, a math teacher must be very knowledgeable about mathematics but the reflective practitioner is awesome also somebody who knows how to teach their particular subject, the **pedagogical knowledge**. So, these two things come together to actually make a professional knowledge and this is an important aspect of professionalism.

Responsibility

Also, an important aspect of professionalism is the accountability that we have as teachers because we are teaching to other people so we have the responsibility to the families and to the children, the people that we work with. In terms of our role and of course, there is in this respect a responsibility to the profession because we all take a particular ethical moral values approach to what we are doing. We cannot do certain things in the classroom, certain things we can do and we need to work within certain ethical and moral frameworks.

Rigour and Relevance

And what all of this means for the reflective practitioner is that we need to develop a particular rigor and relevance in terms of our professional role. This is something that we bring with us into the classroom. We bring with us into the school environment. It's something that underpins the approach to reflective practice. The real issue here is that teaching especially in the modern era is very complex. It's very messy very undefined and therefore the reflective practitioner as a professional is somebody who is working within this very unstable and messy framework. Therefore, if the reflective practitioner is somebody who is simply remaining fixed to theoretical backgrounds, then we are not being responsive and losing that particular rigor and relevance that we need to work effectively with the students.

From Theory to Practice: Good Professional Standing

Moving from theory to practice for the reflective practitioner is about developing a good professional standing becoming seen as a high-performing professional. This means that the reflective practitioner has particular **commitments**.

Commitment to Learners:

One particular commitment is to the learner of course and perhaps this is the starting point for reflective practice. What the teacher must do is maintain professional relationships and take on a professional role at all times with learners. It is simply because of the learners, their families of learners, the people that we're working for, and trust learning to us as teachers. And therefore, there has to be a certain level of respect that we pay to the learners as reflective practitioners.

In terms of the commitment to learners, we also need to acknowledge be aware of and communicate our understanding that all the people we work with are unique individuals. They all have particular learning needs and interests and therefore the reflective practitioner commits to recognizing this and attending to this in the work that they do.

And also, in terms of a commitment to learners, what we need to be aware of as reflective practitioners is that we have a role in terms of motivating students and also encouraging and supporting our students in terms of learning. Learning real learning is actually a very difficult thing for anyone to do and therefore part of our role is to help, support, and therefore motivate students in terms of their roles as learners.

Commitment to Colleagues and Others:

We also have a commitment to the people that we work with. We have a commitment to colleagues and we have commitments to the organizations that we work with and what this translates into is that the reflective practitioner is somebody who works within a community, a professional community, a community that not only supports growth for students but has a place in terms of intellectual, social, moral and ethical growth of everybody working within the community.

We also have a commitment to our colleagues in terms of promoting professional relationships and the collegiality that we have with others, sharing knowledge, and giving feedback to each other on the work that we're doing.

And in terms of having a commitment to colleagues, we need to work as people who will cooperate effectively with others, spend time supporting, sharing resources, and learning from others, and it through that actually raising the standard of the profession.

And along with this particular commitment is that we need to understand that we have a professional role with parents of children families who are bringing their children into the school and therefore we need to ensure that relationships with parents, in their capacity as partners in the educational process, are characterized by respect and trust.

Commitment to colleagues is also concerning a confidential role. As teachers, we have the opportunity to access information. We develop information about the people that we work with and we should see this as confidential and not for the public domain. We should respect confidential information relating to students or colleagues gained in the course of professional practice, unless the well-being of an individual requires disclosure.

Commitment to the profession:

And of course, we have a commitment to the profession as a whole. We need to ensure as reflective practitioners that we are contributing to the profession that we're contributing to review and revision of policies and practice with a view to optimising the opportunities for students or addressing identifies individual or institutional needs.

And in terms of a commitment to the profession, we need to hold the profession in very high standing. We need to maintain its integrity, value it, and assume responsibility for their ongoing professional development needs as an essential expression of their professionalism.

From Theory to Practice: Personal Challenges to Effective Reflective Practice

Building more professional and personal empowerment for the reflective practitioner is about moving from theory to practice. There are certain challenges that we can face as reflective practitioners. For example:

- we can have expectations that are unrealistic or in a way misplaced in terms of what we expect from ourselves and colleagues and from the students that we work with.
-

- Another challenge is that early on in our training we can develop our notions about what education is about, about teaching and learning as a whole. These notions can begin to interfere with our work in the classroom if we are not being flexible and adaptable enough.
- There can be misconceptions or misunderstandings and from what we have learned from theory when we are confronted by the reality of work in the classroom.
- There can also of course be personal preferences which can become a challenge in other words the way we like our classroom to look the way we like students to behave may not be what we experience and of course.
- There are also things like personal egotism and pride which can get in the way of being a reflective practitioner. These can actually become barriers to development.
- We all have we all hold our own beliefs about education and therefore the philosophy that we have about education can be something that can become a challenge for us once we are confronted by the work that we are engaged with.
- And of course, not least is life can actually get in the way. We all have families, we all have a home life, we have friends, and trying to balance the demands that education as the work we have as professionals as teachers can be challenged by the commitment. The responsibilities that we have in our life in general.

Over-coming:

So, the way to try to overcome these things is very personal to each teacher and really, it is about trying to find your own strategy for overcoming particular barriers.

- **Misplaced expectation:** In terms of expectations that may not be realized the thing really to understand is that you need to be honest with yourself about what is possible what is not possible. This can be very difficult because it does require you to be very open-minded and perhaps a good approach would be to have a colleague as a critical friend and with who you discuss particular expectations. They can give you a different perspective on the reality of the work that you are doing.
-

- **Pre-conceived notions:** They can of course be the preconceived ideas about the way we should be teaching, the way students should be learning the way to try to overcome these is perhaps to refer back to the literature, refer to research, develop a broader understanding of what is possible and again share these ideas and explorations with colleagues is a good way to actually develop a better understanding of the notions.
- **Misconceptions** about teaching and learning a good thing to do is perhaps develop and for yourself will develop with colleagues a set of checklists that you can use to guide what you are actually doing.
- **Personal preferences:** In terms of personal preferences, we all have our own style as a teacher and really what you need to do again is be honest with yourself as a practitioner and to realize that what you may need to do is to change your approach and to change your preferences to meet the needs of the learners that will require you to be quite flexible and adaptable in your approach.
- **Egotism:** If you have a personal egotism this can get in the way of learning and what you need to do is try to move away from that a good thing to do is to observe other teachers and to see how they teach. Therefore, to get a bigger perspective on the role that we take on.
- **Pride:** If you develop a certain level of pride this can become a barrier to reflective practice and also be empowered as a professional. The key to this is to have somebody you can confide in, somebody you can talk to about your professional role to stay grounded, to stay realistic about what you are doing.
- **Educational philosophy:** Your personal educational philosophy is something that can become a hindrance. This can be overcome simply by sharing understandings with colleagues about philosophies of teaching and learning, about personal values and beliefs along the lines of education.
- **Life** is perhaps the biggest barrier that can get in the way the key thing is not to try too hard remember that teaching is ongoing. You cannot do everything in your life and sometimes you need to find a better balance.

From Theory to Practice: Questioning

Developing a framework for personal and professional empowerment is about moving theory about reflective practice and to the practical application of reflective practice. A key requisite for doing this is really taking on a questioning role for ourselves as teachers. It is, therefore, useful to explore some key questions that we can relate to our professional role, and through this questioning approach, we can start to develop a sense of personal empowerment. **It helps you to move from theory and discussion, to practice and application.**

Be Backward-looking:

So, one set of questions that we can explore our questions which get us to look backward, look into the past, look at what has happened in the classroom. Questions like how much did you know about the subject before you started teaching it? Questions like have you done similar kinds of work in the past? and also questions like what problems did you encounter when you were teaching these subjects previously? Backward-looking questions provide a foundation for us to explore current experiences. They provide a foundation from which we can build.

Be Inward-looking:

After we have engaged with some questions which get us to look backward, look historically at our practice. We can consider some questions which help us look inward. Inward-looking questions are those which help us understand our reactions, our responses to experiences that we have in the classroom. Questions like what did you or do you currently find frustrating about work? Did you meet particular standards for teaching when you were working in the classroom? and perhaps one of the most important is how do you currently feel about the work that you are doing? By looking inward, the reflective practitioner is taking chance to contextualize the backward-looking responses. And thereby, understanding within different this particular framework how we are working as reflective practitioners.

Be Outward-looking:

So, we have backward-looking questions and we have inward-looking questions and we also have outward-looking questions. This is where we are looking at the context that we are working in as reflective practitioners and trying to develop an understanding about where we fit into the jigsaw picture that is the whole of

education whether that is just within the school, the educational environment that we are working in or whether it's a bigger picture nationally but there are questions that we can ask here for an example: did we teach in the way that other people are teaching in the school? how do we compare with others? and questions like in what ways there are approaches to teaching similar to others? how are they different? and what grade would you give yourself concerning others if you are going to try and measure the work that you are doing?

Be Forward-looking:

So, what we need to think about is how we can put these questions, these categories of questions into a form that we can operationalize. We have a foundation in backward-looking questions and we analyze our role with inward-looking questions and outward-looking questions. So, what we need to do is ask forward-looking questions. In other words, what will we do next? what changes do we need to make? how will we operationalize things of which we are learning? And another question is, what will I change for the next lesson? It is a particular forward-looking question that helps us put into action the learning that we require from the previous types of questioning.

Frames for Learning: Frames, Perceptions and Interpretation**Frames 1**

- Frames help us perceive, understand and describe situations.

What are Frames?

- Frames are particular ways of making meaning of the complexity of situations.
- Frames are structures of beliefs, perceptions, values and appreciations about a situation, most of them underlying conscience.
- From the multiple components of a situation, we select a particular set with which we try to understand the situation and to transform it.

The Process of Framing

- The starting point is a vague image of a reality named by Jon Dewey as a “problematic Situation”.
- This situation is named and framed selecting a few salient /selected features and relations multiple elements of the complex reality.
- These elements receive a coherent organization, and are used for describing what is correct and what is wrong with the situation or what is the problem.
- The direction for the transformation of the situation will come from the elements established in the frame.

Rhetorical and Action Frames

There are two main kinds of frames:

- Rhetorical Frames, and
 - Action Frames
- Rhetorical frames are related with espoused beliefs, and are used for arguing about a situation.
- Action frames are related to policies in use. They are used for dealing with the situation.

Frames and Metaphors

- Frames frequently use metaphors as a way to develop and/or to communicate their particular logic.
- Metaphors are of great utility for explaining and for thinking with a particular frame.
- Based on a metaphor, the actions look more familiar and consistent with the requirements of change of the situation.
- There is a double-sided interaction between frames and interests.
- The teacher’s interests influence the way they frame the situation.
- But the frames of the teachers also shape the way they perceive their interests.
- These both relations tend to be tacit and invisible for the teachers.

Circularity of Frames

- In conflicting frames there is no way of falsifying a frame with valid data.
- When we arrive to situations where different frames are resistant to critique, we are in a situation that is not normal.
- Frames are mostly tacit.
 - “The frames that shape policies are usually tacit, which means that we tend to argue from our tacit frames to our explicit policy positions”.
 - “.... we are usually unaware of their roles (of the frames) in organizing our actions, thoughts and perceptions” (Donald Schon --- Frame Reflection)

Frames for Learning: Frames, Perceptions, and Interpretation

Frames 2

- In most cases, we are not working with problems, but with messy situations.
 - The process of going from the messy situation to the problem is the process of framing.
 - The names we give things are important – they determine how we will work with them in our mind.
 - Then we organize them and we describe what the problem is.
 - In the same way that Schon talks about espoused theories and theories in use , he talks about rhetorical frames (espoused policies) and action frames (policies in use).
 - Rhetorical frames are weapons used to attack other frames and convince people that our frame is a good one.
 - The frame that wins is the one that can expose the weaknesses in other frames, while hiding its own weaknesses.
 - Frames work with metaphors.
 - When we create a frame for a situation, we always use a metaphor.
 - Frames are not things that emerge from scratch, they are always connected to a metaphor.
 - Changing the metaphor is a very powerful way of changing the frame.
-

- The way we frame a problem affects how we perceive our interests.
- It's difficult to discover the limits of our thinking – we always believe our framing is consistent. It usually is not possible to convince someone that their frame is inconsistent by giving them more information, since that information will be processed into the existing frame
- The frame is able to digest information.
- So basically, more data won't ever change frames, and we need to engage people at the level of their assumptions.
- The frame gives meaning to the information by its own logic. Most frames are tacit, we are not aware of the way we frame situations or of the way other people are framing the same situation.
- The way to solve this difficulty is to try to make the frames explicit.
- Finally, when we are discussing other frames, we need to bring their meaning to own frame.

Try this activity in framing a problem.

1. Define the problem?
2. Why is it a problem? (This is about your reasoning in defining the problem)
3. What are the factors that contribute to problem? We can talk here about causes, context, conditions that allow the problem to exist.
4. What do you see as the impact of the problem?
5. Analyze for consistency and completeness.

Frames for Learning : Analyzing Frames 1

Discovering Underlying Assumptions and Values in Frames

Look at the following example:

Naming and Framing the Education Crisis

- **No Licensure:** We need to remove the inconsistencies between different education providers.
 - **Quality of Teachers' Education:** We need to ensure that teachers are being prepared according to uniform standards.
-

- **Common Inspection Framework:** We need to have a system of checking the quality of education across all providers.

Assumptions

- Assumptions are belief about the world
 - 90% of students are happy in our school.
 - Education improves the quality of life.
- The first type of assumption is situational.
- The second type of assumption is theoretical.
- If a belief is tacit in our thinking, we name it as “assumption”, once it is made explicit it becomes a “hypothesis”.

What are Values?

- Values are criteria that we use for deciding if a situation is desirable or undesirable, or if an idea is a good one or a bad one:
 - Being secure or insecure (security)
 - Being treated fairly or unfairly (justice)
- Every time we frame or evaluate situations, we apply values. Frequently values are tacit.
- In the example “**Quality of Teachers’ Education:** We need to ensure that teachers are being prepared according to uniform standard” the quality of teaching is a value.
- In the example “**Common inspection Framework:** We need to have system of checking the quality of education across all providers” how schools provide education is a value.

Frames, Assumption and the “Thinking Box”

- Every time we frame a situation, we base elements of our frame in belief about the world; but we are not aware of them.
-

- The discovery of assumption is easier when we are “out of the box”.
- People who disagree with our frames are more acute for identifying our assumptions.

Frames for Learning: Analyzing Frames 2

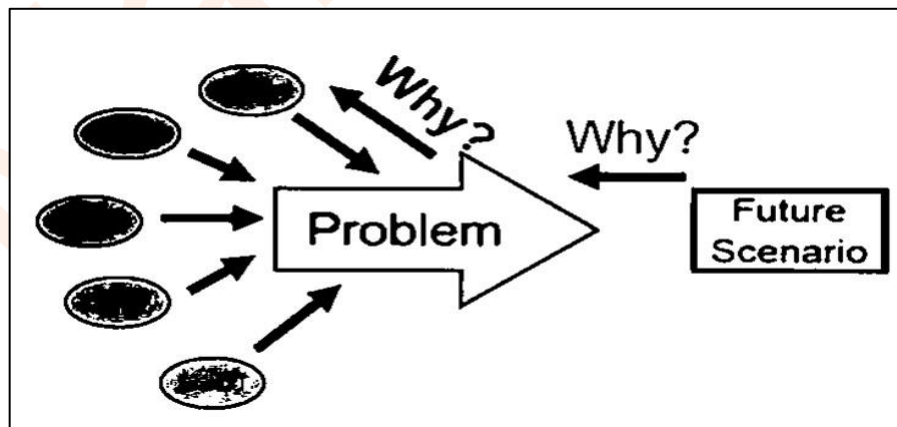
Discovering Assumptions

- Assumptions are beliefs that are behind the causal relation of our frames.
 - If “assumption A”. then “B” generates “C”.
- A possibility for discovering assumption is to go through the components and the causal relations of the frame asking “why” they are stated in that particular way.
- The assumptions should be falsifiable. It means that it should be possible to get information for verifying if an assumption is real or not.

Exploring Assumptions

- If a situation is framed as a problem generated by a group of factors and a scenario that is resulted from the problem, then the exploration of assumptions may go back through the causal relations of the frame. Let’s look at this in diagram form:

Exploring Assumptions



Questions for Exploring Assumptions

- Why the scenario or one component of it, is generated by the problem or a component of it?
- Why the factor A contributes to the problem?
- Which conditions are required for the work of causal relations?
- Why do you think it is a problem?
- Which values are threatened by the problem or by the scenario?
- Which motivations are behind the way the situation was framed?

Frames for learning: Interpolating Concepts and Ideas**Theories –in-use- are means:**

- For exploring concepts
- Developing ideas

Schema of Theory and Action

- A theory of action is a set of interconnected propositions that have the same referent (the subject of the theory).
- For example:
- In situation of S, if you want to achieve consequence C, under assumptions a1, a2, a3, and, do action A.
- Theories are vehicles for explanation, prediction or control.

Theory-in-use

- Argyris and Schon say that theory-in-use is:
“...existing theories people have that already determine practice. We call these operational theories of action theories-in-use to distinguish them from the espoused theories that are used to describe and justify behavior.”
 - Argyris and Schon say that theories-in-use:
-

“.... help teachers remain blind to the actual degree of their ineffectiveness.”

Theory-in-use, Assumptions and Everyday Life

“.... the full set of assumptions about human behavior that function in the theories-in-use constitutes a psychology of everyday life.”

- There is a theory –in-use of building design (for architects).
- There is a theory in –use for diagnosis and treatment of disease (for physicians).
- Theories in-use are means for getting what we want.

How to make Theory-in-use visible

- We can construct someone’s theory in-use from observations of behavior.
- This includes recognizing general characteristics:
 - The governing variables
 - The action strategies
 - The principal assumptions

Clues for discovering the Theory-In-Use

- Reconstruct the espoused theory and /or the action strategy.
- Look for the assumptions through the “Chain of Whys”.
- Look for the relations among the assumptions.
- Discover the governing variables of Espoused theory and /or the action.
- Construct the theory in-use based on these elements.

Requirements for Discovering the Theory-in-use

- Internal commitment to discovery
 - Public testing of theory, assumptions and data
-

- Clear distinction between the self and assumptions (non-defensiveness)
- Look for valid information
- Aptitude for reframing

Promoting PR through Inductive Action Planning

- As a teacher you will be learning every day.
- Knowing what you have learned or what you want to learn from practice and experience needs some planning
- This will be helped by **inductive action planning**.

Essentially this involves:

- Selecting a specific target (e.g., effective rules and routines in class, widening your range of strategies, effective use of ICT etc.)
- Plan a program of discussion, observation and training over a fixed period.
- Evaluate your effectiveness
- Reflect on your own development and select further targets.

Inductive action planning is therefore about:

- Identifying opportunities for you to discuss, reinforce and develop your practice.
- Helping you decide what information/data of your planning and practice you can use as evidence of your development.

An inductive Action Plan will be one that:

- Builds on your career profile
 - is precise
 - Answer the following questions:
 - What is my priority and what do I want to achieve?
 - What am I going to do and when?
 - Who/ what will help me?
 - How will I know whether what I have done has been worthwhile?
 - How will I collect information?
-

In detail, an inductive action plan will:

- Identify your priorities
- At this stage of the year what precisely do I want to be able to do with greater confidence and more effectively?
- E.g., Motivate the class?
- Plan, manage and evaluate the teaching of reading...

In detail, an inductive action plan will:

- Identify what you will do and when
- E.g., I shall discuss school policy and expectation with my principal, I can attend a course which will help me think more deeply about my practice; I shall see school policy in practice by observing colleagues and discussing issues with them.

In detail, an inductive action plan will:

- Identify who can support you
- E.g., I shall enlist the support of my head of department, I shall engage in some background research.

In detail, an inductive action plan will:

- Identify Success Criteria by identifying what has been worthwhile -
- My planning now demonstrates...
- Through being observed and discussing the lesson I am clearer about my strengths and priority areas for development in....

In detail, an inductive action plan will:

- Identify how you can demonstrate your development -
- I have examples of the new lesson planning that I am doing...
- I have examples of the resources I'm using...
- A copy of my appraisal report...

Inductive action planning:

- Responds to, and support, your day-to-day teaching activity.
- It is directed by you.

Frames for learning

Reframing for Resolving Difficult Controversies: The Reframing Matrix 1

The Reframing Matrix 1

- Things look different when you change the perspective.
- When you are stuck on a problem it helps to look at it from another perspective.
- This can be all that you need to do to come up with a great solution.
- However, it is sometimes difficult to think what these perspectives might be.
- Using a reframing matrix relies on the fact that different people with different experience are likely to approach problems in different ways.
- Reframing helps to put you in the minds of different people- imagine how they would face these problems and explore their solutions.

Step 1: Draw the Grid

- Start by drawing a simple four-square grid.
- Leave a space in the middle to describe your problem.
- The boxes are for your different perspectives.

Step 2: Decide on perspectives

- Decide on 4 different perspectives to use in your matrix.
- Use the 4P's approach:

The 4P's

- **The Product Perspective**
-

- Is there something wrong with your teaching?
- Does it respond to the needs of your learners?
- Do you adapt your teaching to the circumstances in the classroom?

- **The Planning Perspective**

- Is your planning at fault?
- Could you improve your planning and preparation?
- How good are your resources?
- What resources are you using?

- **The Potential Perspective**

- How would you improve the learning outcome?
- If the students' grades were to go up how would that affect the problem?
- How are others doing the same work as you?

The People Perspective

- What are the people impacts and people implications of the problem?
- What do people involved with the problem think?
- Why are students not happy?

- These are just some of the questions that you can ask as you look at your problem using these four perspectives.

Reframing 2: The Professions Approach

- This approach helps you to look viewpoint of different specialists, or stakeholders.
 - For example, the way a medical doctor would look at a problem is different to what a civic engineer might use.
 - This approach is particularly useful when you are trying to solve a problem which involves many different types of people.
-

- It is useful when you need to 'step away' from your usual way of thinking so that you can be more creative.

Step 1: Brainstorm Factors

- Brainstorm the factors related to your problems from the four perspectives viewpoints:
 - Product perspective
 - Planning Perspective
 - Potential perspective
 - People perspective

Fill out the matrix.

Example of Reframing

- In the following example you will see how a teacher has used the 4ps approach to explore problems in class.
-

Reframing for Resolving Difficult Controversies

<u>Product Perspective</u> The subject is complex The subject needs some introduction in previous classes The subject is new in the school	<u>Planning Perspective</u> My planning follows a rigid pattern I use the same resources in every lesson I plan to get through a great deal of information in each lesson
Students are not interested!	
<u>Potential Perspective</u> The students are very comfortable with technology Students like to talk a lot and guided discussion would be a useful approach to exploring the subject	<u>People Perspective</u> Students seem generally bored in school Students do not see the need to study Other teachers are facing similar issues

In this example, we can see how a teacher might use the 4 P's approach and the professions perspective in order to reframe their perspective of an issue in order to resolve it. For example, the issue that you might face or any teacher might face is that students are not interested in the content of a lesson. So, by looking at it from the **product perspective**, if we can actually start to deconstruct issues and which students might be faced with in terms of the particular subject. For example, it might be that aspects of the subject are quite complex and difficult to understand. By looking at it from a **planning perspective**, we can then start to think about how we're actually working with students in terms of the subject content and for example, are we planning our lessons in very rigid and strict fixed ways rather than being much more diverse and flexible about it. By looking at the problem from a **people perspective**, we can actually start to understand the problem from other people's point of view, from our students' point of view and for example, it might be that the students don't actually see the need for certain aspects of a subject and in terms of their future goals or indeed their future life. And of course, we can look at a particular problem from a **potential perspective**, it could be for example that the students especially nowadays may be very much inclined to the use of technology in

terms of learning. And, this can help us as teachers to find new ways of working with students and therefore, helping them become more interested in terms of that learning.

Reframing for Resolving Difficult Controversies: Problem Statements

Problem Statements

- The first step is to define the problem statement
- At which level should we formulate the problem, and how widely?
- If we go too deeply and try to look for final causes of everything, we may reach conclusions that are not useful for the problem.
- Select a level where you have enough knowledge about the problem and then act in a timely way.
- As reflective practitioner we should always be working with ‘unsatisfactory’ understanding of the problem.
- The reflective practitioner works with ‘unsatisfactory’ because s/he has to combine the action with improving understanding.
- What method you actually use depends on the knowledge you have, the conditions and other factors (e.g., other people).
- You need to design your own cognitive strategy in each case.

The reflective practitioner has two levels:

- The action level
- The cognitive level
- Consequently, we need to combine knowledge from our own experience with knowledge from other people’s experience.

And in terms of cognitive strategies, the reflective practitioner has two key levels that we can work at. One is the **action level** which is very practical and one is the **cognitive level** where we're thinking and reflecting about a particular problem and what we need to do as reflective practitioners therefore, is to find a

way with a particular problem statement to combine the action level and the cognitive level together so that we have a very full and deepening understanding of the problem.

- Conceptual innovation
- The connections we have in our brains are related to the concepts we use.
- When we change our concepts, we create new areas of relations in our brains.

Therefore, our concepts can change, can evolve, and also can disappear with experience and with thinking through problems.

- Concepts are abstractions of situations – the patterns that emerge when we have seen many situations.
- Without concepts we cannot perceive reality.
- With concepts we can think more deeply about reality.

- With concepts we can:
 - Monitor situations.
 - Understand new situations by extensions of old ones.
 - Work to extend the concepts we are using.

So, conceptual innovation then is very much about working with concepts and their concepts in terms of how we're understanding situations, how we're perceiving situation, how we're monitoring situations and how we're understanding how others are working and experiencing those situations.

- The reflective practitioner needs to be comfortable with grasping and internalizing new concepts.

Reframing for Resolving Difficult Controversies: Concept-based Teaching and Learning

Concept-based Teaching and Learning

Concept –Based Teaching and Learning:

- Enables the ability to transfer knowledge
- Promotes emotional engagement and motivation

Concept –Based Teaching and Learning:

- Thinking which draws on critical, creative, reflective and conceptual abilities.
- Facilitates ‘synergistic thinking’ (factual and conceptual levels of thinking).
- Requires deeper intellectual processing as you relate facts to key concepts.

Concept –Based Teaching and Learning:

- Develops conceptual structures in the brain to relate new knowledge and to illuminate the connections and pattern of knowledge.

Concept –Based Teaching and Learning:

- Facilitates the transfer of knowledge at the conceptual level.
- Provides opportunities for personal meaning making through processes of thinking, creating and reflecting.

Concept –Based Teaching and Learning:

- Motivation for learning which recognizes that emotional engagement is essential to motivation.
- Increases motivation by thinking through the facts, through a personally engaging key concept.

Concept –Based Teaching and Learning:

- Values and respects the thinking of the individual by ‘drawing understandings from’ rather than ‘telling understanding to’.

Concept –Based Teaching and Learning:

- Encourages constructivist learning experiences that are relevant and important.

Concept –Based Teaching and Learning:

- Values collaborative thinking, discussions and problem-solving with the belief that the social construction of meaning leads to quality of teaching and motivation.

Reframing for Resolving Difficult Controversies: Benefits of Concept-based Teaching and Learning

Benefits of Concept-based Teaching and Learning

The benefits correspond directly with the profile of the reflective practitioner:

The reflective practitioner as enquirer:

- A teacher who questions, investigates, explores and discovers.

The reflective practitioner as knowledgeable:

- A teacher who has thorough grasp on their subject and a strong approach of its pedagogy.

The reflective practitioner as thinker:

- A teacher who ponders potential reasons, causes, effects and outcomes.

The reflective practitioner as communicator:

- A teacher who is able and confident enough to express to others their problems, issues and concern, and who is able to listen to feedback, advice and criticism.

The reflective practitioner as open-minded:

- A teacher who holds onto opinion, judgment and conclusion even with multiple sources of information.

The reflective practitioner as reflective:

- A teacher who sees reflection not as a mirror but as a window onto new insights, perspectives and horizons.

Reframing for Resolving Difficult Controversies: Pedagogical Shifts

Pedagogical Shifts

The reflective practitioner relies on:

- Synergistic thinking:
 - ...is essential to intellectual development.
 - ...is a cognitive interplay between the factual and cognitive levels of mental processing.
 - ...assumes that thinking our this factual / conceptual interaction can be shallow.

The Reflective Practitioner is able to:

- Transfer of knowledge and skills.
- Facts do not transfer-they are locked in time, place and situation.
- Knowledge transfer at the conceptual level as concepts, generations and principles
- The ability to use the conceptual level of thinking to relate new knowledge to prior knowledge.

The Reflective Practitioner is skilled in the:

- Social construction of meaning.
- Quality thinking is hard work.
- Reflective thinking requires collaboration to enhance thinking and problem-solving.
- Different minds working together provide support and generate new ideas and solutions.

These shifts in pedagogical thinking lead the reflective practitioner to a number of key components for reflection:

- Focus on the learner (students learning strategies, their interests, their developmental readiness).
- Focus on the teacher (teachers' anxiety, teaching styles, teachers' affective factors/personality).
- Focus on practical aspects (journal writing, surveys and questionnaires, observation, etc.).
- Focus on teachers' cognitive development (during small-scale classroom research projects, attending conferences and workshops).
- Focus on critical/conceptual aspects (teachers' ability to intellectualize, teachers' practice).
- Focus on moral/ethical parameters (listening to students, finding ways to support and challenge students).

Reframing for Strategic Creativity: Framing in Four Steps

- Reframing in 4 steps
- The approach to reframing is a four-step process

Step three is a questioning process and step four is the actual production of the new frame for your reflective practice.

- Reframing uncovers, challenges and overturns assumptions.
- This generates new ideas.
- Allows for innovation.

Reframing for Strategic Creativity: 3-Dimensional Instruction

- The goal of reflective practice is 3- dimensional instruction

This includes:

- Increasing conceptual understanding supported by factual knowledge and skills and the transfer of knowledge across multiple contexts.
- Facilitating student enquiry into important interdisciplinary and disciplinary topics and issues using one or two key concepts.
- Instruction and learning experiences that use concepts along with factual content to ensure synergistic thinking.
- The teacher encouraging group work to facilities shared enquiry collaboration, synergistic thinking and problem – solving across contexts.
- The teacher uses inductive teaching to draw the statement of conceptual understanding from students.
- The teacher focuses on student thinking and understanding and each teacher is aware of each student's ability to think synergistically.

Reframing for Strategic Creativity: Conceptual Learning

- What is conceptual learning?
-

- Conceptual learning is a process by which the reflective practitioner learns how to organize information in logical mental structures.
- Conceptual learning focuses on learning organizing principles the pockets in which the mind organizes facts into ideas.
- Conceptual learning is a catalyst for challenging the reflective practitioner to think at more advance levels.
- Focuses on broad categories of problems
- Fosters systematic observations
- Fosters understanding of relationship among events or conditions that impact situation
- Focus on teaching actions

Tanner's Model of Reasoning

- **Noticing**
- **Interpreting**
- **Responding**
- **Reflecting**

We can talk about a model to help us work conceptually in terms of our learning and the model is called **Tanner's Model of Reasoning**. It looks quite similar to the models of reflective practice that we've been speaking about. So, this is a very convenient model for us to think of in terms of conceptual learning. The first step is **noticing** or in other words describing the experience that we're having. The next step would be to **interpret** or in other words begin to analyze that experience. The third step is what is our **response** to the experience, the emotions, the feelings which are developed, and then this leads to **reflecting** or in other words how we're going to work to resolve those particular issues.

Reframing for Strategic Creativity: Epistemic Stances

Reframing for Strategic Creativity

Reflective practitioner can take different perspectives on their roles and in the discussion around reflective practice we talk about epistemic stances

Epistemic Stances.

- These are ways of knowing and reflection
- These processes are developmental
- The quality of reflection changes as we mature and new epistemic stances emerge

Categorization of Epistemic Stances

Epistemic stances can be categorized into 4 stages

- Absolute knowing
- Transitional knowing
- Independent knowing
- Contextual knowing

We can categorize epistemic stances into four key areas. The first key area is Absolute Knowing.

Two other areas are Transitional Knowing and Independent Knowing and the fourth is

Contextual Knowing. There are four different stances; we can take in terms of the reflective role. **Absolute knowing**

- The teacher who seeks to learn by receiving knowledge from others (experts)

Transitional knowing and independent knowing

- These are intermediate stages

The second and third stage is Transitional knowing and Independent Knowing. The intermediate stage is they move the reflective practitioner from the absolute knowing to contextual knowing. Transitional and

Independent Knowing are simply part way phases that the reflective practitioner is working with, beginning to take facts and figures from others and internalize understanding and begin to own particular skills and knowledge in relation to their own role in the classroom.

Contextual Knowing

- The teacher judges all information on the basis of evidence within context
- Highly critical in pursuing understanding
- Examine both data and own/ theirs' perceptions and values

These processes lead the reflective practitioner to Contextual Knowing, this is where real quality contributions too arose as teachers in the classroom begin to manifest themselves, where teachers exchanges and perspectives exchanged and used productively by reflective practitioner.

- Implications
- Ways of knowing impact the ability to reflect
- More sophisticated types of reflection lead to multiple solutions
- Absolute knowers only find and accept information from some authority.

Reframing for Strategic Creativity: Taxonomy of Reflecting Thought

Taxonomy of Reflective Thought

An aspect of reflective practice we can consider at this point is how we look at reflect thought, how we define reflective thought and, in this way, can begin to look at taxonomies of reflective thought.

Taxonomy

- To examine reflective thinking, researchers and theorists have developed taxonomies to explain qualitative aspects of reflection and have identified critical reflection as the deepest level of reflectivity.
-

Taxonomy of reflective thought

- For teachers critical reflection is the type of reflective thought most closely associated with Dewey's definition, explained earlier and involves reflection on the teaching practices as they relate to moral and ethical issues in society.

So, for teachers, the critical reflection is type of reflection which is most closely associated with models of teaching and reflection which we have looked at in the past. Some of the foundational models we are looking at for example Dewey and Schon, both define processes of reflection which taxonomies tend to reflect quite closely. Essentially taxonomies will look at aspects of teaching and aspect of knowing but within a contextual frame and that contextual frame can be the moral and ethical considerations, the teacher has to confront in the classroom.

- The taxonomy categorizes thinking into levels explaining the different qualitative aspects of the process.

Three levels of reflective thought:

- Non-reflective action
- Reflective action
- Premise reflection

Reframing for Strategic Creativity: The Ripples Model

The Ripples Model

- The Ripples model can be used by the reflective practitioner to consider how to consider how their teaching addresses certain questions:
 - How do you enhance your learner desire to learn?
 - How do you help learner develop ownership of the need to learn?
-

- Do you help learner learn by techniques such as practical activities, trial and error tasks and repetitive activities?

Other questions the Ripples Model can be used for in terms of the strategies that we are using for teaching? Are the learners learning through practical activities through processes where they can directly engage in knowledge and skill development? And learn inductively as well as deductively in the classroom.

- Do you ensure learners receive quick and useful feedback, both from you and from their peers?
- What activities do you provide where learners can make sense of what they have learned?

In the Ripples Model here we can see how these questions can come together in a particular taxonomy. It's called Ripples Model because it is like ripples when you throw stone in water. At the center is what is known wanting our needing aspect of reflective practice process which is focused as what the learners want for learning? What the students need in order to learn? This has a consequence of what we do with them, how we work with them. The strategies and approaches we are using for teaching in the classroom. From this doing in the classroom the reflective practitioner begins to make sense of teaching and learning, starts to create conceptual frames for the approaches which are being employed. From making sense of doing in the classroom, there is the system of feedback or in other words the reflective process that takes place where the information is brought back internalized by the teacher and therefore inform the next level which is teaching the actual process of developing the approaches and strategies. In the Ripples Model what's important to realize is the final stage, the final level is assessing the effect of our teaching. What is actually the benefit or outcome of what we are been doing with the students?

- This model is not a cycle- it is a progression
- Learning by doing is at the Centre

We conceive from this is that the Ripples Model is not the cycle. It is the progression starting from the center point which is wanting and needing and doing or in other words we are learning by doing as reflective practitioners. Through this progression moving through the reflective practice process very systematically by looking at different strategies in terms of reflection.

- True learning and understanding states from the center and radiates outwards like ripple on water encompassing other aspects of learning.

Maha Malik

Reflective Teaching Practices: Criticality

Framing Reflection in External Constraints

- RP does not take place in a vacuum
- It occurs complicated social field with competing habits and constraints that naturally affect teachers in the school environment.

Reflective teaching practices needs to be very critical if they need to be effective and what we want to do is look at different aspects of criticality in terms of reflective practice. One critical aspect of reflection of reflective practice is that we need to be aware of is that reflection does not happen in isolation. It happens within an external framework within an environment. Therefore, being critical about reflective practice means that we need to understand the environment and how those external constraints can affect the reflective practitioner and affect the outcomes of reflective practice.

Reflection with Action

The role the teacher plays within their schools and departments will have an effect on the reflective practice of the teacher.

- There will be certain rules of being a teacher, both written and unwritten, they will constrain or bias the reflective practice that occurs.
 - Teacher do not exist in a vacuum: all their experiences are different because of the school environment they exist in the colleagues they work with and their own habits they bring to their education are all different
 - The colleagues they work with and their own habits they bring to their education are all different.
-

It's the colleagues that we work with, it's the biases we have, the expectations and biases that our colleagues have which come together in the environment which create the reflective practice environment. These actually become constraints or also they can become **empowerers** to the process of reflection.

- To understand teachers' reflective practice there must be an understanding of the social environment they exist in.

Reflective Teaching Practices: Criticality Critical Reflection= Critical Thinking

Critical Reflection and Critical Thinking

- Models of thinking and reflection involve three fundamental processes

Retrospection

- Thinking back about a situation or experience

Self-Evaluation

- Critically analyzing and evaluating the actions and feeling associated with the experience, using theoretical perspectives.

The next level of critical thinking is self-evaluation. For the reflective practitioner it is important to think analytically and in evaluative terms about experiences. Critical evaluation and analysis of our actions and also of the feelings that we have, our responses to experiences we have. It's important to understand all of these from theoretical academic perspectives and begin to make cognitive links **in the conceptual frames**.

Re-Orientation

- Using the results of self-evaluation to influence future approaches to similar situations and experience.
- Part of critically is to transform problems into constructive ideas

Critical Reflection Analysis

- Try to translate the processes of reflection into questions that challenging you further.
- Did the analysis look broadly enough?
- Is the issue recognized by students
- Does the improvement proposal address the original issue?
- Is the analysis and improvement specific enough?

Critical Reflective Enquiry: Critical Philosophy

Critical Philosophy

- From the critical philosophy perspective teaching practice is viewed as a form of social life in which different forms of domination, distortions and misunderstanding are possible.

Becoming reflective practitioner is important in terms of the processes that you engage with from a sort of research perspective in the classroom and what we can consider is Critical Reflective Enquiry in this regard an underpinning idea in relation to the reflective practitioner experimenting, investigating and developing from those investigations is to understand the philosophical underpinnings to reflection and criticality and in this regard we can talk about criticality philosophy or in other words the perspective of teaching that we hold and view as important and we as reflective practitioner can alter the experience of classroom, alter the perception of issues and problems as they arise and therefore the certain distortions and the effects that the philosophy we have can have on our critical reflection.

Hence any study of practice needs to incorporate an emancipatory focus through which social life can be freed from domination and distortions (Herbermas, 1984)

Critical Reflection has a number of key phases.

- Descriptive
- Reflective
- Critical/Emancipatory

Reflective phase

There is then a reflective phase and a product of reflection is our growing awareness of teaching role and the processes we have discussed many times that we engage with in order to do this reflective cycle analysis process.

Critical/Emancipatory Phase

- This phase, moving from the reflective phases, is oriented to correcting and changing and changing less-than good or ineffective practice or moving forward to future assimilation of new innovations emerging from practice.
- Assimilation of new innovations emerging from practice
- It involves discourses about the nature and sources of distortions, inconsistencies between

(a) Values/ Belief and practice**(b) Intentions and actions.****(c) Students needs and teachers actions, which have been identified in the reflective phases.****Critical Analysis**

- One way of deepening your ability to engage in critical reflection is to engage in complex analysis of incidents

There are different levels of analysis

Critical Analysis can occur at number of different levels.

Ethnomethodological Analysis

- Examining taken-for granted assumptions and rules of everyday social behavior.

Hermeneutic Phenomenological Analysis

- Reflecting on interpreting of both the learner's experience and the phenomena being experienced so as to move beyond the partiality of previous understanding.

Discursive Analysis

- Probing how speech and texts construct social truths.

Critical Analysis and RP Models

- Where models are presented, they should be offered in context and it should be emphasized that each is simply one of a range of tools.

Models to Trigger Broader Reflection

- Models should be used to trigger broader reflection rather than an ends in themselves.
 - We should not be presented with just one model, the implication being that this is the way reflective practice is done.
 - Teachers need to grasp that different model engage different levels of complexity and therefore need to be used selectively and carefully.
-

- Students should also be helped to see something of the strengths offered by the different models and methods of reflection

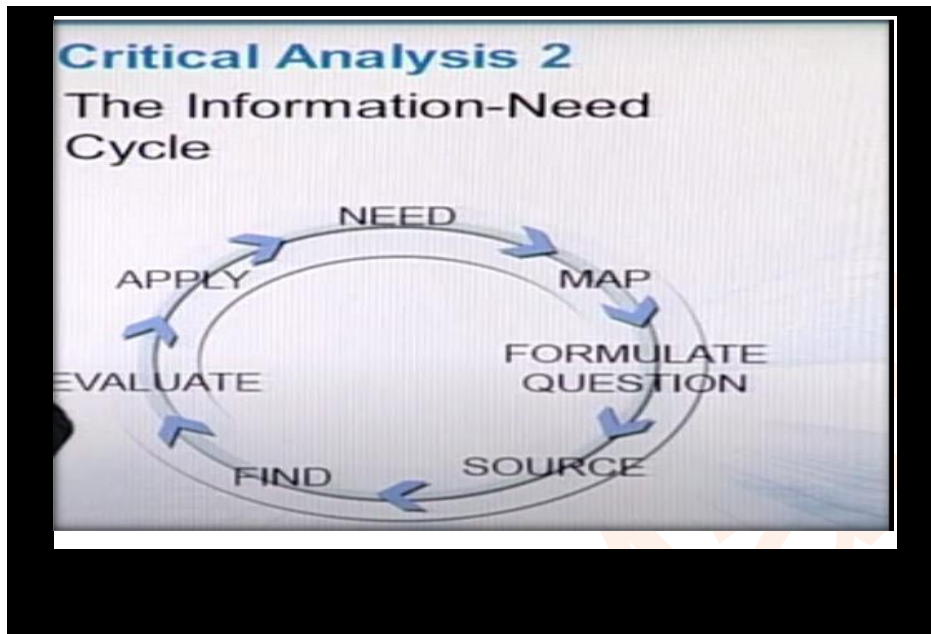
Critical Analysis

- Analysis involves breaking down information into its components parts and examining the relationship within these parts and with the whole.
- By doing this teacher shows the ability to differentiate and distinguish between components or elements of their experience.
- All critical analysis is formed around one basic assumption
 - Applying theory to practice
 - Theory and Practice

The information need cycle

Quite simply the critical analysis for the reflective practitioner can be viewed as a process of applying theory in practice. This is for the reflective practitioner is foundational principle of critical analysis. How to move theory into practical domain and this is viewed conveniently from a point of view of cycle of critical analysis and we can see below this cycle displayed.





First, we will talk about need that arises from an experience of a problem. By critically analyzing we are doing is trying to map, trying to look at the components, the elements of the problem and our thinking about the problem because from these elements we can formulate questions to challenge our conceptual frame. What important in terms of critical analysis is not only form those questions but also to understand to look for sources of evidence for those questions so, that we have some backup some foundation for what we are really doing in term of being critical and from this we are beginning to find to experiment with that thinking to begin to reframe the concepts we have and then to evaluate the change to evaluate how our thinking is changed before making the next application, before applying the change action in our teaching. As a cycle this brings us back to the action and experiences of that action how is the problem resolved or new problems arising.

Need:

- Determined by learning outcomes/ information

So, what we need to do from critical analysis perspective is to look at certain questions and try to refine those questions in terms of determining the outcomes of the information. We could ask a question for example on personal level

a) “I need something on smoking”

Some information, some evidence to work with the students in the classroom about the dangers of smoking. This could lead to another question of

b) “I have an assignment for which I need to find the best evidence to support an intervention”

“I have an assignment for which I need to complete for which I need evidence in order to support the suggestions that I have been making. Another question generating from this could be then that would be

c) I have an assignment for which I need to find research”.

“what research do I need to engage with? What research do I need to engage with? What evidence do I need to find?

Map:

- Map all the possible conceptual pathways to developing a question which will provide an answer.

In the cycle this is about mapping the thinking, mapping our conceptual understanding about the different routes we can take through resolving at particular issues, looking at questions to answer.

- This is a keyword generation, critical creative thinking and language skills.

Therefore, the mapping process is really about identifying the keywords that can drive our thinking that can support the way we are been creative and innovative and the way we are using language around describing issues and the process of reflection.

Formulate questions (the ability to give focus to questions):

- “I need something on group work” becomes
 - “I need information on the teacher role in effective group work”.
-

Formulating questions is about being very critical in the way questions are formed. We have to look at the language, for example a question generated from bad experiences of managing group work, could be about looking at the strategies and approaches teacher is using.

Sources:

- Which information sources will provide the answer to my question?
- Involve critically choosing the appropriate information sources.

Which sources of information are you looking at as a reflective practitioner? In order to support the mapping process are you looking in the correct locations for the evidence?

Find:

- Require the development of a search strategy.
- Strategies will vary according to the information source but developed in line with a standard framework.

This search strategy will help you to develop the framework for reflection becomes much more critical how you are analyzing situations.

Evaluate:

- Does the information found answer the question?

The final part of the cycle is evaluation and application. Evaluation is doing the information; does the research I have done help me solve the problem help me answer those questions

Apply:

- How can the information be used to answer my question? What conclusion can I draw?

And when you are applying, what conclusions can you draw from that application?

Critical Analysis

Influences on Critical Analysis

Critical analysis will be influenced by:

- Level of support in the work environment
- Time for critical analysis
- Tools available for critical reflection

Critical Analysis Strategies

- Different forms of reflection in different context
- Busy practitioners might rely on reflection-in-action
- Work in a dialogical team context (to hear different perspectives)
- Use critical incidents, case studies, reflective journals, practical exercises

Competencies of Critical Analysis

Four competencies that characterize the critical thinker:

Clarification

- It means to identify the focus, analysis arguments, ask and answer questions of clarification, define terms, judge definitions, deal with equivocation and identify unstated assumptions

Basis

- Judge the credibility of a source, make and judge the observations.

Inference

- Deduce, judge deductions, induce, judge inductions, make and judge value judgements.

Metacognition

- Reason from premises, assumptions and positions with which one disagrees or doubts without inferring with one's thinking and integrating the other abilities in making and defending a decision.

Ways of Knowledge Generation: Nurturing Relationships

Nurturing Relationships

Becoming a reflective practitioner requires us to explore ways of generating knowledge. One way to do this through nurturing relationships that we have in our professional roles. Critical reflective learning is actually nurtured by relationships between teacher and learner, there for becomes a much more reflective process. What we need to consider is the nature of these nurturing relationships.

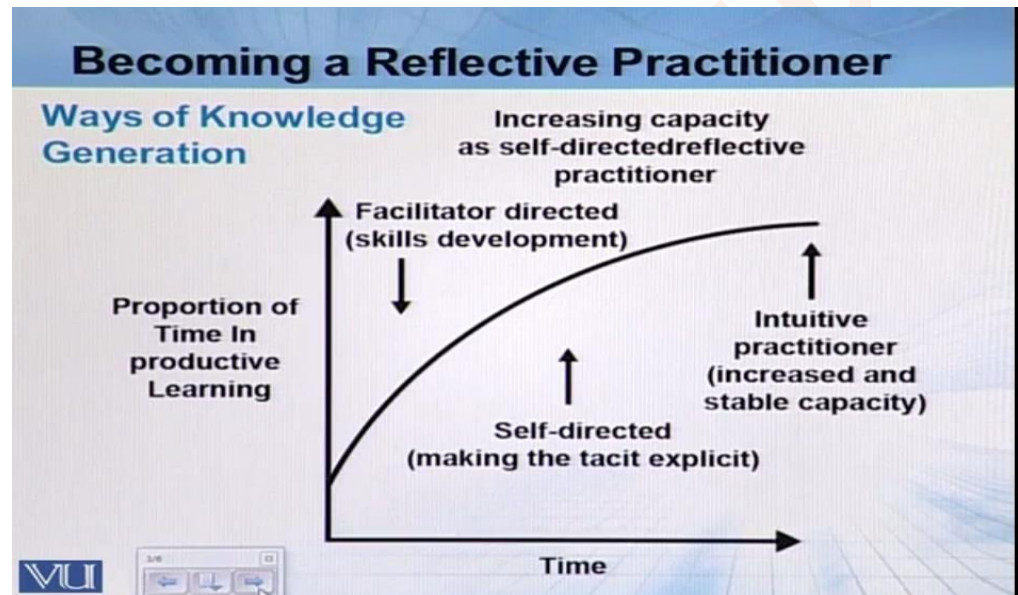
Key Aspects of Nurturing Relationships

There are some key aspects we will consider.

- **Mutual:** Any sort of positive constructive relationship is mutual for everyone involved, in other words everybody benefits from it.
 - **Open:** There is of-course open us in relation to way people are interacting with each other.
 - **Challenging:** Within the domain of reflective practice they also need to be challenging relationships, in other words critical but supportive.
 - **Contextually aware:** The relationships we have in order to help us generate knowledge also need to be representing properly in the context we have.
-

- **Characterised by dialogue:** Perhaps fundamentally relationships are defined by the language we use, the dialogue, the context of the dialogue that we have with others.

We can look at this from the perspective of a simple chart



The increasing capacity of reflective practitioner to be more self-directed in terms of their reflected practice to develop **their** knowledge and to generate knowledge about their roles is in terms of the amount of time that they have within their roles and also the amount of time they give, they are able to give to the process of development to the activity of learning. For teachers who enter newly into the profession of teaching, we conceive that very much knowledge is much lower and therefore needs to be built. Therefore, the activities we have as reflective practitioner as teachers in the classroom is very much focus on the skills we need to survive in those environments. Therefore, it is very much directed by this particular need but with time and with experience and with opportunity to give to reflection and to learning the teacher becomes or can become much more self-directed. In other words, learning is moving from being tacit to much more explicit to being sort of very vague and be known to something we not know very clearly.

As we mature in our roles, we become much more intuitive in terms of the practice of teaching we are doing and this where we experience a level of emancipation as reflective practitioners. The amount of knowledge generation that we can engage with and the amount of knowledge we can develop become much greater at this level because we are not focusing solely on surviving and developing our skills but actually broadening and deepening our understanding of teaching learning. So, nurturing relationships is a very useful strategy for the reflective practitioner to engage with in terms of developing and generating knowledge.

With time nurturing relationships can actually help us to develop as practitioners and essentially the relationships **that** we have with our students and the relationships **that** we have with our colleagues in the work environment. These relationships can help us become much more self-directed as learners and self-directed as reflective practitioners and it really as we conceive from the job moves us from focusing things purely like skill development to be much more intuitive and much more investigations of the teaching that we are doing.

Self-Directed Learning

Self-direction is quite powerful. With self-directed learning we generate knowledge in a way which is responding to the learning needs and also is part of the responsibility of being a reflective practitioner.

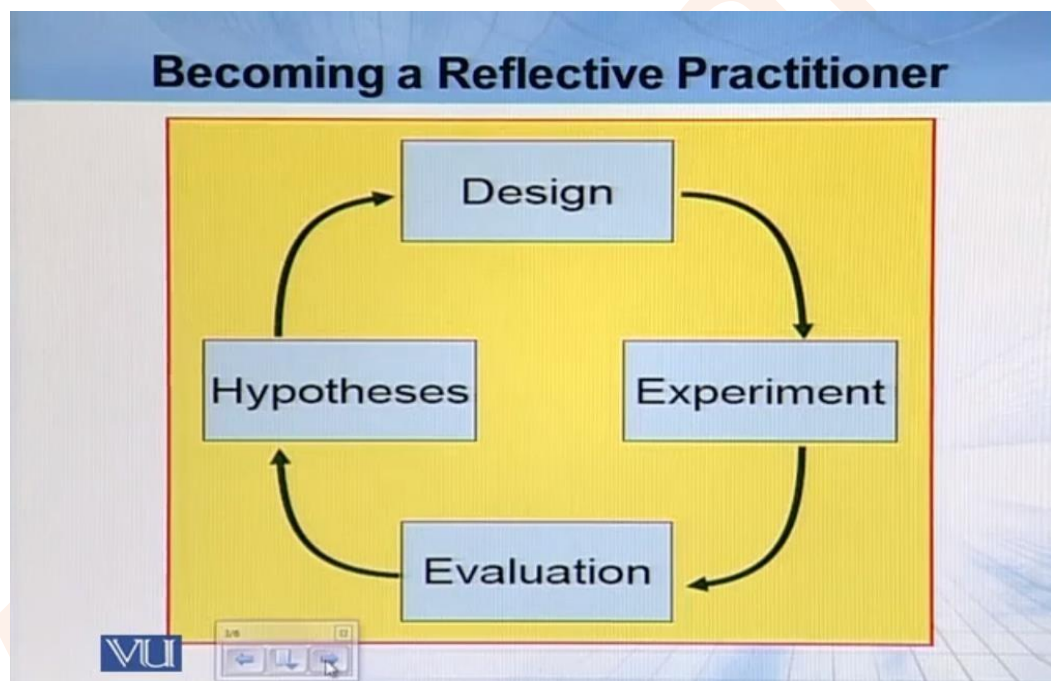
The individual teacher is when they are being very intuitive very self-directed is empowered to select what knowledge is powerful and useful and to manage that knowledge in terms of application in the classroom. Assessing the quality of that knowledge and how it is used. Perhaps the key thing to be aware of in terms of self-direction as a reflected practitioner is that motivation drives the whole process.

As a teacher, as a reflective practitioner if you are not motivated to learn, not motivated to develop then the reflection is not going to help that development at all. Therefore, independence is significant part of self-directed learning as well as the ability to set goals to define what is worthwhile to learn, teachers have to focus on the particular aspects of experiences in the classroom and then to look at those people who can support the process whether they are colleagues' or critical friends.

Ways of Knowledge Generation: Scientific Method

Scientific Method

A way of generating new knowledge for the reflective practitioner is through the scientific method. The scientific method has proved essential in terms of knowledge building through history. Essentially it is based on identifying certain hypothesis and trying to experiment in order to see whether these hypotheses stand up to experiment or fail in terms of the experiment. **It is possibly the single most important cognitive innovation in history and** we can look at the scientific method from a nice simple design where the reflective practitioner is abstracting thinking, designing their concepts and experimenting on these concepts.



Experimentation

For the reflective practitioner teacher this is looking at issues and experimenting to overcome those issues. From that experimentation we look at the results and we evaluate the effectiveness of the experimentation and then refer the evaluation back to the hypothesis. Is our hypothesis correct or is it false. If it is false, we need to re-hypothesis and re-design in order to experiment again and re-design and refine thinking about the

issues. This cyclic process which is very useful for the reflective practitioner although of course, it can be argued that teaching is not a scientific process but the scientific method is very useful in providing a structure for the experimentation where experimentation can be seen as trial-and-error process. The scientific method gives experimentation a much more scientific structure and a much more valued structure.

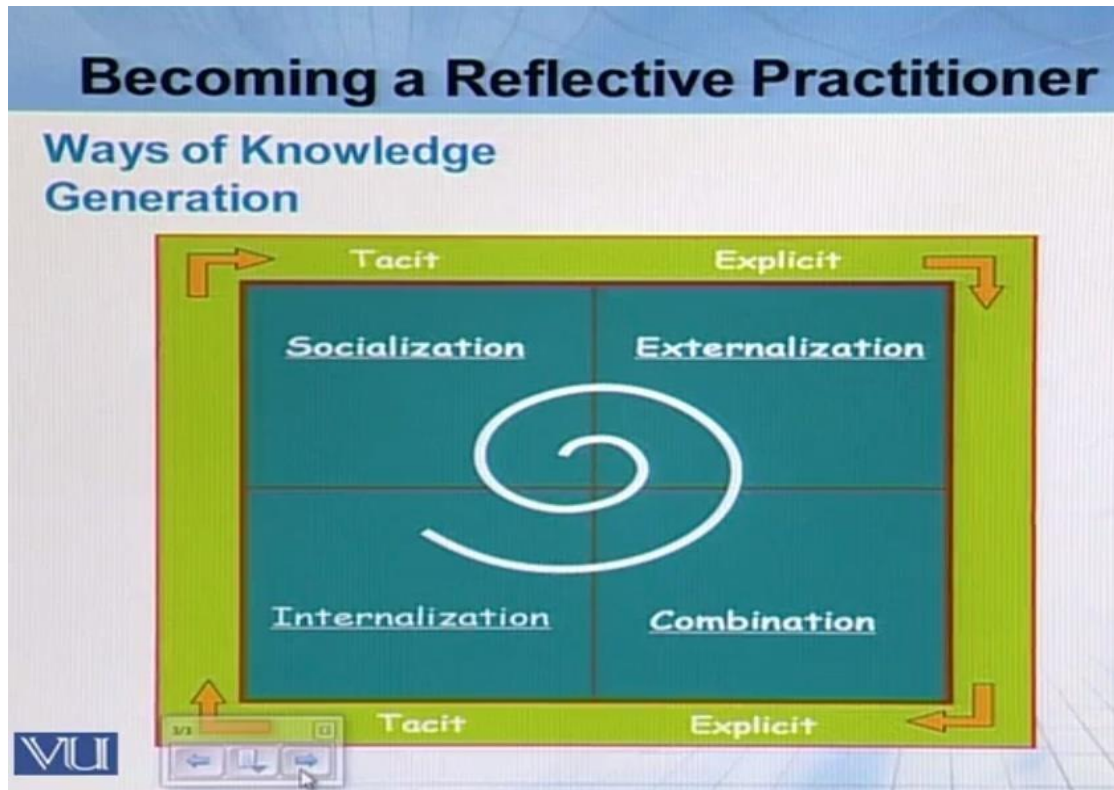
Statistical Analysis

What plays a important part in scientific method is analysis and what we bring to that is essentially is systematic evaluation of the information we are gathering. The analysis phase is very much about looking at the variables which we set in the hypothesis and then seeing if we can verify those variables, seeing if these stand up to the scrutiny and looking at the within the whole context of teaching. The data results that come from this is observation of reality is very much not about changing our experience but trying to be reflective about the experience without any change taking place. Statistical analysis from this scientific method perspective for the reflective practitioner is very much about possible to experiment in the classroom. To make that experimentation the object of the research, we had to remember that statistical analysis is quite key when dealing with human beings simply because working with variables constantly rather than fixed items we can catalogue.

Ways of Knowledge Generation: Cycle SECI

Cycle SECI

In becoming a reflective practitioner, we are looking at ways of generating knowledge. One way to do this through the SECI cycle. SECI cycle was developed by Hirotaka and Takeuchi. A way to generate knowledge by transforming it from tacit knowledge to explicit knowledge and then back to tacit knowledge. This is represented in the diagram where we can see that knowledge transforms through number of phases in a never-ending cycle



Socialization

Looking at this in more detail what we can see in model is that if we can look at the bottom left-hand corner knowledge is generated and internalized by the teacher by the reflective practitioner. As an internal process it's quite tacit and perhaps unknown to us, it's just acquired experience but through discussion with colleagues through further experience with the context of the work environment. The knowledge through socialization starts to become much clearer to us.

It is still tacit knowledge, it's still little bit unknown and vague but we are starting to realize that it's not just our knowledge but there are components that are shared by others in the context in which we work.

Externalization

From this the reflective practitioner through dialogue through the process socialization begins to externalize that knowledge. In other words, it becomes explicit within the context of the work. Externalizing that knowledge begins to make it more clear and helps us in understanding it in greater depth and detail and as we think more about it and externalize that knowledge through the socialization process, we begin to find points of comparison and contrast with that knowledge.

Combination

Comparison and contrast with theory and comparison and contrast with other people experiences. So, we start to combining pieces of knowledge together in a very explicit knowing way in order to rationalize the experiences we are having. From this process of combining the components of knowledge, we start to internalize into the conceptual frame so that it becomes much more tacit to our experiences.

Internalization

We do not think explicitly about this new knowledge, about this new learning instead we simply apply and understand what we have learned in a very intuitive way. This leads to new experiences, new internalizations which are again tacit but through the process of socialization we begin to develop tacit knowledge into much more explicit understandings. From socialization again we move to externalization where we are sharing that knowledge with others and others are sharing that knowledge with us.

Ways of Knowledge Generation: Modeling and Conceptual Innovation

Modeling and Concept Mapping

Another effective tool for the reflective practitioner is through Modeling and Conceptual Innovation. Modeling is used essentially for two key purposes and that is to confirm theories that we are building and also to make prediction about the results of actions that we are undertaking. We can look at modeling and concept innovational concept mapping. In the literature, there are different models we can look at for different

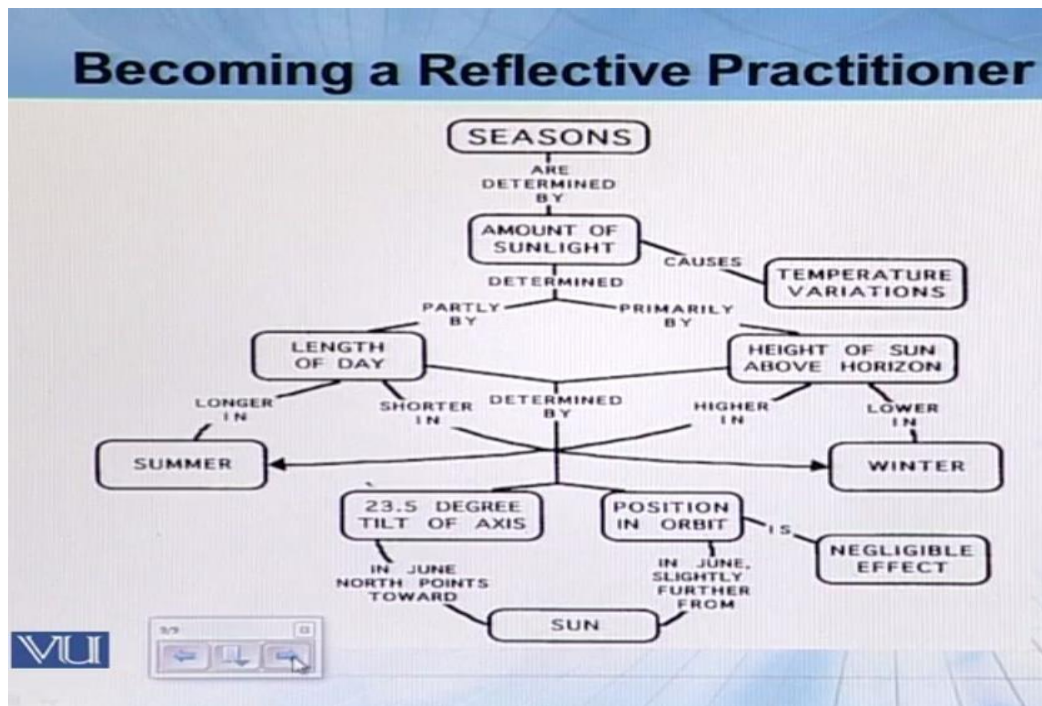
things from science through to nature. In “A New Kind of Science” and “Mathematica” Stephen Wolfram uses modelling for exploring patterns in nature.

For the reflective practitioner the notion of modeling and concept mapping is very new. Essentially modeling is something which is going to grow in the future, as a process of knowledge generation for teachers and as a process of generating knowledge the reflective practitioner is using models, using concept maps in order to build connections and to develop understanding between components of knowledge. In order to make sure that the knowledge is constructed and through this process of reflective practitioner is looking to build theories of practice from theories of teaching and learning and moving the theory into action.

Using Concept Maps

Concept maps are very graphic form for the reflective practitioner to use to represent the knowledge which they are acquiring and building over time. Concepts maps provide number of insights for the reflective practitioner into not only the concepts they have and the connections between those concepts. Concepts maps therefore are therefore comprised of two key components. The first component is the actual concepts that we are engaging with. The other component is linkages between those particular concepts. The linkages can be in form of linking words or linking phrases which connect one concept to another. Very much they are about the relationship, specifying the relationship the type of relationship between those concepts in the knowledge.

Model of Concept Map



We can see above a model of concept map for seasons. It's a very simple model to look at from the reflective practitioner point of view because it can help us to understand how a concept map can be use. **Essentially** a concept map can have a number of highlighted features of concepts that the reflective practitioner is dealing with. In this example we talk about amount of sunlight and temperature variations.

Ways of Knowledge Generation: Design

Design

Reflective practitioner can generate new knowledge through the process of design. Design is a way for the reflective practitioner to gather artifacts which are outcomes of experiences and use this overtime to build up bodies of knowledge. **Designs are artefacts that contain great amounts of knowledge embedded in them.** These artefacts automatically part of the design and then become huge amount of knowledge which the reflective practitioner is holding. The process of design is very much a dialogic process about reflecting with self through the reflective journal. It's also about dialoging with other colleagues, other stakeholders, **parents**

and also with students in order to build up the framework we are using for the design which is the **basis** of knowledge generation.

The design process therefore is very interactive and as an interaction process **between** the ideas we have, the values and beliefs embedded within those ideas and also recognition of the context in which we are working, the situation which influences those beliefs and ideas and of course the expectations and the impositions of the others involved in the situation, in other words the colleagues or stakeholders. So, design is a mean for these three components to come together to **interact and** inform **one** another. As such the knowledge generation process through design is very much about making visible the process of **generating** expertise. As a reflective practitioner it's therefore about experts and experienced practitioner sharing knowledge within the particular discipline. In order to create that design and also its sometimes perhaps complex it's a very detailed activity for the reflective practitioner to engage with to generate knowledge.

A key thing to remember here is that design as a way of knowledge generation is a socially constructed process rather than an individual process. It's through this explicit social process where perhaps the power of design really comes through. It is important **to remember** for the reflective practitioner to remember that the process of design for generating knowledge has a very key foundation and that key foundation is that the reflective practitioner is able to engage with conceptual frames for learning and be able to reframe concepts and reframe understanding in order for the knowledge generation to take place. Without reframing of concepts, without reframing of our understandings and reframing the way we approach our understanding; design cannot work as a means of knowledge generation. So, the key framing of reframing in terms of design is what will generate as a new knowledge for the reflective practitioner.

Theories, Knowledge and Practice: Teacher Professionalism

Teacher Professionalism

Becoming a reflective practitioner is a process of combining theories, knowledge **and practice** into coherent whole which is meaningful for the teacher and this all comes together to generate a **concept** which

we can think of as teacher professionalism. The fundamental question we have to ask ourselves here is good teaching is a direct consequence of reflective practice?

“Reflective practice is the habit of inquiring and investigating a problem situation in order to understand how to frame a solution”

Donald Schon, 1983

As we have seen reflective practice is very much a process of inquiring into a problem of investigation and understanding a problem in order to come to some sort of solution to drive processes to solve that problem. What this means is that we need to think of reflective practice from certain viewpoints in terms of professionalism. For example, we need to think of how reflective practice can boost teacher professionalism and improve and sustain teacher professionalism.

Assertion: Reflective practice Boosts Teacher Professionalism:

Caliber

One thing we need to look at here is the caliber of teaching, the quality of teaching. The caliber of teaching, the caliber of reflective practice is very much about challenging the assumptions we have about the practices of teaching about the pedagogical practices **that** we are using. Using these to establish for us the principles of good practice. In other words, to see teaching as an art and the process of reflection is to refine and to improve the quality of the art of teaching.

Discourse

Teacher professionalism through reflective practice is also about discourse. Discourse dialogue helps us to cultivate for ourselves **as professionalism** a level of confidence in our abilities and discourse also helps us to describe those abilities to others to articulate the level of knowledge, the level of expertise, the level of professionalism we have.

Knowledge Creation

We also know that teacher professionalism from the reflective practice standpoint is also about knowledge generation, knowledge creation because what the reflective practitioner doing is thinking and observing critically and using this criticality to frame changes in terms of our frames of understanding in order to develop for ourselves new models and theories of teaching.

3 Levels of Teacher Knowledge and Inquiry (cochran-Smith & Lytle, 1999)

We have got the level of Knowledge for Practice acquiring and generating knowledge for improved practice. We have also got Knowledge in Practice or in other words knowledge as applied in the classroom and also of Knowledge of Practice or in other words, our metacognitive processes in relation to the work we are doing.

What we can do with this is begin to think of a diagrammatic representation for this knowledge and practice generation.

1. Knowledge for Practice

This format we can see that knowledge for practice is the foundational stone of teacher professionalism. It's where the teacher is learning content and learning pedagogical knowledge from theories, from researches and beginning to apply this in practice and although we can consider it a very basic level of professionalism. It is rather a foundational level of professionalism equally important to any other level without it we cannot call ourselves professionals.

2. Knowledge in Practice

This is where teachers are acquiring practical teaching knowledge from the experiences they have of teaching over time. From the reflection of that experience and inquiring into experience from day-to-day basis. Knowledge in Practice is ongoing and forms perhaps the most significant part of teacher professionalism.

3. Knowledge of Practice

This is where teachers overtime developing skills sufficient enough to be able to generate knowledge which is coming from the experiences of the classroom and the school and whether teacher is seeing such contexts as places of learning for themselves, as process of inquiry into the process of teaching. It's a way for the teacher to make cognitive connections between theory and practice, between the issues that they are having personally with the bigger issues concerning education as a whole.

Knowledge of practice is therefore very much taking a very critical perspective not only of theories but also of experiences of others and their own experiences in order to bring these three things together into a new generation of knowledge and understanding. From this generation of knowledge that we have start to develop the profession and also to develop our own roles as a professional within the context in which we work.

So, the three levels of thinking here and knowledge generation fundamental to the process of professionalism. The foundational level knowledge for practice is not insignificant it's a foundational level and therefore as important of knowledge of practice which is generating knowledge over time.

Topic 134

Theories, Knowledge and Practice: The Moral-Ethical Dimension in RP

Open-Mindedness

Our attitudes include things like open mindedness. This is important for the reflective practitioner because it underpins beliefs about the **diversity** of our students. The intellectual, the social and the other dimensions of diversity that we can be working with when we are in the classroom working with the groups of people. Another aspect of open-mindedness that we need to embrace as reflective practitioner is **humility**. We must remember that teaching for the reflective practitioner is the process which is not one way but two ways.

The reflective practitioner is learning from and with the learners and with the students as much as they are learning from the teacher. Therefore, this does require open-mindedness and humility in order to be honest with ourselves about the mistakes we are making and about the direction that we are travelling in. Another attitude which is significant for the ethical and moral dimension of reflective practice is that we are very caring about **our students** and we hope for them in terms of their learning and in terms of their future. What we are doing with them in the classroom is preparing them for their future for five years, for ten years, for fifty years hence. Therefore, that's a very great responsibility for us and part of the attitude which underpins the process of reflective practice.

Responsibility

Another attitude which is significant for the reflective practitioner needs to be aware of this moral and ethical dimension is the sense of responsibility that we have. As reflective practitioners we need to **consider the consequences** of our actions **in the classroom**. Our actions and experiments with our practices, the approaches and strategies we use are always for the good of students not for the sake of personal development. Therefore, the responsibility is that we are always looking at the consequences of our actions. The consequences should always be positive and good and there is also responsibility that we are developing from the experiences we

are having. We are **making meaning** from them for ourselves as professionals in terms of teacher learner interaction and how this comes together for the reflective practitioner as a professional.

Engagement (Whole-Heartedness)

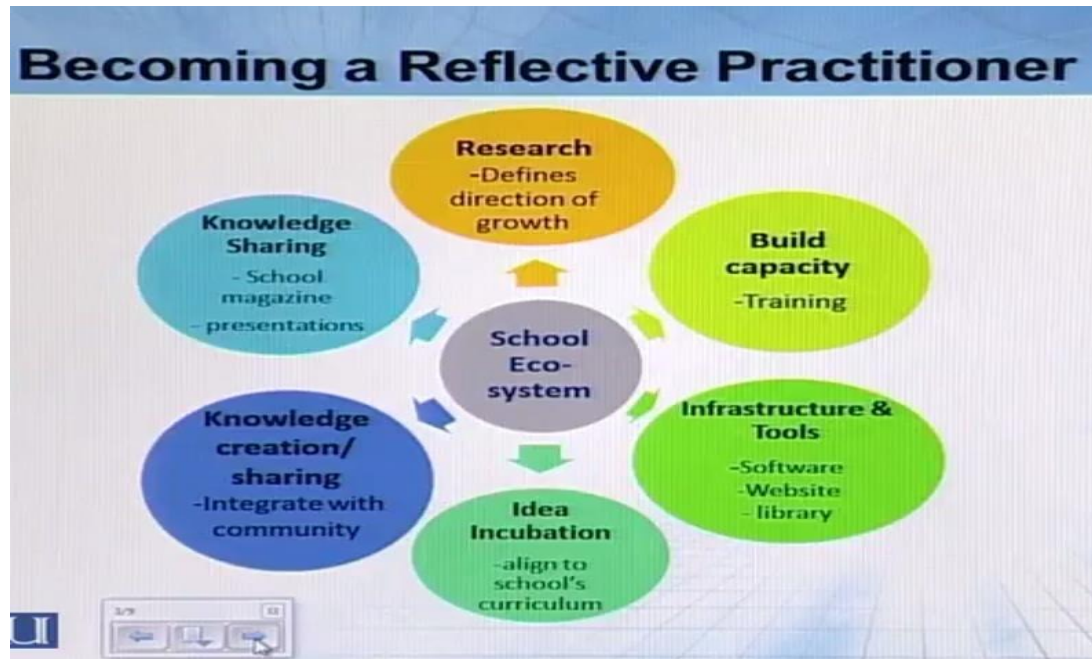
There is also an attitude of engagement or in other words whole-heartedness. Reflective Practitioner is somebody who is being very critical, asking questions about their role as teachers and what this means is that we need a certain bit **curiosity about subject** that we teach, our professional role, about the pedagogy of that subject **and about** the **impact of approaches on teaching and learning** and on students and therefore this whole-heartedness is part of the moral and ethical dimension of reflective practice. We need to remember for the reflective practitioner is that the teaching is very much an art individual to the teacher but is far reaching in terms of its impact. Therefore, it can be directed and guided in any particular way by the reflective practitioner so therefore it is something which is very much developed over time and evolves **along** with the teacher. So, this moral and ethical side of reflective practice realizes that the teacher the reflective practitioner is a growing and evolving practitioner.

Theories, Knowledge and Practice: Teacher Knowledge and Enquiry

Becoming a reflective practitioner then is about combining knowledge and theories and practice.

It's easy to think of this in terms of teacher knowledge and teacher inquiry. We know that the process of teachers in terms of becoming reflective practitioner is very much about thoughtful and reflective process and the purpose of this for the reflective practitioner is to be active contributors to community the work, the educational community. What we need to think about this is where this all comes together in terms of developing as a professional, developing as a reflective practitioner and we can look at this in terms of very simple model for becoming a reflective practitioner based on knowledge development.

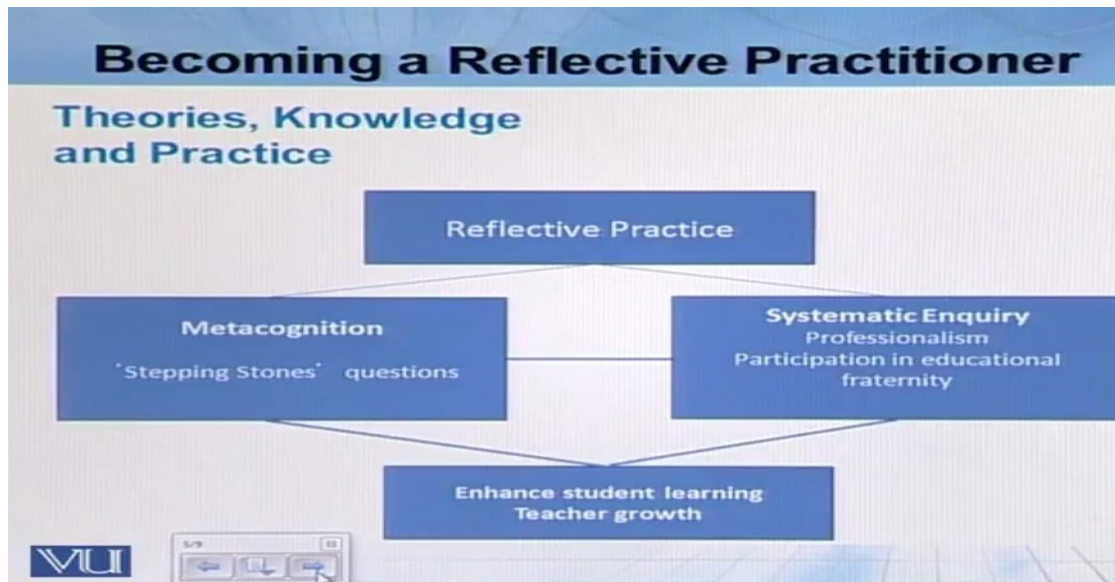
Approaches to build Professionalism



We as teachers are working within a community. We can think of it as an eco-system from this we can use different approaches in order to build knowledge and therefore professionalism. For example, we can look at research and this personal research defines the direction of personal growth. So, think of it in terms of training, capacity building and this approach very often taken by the schools in terms of building professionalism in teachers. We can also look at in terms of tools that teachers can use and infrastructure which schools provide and this could be things like internet access or libraries where teachers are able to access information.

Perhaps the new idea is thinking of school as an eco-system where people, professionals are working together to order to explore, define and re-define the components of teaching profession. In other words, it's like idea incubation where ideas are defined and tested and then built on and of course within a collaborative environment we can think of the school eco-system as a place where knowledge creation and sharing occur, where colleagues are working together. There are more pure forms of knowledge sharing through things like **school magazines** and presentations and so on. Within this model we have to think of over selves, where does

this reflective practice fitting to this knowledge acquisition process. We can look at another model define this, where reflective practice has two key components.



Systematic Inquiry

One component is about inquiry about investigation and its inquiry into professionalism in terms of teaching and its inquiry into have an individual is not only participating but contributing to the educational fraternity to the community as a whole.

Metacognition

The other component of reflective practice from this knowledge point of view is essentially metacognition or in other words how we are thinking about the process of professionalization and participation. The questions we are asking ourselves and challenging ourselves in order to move towards the professional identity.

It's these two things which come together within the profession to enhance student learning and promote teacher growth. So, we see that these two comes together what we can call Teacher Growth Model.

Teacher Growth Model

The Teacher Growth Model is therefore **multi-faceted** it comes from many perspectives from it in terms of knowledge creation and knowledge generation through the system, also in terms of becoming experts in pedagogy and the subject pedagogies which will move the profession forward in the 21st century. **Due to the multi-faceted nature of teacher's work, this implies that having a repertoire of strategies and content mastery is not sufficient for a teacher to be competent professional in the 21st century.** This Teacher Growth Model that we can talk about therefore is significant for teachers now to engage with because it's through this process of multiple mode of learning that the reflective practitioner actually through research, capacity building and through knowledge acquisition becomes a true professional in the 21st century.

Theories, Knowledge and Practice: Communities of Practice

Collaboration

Another aspect we need to think about from this point of view of this community of practice is the context, the domain in which the community occurs because the **context** very much provides the identity to the group. It actually drives the focus of sharing the experience and the insights the group engages with. Of course, part of **community of practice** is actual notion of practice or in other words the methods we applying, the knowledge about those methods and the notion that we bring to the community of practice of expertise and professionalism. These all provide the defining criteria in which the community of practice operates. The community of practice is really in order to bring **value** to its members. Value in terms of the quality of learning, the extent of learning and the depth of learning. Therefore, all members of community of practice are willing participants in the process of collaboration of working together in the process of contributing knowledge and sharing knowledge around those within the community

Why establish a community of practice in school?

We have to think about why establish a community of practice in the school.

- **A mechanism to promote rapid sharing of knowledge and expertise across diverse interest groups.** It's a very good mechanism; it's a very good means for developing rapidly because the individual teacher is not reflecting individually. In community of practice all reflective practitioners are going through the process of reflection and sharing, the experiences, the new insights and the new understandings with others. Therefore, knowledge and expertise grow rapidly in comparison to individual reflection.
- **Provides** a forum for exploring and testing ideas. The community of practice move the professionalism and the profession of teachers much more quickly towards good practice.
- Community of practice is a place where the new knowledge is more easily originated, where new understandings are created much more readily and explored and considered in more detail and therefore can develop much more strongly over time.
- Most importantly is that knowledge development for a reflective practitioner through a community of practice is very much responsive to the context in which teachers are working. The context will have unique issues associated with it, new problems arising all the time which are individual to the context. Therefore, the reflective practitioners within the community of practice have the opportunity to respond uniquely to those issues or problems as they arise within the context in which they occur.

Theories, knowledge and Practice: Communities of Practice

Communities of practice are useful for teacher becoming a reflective practitioner. Communities of practice support the individual professional's ability to generate knowledge and to manage **acquired** knowledge from experience.

Communities of Practices and Knowledge Management

- Communities of practice are knowledge management therefore is essentially about solving known problems **with** known solutions.
-

- A new teacher can present the problem to the community which is experienced by somebody else and therefore the solution to that problem is readily available. It's therefore about sharing about transferring know-how what we already know about the problems and how to solve them.
- It's very much about applying principles of good practice to what we are doing in the key areas of learning. The community of practice is a place where good practice has been stored and therefore can be transferred to the other shared and the community. Therefore, good practice by all members of the community everyday occurs.
- It's very much a process of sharing trust and building relationships with colleagues in order for this knowledge transfer and knowledge sharing to occur.
- It's a process of making knowledge acquisition much easier and because this is easier for the reflective practitioner for a teacher to find somebody else with answers to problems which they are experiencing.
- It's a very easy way to access that knowledge to organizational group intellect that available and therefore it's a very powerful tool for the reflective practitioner to use.

Types of Communities of Practice

There are different types of communities of practice and these are based on the strategic intents of the community of practice.

Helping

Helping in terms of its forum for helping an individual or individual solve problems and work with solutions that manage day to day classroom experience.

Best practice

It's very much about making sure that good practice becomes best practice. In school or some other educational environment, it is an opportunity to develop and disseminate to share around the best approaches,

the best strategies that are already known by others with those particular students with whom we are working. Therefore, we can say that this becomes the best practice **in that context** rather than good practice

Knowledge Stewarding

Another aspect of community of practice is about being custodian of knowledge, knowledge stewardship. In a community of practice knowledge is organized, managed by the group and held carefully like a repository where knowledge can be drawn upon by any particular member at any particular point and therefore a community of practice **becomes** a key point of knowledge input for the reflective practitioner.

Innovation

Another aspect of community of practice is innovation. It's a way to explore with other professionals' new ideas, break through ideas, new innovations to create and generate new practices which can **actually** become highly successful in terms of managing students learning.

Critical Success Factors of Community of Practice (CoPs)

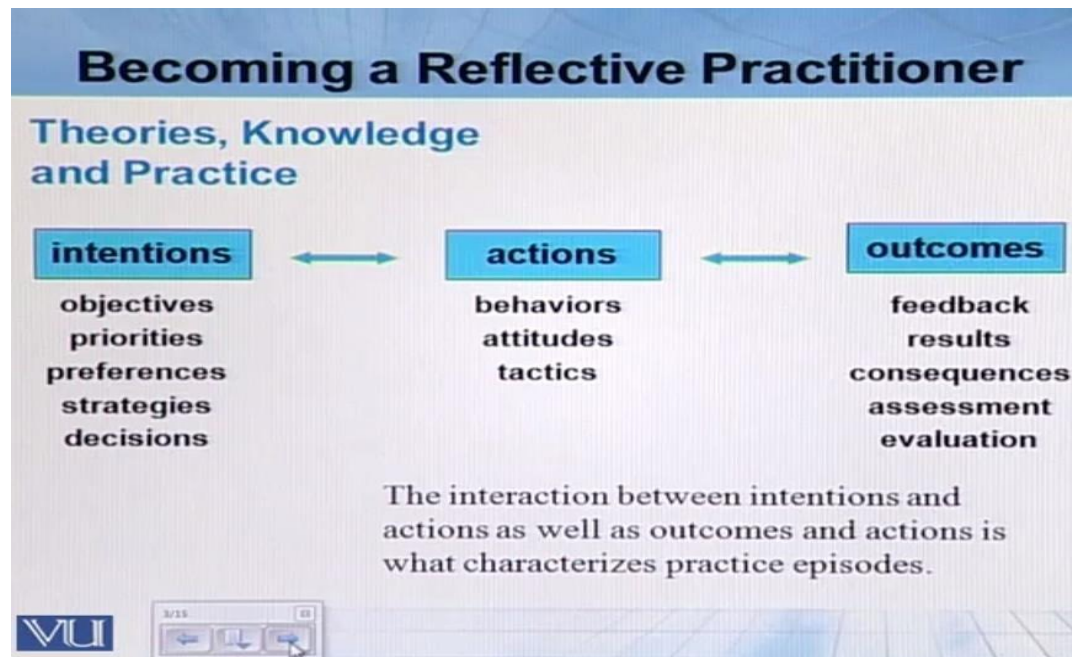
- The critical factors for the **success of the** community of practice are that it does reduce the numbers of hours that any individual needs to give in solving problems or issues that they are experiencing because answers may already be within the community.
 - It decreases the extent of the learning curve. Learning for reflective practitioner happens much more quickly, much more rapidly because of the knowledge which is being shared.
 - It cuts the amount of experimentation which is needed by the reflective practitioner. There is no need for re-working and re-inventing when somebody has already done that.
 - It's also about increasing the amount of innovation that can take place **because** innovation is supported by the community.
-

- Because the community is at to provide support, it means that the number of mistakes and errors that can occur will reduce because people will anticipate them and will have solutions for them.
- Because the community is there to support reflective practice. It means that any changes happen very quickly rather than taking time because of the reflective process.

Theories, Knowledge and Practice: Practice Episodes

Practice Episodes

Practice Episodes are a useful means of an individual to be a reflective practitioner. Practice Episodes are made of three components. Antecedents this is very much related to context in which the reflective practitioner is working. Theories of Practice that we are applying to teaching and something which we called practice episodes. Practice Episodes are comprised of three components. Our intentions within our teaching that action we can take and the outcomes of those actions. The intentions that actions and the outcomes form are a cycle that we can use as a reflective practitioner which brings substance which built the quality of the experience in the practice episode.



Intentions

We can see in this diagram the intentions that we bring in this practice episode. Things like our objectives, our priorities, preferences we have about the way we teach and the way we interact with students, the strategies we employ and the decisions we make in a lesson and about lesson. This interacts with the actions we take and the actions are based on our attitudes, behaviors and the strategies that we are directly applying.

Actions and Outcomes

The actions interact with the outcomes or in other words their consequences on the students learning, there results that comes out of the actions, their feedback we get and how we can use feedback for evaluation and assessment. The interaction between these components, between the intentions and actions, between the actions and outcomes is really what categorizes a practice episode. From this we can say that practice episodes are actually very much dynamic. This very much about the relationship between the intentions and actions on the comes. How the actions reflect the intentions and how the reality reflect the actions and of course how the intention reflect the outcomes, how these three things come together and interaction with one another.

Within this interaction which occurs within a specific context, practice episodes occur within a specific cultural context which is the antecedents' part of the framework.

Cultural Milieu (Antecedents)

The cultural context, the social context, the pedagogical context, the religious context the so many different contexts which provides background in which practice episodes occur. This context or antecedent component of practice episodes influences directly and quite powerfully the practice episodes that the reflective practitioner is engaging with. Therefore, we can understand the dynamism within the practice episodes not within the episodes but within the antecedents which impact on the practice episodes. Therefore, these can intrinsically affect the theories of practice in relation to the work that they are doing. So, practice episodes can feedback to a teacher understanding of theories as they are applied to practice. Understanding of the theoretical foundation that we are applying. Practice episodes inform us about the practice of theory. This is really not enough for the reflective practitioner. While integrating the theories of practice and the practice episodes, we need to do as reflective practitioner to add in the antecedents to this because they are powerfully influencing what we are doing and also incorporate all of this into decisions that we are making about the practice on the actions.

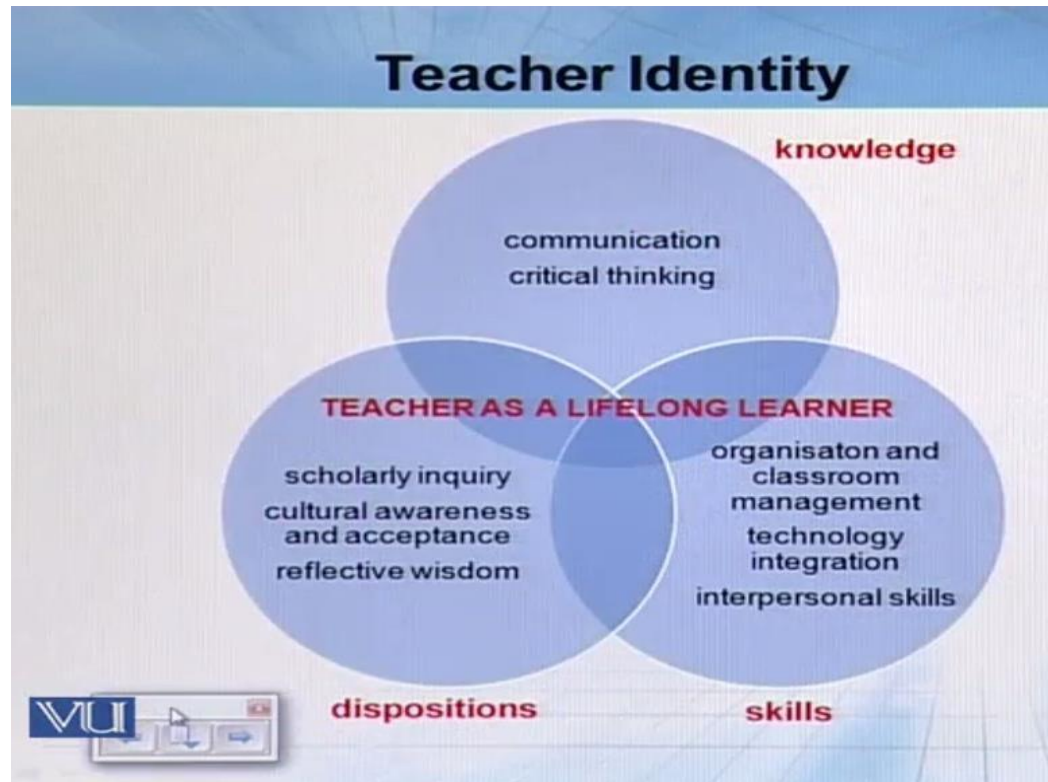
Reflective Practice

Therefore, we need to build antecedents into the model where the practice episodes and the theories of practice that we have interact with one another and the antecedents' than inform that interaction. In other words, we are getting information which we are feeding into the practice episodes from the antecedents which inform the interaction between the components of the practice episode. Therefore, it creates an opportunity for the development of professional knowledge for the reflective practitioner in the way we are linking facts and interpretations into a combined whole which we can call professional knowledge.

Reflecting and Connecting with Lifelong Learning: Teachers as a Lifelong Learners

Teachers as lifelong learners

To develop teacher identity, reflective practitioner needs to connect with lifelong learning. It's important for us to consider teacher as a lifelong learner. For reflective practitioners there are certain areas of learning that we need to engage with as a lifelong learner. There are certain proficiencies within these domains that we need to develop as reflective practitioners because they really guide us as lifelong learners. We can see here that the reflective practitioner is working with three domains of learning. Knowledge, skills and dispositions.



Skills

In terms of skills the teacher as a lifelong learner is working with classroom management and organization of management and technological integration for example, interpersonal skills within the context of working. In terms of the dispositions the reflective practitioner is someone who is working to develop certain wisdom in terms of reflective practice and to also a cultural awareness and acceptance of values within the school environment. As well as a confidence and a competence with inquiry into the process of teaching and reflection.

Knowledge

In terms of knowledge teacher identity is developed through communication and through critical thinking and through these three domains we can consider particular aspects of the process of reflection. Knowledge

definitely includes courses of education in order to develop us intellectually but also developing practical skills whether written or oral. Its written or oral communication which is a means of developing knowledge, quantitative literacy of what we are doing and also the ability to think what we are doing and also the approach to manage successfully as teachers. In terms of knowledge, we must be highly performing teachers very well versed in the subject we are teaching in the pedagogy of subject we are teaching. In essence we must be expert in the field and perhaps this is one of the defining characteristics of this domain of teacher identity. As reflective practitioner we must be able to demonstrate the ability to apply that knowledge to apply content and to also apply to apply the teaching of that content to the students and be critical about that within the educational setting.

The knowledge domain for the teacher is about possessing knowledge and the materials we need to acquire, the materials we need to find in order to access the knowledge especially within the subject we are teaching. There is the skills domain also which is very much focusing on the practical abilities that we are bringing to the roles of teaching and it can be very broad and expand across the school environment but can also develop from teacher beginning phases, teacher preparation through to experience practitioners. With experience skills domain become very powerful area for teacher skill development because through this our metacognition becomes significant through emancipation processes.

There are examples we can draw on in terms of the skill domain and what this means that we are looking at the skills within the context within the climate of the school. The micro politics within the school and the issues around the politics that occur and how this impacts our motivation to teach, the passion we have as teachers. This also generates for us the curiosity that we have about our professional identity and how we grow as professional selves.

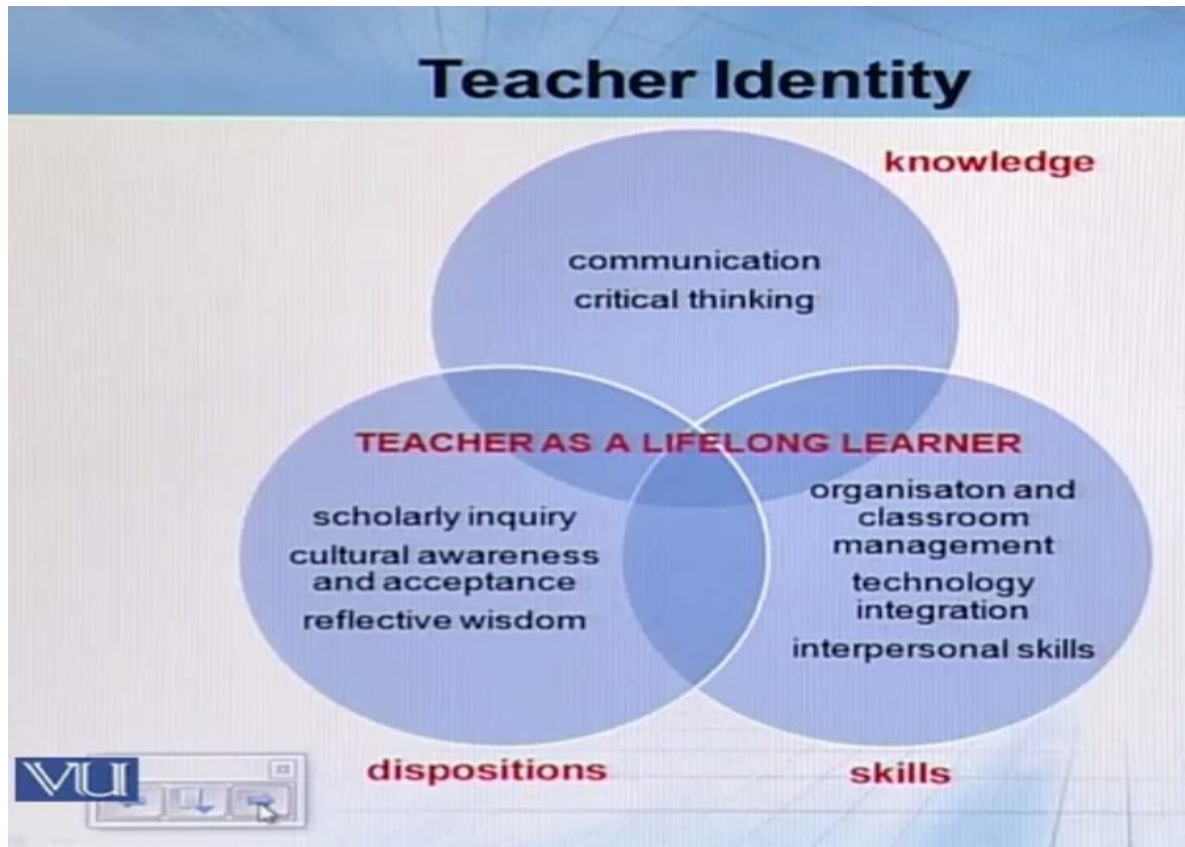
The skills domain is also significant for building confidence, confidence as teachers and confidence as experimenters in reflective practice. Experimentation involves certain amount of risk. We need to be confident to deal with risk. There is also an underlying premise for the teacher as a lifelong learner in the skills domain and that is we believe that all students can learn and learn very well and achieve well and this belief underpins the skills domain so that it becomes a key component of our identity.

Dispositions

So, the teacher as a lifelong learner is always extending his knowledge, the practical knowledge, the subject knowledge and the experiential knowledge they are acquiring over time. Through this teacher identity really evolves into something quite powerful and of course there is dispositions domain the attitudes, the beliefs and the assumptions that an individual reflective practitioner has about teaching and about the patterns of professional selves. The teacher therefore within this domain, the teacher who has grown identity is somebody who can demonstrate behaviors consistent with ideals of best practice and bring these to classroom in terms of fairness and openness and honesty so that all students have best opportunity to learn.

Reflecting and Connecting with Lifelong Learning: Teachers as Lifelong Learners**Teachers as Lifelong Learners**

We have seen that there are three domains of learning and there are certain proficiencies and we need to understand and explore these particular proficiencies within these domains



Knowledge Communication

In terms of knowledge domain there is communication, how the reflective practitioner uses communication skills effectively in terms of teaching and in terms of their reflective practice.

The reflective practitioner is using appropriate verbal and written communication, articulation of ideas in order to develop their practice and develop the practice of others and it's not about necessarily communication but it's about using language to build an understanding of ideas and to bring experience together with theory. Additionally, it's also about how the reflective practitioner uses different media different means to communicate ideas whether it's written to journal or through magazine a professional magazine or through other medium may be a blog or through other writing. Communication therefore brings clarity to the reflective process and it provide opportunity for feedback and developing ideas and goals for further development. This very much contributes to the ability to structure and restructure knowledge and to reinforce their learning.

Communication skills also allow the teacher to achieve the goals they have set for themselves because it is a means for clarifying those goals and therefore motivating towards the goals and therefore motivating towards the goals provide enthusiasm about the subject and about the learning of that subject. Teachers as lifelong learners are therefore continually in the process of using and refining communication skills with themselves with colleagues with students in order to develop the teacher identity

Knowledge: Critical Thinking

Another component of knowledge is critical thinking and the teacher specifically needs to engage specifically with critical thinking because this focus understanding of key aspects of professional practice and professional identity and informs the creation and generation of the skills for the profession in a very reasoned and logical and informed way which can be trusted by teachers and by communities of practice. Critical thinking is very powerful model for the teachers to use in terms of the process of thinking through teaching in the classroom and working with students and therefore, it does contribute to the context of teaching and even contributes to the wider society

Skills: Interpersonal Skills

In terms of domains of teaching interpersonal skills becomes very significant for the lifelong learner. The capacity to interact with people to sympathize with others to share sensitivity towards other people in the community and to work with both adults and children very productively very positively in caring and meaningful ways and therefore producing environments which are good examples of these qualities. Interpersonal skills we possess therefore in terms of the skills we need to foster in terms of peer collaboration. The teacher therefore must be willing to work with others to seek help from others and advice and guidance in order to develop themselves as lifelong learners. Therefore, interpersonal skills form a basis for this continuation process.

Skills: Integration of Discipline

Integration of discipline knowledge is very important. The discipline the subject we are working in is significant for lifelong learning because it's the subject we are teaching, we need to integrate the different perspectives of the discipline into our views of the subject and how we intend to teach that subject to the students and how the teacher brings all of these together actually helps with teacher identity.

Skills: Technology Integration

Technology integration in the classroom or in the school environment is useful for constructing knowledge because it's through this process which brings reflective practice into the 21st century

Skills: Organization and Classroom Management

Organization and classroom management is another proficiency that the teacher is engaging with, because classrooms are very complex environments. Teaching engages with very complex activities and therefore there needs to be very careful supports mechanism for the reflective practitioner in this regard with relation to reflective practice. We need to therefore adopt strategies which are very productive in terms of developing skills for classroom management and organization management and which support the organization as well as our own positive expectations of teaching and learning.

Dispositions: Scholarly Inquiry

Scholarly Inquiry is a very fundamental principle of reflective practice. Teachers need to be well versed with experimentation, exploration and also be knowledgeable about trends and issues which are emerging in teaching and how can we bring these to our own teaching through research and through experimentation within the academic environment.

Dispositions: Reflective Wisdom

Through reflection teachers gain more understandings of their roles and the knowledge they bring to the teaching and therefore this can help teacher become much more effective as a teacher, develop the teacher

identity and this is very useful for lifelong learning. Reflective wisdom is ongoing it's not something that we acquire. It is ever evolving and ever changing. It is something that the reflective practitioner will grow on continuously as a means of ongoing learning and development

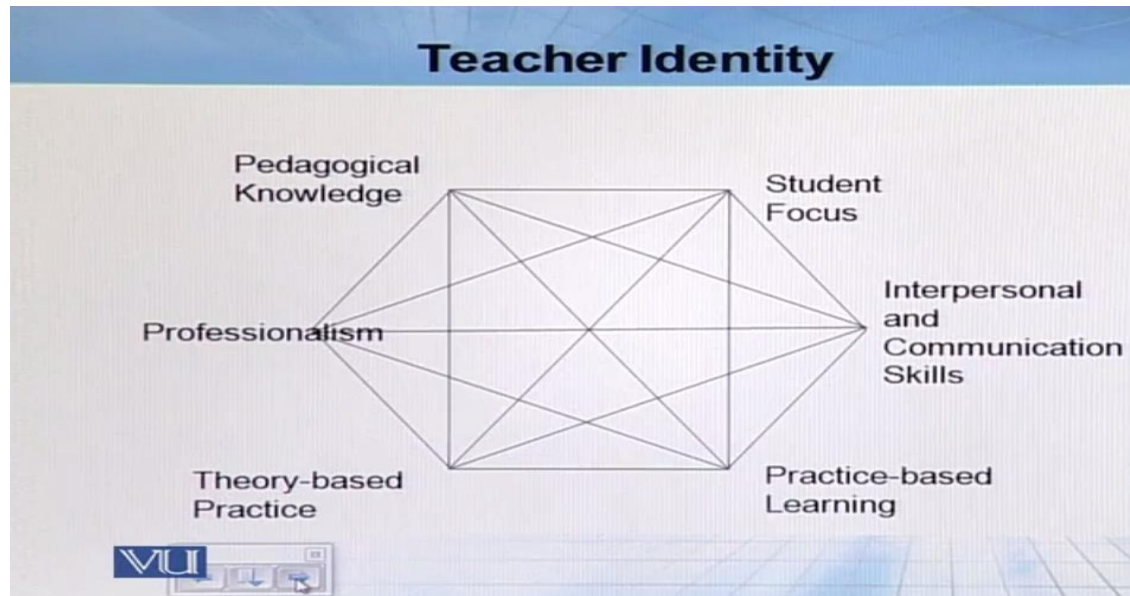
Dispositions: Cultural Awareness and Acceptance

And also, the cultural awareness and acceptance, the context in which the teacher is working and the diversity of personalities and experiences the teacher is working with. The teacher must provide educational opportunities for everyone, themselves and the students they are working with and their colleagues because this will strengthen everybody's appreciation through the diversity of what is potential and what is possible.

Reflecting and Connecting with Lifelong Learning: Competency Framework for Lifelong Learning

Competency Framework for Lifelong Learning

To develop teacher identity, we can look at the competency framework for lifelong learning. This is a framework which enables the reflective learner to consider components of their practice which will help them develop their identity as teachers. The competency framework which is about lifelong learning is a framework which is continuous and supportive and most importantly it is also impairing for the individual. The competency framework therefore evolves overtime. It can be added to and is used long term. It lasts for the life span of the teacher. It is not theoretical; it is very much an applied framework for the teacher and is useful in terms that it builds confidence within into the teacher identity development process. Any competency framework that you use is very useful for the reflective practitioner because it helps define not only the components of teacher identity but also the interaction between those components of teacher identity.



For example, in this framework we can see that interpersonal and communication skills form parts of teacher identity as well as learning from practice and the theory-based practice. These are three components of framework for competence. Also, another component which we can think of in terms of professionalism as well as pedagogical knowledge the knowledge we have about teaching the particular subject and then the final component of the competency framework could be something like our focus on the learners we have and the students we are working with. So, we see a number of components of teacher identity and there can be connections between any of these components. For example, how does interpersonal skill interact with professionalism? How does interpersonal skill interact with pedagogical knowledge? We are not only identifying the components of framework but how these components interact with one another. So, the competency framework then is very much about active participation in the process of teaching and the process of reflection. It very much helps us define the goals and how to accomplish those goals for teaching. It helps us set goals which are very specific, very measurable that we can define and realize the outcomes of our actions. They can set the timeline that we are working with in terms of development and also help us to identify the resources we need in order to help us develop as reflective practitioners. It's also senses about defining the goals, setting the goals. We need to consider that the particular goals can be set in terms of teacher identity as a reflective practitioner.

Essential Goals

There can be essential goals, dictated perhaps by the school community or by wider society.

These are goals that are essential for us that we do not have a choice about.

Problem-solving Goals

We also have problem solving goals. We experience issues in the classroom, problems in the classroom and therefore we have goals which we set to overcome those problems.

Innovation Goals

Then we have innovation goals. Goals which are about change, development and about professional growth which all come together to lead to professional teacher identity and we can look at this in terms of smarter framework

RP and the Socratic Method

- Socratic method aimed at understanding.
- Socrates questioned and examined and cross-examined his fellow citizens, not to convey a new truth to them in the manner of an instructor but only to point out the path along which it might be found.

A useful strategy for developing teacher identity through reflective practice is the Socratic method. The Socratic method is essentially aimed at developing understanding. Socrates developed an approach to questioning reality based on a thorough understanding of skills for questioning and exploring or cross-examining evidence in order to arrive at the truth of a situation.

The Socratic method:

- Questioning in order to reveal the ‘not-knowing’ (Socratic attitude).
- Gathering of examples from ordinary life.
- Reason from examples to most general presuppositions.
- Planned investigation into concepts.
- Investigate the concepts as such (definitions).
- Make yourself think.

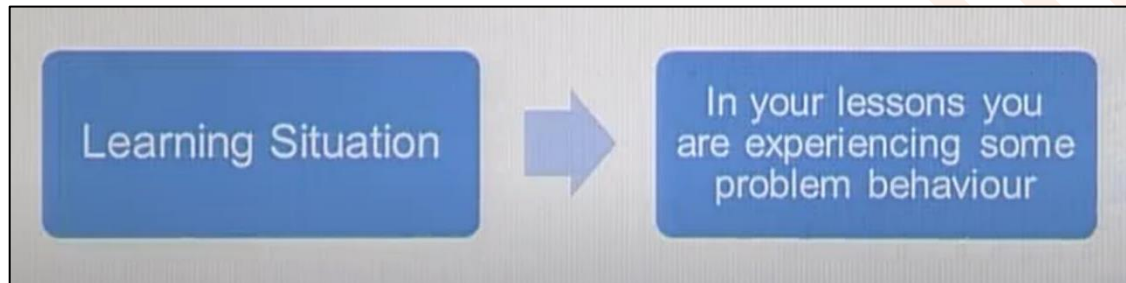
Six Steps in Socratic Questioning

The six steps in Socratic Questioning are:

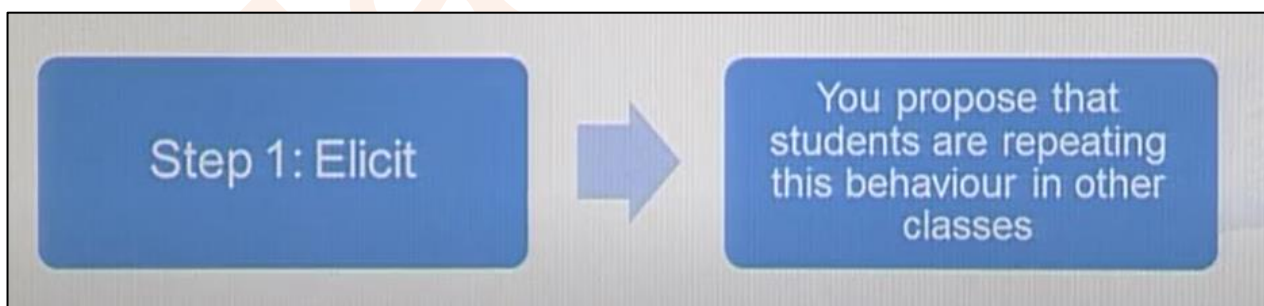
- Choose a topic and a question
- Give examples from own experiences
- Which example suits the topic best?
- Write out the example in detail
- Examine underlying principles
- Generalize answer (regressive abstraction)

The 4 Steps of the Socratic Method

- The ultimate goal for the Socratic Learning Method is not to help you to come up with a proposition that you can rest safely with.
 - The true goal of the Method is to help you examine your own beliefs and new information you encounter.
 - In frequently exerting the Socratic Learning Method, you should become independent as a reflective practitioner with curiosity and sensitivity toward new information, and gradually develop a mental habit of active inquiry and vigorous thinking.
 - The four steps of the method can be represented diagrammatically:
-

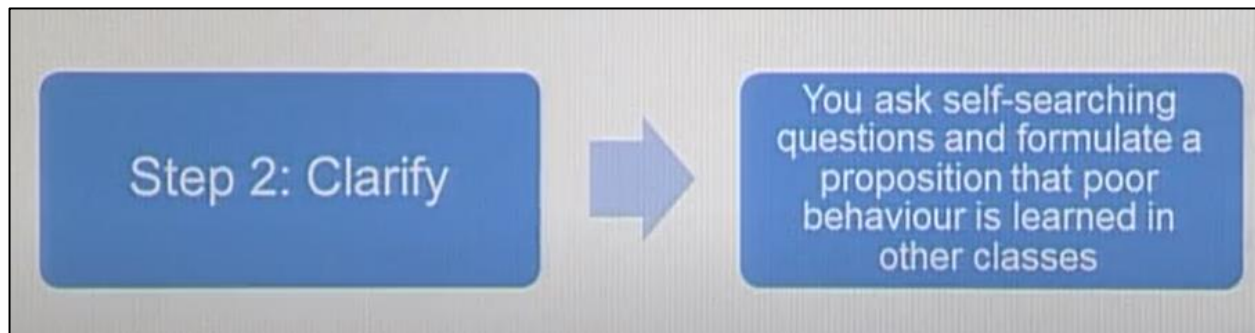


Here we can see that the Socratic method is defined as a sequence, moving from a situation to a decision. Let's look at an example: the Learning Situation in terms of the Socratic method, it might be that you as a teacher experiencing some problem behaviour in the classroom, note how this is framed differently not as a problem, not as an experience, but as a learning situation, and this is quite key for the Socratic method. The experiences that we have, the problems that we're experiencing are not spoken of in terms of problems, but in terms of potential for learning.



The **first step** in the Socratic method is to **elicit** meaning and to elicit the detail from the learning situation. And in the example which we are looking at, it could be that the teacher might propose that the students are

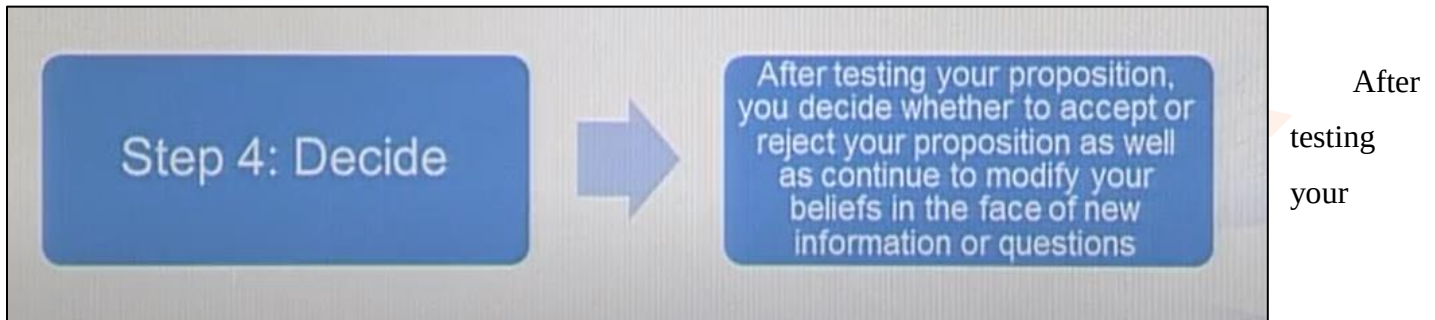
misbehaving in all lessons and not just in their own lesson, and that might be a starting point for further questioning and further exploration.



The **second step** is for the reflective practitioner to **clarify** the learning situation to get a better appreciation of the components which are creating the learning situation. So, at this point the teacher may be asking some questions about their own role in relation to the students' behaviour, and trying to come to some sort of proposition about why the behaviour is occurring. It might be for example that the students are misbehaving in all of their lessons, and therefore, this is some sort of learned behaviour which is carried through the whole of the school day.



The **third step** is about **testing** the propositions, it could be speaking with colleagues, it could also be about observing behaviour in the other classes, just to see for yourself whether the proposition about poor behaviour across all of the lessons is actually true.



proposition. You decide whether to accept or reject your prop

The **fourth step** is making a **decision**. Deciding whether the proposition is true and whether the experience needs to be questioned further. At this point, you're actually testing whether what you believe is true and making a decision to either accept the proposition or to decline it. And therefore, to rethink the initial idea about what is the cause of the problem, in order to then begin questioning again to find a better solution.

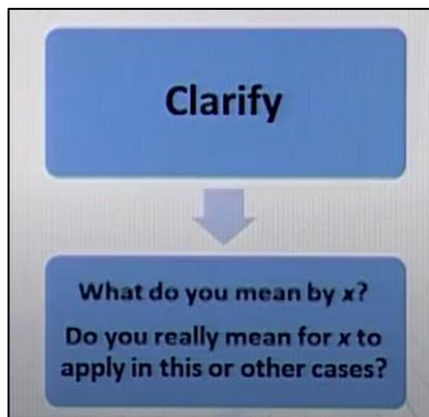
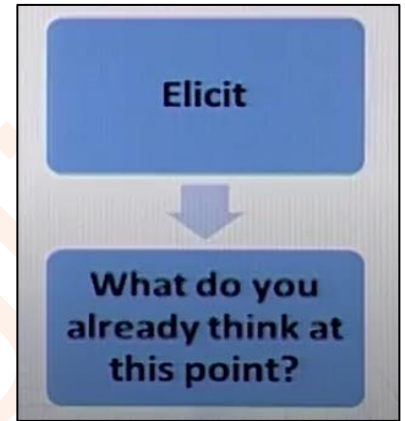
RP and the Socratic Method: Questions for Each Step in the Socratic Method

Questions for Each Step in the Socratic Method

- There are some questions you can ask yourself for each step of the Socratic Method.
-

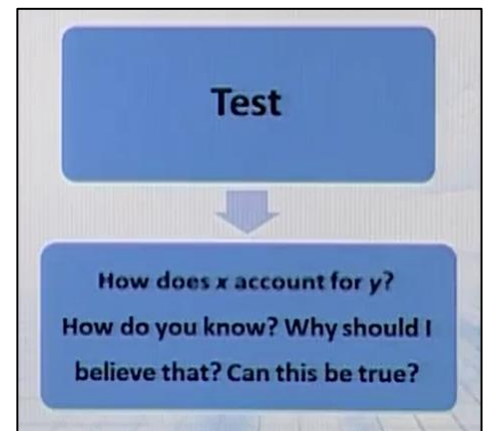
We have explored how the reflective practitioner can use the Socratic method to develop their identity as a teacher, and we will look at the questions that can be applied to the Socratic method. What is useful at this point is to explore specifically the questions for each step of the method.

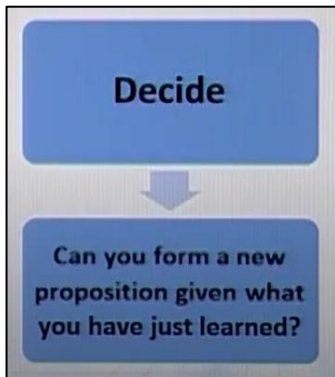
The first step of the Socratic method is called elicit: Elicitation of facts and understanding from a situation. And a question that the reflective practitioner can ask themselves at this point is what do they already think about the situation, what are they already know about the experience, and what other experiences are informing them at this point.



The next step of the Socratic method is clarifying understanding about a situation or experience. So, questions which are useful to ask at this point is: what do I mean by X, what do I mean when I say this thing, what do I really understand about the situation as I experience it and as I perceive it.

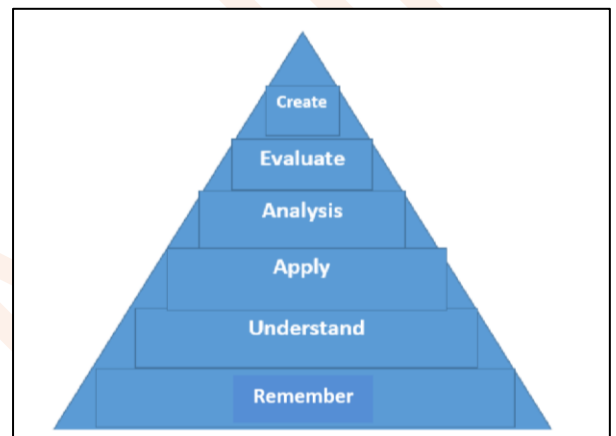
The third step of the Socratic method is testing our understanding, testing the assumptions and the presumptions that we are building as reflective practitioners. So, questions like: how does X relate to Y, how do I know what I'm saying from this experience, can it actually be true and can it really be true what I'm saying is happening, or is there another way of saying this is another reason.





And then **the final step of the Socratic method is decision making** for the reflective practitioner. So, the questions like: can the reflective practitioner form a new proposition, a new understanding of the situation based on the questioning process.

- **Link your questions to Bloom's Taxonomy**
- **Move from lower-order questions to higher-order questions.**
 - **Remember:**
 - 'Description',
 - 'Interpretation',
 - 'Outcome'



RP and the Socratic Method: Questions for Each Step in the Socratic Method

Questions for Each Step in the Socratic Method

Question Types

- Questions of purpose force us to define our task.
 - Questions of Information force us to look at our sources of information as well as the quality of our information.
 - Questions of Interpretation force us to examine how we are organizing or giving meaning to the information.
 - Questions of Assumption force us to examine what we are taking for granted.
 - Questions of Implication force us to follow out where our thinking is going.
-

- Questions of point of view force us to examine our point of view and to consider other relevant point of views.
- Questions of Relevance force us to discriminate what does and what does not bear on a question.
- Questions of Accuracy force us to evaluate and test for truth and correctness.
- Questions of Precision force us to give details and be specific.
- Questions of consistency force us to examine our thinking for contradictions.
- Questions of Logic force us to consider how we are putting the whole of our thought together, to make sure that it all adds up and makes sense within a reasonable system of some kind.

RP and the Socratic Method: Questions for Each Step in the Socratic Method

Questions for Each Step in the Socratic Method

- The mnemonic PAPER CLIP can be used to construct a sequence of questions which stimulate deeper thinking.
- Three types of questions are often used: exploratory, spontaneous and focused.
- Exploratory questions show us how much we know and may be used to explore a new topic or review past discussions that may relate to the current issue.
- Spontaneous questions can be used to explore our beliefs and assumptions; they allow us to reflect on the issue at hand.
- Focused questions narrow the discussion, stimulating us intellectually.

Questions are perhaps the most useful component of applying the Socratic method to reflective practice, and we've seen how questions can be used. There's a very simple mnemonic that we can use to help or guide the questioning process, and that mnemonic is **PAPER CLIP**. This helps move thinking from low-order to higher-order, and uses questioning to guide that process.

There are three types of questions: exploratory, spontaneous and focused. **Exploratory questions** are those initial questions helping us to understand the situation. **Spontaneous questions** are about our response to a situation. **Focused questions** help us get detail from the situation.

Exploratory questions are very much about the reflective practitioner using the Socratic method, and using this particular mnemonic PAPER CLIP to get a thorough understanding, gather all the information required to get very much clarity about an experience of they're having.

Spontaneous questions which are very much related to the immediate response, and the teachers' assumptions in relation to the situation that they're experiencing.

Focused questioning where we narrow down the discussion from a broad topic and get more clarity, or in other words, we are moving from broad ideas to very specific and detailed ideas.

So, using the mnemonic PAPER CLIP can help guide this questioning process.

P	Precision questions .Can you be more specific?
A	Accuracy questions. How could we test that?
P	Perceptive question. Is there another point of view we could examine?
E	Equity Questions. What conflict of interest exist here?
R	Relevance questions. How does this relate to the problem?
C	Complexity questions. What make this difficult question to answer?
L	Logic questions. Does this all make sense tighter?
I	Importance questions. What is more important issue on which to focus?
P	Perspicuity questions. - What do you mean?

The mnemonic PAPER CLIP identifies particular types of words to ask at any particular point. Using PAPER CLIP, we start off with **precision questions**, or in other words, questions which are helping the reflective practitioner become more aware of the situation and the details in the situation. For example, as a reflective practitioner you might ask yourself: can you be more aware of what is going on, can you be more specific about what is going on.

Accuracy questions: how could you test that particular assumption, how could you get a better understanding of whether the assumption is true or false.

The next letter in the mnemonic is about **perspective questions:** is there a different way of seeing the situation or the problem, is there a different perspective that you can use to examine the situation.

Equity questions: what are the conflicts of interest, or what are the biases that may exist within the situation which can affect the way it's being perceived.

The **'R'** in PAPER CLIP stands for **relevance questions:** how does this relate to the problem, how does X relate to Y, how is all of this understood as a logical whole.

The **'C'** in PAPER CLIP stands for **complexity questions:** what makes this a difficult question to answer. Remember, as reflective practitioners, we're always working in messy situations, and always trying to set unsatisfactory questions, so that we're always questioning rather than just finding solutions.

The **'L'** in PAPER CLIP stands for **logic questions:** does this all make sense, how does all this come together, how do all the component fit together into a logical whole.

The **'I'** stands for **importance questions:** what is the most important issue here, what is the most significant component of this situation.

And then the **'P'** letter in PAPER CLIP stands for **perspicuity questions**, in other words, what does this mean to me, what does this actually mean as a whole, how is all of this translated and perceived by all the actors in the situation.

So, we can see that this mnemonic **PAPER CLIP** provides a very sound structure, a list of trigger words that we can use to question a situation from the perspectives of description and through to a decision for the reflective practitioner.

Socratic Questioning and Critical Thinking

- Thinking is driven by questions: No questions mean no understanding.
-

- The key to distinguishing *Socratic Questioning* from questioning is that Socratic questioning is always systematic and deep, focusing on complex concepts, principles, theories, issues or problems.
- Socrates was fundamentally concerned with the soundness of reasoning, with getting closer and closer to the truth in any given situation. He was more interested in the process of learning – the questioning process- than in reaching conclusions.
- The Socratic method involves two primary processes- the destructive and the constructive process.

- **The Destructive Process**

In the destructive process, ideas formerly held are shown to be illogical or otherwise unsound. In other words, the practitioner comes to recognize the flawed nature of the reasoning.

- **The Constructive Process**

In the constructive process, the practitioner is encouraged to replace the flawed thinking with logical or justifiable thinking.

- **Critical thinking and Socratic questioning both share a common end.**

Critical thinking gives a comprehensive view of how the mind functions and Socratic questioning takes advantage of that overview to frame questions essential to the quality of that pursuit.

- **Critical Thinking**

The goal of critical thinking is to establish an additional level of thinking to our thinking, a powerful inner voice of reason, to monitor, assess and re-constitute in a more rational direction- our thinking, feeling, and action.

- **Social Discussion**

- Socratic discussion cultivates that inner voice through an explicit focus on systematic, deep, disciplined questioning.
- The Critical Thinker considers the elements of thoughts.

The goal of critical thinking is essentially to establish levels of thinking about our thinking, and creating a questioning reflective, in order to take the situation re-constituted in a very meaningful way for ourselves, develop a much deeper understanding and understand our responses to that situation.

And in a way, Socratic discussion is a process of dialogue, whether it's dialogue with self or dialogue with others in relation to a similar explicit, and deep appreciation of the experience that we're having.

The critical thinker is looking at particular elements not only of the situation but elements of thought, which is also part of the process of Socratic questioning. What we can end up with at this point is a sequence, a cycle of thinking and questioning. When we bring these two things together, we are looking at the goals of

the process of thinking, where exploring issues through questions, gathering information, eliciting deeper understanding to make interpretations and inferences about the experience.

Then from this, we're divining concepts, bringing clarity for ourselves about theories and principles which lead to assumptions for ourselves as reflective practitioners, and which help us develop new

frames for thinking, and this ultimately leads to new goals and objectives as part of the reflective practice process. So, we can see that critical thinking and Socratic questioning share very common processes and very common goals.



How to be Reflective

- For the busy teacher, it can feel like there is not enough time for RP.
 - There are many tools to assist you.
 - **Feedback**
 - Feedback is central to the process of reflection.
-

- Can be written or verbal.
- We receive feedback about our behaviour, our skills, our values, the way we relate to others.
- Key question: “How do I know that I have accurately perceived what I have seen and what I have heard?”
- We can ask others for clarity.
- We should ask others how they see us.
- Ask this simple question: “will you give me some feedback on what I did?”

Who you should ask?

- Someone you trust.
 - Someone with more experience than you.
 - Ask more than one person.
 - Be as specific as possible and request specific feedback.
- **What I have learned?**
 - Ask yourself regularly, “what I have learned today?”
 - This is a positive approach to processing information.
 - **Valuing Personal Strengths**
 - Take time regularly to review all the things you have achieved.
 - When you think of something that needs improvement, think of 5 things you have achieved.

Viewing Experiences Objectively

- Recall an incident.
 - Imagine the incident on a stage with you watching in the audience.
 - Look carefully to see what is happening and what people are saying.
 - This is called a ‘participant observer’ of your own experience.
- **Empathy**
 - Try to see, hear and feel what the other person is experiencing.
-

- **Keeping a Journal**

- Keeping a journal to log your reflections.

When you read your journal, ask yourself: “what do I see when I do this review?”

Who Can We Reflect With?

- Personal or individual reflection
- Shared reflection (with colleagues)
- Families
- Students
- **Personal or Individual Reflection**

Some critical reflection is deeply personal and would not be shared until some time has elapsed and you feel ready to talk about what you learned about yourself.

- **Shared Reflection**

Critical reflection cannot be just by yourself. Just saying something out loud to someone else can help fill in holes and can sometimes help you to find answers.

Facilitating shared Reflection

Conversation and debate with:

- Colleagues
- Other professionals working in the same setting
- Professionals working in another setting
- A professional network (peers on a course)

Reflecting with families on:

- Their values, culture and traditions and what that might mean for practice.
-

- Their understanding of their children's interests, strengths and abilities.
- Their priorities or aspirations for their children and how a teacher can support these goals.
- Their children's progress.
- Their perspectives on play and learning.
- How they and professionals can work together to progress the children's learning and development.
- How they experience the school.
- Opportunities for learning in everyday experiences.

Using reflecting questions with students.

- Reflective questions are open, requiring answers other than 'yes' or 'no'.
- Can motivate them to persist as well as encourage them to try new strategies and experiences.
- Displaying or documenting children's work can help to make children's reflection visible by including their words about how they learned, what they learned and who or what helped them to learn.

The Practice of Reflection: Bloom and Professional Careers

Bloom and Professional Careers

- **Recognizing Bloom's Domains of Learning;**
 - Cognitive domain: knowledge based
 - Psychomotor domain: skills based
 - Affective domain: attitudes, feelings

We've been looking at how to support reflection especially when teachers are very busy. A useful scaffold for engaging with reflection is 'Bloom's Taxonomy'. Bloom's taxonomy is divided into three domains. The first domain is the **cognitive domain** which is knowledge-based, the development of an acquisition of knowledge for the reflective practitioner. The second domain is the psychomotor or the skills-based domain, the third domain is the **affective domain**, or in other words, their feelings, attitudes and values that the reflective practitioner has.

- **Cognitive Domain:**

- Lowest level-Knowledge- recall of data
- Comprehension
- Application
- Analysis
- Synthesis
- Highest level-Evaluation-critiques, judgements about the value of ideas.

- **Relevance to Career Path- Employers look for self-reflective people with:**

- Skills of analysis
- Interpersonal skills
- Self-development
- Critical reflection on performance

Setting Objectives

- An objective is a clearly defined target that has the following characteristics:
 - Specific
 - Measurable
 - Action-based
 - Realistic
 - Time-Bound

The Practice of Reflection: The Professional Development Plan (PDP)**The Professional Development Plan (PDP)****What is PDP?**

- A way of recording, reflecting on and writing about the practical outcomes of experiences.
 - A personalized record of one's career or experiences.
-

Why keep a PDP?

- It provides you with a record of your professional development.
- Useful as a means of updating a Curriculum Vitae.
- It personalizes learning by encouraging the teacher to own and value their development and achievements.
- It enables the individual to demonstrate coherence in their professional development and plan for the future.

Example of a PDP

Becoming a Reflective Practitioner	
The Practice of Reflection	
Example of a PDP	
What am I good at?	What am I not so good at?
What practices I want to continue?	What areas I wish to improve in?

A simple format for the professional development journal is to use a framework of questions. An example here of questions in a PDB could be: 'what am I good at' that would be a good starting point for the reflective practitioner in terms of identifying strengths and interests, and in connection with this it would also be good to ask: 'what am I not so good at'. To identify those **development needs**, **another simple question to add into** a PDP could be: what practices do I want to continue, what ones do I want to develop in more depth and detail, and in contrast to that what areas do I wish to improve in, what weaknesses do I need to overcome. So,

we can see that a professional development journal can provide a very simple framework for reflection, and engages a reflective practitioner in a well-defined process which can be very efficient and supportive.

The Process

- How are you going to develop the skill?
- What is the time frame for this development?
- How are you going to check whether you have been successful at developing this skill?

Reflecting on your PDP

- Do you feel you have succeeded in your personal/academic development? If so, why? if not, why not?
- What changes would you make to the Personal/Academic Development Plan if you had to do it again?
- What is the main thing you learnt from the process?
- What other things have you learnt from the process?
- What ideas have you had for developing other areas of your own practice?

Why Be a Reflective Practitioner?

- To enable professional development:
 - The 'knowledge' economy.
 - Undertaking continuous professional development is a requisite of working as a teacher.

To maintain quality

- Reflecting on what went wrong and why=improvement.
- Reflecting on what went well and why=good practice.

It's important to be a reflective practitioner because as a very simple process, it enables us to ensure the quality of what we're doing, and we can focus on why things are going wrong, and why they're going wrong, and this gives us a focus for improvement. And at the same time, we can focus on why things are going well, and

how we can not only maintain that good work but also develop that expertise. In other words, it enables us to achieve what we would call good practice.

- Reflection then includes hard systematic thinking and a soft initiative insight, leading to a plan of action based on critical evaluation of all the available evidence.
- Do not just think about what went wrong/well and why – also plan to improve skills if necessary, or to use those skills in future events.

So, the reason why we reflect is not just to reflect on what went well and what went wrong, but actually to look at the whys and the hows of these questions, and to begin to really think about these things within a broader context, and that broader context is about quality and about improvement.

- Enhance your self-awareness
- Develop creative answers to difficulties
- Enhance your problem-solving skills
- Enhance your ability to evaluate
- Enhance your ability to action plan for success
- Make you an effective manager

Ghayle's Process of Development

- Good evidence is important
- Carefully selected
- Represent the standard
- Demonstrate competencies of standard
- Personally meaningful
- Paired and relevant
- Insightful and relevant

A level of the reflective process is about looking very closely and internally at our own experiences what Ghayle calls it '**Navel gazing**'. This is simply about being honest, and developing our self-awareness, and

considering the process of reflection as about improving on what we've done and learning from the experience, and learning from that improvement. It's about finding connections and re-assembling what we're doing to make more sense adjusting and manipulating it. It's about developing an individual identity and not just fitting in with what everybody else is doing. It's something which is part of the reflective practice process and in a way therefore, it supports the reflective practitioner in finding justifications for their actions in the classroom

Another level of the process is about understanding and learning from the day-to-day experiences which Ghayle calls as **'Learning from the day's chaos'**. This leads the reflective practitioner to do things like: to find reason through the experiences, and to find connections. And in this way the reflective practitioner is developing their work awareness and becoming more **aware** of their role and other people's role in the experience. It's therefore based in things like questioning of self and questioning of others, and this can lead to increase in confidence of the reflective practitioner in the classroom which ultimately leads to professional growth.

The third level of Ghayle's process is **'Talking about you do with others'**. This level is about sharing and learning from collaborating with others, and so at this level, the reflective practitioner is describing, remembering the experience, and seeking feedback which can be constructive in terms of the reflective practitioner's development. And so, the reflective practitioner is understanding feelings, understanding their personal response to experiences and where their response may be influencing the outcomes. And so, it might be about focusing on mistakes, but this is from a very developmental and improvement focused point of view.

- A 'reflective practitioner' is someone who does something and is automatically reflective.
 - Reviews what happened and asks 'why', 'how', 'where', 'what'?
 - What skills can you take forward/need to improve/develop?
 - How will I do this?
 - The best practitioners are constantly learning, evaluating and refining their practice, even after years of experience.
 - The next stage is to apply what you have learned from one situation to the next.
-

The Nature and Purpose of Professionalism

- There is great importance placed on teachers meeting a wide set of professional commitments and obligations.
- What values and practices should we reflect on which comprise 'being professional'?
- For reflection purposes, we need to think about 'professionalism-in-action'.
- What is it?

Professionalism-in-action

- Relating to pupils
- Relating to colleagues
- Relating to other professionals
- Relating to parents
- Relating to the school in its particular context

Teaching happens within a context, and so part of professionalism-in-action is the reflective practitioner relating their professional skills and their development of professional skills to the school context in particular.

- The professional role of the teacher is multi-faceted.
- There are different dimensions to this role that require reflection.
- The teacher's pastoral role.
- The teacher's academic (subject) professionalism.

This leads to the teacher:

- Reflecting on the reasons behind poor performance (for example).
- Reflecting on planning to integrate isolated/shy students into group activities.

The Nature and Purpose of Professionalism

- In addition to the academic and pastoral roles which need to be reflected on, different teachers may also have an underlying tendency towards a preferred mode of professionalism.
- This point is illustrated in the following diagram.
- Here we see the locations of two teachers (T1 and T2) on a matrix, where the horizontal axis measures degree of academic focus and the vertical axis shows degree of pastoral focus.
- Teacher (T1) tends to emphasize the pastoral aspects of the teaching role, whereas Teacher (T2), while mindful of the pastoral dimensions of teaching, tends to stress rather more its academic focus.
- The ability to be flexible and to adjust your own preferences to circumstances and requirements is an important skill to develop. It is a complex and subtle skill, however, and one which will continue to be developed as you continue to reflect on your professional role.
- Finally, your reflections on your professional role as a teacher will also encompass your contributions to extra-curricular activities, to whole-school initiatives and to the corporate life of a school.
- The idea of living out the ethos of a particular school, and helping to shape it, is at the heart of the rather special role of being a reflective practitioner.
- What is central to this distinctiveness is the idea that the reflective practitioner identifies and adheres to a code of professional values and practice.

Reflecting on Professional Roles: Reflecting on Notions of Professionalism

Reflecting on Notions of Professionalism

- **Reflecting in order to develop subject expertise is not sufficient.**
- **Other things are also vital:**
 - The imaginative mediation of content
 - Varied activities
 - Concern for all pupils
 - Fairness

We can look at the reflective practitioner's focus areas when exploring their professional roles and identity.

Reflecting on Professional Roles: The Reflective Practitioner-Professional Relations with Pupils

The Reflective Practitioner - Professional Relations with Pupils

- What is regarded as 'professional' must be linked with the in-action responsibilities and reflective practice of the teacher.
- Reflective practitioners must be seen to treat pupils with respect, fairness and consistency, that is from an unconditional positive stance.
- This requires you to operate with a knowledge of pupils' cultural and social backgrounds, their experience and their interests.
- This is not simple since the knowledge we possess of any pupil is partial and it may be the next piece of information we get about them that makes a difference to our interpretation of a situation and how to tackle it.

These concerns direct our focus as reflective practitioners onto key roles we have with students:

- Teacher as mentor
- Teacher 'in loco parentis'
- Teacher as observer
- Teacher as carer

For the professional, this leads reflective practice into an area of focus for reflection, and what we need to be doing in terms of our professional relationships with students is reflecting on particular roles that we take on. One particular role is the teacher as a **mentor**. A mentor is somebody who is working alongside students in the classroom, supporting and encouraging whilst challenging and cheerleading the successes of the students.

The teacher is also within a professional role **taking on some responsibilities as a parent** in the classroom. The teacher with students in the school environment must realize and also reflect on their responsibilities of care, nurture and support, that they have within that parental role in the classroom.

Another role is teacher as a professional in the classroom need also to work as a **participant observer** of their own practice. In this way, their professionalism can grow and their objectivity about their professional role can support the professional relationships that they have with students.

Also, there's the teacher as **carer** in the classroom. The professional relationship that we have with students is such that we are not simply there to teach the students a subject or for the students to show that they have achieved certain facts and figures, but we need to care for the students on a personal level as well as a professional level, and caring for students leads to better learning process.

Reflecting on Professional Roles: The Reflective Practitioner-Professional Relations with Colleagues

The Reflective Practitioner-Professional Relations with Colleagues

- When you are in school, you are not on your own, you are a member of different teams in the school.
- You need to reflect on how you are regarded as team-player in all these different teams.

Your professional relations with colleagues are an opportunity to extend your thinking.

- This is in contrast to solitary reflection.
- Reflecting with colleagues opens up a world of possibilities.

We can see that the professional relationships that we have with colleagues open up a number of worlds to us, open up our professional roles to opportunities that we have in terms of working with others. These opportunities are many and varied in terms of how we can develop our professional roles. We can think of our professional identity in terms of colleagues through any number of approaches and relationships, for example, things like **specific interest groups (SIGs)** are very useful to build up in a school environment working with colleagues, sharing with colleagues and working with others.

We can also think of contributing to the school and contributing to our colleague through things like **working together**. So, we're always therefore building the relationships that we have within the team, and learning from each other and sharing with each other as part of a team.

From a very simplistic point of view, we can think of their professional relationships we have with colleagues in terms of **staff meetings**, and even informal meetings that we might have in the classroom or in the school at any particular point. So, we can see that there are many different ways that we can engage professionally with colleagues, and with other professionals in the school.

Reflecting with colleagues is very important for us because it not only help us integrate into the school and feel valued with other professionals in the school, but it really is an opportunity to grow professionally with support and encouragement from others.

Reflecting on your Contribution to the School

- **Contributing to the life of the school is an indicator of a vibrant, inclusive school, generating gains for both skills and community.**

Reflecting on Your Contribution to the Life of the School

The definition of contribution includes:

- Cross-curricular activities
- Participating in projects
- Contributing to the distinctive 'presence' of the school.

School effectiveness can be viewed in terms of outcomes:

- Attendance
- Student attitudes
- Student behaviour
- Scholastic attainment

Reflecting on Your Contribution to the School

- The reflective practitioner needs to reflect on their role in uncovering what makes the school effective.
- How do you contribute to the 9 characteristics of high-performing schools:
 - A clear and shared focus
 - High standards and expectations for all students
 - Effective school leadership
 - High levels of collaboration and communication
 - Curriculum, instruction and assessments aligned with highly regarded standards
 - Frequent monitoring of learning and teaching
 - Focused professional development
 - A supportive learning environment
 - High levels of family and community involvement

Another characteristic to look at in terms of a high-performing school for the reflective practitioner to focus on is the curriculum, the quality of the curriculum, the instruction and the assessment associated with the curriculum content and the curriculum objectives, and these all need to be **aligned** together in terms of high-performing standards.

The Impact of Organizational Dynamics: What are teams?

Groups of two or more people who interact and influence each other are mutually accountable for achieving common objectives and perceive themselves as an social entity within an organization.

Groups vs teams?

- All teams are groups
 - Some groups are just people assembled together
 - Team has task interdependence whereas some groups do not (e.g. group of teachers enjoying lunch together).
-

Types of teams?

- Team-based department
- Team-based organization
- Quality circles

Temporary teams

- Task forces
- Temporary teams that investigate a problem

Some teams are temporary which tend to be very short-termed and are focused on achieving a particular goal. We can see the impact of organizational dynamics by looking at the categories of organizational dynamics. For example, the teams operate within an environment and the teams creates its own environment, its own identity. And therefore, they are imposing factors on the reflective practitioner from this category. For example, how communication occurs between teams and the organizational space and structure for the team and its work there is also the team design how big is the team and people involved in the team. This is another category that the reflective practitioner need to be aware of because of its potential impact that the team processes are very important to be aware of. The team processes can affect the functionality of the team and automatically all of these lead to team effectiveness and in other words how the goals are being achieved to what extent the objectives of the team are being met.

Teams' elements to reflect on:

Task characteristics

Better when tasks are clear

Easy to implement

Task interdependence

Share common inputs, processes of outcomes

So, we can see from this that the impact of teams and the impact of dynamics of teams for the reflective practitioner can be quite considerable from both a positive and negative point of view. So, there are certain characteristics to be looked at in terms of the impact of teams. The characteristics for the reflective practitioner are that the task should be accomplishable and it should be relevant to the team. And the reflective practitioner within the team can actually consider these imposing factors.

Team size

Smaller teams are better but large enough to accomplish task

If the team is quite small but large enough to complete a task this can have an effect on how the reflective practitioner is undertaking their role and undertaking the activity of reflection.

Team composition

- Members
- Motivated/competent to perform task in a team
- Environment
- Team diversity

Team composition is who are the people who comprise the team are they specialists or are they an accumulation of different specialized skills.

Homogeneous Teams	Heterogeneous Teams
• Higher satisfaction • Less conflict • Faster team development • More efficient coordination • Performs better on simple tasks	• More conflict • Slower team development • Takes longer to agree on norms and goals • Better knowledge and resources for complex tasks • Tend to be more creative • Higher potential for support outside the team

So, what this leads to is an understanding that teams are really homogeneous or in other words their people are supporting each other working together and developing as an entity in their own right there's a higher degree of satisfaction in teams than there is in groups and considerably less conflict whereas in a group of individuals who are not functioning as a team we're potentially looking at more conflict and also a slower pace of development. So, the reflective practitioner can look at these two components and understand whether they are a team or a group and. This is important in terms of reflective practitioner to understand the impact that working within an organization and the dynamics of the organization can have on reflective practice.

The Impact of Organizational Dynamics: RP and the Stages of Team Development

While we're looking at the impact of organizational dynamics on the reflective practice, we need to understand that a group of people working as a team go through different stages of development and functioning, **We can represent these stages diagrammatically.**

Here we can see that the group dynamics of a team go through different levels of functioning, at the initial stage of functioning is something called **forming**.

This is where the group comes together and starts to get to know other members of the group. This is the first stages of team development and essentially sets the tone for how the work occurs within the group.

The next stage of the group is called **storming**. This is where there can be conflict arising in the group because people have different ideas about how the work of the group should be achieved. Different people could try to lead the group towards achieving those goals. This can impact on the effectiveness quite severely of the group.

Once the group gets past the storming stage this point of potential conflict the group is known to move into a stage called **norming**. This is where the individual group roles are very clearly defined and people are undertaking those roles efficiently and effectively supporting each other in those roles. This is where the group of people starts to become a functioning team to the point where the team ultimately starts to perform at a high level.

At the **performing** stage everybody supporting each other working together and understanding how that work supports the achievement of the goals.

In permanent teams these four stages can be transitioned through backwards and forwards the group can always go back to storming after it's been performing some for different reasons. It could be that a new member joins the group or there is some change in the functioning of the group. It's certainly not a one-way road. It is a journey backwards and forwards potentially that the group goes through.

If it's a temporary group then there will be a point where the individual members leave the group. The task is completed and this is known as **an adjourning stage**, at this stage the group members are leaving each other the work is completed. There's no reason for them to see each other or to be involved with one another again.

Team norms

- Informal rules and expectations used to regulate behaviors.
 - Norms develop through-
 - Critical events in team's history
 - Explicit statements
 - Initial team experiences
 - Beliefs/values teachers bring to the team
-

Reflecting on team norms

- How to introduce norms when forming teams
- Selecting members with preferred norms
- Discuss counterproductive norms
- Rewarding of behaviors representing desired norms
- Dysfunctional norms

Reflection on group Stages

- How to move the team from 'forming' to 'performing'
- How to deal with 'storming'
- Your role in helping the team 'perform'
- What do you observe are the causes when the team slips backwards?
- What ideas do you have to the team move forward?

The Impact of Organizational Dynamics: Team Cohesiveness and Performance and the Reflective Practitioner

There are a number of causes/influences on team cohesiveness:

The reflective practitioner needs to consider the members of cohesive teams (including themselves):

- Want to remain members
- Are willing to share information
- Have strong interpersonal bonds
- Want to support each other
- Resolve conflict effectively
- Are more satisfied and experience, less stress

Team performance

The reflective practitioner needs to also consider the performance of the team and their role in the team.

What do you need to reflect on?

- Individuals better/faster on some task
- Cost of developing and maintaining teams
- School does not support best work environment for team dynamics
- Social loafing

Conditions for social loafing

- Low task interdependence
- Individual output not visible
- Routine, uninteresting tasks
- Low task significance
- Low collectivist (group) values

Reflect on the school and your role/others' roles in the school.

The Impact of Organizational Dynamics: Task Interdependence

- Where do you as the reflective practitioner, fit into the team?
- What are you reflecting on when you are part of a team?

Task interdependence

What is task interdependence and where is the reflective practitioner in this?

- As teachers we are all dependent on each other for our work.
- You can reflect on how interdependent the team is and your role in supporting this

Team composition

- Effective team members must be willing and able to work in the team.
 - Effective team members are good at the 5 C's:
-

- Cooperating
- Coordinating
- Communicating
- Comforting (psych, support)
- Conflict resolving

Team interdependence depends on:

- Interpersonal knowledge and trust
- Understand roles
- Appropriate behaviors
- Learning to coordinate with each other
- The team and your role within it is a reach area for the reflective practitioner.
- Remember, you may be reflecting on yourself but you work within a context and with others – this also needs reflection.

The Impact of Organizational Dynamics: Behaviour in Organizations

- The behavior of groups/ teams in organizations can become a barrier for the reflective practitioner.
- The mode of thinking that persons engage on when concurrence seeking becomes so dominant in a cohesive group that it tends to override realistic appraisal of alternative courses of action.

Symptoms of “Groupthink”

- An illusion of invulnerability
 - Collective construction of rationalizations that permit group members to ignore
 - Warning or other forms of negative feedback
 - Unquestioning belief in the morality of the group
 - Strong, negative stereotypes views about the leaders of ‘enemy’ groups.
-

- Rapid application of pressure against group members who express even momentary doubts about virtually and illusions the group shares.
- Careful, conscious, personal avoidance of deviation from what appears to be a group consensus.
- Shared illusions of unanimity of opinion.
- Establishment of ‘mind guard’ – people who ‘protect’ the leader and fellow members from adverse information.

Impact of structure on behavior

Each school/organization has structure that:

- Define the unique ways that people are divided
- The behavior of groups/ teams in organizations can become a barrier for the reflective practitioner.
- How specialist roles and functions are coordinated
- How information flows among people and groups
- How the system of controls (task measurement, evaluation, and change) is to work
- Structure is only one of the forces that affect behavior. Others include peer group pressure, group norms, social and technical aspects of work tasks, and internal and external cultures.
- The reflective practitioner is someone who works within organizational structures whilst simultaneously reflecting on those structures.
- The reflective practitioner will reflect on function as well as dysfunction.

The Impact of Organizational Dynamics: Reflecting on Motivation

Reflecting on Motivation

- The Hawthorne experiments demonstrated how work places are predominantly social institutions.
-

- Assumptions about behavior can be self-fulfilling prophecies.
- However, public organizations have difficulty developing coherent philosophies because of conflicting goals and objectives.

Self-Reflecting for Motivation

- In motivation, choice (especially autonomy and control over situations) does not have to be central to reflection.
- Self-regulation requires some degree of choices/ intentional selection of strategies designed to help you achieve a goal.

Self-regulated teachers

- Self-regulated teachers therefore people who are aware of their strengths and weaknesses and this is something the teacher to reflect on.
- Self-regulated teacher is also able to think about the way they are thinking and look at the why they are learning and reflect in a metacognitive sense on their role in the team or in the school strategies: e.g., questioning one's learning and monitoring one's learning, to approach academic tasks.
- Self-regulated teacher also has certain attribute around success or failure, and this affects how much control they feel they have over their work role.

Self-Regulation as a Process

- Self-regulation is a cyclical process. Teachers who are motivated to reach a certain goal will engage in self-regulatory activities they feel will help them achieve that goal.
 - Self-regulation promotes learning, which leads to a perception of greater competence, which sustains motivation toward the goal and to future goals.
-

Dimensions of Self- Regulation

Researchers identify three critical dimensions or characteristics of self-regulation:

- Self-observation
- Self- judgement
- Self- reactions

Self-observation

- Refers to the deliberate monitoring of one's activities. Self observation may take the form of recording frequency, duration or quality of a behaviour.
- Self-observation aims at being very critical to the regulation of performance.
- Self observation may also lead to higher motivation.

For example: If you realize your preparation habits were causing you to perform poorly, you may adjust the way you prepare, leading to higher satisfaction and more motivation to continue to improve your work habits.

Self- judgement

Self- judgement refers to the extent to which you can evaluating one's current performance levels compared to the goal level.

Self- reactions

Self- reaction refers to one's behavioral, cognitive, and affective responses to self-judgments. Self- reactions can be motivating if you believe you are making progress towards your goal.

Models and Mentors: Peer Mentoring

Peer Monitoring

- A shared relationship, involving both giving and receiving.
- It is an equal relationship between two people who value and respect each other and believe each can enrich the other.
- The effectiveness of these relationship is because of unrestricted openness, trust, commitment, sense of responsibility for each other and mutual accountability
- Which is already present in good friendships. It can be developed further when there is an agreement to be more purposeful.
- Unfortunately, this is so readily available, and most effective form of mentoring is overlooked by many or not developed to its full potential.

What is important for the peer mentoring is a professional relationship and not just two friends talking.

- Begins with a friend
- Set aside regular times to be together
- Maximize your times together
- Maintain a balance in your relationship
- Seek to be transparent with each other
- Focus on major areas of concern
- Ask good questions
- Be mutually accountable
- Show mutual respect

The Best Peer Mentoring is Characterized by:

- Honest, two way exchange
-

- Respectful debate on differencing views
- Critical assessment of ideas
- A leveling of the power gradient
- Mutual and unconditional acceptance
- Respect for limits and boundaries

Benefits of Peer Mentoring

- Creates sense of belonging and support
- Contributes to work success
- Contributes to relationships/networking
- Contributes to sense of community
- Contributes to culture of success
- Builds awareness of resources

What does a Research say about Peer Mentoring?

Ferrari (2002)

- Improved academic performance
- Improved academic self-efficacy
- Improved satisfaction with academic programs

Brown, David and McClendon (1999)

- Ability to make a career choice and increased persistence to achieve goals
-

It also has been shown that peer mentoring helps people develop sense of autonomy in terms of choice and this leads to increased persistence to achieve goals (Brown, David and McClendon, 1999).

Sosik and Godshalk (2005)

- Improved interpersonal communications
- Improved psychosocial support

Making Good Use of Your Peer- Mentor

The Roles of the Mentor are to:

- Listen to the reflective practitioner
- Share experiences and understanding about practice
- Work together to mutually develop skill for teaching and working in school building
- Work together to develop network around the school environment
- Be one of many influences.

Reflective Practitioner Roles

- Active participant
- Retain critical faculties
- Seek new capacities, not just knowledge
- Remain open to multiple influences
- Own responsibility for success

Models and Mentors: Mentoring Skills**Mentoring Skills**

A mentor is important for a reflective practitioner because:

- Helps provide professional development
- Demonstrates the employer's recognition of knowledge, skills and abilities of the mentee.
- Usually helps to advance career more quickly
- Increases confidence
- Develops creative and independent thinking
- Helps acclimatize a new staff member to his or her job and school culture more quickly
- Helps teachers feel more in touch socially and professionally
- Results in a greater awareness of organizational politics and culture
- Provides an appreciation and effective use of networking
- Develops proactive approaches to tasks and projects
- Movement toward "expert" status

What Can You Expect from a Mentor?

- In the role of coach, a partnership with an colleague is developed in order to strengthen and expand the reflective practitioner's skills in specific areas of job performance.
 - *Trust*- the glue of all relationships. When we work in a climate of trust, we know that we can predict the positive actions and attitudes of another person.
 - We know that she/he cares about us and support our efforts.
 - *Feedback*- positive recognition of a job well done or the constructive criticism of a skill or task which could use improvement.
 - *Learning* about desired behaviors- the actions or skills that provide the most effective completion of a task.
 - Learning about replacement behaviours- new behaviours to replace ineffective or inappropriate behaviours.
-

What to Expect from Constructive Criticism?

- Is specific and focused on the behaviour, not the person.
- Describes the significance of behaviours for the team or team goals.
- Remains calm
- Listen to your perspective of the behavior and/or situation.
- Identifies the benefits of improving the behaviour.
- Balances positive and negative feedback.

Models and Mentors

- The hierarchy gap (“speaking truth to power”)
- Political correctness (over compensating)
- Same words/ different meanings
- Avoid exploring own feelings, attitudes and values
- Problem denial

Clarity of Purpose vs Rapport



Models and Mentors: The Skills of Dialogue

The Skills of Dialogue

- Managing stereotypes
- Permission to explore
- Fearless questions
- Analyzing assumptions, behaviors and values
- Achieving clarity
- Valuing the insights from different perspectives

Four Key Issues Relating to Difference in Mentoring

There are 4 key issues relating to difference in mentoring which will impact on the level of dialogue and the quality of dialogue between mentor and reflective practitioner. These four issues are:

- **Perspective:** the ability to realistically look the environment
- **Networking:** ability to make connections between the individuals may be departments within the organizations.
- **Power:** sense of power between the mentor and reflective practitioner there should be an equal relationship and one person is more powerful.
- **Dependency:** mentoring relationship is very supportive, but it must be clear that the reflective practitioner should not become dependent on the mentor for that work.

How the Reflective Practitioner Views Differ in Mentoring?

Issue	Same group	Different group
Perspective	More empathy	Wider perspective
Career	Less likely	More likely
Network	Sharing same concerns	More influential
Power	Easier rapport	Insight into politics
Role modeling	Occurs more naturally	Promote wider range of responses.

The Value of Difference in the Mentoring Relationship

- Encourages RP to set and work towards ambitious career goals, increasing their confidence.
-

- Provides RP with visibility and influence and senior levels and with insight into management thinking and organizational politics
- Encourages RP to recognize their value and enhance their self confidence
- Stimulates more effective and wider networking.

Your Mentor Should:

- Establish confidence by beginning with work- related issues.
- Identify common interests and values
- Make efforts to learn about each other
- Show empathy
- Be clear about needs and expectations
- Avoid stereotypes and untested assumptions
- Provide ‘risked discomfort’ to make the relationship work

Areas of Dialogue for the Reflective Practitioner and the Mentor

- Universalism v particularism
- Individualism v communitarianism
- Affective vs neutral
- Specific vs diffuse
- Achievement v ascription
- Sequential v synchronous
- Internal v external control

Reflective practitioner and the mentor have the certain areas of dialogue to consider:

- In terms of content the dialogue between the reflective practitioner and the mentor it can focused on very general/ universal aspects of working in an organizational setting or it could be focused a particular minute area of practice.
- It could also be related to the individual is settling working in group environment or it could be focused on team community within a school.
- The dialogue can also be quite neutral in terms of description, explanation, and analysis but it also focused on emotion, feelings, and reactions. The dialogue goes on between reflective practitioner and the mentor can be very specific or can be quite nebulous, diffuse, and varied in terms of ground covered.
- It focused on achievements or ascribing standards to the work that the individual is doing.
- The dialogue can be on going its synchronous with the work of reflective practitioner it could be in response to after an event, in other words sequential to the work of reflective practitioner.
 - In terms of control dialogue can go between internal control in other words autonomy of reflective practitioner as well as external control or in other words control by the school or by line manager so the dialogue between the reflective practitioner and the manager can vary between extremes in whatever area the two are looking at.

Models and Mentors: Situated Learning

Situation Learning

- Seeing to integrate formal and work place learning
- ‘Learning to teach’ and ‘teach to learn’ (Loughran, 2006).

Learning and development for the reflective practitioner does not happen in isolation and there is a notion around modeling and mentoring in school where learning is situated. This is in a sense looking at formal integration of learning from the workplace. We can think of it as ‘learning to teach’ and ‘teaching to learn’ (Loughran, 2006).

- Student as teacher and learner (think critically, question practice and explore new principles) (Taylor, 2008).

Teacher is not only teacher but also a learner and as such learning to think critically, question practice and to explore new principles (Taylor, 2008).

- ‘Transfer of Learning’ an inadequate metaphor
- ‘Becoming’ -within a transitional process of boundary-crossing (Heger & Hodkinson, 2009).

The Reflective practitioner and Situated Learning

There are certain stages that considered about situated learning that support reflective practitioner.

Stage 1: Implicit modeling of strategies and values at this point the individual is practicing and behaving certain ways and through their behavior expressing values which are reflective in an organization.

Stage 2: Explicit modeling of reflective learning and change in practice.

Stage 3:

- Linking practical wisdom to abstract theory

Stage 4:

- Reconstruction by teachers in the classroom

Reflective Practitioner as teacher and learner

Situated learning focuses in a holistic way on teacher learning-enabling teachers to think critically and originally about the work they are engaging in, question existing practices and exploring new principles -this has resonance with Loughran’s learning to teach and teaching to learn. We can look at this as a powerful tool for the reflective practitioner.

Models and Mentors: Social Learning Theory: Elements of Social Cognitive Theory

Elements of Social Cognitive Theory

- Social learning theory influences the mentoring relationship and the process of reflective practice in an educational environment.

Social learning distinguishes between the acquisition of knowledge (learning) and the observable performance based on that knowledge or (behavior).

- Both internal and external factors are important because they are interacting in the process of learning for the reflective practitioner.
- There is an idea of reciprocal determinism. An explanation of behavior that emphasizes the mutual effects of the individual and the environment on each other. In this sense the distinction between enactive and vicarious learning.
- Enacting learning is learning by doing and experiencing the consequences of your action.

Consequences are seen as providing information about appropriate actions, creating expectations, and influencing motivation.

Distinction between enactive and vicarious learning

- Vicarious learning is learning by observing others. Emphasises the powerful effect that modeling learning and imitation can have on learning.

The emphasis is on other's modeling, behaviors and attitudes and therefore the reflective practitioner imitating those behaviors and those values and learning occurs through this imitation.

Learning by Observing Others

Two main modes of observational learning for vicarious approach are:

- First, observational learning can take place through vicarious reinforcement.

vicarious reinforcement is important because it happens automatically.

- Second imitates the behavior of a model even though the model receives no reinforcement or punishment while the observer is watching.

The second notion is that learning by observing others imitates particular models (behavioral models) even though the models receive no reinforcement. Just by watching we learn and we can take on particular behavior.

Second, learning by observing others imitates the behavioral model that even receives no reinforcement or punishment while the observer is watching.

- Observation can be a very efficient learning process.

The other aspect of observational learning is that observation can be a very efficient learning process there is no effort in terms of having right it just understanding and observing.

- Social cognitive theory distinguishes between acquisition and performance of learning.

Four important elements to be considered in Observational Learning

- **Attention**-pays in watching a role model
- **Retention**-understanding and learning from that observation
- **Production**-individual putting in practice what has be learned
- **Motivation and Reinforcement**- reflective practitioner uses his observations to motivate themselves and to reinforce new behaviors.

Observational Learning for reflective practitioner has five possible outcomes.

1. **Teaching new behaviors.**

- It teaches new behavior. Modeling can deliberately be applied to practice once it been explored. And this can be very powerful for the reflective practitioner.

2. **Encouraging already-learned behaviors.**

If you can see another teacher already doing the same things that you so that reinforces, encourages the behavior you engage with.

3. **Strengthening or weakening inhibitions. The ripple effect.**

4. **Direct attention**

5. **Arousing emotions.**

In OL you may develop emotional reactions to situations you have never personally experienced.

The Dialogic Process in RP: Dialogue as a Self-Assessment Tool

Dialogue as a Self- Assessment Tool

To assess knowledge

To assess practice

To evidence reflection on practice

To assess professional learning

We have seen that reflective practice can be a dialogic and as such dialogue process and as such dialogue can be self-assessment tool for the reflective practitioner. It is useful assessing knowledge as well as practice but in terms of evidencing reflection and development for reflective practitioner.

Characteristic elements of dialogue for reflection

Trust: empathy, honesty, sincerity, openness, learning, self-awareness, growth, mutual responsibility, listen, challenge, interrogation, inquiry exploration, reflection and critique, shared knowledge, process, multiple levels of communication.

There are some critical elements to the dialogic process in terms of reflection and these aspects of trust; empathy; understanding and openness to learning and in terms of reflection taking responsibility and challenge to develop.

It involves exploration as well as critique of experience and is a process of communication at many different levels from a very descriptive level through to an analytical and evaluative level.

Defining Dialogue Reflection

- Conversation with a purpose
- It has to involve moving forward
- Learning should emerge for all
- Care should be given to the different purposes of dialogue

What we need to do then for the reflective practitioner is trying to define what would be included in terms of dialogue for reflection it is indeed conversation with a purpose it is not just general conversation, and it involves the reflective practitioner in moving forward professionally. It is learning which should emerge over time dialogue helps learning develop rather than just become instantaneous and therefore the reflective practitioner needs to take care in terms of the professional dialogues that takes place.

- There is an inherent power differentiated due to the different roles

There is of course an inherent power differential in any dialogue that a reflective practitioner engages with and this power differential can be quite useful but needs to be fully understood by the reflective practitioner.

- Assessment involves: quality of dialogue, evidence of understanding
 - Process verses product, exposing of the process
-

- Assessment for learning

And there is an understanding that there for the reflective practitioner should access the quality of the dialogue that is being engaged within the evidence of the understanding that's growing from that and as a tool for the reflective practitioner to use. It is about process as product it's as much about learning evolving over time from engaging in dialogue rather than just outcomes of the dialogue.

- A shared purpose
- Genuine questions to seek unknown answers

Dialogue has a shared purpose it's not just one way it is a two-way street and there are genuine questions which are based in experience and develop the reflective practitioner to higher level of performance.

- Formative for the participant (backwards and forwards)
- Power relationship is two-way

For the reflective practitioner it is formative in the sense that dialogue can move backwards as well as forwards reflecting on previous experiences as well as looking towards the future and the power relationship the two-way power relationship enables this backward and forwards movement as part of the learning process.

- Consideration of influence of assessment criteria/ learning outcomes
- Dialogue is about learning evolving
- Assessment is about summative element

Consider that criteria especially assessment criteria learning outcomes for teachers may have on the dialogic process they may influence or guide the dialogue, and this affects the quality of the dialogue and can be beneficial to reflective practitioner.

- Dialogue involves complex interaction, compare with monologue
-

- Mutuality, turn taking
- Different expectations can be an issue

The dialogue is quite complex and involves complex interaction with another. There are not only needs to be speaking but also listening. Effective dialogue in terms of professional conversation engages in turn taking a mutuality in terms of sharing information and working with understanding and there of course is based on different expectations and intensions for the dialogue.

- Notion of process is critical
- Emphasis is on growth
- Would students know when dialogue is in process?

Reminder: Dialogue as a Self-Assessment Tool

The Dialogue Process in RP

Reminder: Dialogue as a self-assessment tool

- To assess knowledge
 - To assess practice
 - To evidence reflection on practice
 - To assess professional learning
-

The Educational Environment	
Written Reflection	Dialogic Reflection
Static, superficial, write to order	Multi-layered Flexible
Writing reflectively	Easier to reflect
Emergence of values	Engagement easier to show and explore
Word limit	Opportunity to probe, question
Limited opportunity to probe, question, explore	Depth of learning
Evidence all present	Natural when talking about self and practice
	May need back-up evidence

The Outcomes of Dialogic Reflection

The outcomes of dialogic reflection include:

- Assessment of informal professional learning

An assessment of informal learning which is engaged within the school environment and from experiences in the classroom.

- Creation of a learning space

It is useful because it creates a space for learning between individuals and regularity in terms of reflection and therefore learning and because it is a process of interacting with others.

- Re-profiling ideas of a teaching professional
- Reconfiguring teaching and learning

It is a way of revision approve and professional roles and refiguring notions about teaching and learning.

- A career development and CPD tool

Dialogic process is very useful as a professional development tool and a career management tool for individual teacher.

- Means of organizational change

It is a process of sharing and working with others and it is useful means for an organizational change and development.

Asynchronous Discussion-The Theory

- Content based (theory oriented): Asynchronous discussion (AD) allows teachers to have more time to reflect on and produce their contributions.

The dialogic process for reflective practice can be asynchronous. Asynchronous dialogue occurs sometime after experience and between experiences. Asynchronous discussion tends to be very theory oriented and allows teachers to have time to reflect and to produce their collaboration and contributions with others.

- AD enables teachers to develop higher order thinking skills (e.g. to gather, apply, analyze, synthesize and evaluate information).

Asynchronous discussion enables teachers because of the time to think and reflect and to be critical to develop higher order thinking skills for the evaluation and synthesis and creation and these add to the reflective practitioner our bags of skill.

- AD allows students to conceptualize a topic/issue/theory from multiple viewpoint.

Asynchronous discussion allows the reflective practitioner to conceptualize ideas and issues and theories from different viewpoints to understand different perspectives in things.

- Asynchronous discussion promotes collaborative learning – the act of collaboration is an act of shared creation and /or shared discovery (Schrage,1995).

Theoretical Framework

Vygotsky's sociocultural learning theory

Learners' experiences at the sociocultural level (i-e; experiences communicated and shared with a community of professionals) and this essential for the internalization of knowledge. There is a constant interplay between social processes and individual cognitive development.

Dewey's experience-based learning theory

Experience is shared and reconstructed through reflection. Reflection is the heart of the knowledge-constructing process. A reflective practice is framed by a perplexing and confused experience initially and a unified or reserved situation at the end. In this sense it is very much about integrating deliberation and action.

Halliday's social-semiotic perspective on learning

Language simultaneously performs three macro-functions: ideational (experiential), interpersonal and textual functions in terms of how people are interacting. Language's use is a socially shaped and shaping action but also shaping our experience and there is an interplay between language and dialogue.

Asynchronous Discussion- The Practice

- Asynchronous dialogue needs to be designed
- Designed by the other (mentor) asking the teacher to reflect on how the theories they have learned about teaching.

Asynchronous discussion as a process of dialogic reflection needs to be designed it does not happen instantaneously and it can be designed by some other individual perhaps a mentor rather the reflective practitioner. It is very much about asking teacher to reflect on how the theories they have learned about teaching is being understood and explode/ applied in different situation.

- Asynchronous discussion (dialogue) occurs over time rather than spontaneously providing time for thinning about events.
 - Can be via email/text
-

- Care needs to ensure that the level of interaction remains high.
- Monologic discourse compared to dialogic discourse.

Asynchronous discussion (dialogue) between people dialoging occurs over time rather than spontaneously, providing time for thinking about events. It can occur via email/text of course and this is the remains high for asynchronous discussion because their time lag is involved in that written exchange, but care needs to be taken by the reflective practitioner that the dialogue remains as a dialogue and not a monologue they should be a two-way exchange of information and understanding.

Interpersonal discourse:

It is about showing/giving admirations, understanding, compliments, supports, encouragement, appreciation, eager for success, consoling people about difficulties and finding solutions, agreements, offering suggestions, clarifications, invitations for more discussion.

Asynchronous discussion is a process of questioning but engaging symbol tenuously, expressive, but responsive, skeptical but respectful and challenging but supportive.

- The other may need to model or explicitly “teach” desired discourse used in asynchronous discussions.

Fabro and garrison (1998) revealed the established presence of a ‘significant other’ in dialogic discourse.

Dialogic Diaries

- Dialogic diaries are a type of reflective journal in which teachers (reflective practitioners) reflect on their learning and a ‘significant other’ writes a response.
- The diary process regularly between the two.

Stage 1: Uncertainty- At the initial stage of writing a journal is uncertainty. At this point reflective practitioner is engaging in maintaining a dialogic diary. There is uncertainty about what to write and how to write it the quality of that writing is good enough for reflection. It helps individual develop their practices through reflection and then this moves us to another stage the process which is about enthusiasm.

Stage 2: Enthusiasm –Once the level of uncertainty has been overcome the reflective practitioner can find that writing in a journal and sharing the journal with another who also writes in it responds to those reflections become quite an engaging process and can develop a lot of enthusiasm by the reflective practitioner.

Stage 3: Lack of Enthusiasm- Once the dialogic diary has been engaged with for some time it becomes a little bit habitual and therefore it can lead to a sense of apathy or lack of enthusiasm for maintaining the writing and indeed at this point journal writing for the reflective practitioner could just become things like simple words rather than reflective and explorative sentences and paragraphs however if this stage is overcome it does become a habit and learning can develop very easily very quickly from that habit.

Stage 4: Habit and Learning- This requires motivation and for reflective practitioner its worth at this point mentioning being aware of this and mentioning this to the significant other. It is quite a powerful tool because learning can be ongoing, and the process can continue because the habit is there to maintain it.

There are certain issues with the dialogic diary process which we become aware of in terms of maintaining a diary:

Time:

- For teachers to write the diaries?
- For 'the other' to respond?
- For the diary writing process to continue?

Time is needed for the individual to write in the diary and significant other to respond to reflections in the diary. And needed for the diary writing process to continue long-term. This is a very significant factor because over the long term of lack of enthusiasm can take hold and therefore diary writing starts becoming unproductive.

- Writing:
 - Writing v. Drawing
 - Blogosphere?
-

It could be via the internet or maintaining a blog and teachers interacting with each other; the blogosphere and of course it does not have to be about illustrations and drawings.

Ethics:

- The ethics of 'diary writing'

Shared writings we are discussing issues may be delicate or sensitive we must be aware of the ethics of putting this into permanent form.

- Political process (writing about roles)

Writing reflections is a political process in the sense it does raise questions about roles and responsibility that we have in the classroom and in the school.

The Dialogic Process in RP: The Academic Portfolio**The Academic Portfolio**

- A fundamental feature of the (self-evaluative) academic portfolio is the intentional focus on learning and assessment.

The academic portfolio can also be used for engaging with the dialogic process. It is self-evaluative mechanism which has an internal focus for the reflective practitioner focusing on learning and assessment.

- The deliberate and systematic attention not only to teaching but also action research skills.

Therefore, it is very deliberate and systematic attention not only to but also action research skills in terms of reflecting on experiences in the classroom.

- A fundamental feature of the (self-evaluative) academic portfolio is the intentional focus on learning and assessment.
-

This focus is very different to things like dialogic writing the academic portfolio is an evolving collection of experiences and experiments over time.

- To a teacher's self-reflective, meta cognitive appraisal of how and more importantly why learning and teaching.

It is very much about the teacher's self-reflective, metacognitive stance on their stance appraisal of how and more importantly, why learning and teaching from their experiences.

Benefits of the Academic Portfolio for Dialogic Reflection

Increased understanding of

- How you learn and what are your learning outcomes.
- What you have and have not learned)

There is an increased understanding of how we are learning and what are our learning outcomes from reflection. Also, about what we are not learning.

- What do you value?
- Role in learning how to articulate in writing one's thoughts and ideas.

With increased understanding we can explore what aspects of our work and our professional role we do value. The role in learning how to articulate in writing those values and our understanding of those values.

- How others view your work?

Another benefit of the academic portfolio there is an increased understanding of how others view the work that we do what their ideas are about our professional role.

Enhanced ability to make corrections.

- Among courses thought and in the context of teaching 'without boundaries'.
 - Among research experiences and research collaborations.
-

- Increased sense of learning over time.

What to include in the Academic Portfolio?

- Documentation of steps (analogous to keeping a log for lab research)
- Commentaries (as for solutions for math problems)
- Evolution of a course, of a speech, or of a paper.
- “History” of a piece of artwork.

It also evolves over time like a speech on paper and there is a history to it you will be able to look back months and see progression of your learning and the journey that you have been on.

Four Basic Processes for the Creation of an Academic Portfolio:

What we need to consider then is their certain processes involved in constructing an academic portfolio.

- **Collection:** Firstly, we must consider what we are collecting in terms of material to include should enable us to explore and to reflect not just inclusions for the sake of putting them in academic portfolio.
- **Select:** we need to select those materials quite carefully because part of creating of portfolio is to consider who will read it and who will benefit from it and purposes for which they will be reading.
- **Reflection:** we need to understand that the academic portfolio is a tool for reflection and therefore we need to make sure that the inclusions are there to help us learn until develop our teaching and we are simply part of research of experience that we engage with.
- **Connections:** Finally, the academic portfolio a mechanism that contains materials that enable us to make connections between different aspects of our professional role.

A Document that Evolves Over Time

- Remember: the portfolio is a living collection of documents and materials which change overtime.
- New items are added, others are dropped.

Teachers Collaborating

Every teacher is an active participant in their reflective practice and there are definite benefits from collaborating with others in a process of continuous learning and we can see that continuous learning through teacher collaboration actually forms part of a cycle focused on particular aspects within the process of observation and we can see that step-3 of this 5-step cycle. It is about observing and collecting of evidence and this is the key point where teachers collaborating come together. It's at this point where evidence is collected of processes and practices where observation is used as a means for that collection, and it continues until the end of the cycle as part of the process of reflection and evaluation.

Principles of brief

There are certain **principles of brief** that we need to consider in terms of collaborative observation as process for reflective practice and these are that **observations** must be frequent so that learning is always **ongoing**. They must be **focused** on aspects of practice rather than just general observational. They must be **varied** as well as so that they do not fall into a habitual process. They must include **useful and timely feedback**.

Collecting Evidence through Observation

What can happen in observation is that we take notes as reflective practitioner and focus on particular aspects of the teaching using a quick shorthand to explain or define aspects of our observation and it's through this shorthand that we get focus through the writing on things that we need to consider.

Here the RP has been used abbreviations such as: T for teacher, S for student, FFL for focus for learning, LD for lesson delivery, as well as time stamps at the bottom.

Getting Feedback from others on Observation:

Focused: helps focus attention on the content of that paraphrasing so teachers collaborating is a means for the reflective practitioner to get feedback.

Evidence-based: It is important that feedback is focused on aspect of the observation it also needs to be evidence-based. It should be grounded in what has been observed in the evidence for that.

Constructive: Feedback should reinforce effective practice and identify areas for continued growth.

Timely: feedback should be provided shortly after the observation so the connections can be made with the information from the feedback process and the activity in the classroom.

Collaborative Observation and Feedback: Peer Observation

Peer observation

There are number of benefits to the reflective practitioner of using peer observation:

- ✦ Peer observation helps the individual maintain and enhance teaching quality and therefore improving the quality of students learning experiences.
 - ✦ It is a process of building awareness for an individual about variety of practices that they can be using in the classroom.
 - ✦ It is a way of reflecting on the various aspects of their approach and their teaching practice that they are aiming to master.
 - ✦ It recognizes and identifying good practice in relation to others.
 - ✦ It is reflective of development needs for an individual practitioner's
 - ✦ It as a process provides evidence of quality teaching practice for promotion applications.
-

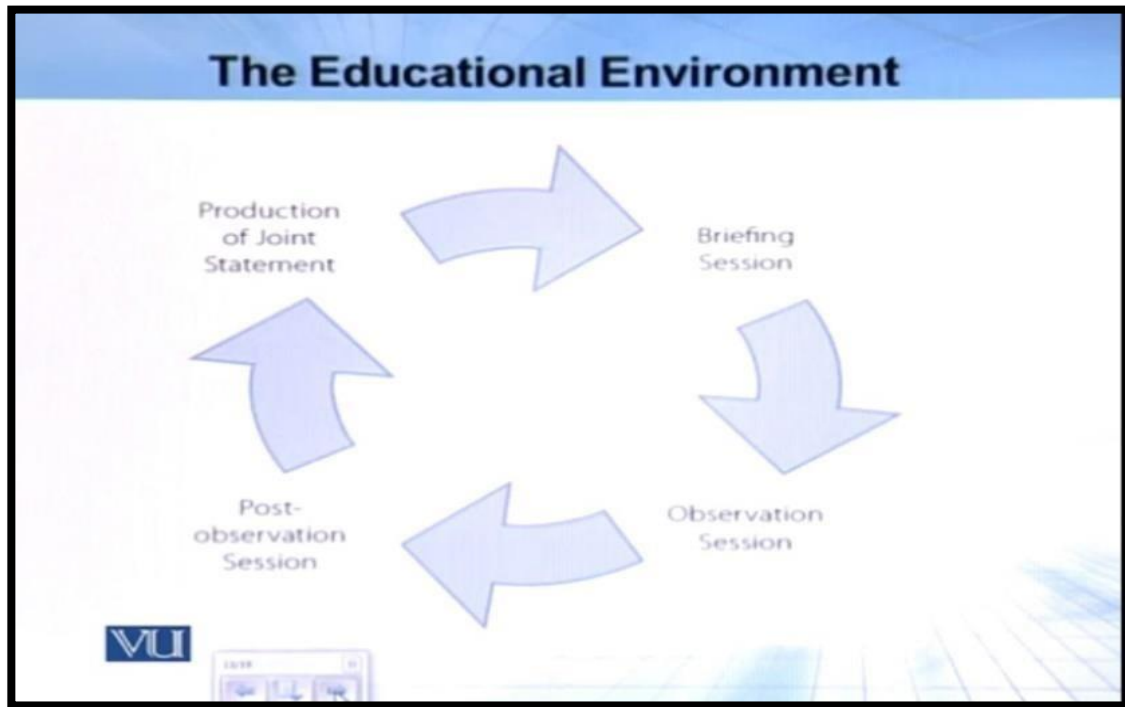
- ✦ As a means for development, it identifying and promotes good practice and innovation in teaching and learning.
- ✦ In this way it can help reflective practitioner deepening understanding of the work of colleague in and across teams, departments, and faculties.
- ✦ Because peer observation involves our colleagues in the process reflection then it is good opportunity for networking and developing connection with colleagues to discuss various learning and teaching issues.
- ✦ Increasing the sense of collaboration and enhanced trust through allowing colleagues that we have in schools **in order** to help build the practice that we have the ability to observe and comment on other people's teaching becomes very powerful here.

The Process of Peer Observation

For the Reflective practitioner process has four main stages:

Briefing session before observation where a focus is given to the observation then the **observation** followed by post observation. In **post observation** there is reflection and feedback from the observer and then there needs to some recording of the observer.

We look at reflective cycle where the briefing session occurs before observation which leads ultimately through a statement a record observation which can lead then into another pre-observation session in terms of development. This becomes very useful very simple approach to peer observation linking it to reflective practice.



Briefing session: the reflective practice being observed will set out the context of the observation and draw the observer's attention to anything that they particularly want comment on.

Post-Observation session: this is a collaboration reflection (practitioner and observer) and will involve appraisal of the session by the teacher who was observed and comments from the observer

Collaborative Observation and Feedback: Peer Observation

Peer Observation

There are four key focus areas for peer observation:

- Planning/Organization/ content
 - Teaching Strategies/resources
-

- Presentation/management
- Assessment/monitoring

Planning/Organization/ content

- Focuses on the teaching purposes are clear and stated in appropriate term, such as aims, objectives and outcomes.
- Teaching purposes are appropriate to the needs, experience, and abilities of students.
- In terms of observing their planning and content it would be good for observer to look at how the lesson is structured in term of introduction and content development of ideas and learning how the lesson is concluded and evaluated.
- Content is appropriate for the levels, abilities, needs and interests of students.
- Content is well researched and up to date.

Teaching Strategies/resources

- Observer is checking that the methods are appropriate to purposes of session.
 - Methods are chosen about students' abilities, needs and needs of students.
 - Methods are chosen to gain interest and participation.
-

- Class management is effective and appropriate in terms of the number of people in the classroom and the activities they are engaging with.
- Resources are used effectively and complement content, methods, and purposes they support clearly the aims and outcomes.

Presentation/management

- Observer is looking for things like effective communication skills and presentations of ideas and concept
- Individual teacher is questioning students and asking effective questioning to monitor and promote understanding.
- Student contributions and participation are encouraged in a conducive atmosphere.

Assessment/monitoring

- The observer wants to see that individual teacher is monitoring learning throughout the individual lesson and also looking at a suitable method are used to identify and monitor students' progress.
- Part of this aspect of teaching of course is about the teacher continual use of feedback to the students in a constructive way so this enhances learning.

Collaborative Observation and Feedback: Peer Observation

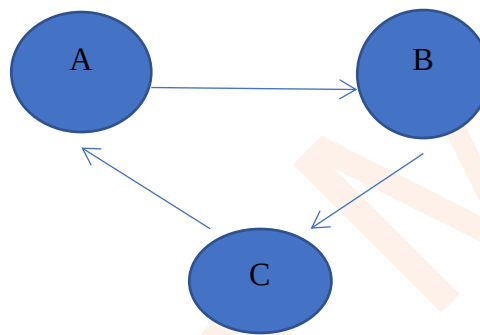
Peer Observation

The process of peer observation of teaching (POT) can be undertaken in several different ways:

Buddy System

We can consider it does not just have to be with two people it can be with more, but with two people involved in this process of observation it is important to agree to act as observer **and the one being observed** and share these two roles.

Circus



We can also consider observation in terms of several people observing for example there could be three teachers three reflective practitioner engage in the process of observing each other and we can call this appear observation circus this is where colleague A would observe colleague B , colleague B would observe colleague C and colleague C would observe colleague A, and this can go around the group continually and can be in reverse order so that all participant observed each other. It not only builds a sense of team work, but it does mean that an individual gets observed by two different practitioner and more practitioners involved in a peer observation circle you can get more feedback perspectives.

When you are the observer:

- You need to make sure that you discuss all aspects of the activity with the person being observed and ensure that feedback is structured around this and provides the sense of exploration that the observer needs.
-

- It should be very efficient and not be too time consuming and should be very clear in terms of instructional value.
- During observation don't be distracted by the content if you are there to observe teaching strategies, for example, it could be looking at teaching strategies and rather than resources.
- You should also allow the one being debriefed to say something about the observation before you give your feedback.
- Focus comment on the behavior not the person in the classroom rather than products.
- In terms of feedback, it is very important that the observer follows a very simple strategy and that it should be clear and owned, balanced, brief, and specific.
- The important thing to remember is for you as an observer to give feedback as soon as possible after the observation otherwise any feedback you give can lose its sense of urgency it makes it very difficult for the observer to make connections between what is your experience and their experience in the classroom.
- What you want to do in term of giving feedback is to priorities your comments. Focus first on what is most important and then work down from there to what is not so important, and this gives much better focus to the feedback from the observation.

Collaborative Observation and Feedback: What is Lesson Study?

What is Lesson Study?

Lesson study is another collaborative observation technique that the reflective practitioner can use. Lesson study is a professional development process that Japanese teachers engage in to systematically examine their practice.

The goal of lesson study is to improve the effectiveness of the experiences that the teachers provide to their students.

The core activity in lesson study is for teachers to collaboratively work on a small number of

“study lesson”. These lessons are called “study” lessons because they are used by the group to examine the teachers’ practice.

Working on a Study Lesson

1: Research and preparation: The group of teachers jointly draw up a detailed plan for the study lesson. The lesson plan is not constructed by an individual but by the group.

2: Implementation: A teacher teaches the study lesson in a real classroom while other group members look on.

3: Reflection and improvement: The group after the lesson delivered comes together to discuss their observations of the implementation of that lesson.

4: Second Implementation and reflection: This is optional but after first discussion and reflection phase and the teacher may go back and teach the lesson again or another teacher may teach that lesson.

The Lesson Plan is the Backbone of the Study Lesson

The lesson plan supports the lesson study process by serving as a:

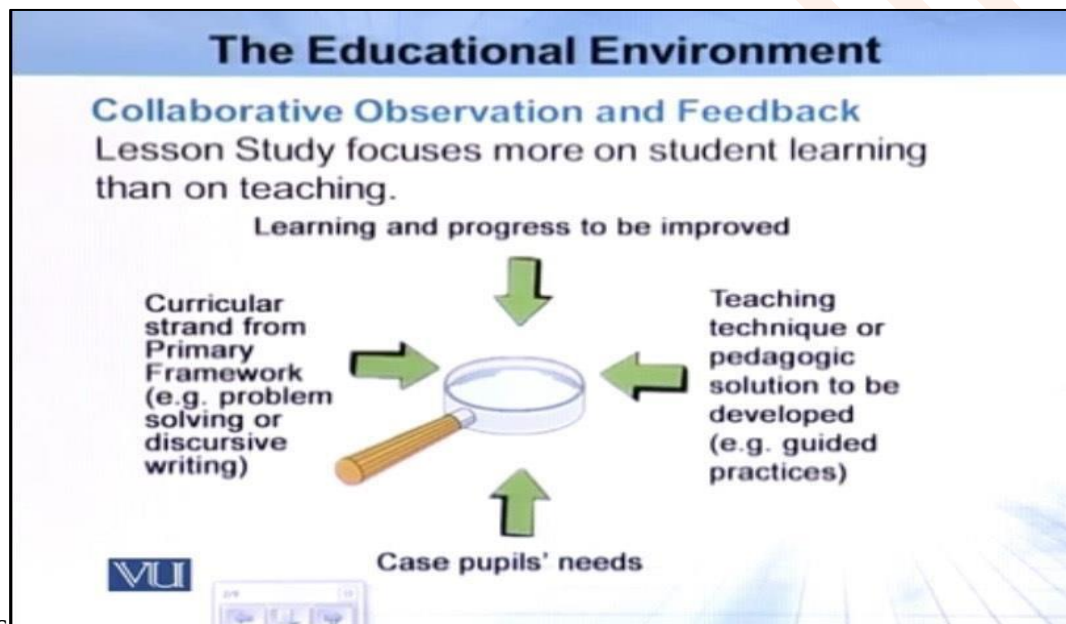
- Teaching tool—it provides a very well define script for the individual teacher to follow in terms of activities and resources used in the lesson.
- Communication tool—it conveys to other members of the group, the thinking of the teacher who planned the lesson.
- Observation tool—it guides what to look for in the lesson; it is a place for the observers to record/share observations.

And so all together this process is very collaborative very systematic and can lead to high levels of reflection and development

Collaborative Observation and Feedback: How do RPs Share Learning form Lesson Study?

How do RPs Share Learning from Lesson Study?

Lesson study is a powerful tool of collaborative observation for the reflective practitioner and what is important to think about is how the reflective practitioner shares what is learned from the lesson study process. There are number of ways for it, it could be very formally through things like **reports and publications** but perhaps it could be more informal through **lesson study open house** or **rotating teachers** through different grade **levels** or just through informal mechanism like **structural supports for teachers** in the school.



What we can do is focus on the passing on of that knowledge by remembering that **lesson study focuses more on student** learning than on teaching. What are the outcomes of teaching and in this sense the lesson study has several imposing factors which is down to do like students' needs as well as the pedagogic approaches used to respond to student's needs?

There is of course a demand from the school in terms of the content that needs to be taught on the timetable that an individual teacher is working to. So, the lesson study is a collaborative process for the teachers focusing on students learning as a center of reflection.

Sharing the outcomes of lesson study can be very formal like in things like:

Reports or publications- what is important to remember is that this type of sharing of knowledge is not just about including materials or lesson plans but there needs to be some sort of reflection there are of the discussions around the construction of those lessons plan. The motivations and the goals behind those on the intended achievements even challenges that were expected **behind each lesson study process.**

Outside observer- It can be useful for the lesson study group to use an outside observer. There are three basic benefits to using an outside observer:

1: an individual can give a different perspective when reacting to the lesson study.

2: an outside observer can provide **additional** information about content/new ideas.

3: observer share the work of other lesson study groups.

Open House- In Japan they use something called open house this is where individual teachers can share their lesson study work with other schools. **So that good practices share around the system.**

Rotation of Teachers- Teachers can be rotated through grades levels within their schools. **It help people understanding the issues that the group is experiencing.**

Structural Supports for Teachers- There can be structural support in the school itself teachers in terms of sharing the learning from lesser study.

Collaborative Observation and Feedback: Group and Peer Supervision

Group and Peer Supervision

What we need to consider are **roles and approaches** to supervision **as part of collaborative supervision.** Within collaborative observation there is a **focus person** the reflective practitioner and a **mediator** and **supervisor.**

Looking at these three roles and how they combine the focus person is the subject of the dialogue essentially, they provide the subject of discussion within the group.

Collaborative Observation and Feedback



The supervisor is the individual who listens actively to the dialogue, ask questions in an appropriate way to build confidence and selfesteem and supports the individual, using questions to investigate further and deeper in terms of their experiences.

The mediator is one who listen actively to the dynamic process between the focus person and supervisor and work to support the process of dialogue.

The Roles

There are particular roles which each individual in the process of collaboration take on:

Focus person: Clearly defines theme and subject of the dialogue.

Supervisor: Is the individual who asks questions about what has been seen and heard in an investigative, exploratory, and appreciative way.

Mediator: is the individual that keeps a check on the dialogue and gives feedback on the process of the dialogue, its structure and depth to the focal person. Mediator listens carefully and then sums up the content and the process of the dialogue between the two.

We can see that the three together make a reflecting team. The reflecting team is made up of the mediator the supervisor and the focus person working together regularly over time.

The Role of Reflecting Team

- The role of the reflecting team is about listening to the dialogue without interrupting so that full understanding can be required.
- Reflecting team take notes on the dialogue to record the content and to give a focus to the individual concerns.
- And notes will also help the reflecting team finds connections between experiences and to understand the associations between different aspects of the process.
- The reflecting team will discuss what they have seen in a very non judgmental way and supervisor can make suggestion on proposed questions to help the focal person understand better the experience and get more clarity about how to move forward.

Group and peer Supervisor: The Rules

There are rules that should be followed:

- The roles within the group need to be very clearly defined theme structured by the focus person.
 - The approach should be very appreciative of the focus person rather than constructive.
 - The feedback needs to be clear and precise.
 - Supervisor should engage in useful questioning technique which help to develop depth and detailed.
-

- The whole process should be about building confidence for the individual.

Constructive Feedback

- ✦ For the reflecting team constructive feedback should be respectful to the focus person and acknowledge the insights of others reflecting team, their values, opinions and their ideas.
- ✦ Show understanding and recognition of the other's feelings and views within the group.
- ✦ Help all members of the team but particularly the focal person to be better understanding, strength, weaknesses, and potentials that they have.

Questioning technique is an important part of group supervision and its important that open questions are used as these promote examination and exploration for example questions like can you give me more details about? And questions to make the other elaborate and argue; like why use this activity...? Why did you do this thing...? this get the focal person to really think critically about what they have done and questions by the supervisor should also challenge the focus person; like What did/do you intend the students to learn?

Questions can also be used to **Direct** (which resource are to be used in this activity?) attention to particular area of the lesson but **In Direct** (how do you think the students will interpret the result?) also to get the focal person to reflect on other person expectations and experience.

Using close questions as part of the reflecting team for validation and interpretation so for example supervisor could say something like Is it correct when I understand this particular thing? because the answer is yes or no and that's provide clarity in terms of reflection.

Supervisor can ask particular probing questions (what exactly?) and funnel questions can make the other focus.

Developing Dialogues

The process of collaborative observation where there is a focus person, and a supervisor has a number of dimensions to, it can be used to inform the group to particular aspect of practice and of course, it can be used to influence practice or the focus person and there can be very linear assumptions about the practices which are being employed as well as circular assumptions or in other words the ability to look back and then to change practice from what has been going on.

So, there are particular roles to be taken on within the process of collaborative observation for example the focus person and supervisor can see themselves as detectives focusing on facts and looking for future understanding can be as an advisor looking for solution in relations to issues and problems. It can be about just exploring , understandings and getting much better appreciation, or it can be about looking better perspective from different viewpoints facilitating understanding.

The Rules:

The rules which need to be follow are very precise but the supervision of a focus person as part of collaborative observation is a very useful technique for reflection on practice.

- A clearly defined theme
- Appreciative approach
- Clear feedback
- Questioning technique
- Confidence

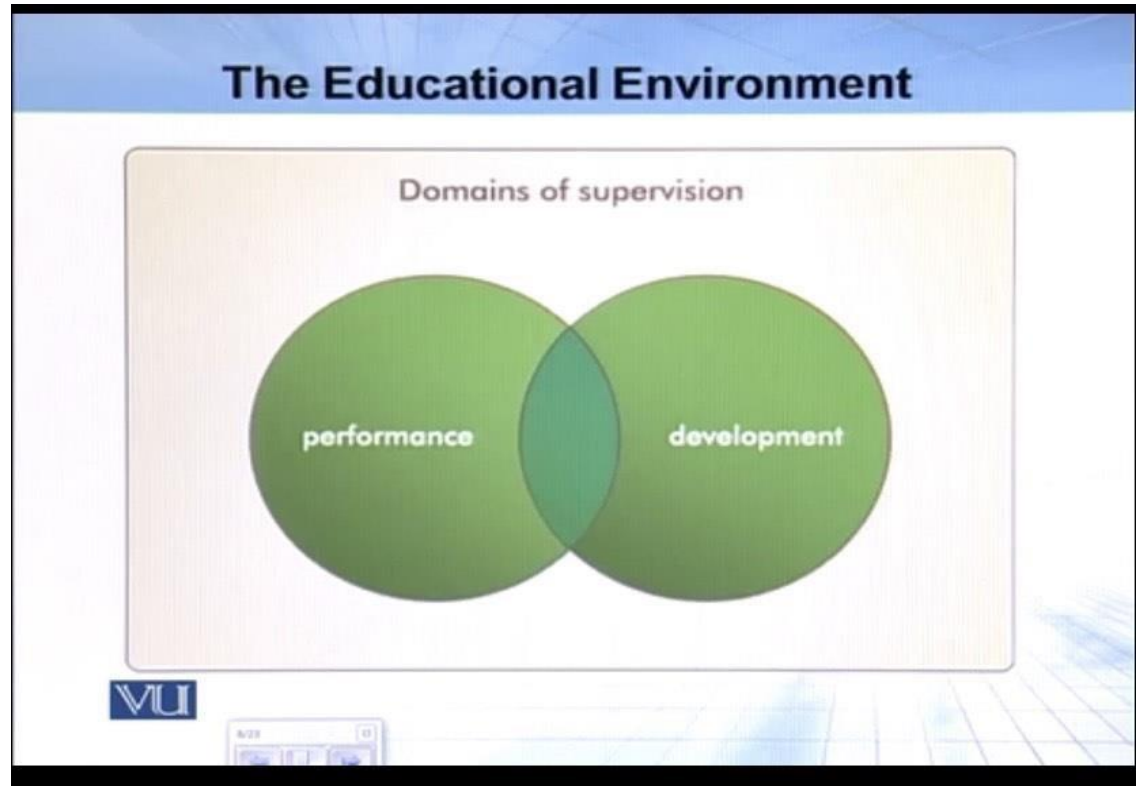
Collaborative Observation and Feedback: Supervision: What Is It?

Supervision: what is it?

Supervision in education is made up of collaborative observation and feedback and both of these aspects are important and relevant to varying degree in supervision, depending on the context in which they are conducted. It can be helpful to think about supervision both in terms of development (which related to ongoing professional learning) and performance (which is related to clinical governance and standard setting.) of and individual in an organization.

Domains of Supervision

We can look at this diagram as domains of supervision in terms of performance and development think about some different context in which supervision can occur in school, it could be through things like; peer supervision, team teaching, multidisciplinary team meetings. Therefore, it's worth considering in which part of the diagram these supervisory conversation can take place, are they taking place in performance domain or are they taking place in development domain? Although many aspects of supervision are common across all context, in the critical setting it is useful to tease out some of the specific aspects of reflection and teaching; not least because there is an emerging distinction being made in education between the two closely related and overlapping activities of critical and educational supervision.



So, in which part of the diagram do these supervisory conversations take place? Because in this diagram it is very useful to **understand this as much as possible** and these domains interact with one another performance and development, so they are interlocking characteristics of the domain.

Principles of supervision

- Be clear about why there is a need for supervision and who has asked for it.
 - Set a time frame for the supervision session; even a few minutes of focused time can be worthwhile.
 - Protect the time and space where possible and appropriate; try to ensure there will be no interruptions and that there is privacy.
 - Ensure that there is confidentiality.
-

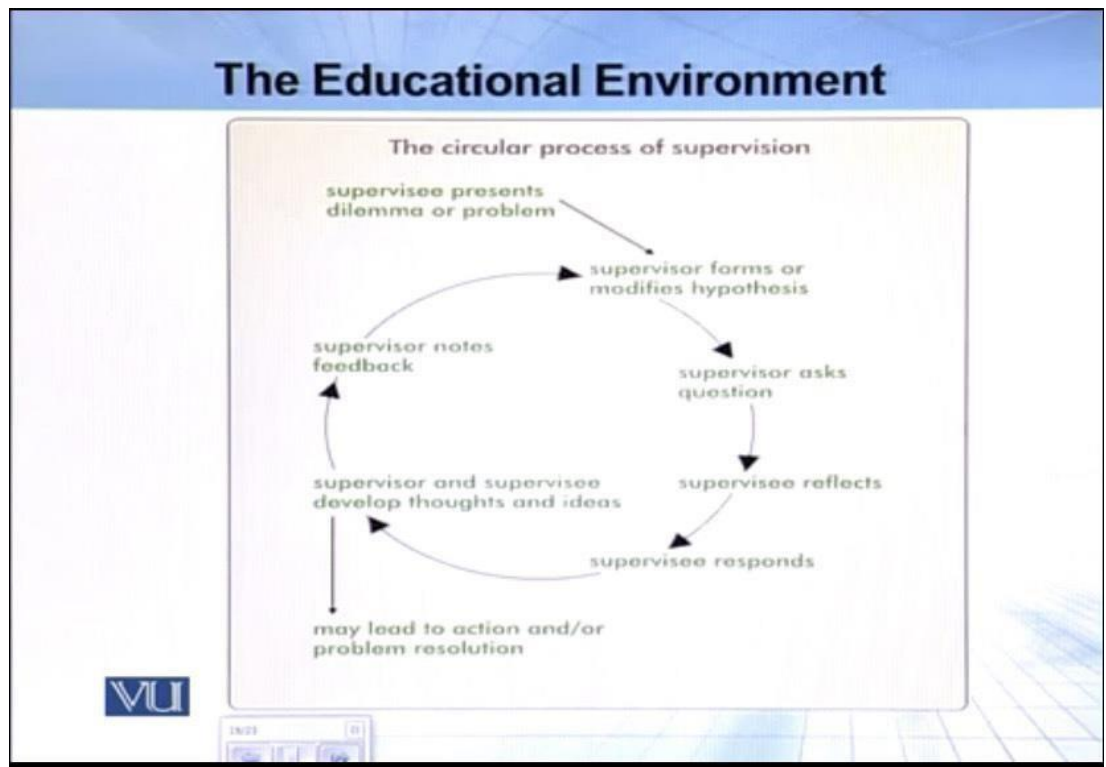
- Be transparent about the extent to which the supervision is about development or performance; this may need to be renegotiated or stated during the session.

What is Supervision For?

Essential for improving experience of students it also available for improving performance within the context. It is also about ongoing development for the teacher especially in terms of career development.

Using question in supervision

- The RP is given the opportunity to reconstruct their view of a particular issue or difficulty. This is achieved by the supervisor asking them questions. They are trying to help them see things from different perspectives and in different context.
 - To help the RP come to conclusions and solutions, the supervisor may wish to refrain from giving any advice until towards the end of the supervision conversation.
 - To identify those solutions, we can represent this as a circular process.
-



Supervision occurs within an ongoing cycle rather than as a linear process beginning to the end. The supervisor and supervisee meets and identify issues which need to be questioned. The question help the reflection of the process to which they are responses, it can lead to thinking through and development of ideas around issues which may lead to action or a solution of a problem and this then needs to be recorded by the supervisor and then this can actually lead into new notions of change and development, new question can be raised.

Constraints and Barriers

In terms of supervision there can be particular constraints and barriers to supervision being effective.

- The most obvious barrier would be amount of time that supervision would take.
 - They can also be worries about the possible enormity of the problem, opening a ‘can of worms’,
 - There is also need for experience or training to carry out supervision.
-

- Embedded cultural attitudes within the school about the process of reflection, for some teachers there is a tradition of working alone.
- Fear of showing areas of weakness or need.
- Anxiety about professional revalidation.
- Attitudes about policing the profession.

Collaborative Observation and Feedback: A Question-Based Approach to Supervision: The 7C's

A Question based Approach to Supervision

To conduct supervision, we can look at a model which is describe as the 7C's model of supervision:

1. Conversation

Within the question-based approach this implies that language and conversation itself is the working tool. Effective conversation can creates new understanding, both for the focus person and supervisor, it s a way of understanding, unpacking reality and getting much more components of reality through “rethinking” and reconstructing stories in order to make sense of them.

2. Curiosity

This is used to develop story about students, colleagues, and oneself. It involves paying close attention to both verbal and nonverbal language. This approach of supervision is very useful because it's holistic.

3. Context

The reflective practitioner is working in a very particular environment and therefore both the supervisor and the reflective practitioner should understand the network, some cultures, faith, values, history and beliefs, community and how these affect teaching. All of these can play a significant part in unpacking the experience.

4. Becoming more interested in interactions

Becoming more interested in interactions/patterns between people and enjoying the process of supervision and benefiting from that process in a very personal way by exploring the richer stories

5. Creativity

Finding ways to create an account of reality that make sense. It implies the creativity process of jointly constructing a new version of the story of process of supervision.

6. Caution

Looking for cues from teaching. It involves working with a level of challenge. So, in order to conduct effective supervision it is important that a lot of care is taken so that challenge is not too much, because too much challenge can actually become stress.

7. Care

This encompasses being respectful, considerate and attentive. It means ensuring reflective practice is carried out within an ethical framework. Therefore, the whole supervision cycle and certainly the work of the supervisor need to be conducted in a way which is very ethical and considerate and understanding the confidentiality.

Collaborative Observation and Feedback: Differentiated Supervision

The Premises

Using supervision for reflective practice is a useful tool. What we can also consider is the notion of differentiated practice and there are certain understandings and premises that we need to explore in order to use differentiated supervision.

- There are lot of very good teachers in education.
- Teachers are the best determines of their own professional development needs rather than school or any body else.
- Not all teachers need to be observed continuously and this is the idea of differentiated supervision that some teachers will need more supervision.
- Therefore, a collaboration and professionalism can breed from creativity and motivation from differentiated supervision because it becomes much more flexible for the individual in terms of how they use it.

Four modes of differentiated supervision

There are four modes of differentiated supervision:

- **Focused Assistance** which is very much directed at meeting very specific needs.
 - **Technical Supervision** around some practical aspects.
 - **Peer coaching**
 - **Self-directed professional development.**
1. **Focused Assistance:** To solve a very specific problem it identifies that problem and then helps provide direction for teachers in order to solve that problem. It is very useful as a supervisory approach for those just entering in teacher profession.
 2. **Technical Supervision:** Teachers not in focused assistance can choose this form of supervision. All reflective practitioner benefit from participation in this phase. Technical supervision will focus on specific aspects of work and therefore , it's not just for new teachers for also those teachers who are teaching from some time to improve habits or develop new approaches.
-

3. **Peer coaching and self-directed professional development:** Teachers not in focused assistance can choose either of these options for continued professional development.

What is Peer Coaching?

It is where two teachers working as a team to collaborate around individual development goals in a non-threatening supervisory model.

What is the Process of Peer Coaching?

- To select a partner, somebody who have think you can learn from and you can help them in their professional development.
- Set goals for practice which meet both of your interests and needs.
- From this goal setting process, you both operationalize those goals, you put into practice certain activities in order to achieve those goals.
- Then develop and collect data to look at the impact of changes in relation to those goals.
- Therefore, the process of peer coaching will incorporate things like observation as well as sharing data from individual teaching practice and also modifying practice then looking at the record of those modifications.

Collaborative Observation and Feedback

Challenges of Inter-Professional Working for Reflective Practice: Drivers of Inter-Professional Learning

There are certain challenges for reflective practitioner in terms of professional working. What we need to consider are the driver of interpersonal learning is:

- Need to create some changes to modify attitudes and perceptions.
-

- For making change remedying certain aspects of the teaching practice between professionals in the learning environment.
- To reinforce collaborative competence to secure collaboration.
- To implement policies
- To implement new services
- To effect changes
- To enhance job satisfaction and ease stress.
- To create more flexible working conditions
- To counter reductionism and fragmentation as the teaching profession proliferates in response to technological advances
- To integrate teaching approaches and changing learning preferences
- Economic drivers also support Inter Professional Learning. This is especially for teachers who are working in rural/remote areas where resources are quite sparse.

Therefore, these situations can create environment where inter professional learning can be effected and therefore, economic consequences, the resource allocation, the funding for resources can be a key driver for the reflective practitioner.

Collaborative Observation and Feedback

Challenges of Inter-Professional Working for Reflective Practice: Principles of Inter-Professional Learning

When we are looking for the challenges of inter-professional learning for the reflective practitioner, we need to consider certain principles involve in IPL.

Principles of Inter-Professional Practices:

- Improves the quality of teaching
-

- It helps the reflective practitioner to focus their attention on the needs of students
- Encourages professionals to learn with, from and about each other.
- Respect the integrity and contributions of each professional. It enables an individual teacher to feel a part of professional community.
- Improves professional satisfaction and begin to recognize their developing professional identity.

Principle of IPL

There are principles that we really want to look at in detail:

- **Improves the quality of teaching-** IPL underpins the reality of the complexity of education. Individual professionals working in isolation does not develop expertise.
- **Focus on the needs of students-** IPL puts the needs and interests of students at the center of the practice. Therefore, provide a very strong focus not only for their approach to the work they do but also their metacognitive processes about the professional and teaching as a whole
- **Respect the integrity and contribution of each professional-** participants in IPL are seen as equal learners.
- **Increase professional satisfaction-** This is achieved through mutual support and guidance and encouragement or the work they do. It is through this process that the individual teacher can discuss with others about their roles and responsibilities and benefits about collaborative practice.

Challenges of Inter-Professional Working for Reflective Practice: Effective Inter-Professional Team-Working

- Effective schooling requires coordinated and concerted efforts from teachers
 - Effective care requires the concerted and coordinated activities of multiple professionals
 - Team work can contribute to job satisfaction
 - Teamwork can assist in the development and promotion of inter-professional communication.
-

- A clear understanding of one's professional identity, likely role within a team and the ideas about teaching as a profession are tested and developed.
- The perceptions of one's own professional identity and others assumption about the professional identities of other groups may not align.
- There is an expectation that in professional settings a grouping of individuals will have the capabilities required to adjust their own practice.

Differences in career histories

And interestingly in terms of working in a team all teachers will have different experiences in their professional role and in other words different career histories and these provide a foundation for exploring different approaches and different successes and different failures people have had in learning from those.

Varying levels of preparation, qualifications and status

And what therefore is possible to see from this is that the reflective practitioner is able to use all of these aspects of working within the team to improve what they are doing to work in a way which helps them develop their professional status and improve on the qualification that they are developing as they go and this helps lift the quality of teaching within the school environment.

Collaborative Observation and Feedback: Communities of Practice and IPL

The learning that comes from being involved in a community of practice (COP) is often not formalized and usually unacknowledged by the workplace as a legitimate way of learning.

Inter-professional learning from team work can help the reflective practitioner become a part of community of practice. A community of practice is a very significant body for the reflexive practitioner to be involved with and it is often not a formalized process but is actually very informal but very collaborative and supportive and the Idea of a community of practice has been acknowledged as very significant in the school environment to manage the quality of teaching and for the individual to manage what they do.

The concept that learning is a social phenomenon (Wenger) leads to informal learning experiences which occur at all levels within a work environment.

The concept of community of practice was developed a number of years ago and relates to the informal learning that an individual that a reflective practitioner can gain from being within a team, being within a community of practitioners. Learning not only from their insights but also from their experiences and this can occur at any and all levels within a school.

New knowledge allows teachers to act as change agents as they find new ways of doing things and have the opportunity to challenge practices, which ensures that best practice is incorporated into the workplace on an ongoing basis.

A community of practice then is a way for teachers to consider knowledge to build new knowledge from experiences from insights and ideas and work with these ideas and insights, work with the new knowledge to explore the quality of it, to use the opportunity but also to challenge existing practices so that constant improvements can be found constant change can enable the reflective practitioner to reach heightened levels of quality in terms what they do.

If the working environment allows and encourages this type of learning experience, opportunities for professional and practice development are greatly increased

If the working environment therefore promotes the idea of teachers sharing experiences and learning from other people's experiences opportunities then the opportunity for developing practice is very greatly increased and so community practice for the reflective practitioner is an extremely powerful supportive tool to manage the quality of what they do.

A community of practice can be described as a group of people who work together to achieve a common goal.

So community of practice is significant for inter-professional learning and a community of practice we have seen is actually very informal and the boundaries of the work of a community of practice are somewhat fuzzy undefined in terms of the normal structures and hierarchies that we experience in organizations and however there is a common goal for people involved in the community of practice and that common goal is about developing skills and experience developing knowledge and by sharing working together and collaborating.

The process of working together and sharing knowledge and resources can lead to an enriched learning experience as people are exposed to new ways of thinking and problem solving.

And collaborating the process then of working together and sharing knowledge and also sharing resources is an opportunity for teachers to enrich not only the quality of their teaching but enrich the learning experience of the students and of course it also enriches the learning experiences of individual teachers as they progress in the development of their skills and the progress in the ways of thinking around activities that they undertake in the classroom it's also of course a means for reflective practitioners to find supportive and new ways to deal with those interesting and significant and a problematic issues that we face in the classroom

We have seen that a community of practice has a specific criterion which makes it so. There are differences between a CoP and, say, an informal network.

So we can see that community of practice then has some specific criteria and it's these criteria which define the community of practice and which make a community of practice different to a team of people working together in an organization even if that team is working quite uniformly

Communities of practice share three specific domains

- **Knowledge – a common body of knowledge within the community.**

Community of practice share certain attributes and the first key attribute for the reflective practitioner is about knowledge sharing for reflective practitioner practitioners within the community of practice knowledge becomes a shared commodity there is a body of knowledge which is contributed to by all members of the community and is therefore a benefits to all members in equal portion

- **Community – commitment to forming a group for networking.**

Another key criteria that helps us define what is a community practice is the fact that it is a community and the community exists because there is a commitment to not only learning from others but learning with others and maintaining the lines of communication and networking between people within the community and there's also a commitment for the reflective practitioner and to share practice what this means is that all teachers are learning from their experiences in the classroom.

- **Shared practice – sharing of ideas, resources and strategies**
-

The community of practice is a forum where teachers can come together share those experiences, share understandings and knowledge which is developed from those experiences even come together to share resources and materials which are developed over time and also strategies which are useful and strategies which have been shown to fail in the classroom all become part of the content of what the community of practice aims to engage with and this extreme engagement with all aspects of the experience that teachers have is what provides the community of practice with real strength in terms of supporting learning and supporting the reflective practitioner in that teaching.

The RP ensures that IPL is effective at many levels: The level of the curriculum (its design and balance of activities); timetabling; allocation of resources; relationships between different academic groups; selection of activities for IPL.

So what is the role of the reflective practitioner in inter-professional learning the reflective practitioner ensures that into professional learning is effective at many levels in the organization for example at the level of the curriculum which is being used in terms of its design and the balance of activities which take place the reflective practitioner is ensures that inter-professional learning happens at the level of the timetable and allocation of resources within the timetable and curriculum. The reflective practitioner is also concerned about the relationships between individual groups are within the organization and the selection of activities also which support inter-professional learning.

Once higher-level decisions have been made to implement IPL activities, the RP is also responsible for what goes on in the learning environment itself – the micro- culture of the classroom.

So once a decision has been made that inter-professional learning can occur in an organization, the reflective practitioner is responsible for being involved in terms of what goes on in the learning environment itself but also within the micro culture of the classroom where they put into practice the skills and knowledge that they are developing.

Guidelines for School Management of Inter-Professional Groups

- Encourage learning from ‘rather than learning with ‘one another.
-

There should be an encouragement for individuals to learn from each other rather than just learning with each other the idea about teachers attending workshops or courses is that they will be learning with one another, from some expert the idea of inter-professional learning is that people are learning in collaboration with one another and learning therefore from experiences that people are requiring a knowledge that they are building up over time.

- Make sure there is adequate, diverse and equal mix of experience.

The reflective practitioner needs to make sure that within the group there is a sufficient diverse experience to be drawn from a way there is a wide breaths of knowledge and experience and this adds to the quality of learning within inter-professional groups.

- Ensure the majority of collaboration has relevance to all.

It is important to make sure that collaboration within the group occurs for everyone and is relevant to everyone in the group so that everyone benefits and everybody feels the value in sharing.

- Utilize the skills, knowledge and expertise of all the participants.

It's important to use all of the skills all of the experience and expertise which is built up all of the knowledge which has been acquired by people over time and to benefit the group and benefit all participants within the group.

The Role of the Reflective Practitioner in IPL: Learning Theory

- The rationale for IPL is also underpinned by learning theories.
 - Various theories focus on effective learning happening in the gap (or disjuncture) between what someone thinks they know and what they think they need to know. So, the rationale behind inter-professional learning is really based in learning theory and there are a number of theories which identify learning as occurring within a that gap in knowledge and experience and the idea is that learning occurs when we tried to close the gap between what we think we know and what we actually know
-

In theory, slightly unfamiliar contexts, such as IPL, create disjuncture, revealing learning needs and motivating RPs to close the gap.

And therefore, inter-professional learning is an opportunity to look at that gap and for the reflective practitioner to be motivated by the need to fill that gap in knowledge or their skills

Skillfully facilitated and planned IPL can utilize constructive friction, creative conflict and the learning edge 'to promote change, stimulate debate and discussion, and promote professional development.

And inter-professional learning is used productively where it's planned and it's structured it is an opportunity for the reflective practitioner to identify gaps in knowledge and experience and skills and to see these as opportunities for constructively managing learning in a sense it's a conflict or a difficulty which the reflective practitioner wants to close and wants to promote some sort of change and this gap which needs to be closed and can stimulate this debate and discussion within the group and it's this debate and discussion collaborating and learning with people where professional development occurs.

What is Grounded Theory?

Focus on generating theoretical ideas (or hypotheses) from experiences rather than having these specified beforehand. Grounded theory is inductively derived from the study of the phenomena it represents.

Grounded theory is a way for a reflective practitioner to focus on generating ideas theoretical ideas or in other words hypotheses based on their experiences rather than having these pre-defined in texts and by others and therefore grounded theory is derived inductively by the reflective practitioner or simply by looking at the phenomena and trying to understand experiences from a cognitive and metacognitive point of view.

Core Elements of Grounded Theory

- Inquiry shaped by the aim to discover social & social psychological processes **with in the school.**
 - Create analytic codes and categories from the data **therefore helps us to drive the theory that explain the processes would be using.**
-

- Data collection and analysis proceed simultaneously.
- Analytic process employed prompts theory discovery and development rather than verification of pre-existing theories. In this sense grounded theory is an Inductive process.
- Theoretical sampling refines, elaborates and exhausts conceptual categories.
- Systematic application of grounded theory analytic methods will progressively lead to more abstract analytic levels.

Data analysis begins o develop theories(explanations) that suggest further cases to sample.

- Use these to elaborate and refine emerging theoretical categories
- Develop properties until no new ones emerge
- Involves comparison of people, places, events, conditions, settings etc.

Purpose

- To develop theory about phenomenon of interest which are grounded or rooted in observation
- Allows the reflective practitioner to take a researcher role in order to seek out and conceptualize latent social patterns and structures through constant comparisons. As teachers we are always seeking new knowledge and conceptualizing our understanding around occurrences and around patterns of behavior that we experience in the classroom.

Constant Comparisons

- involve comparing one segment of data with another to determine similarities and differences
- provide abstract conceptualisation to avoid descriptive interpretations
- help facilitate the discovery of patterns in the data

Coding

While you do comparisons, you will be taking notes and coding

- Coding: identifying categories and properties
- Can be done formally or informally

Grounded Theory: Stages in Developing a Grounded Theory (GT)

Three Stages

1. **Open coding** - a procedure for developing categories of information
2. **Axial coding** - a procedure for interconnecting the categories
3. **Selective coding** - a procedure for building a story that connects the categories producing a discursive set of theoretical propositions.

Open Coding

- Examine the text for salient categories
- Applying codes to the text is labeling phenomena.
- Key is to avoid mere description. e.g. “conferring” not “talked to a manager” “Information gathering” not “reading the schedule”. The important thing for the reflective practitioner to remember is not just to describe experiences but to take on the role of trying to understand experience much more and therefore clarity is very important in terms of looking at words and how using words to describe experience
- Use constant comparative approach in an attempt to saturate

Saturation = look for the instances that represent the category and continue looking until new information does not provide further insight into the category

Saturation = look for the instances that represent the category and continue.

Coding

- Stress on ACTION Not what does this represent, but what is the person doing? What are they trying to achieve by doing a particular action? What strategy are they using?
- Code social and psychological processes.
- Coding process is iterative. Builds up gradually, based on early coding.

Axial Coding

- Explore the relationship of categories, making connections between them.
- Then apply a model to this.

Model:

Causal conditions => Central Phenomenon => context => intervening conditions => Action/interaction strategies => Consequences.

Look for

- **Causal conditions:** what influences the central phenomenon, events, incidences, happenings
- **Phenomenon:** the central idea, event, happening, incident about which a set of actions or interactions are directed at managing, handling or to which the set of actions is related.
- **Strategies** for addressing the phenomenon. Purposeful, goal oriented.
- **Context** - locations of events.

Grounded Theory: Stages in Developing a Grounded Theory (GT) 2

Three stages

Selective Coding

The third stage of developing grounded theory is a selective coding this is where the reflective practitioner constructs a story around the data in order not only to explain it but to conceptualize the thinking which underpins the data

Story line: The conceptualization of the story, the core category

Selective coding: Systematically relating the core category to other categories and filling in categories that need further refinement.

Through selective coding grounded theory then emerges over time by that process of constant comparison always checking and re-checking searching for meaning in the data and in this sense it's not a forced process it's an iterative and informal and informative process and therefore theory emerges over time and becomes a something which it grows with experience. It is then in a sense for the individual a prescriptive process and develops categories in terms of trying to prescribe understanding to the data.

In this sense the individual categories coming together and helping the reflective practitioner developed theory are constructed and in this sense we can understand that this whole process is constructivist from a learning perspective. Selective coding then examines how that experience is constructed over time how the individual is working with experience and starting to construct knowledge from that experience.

Coding not a description, rather it gets to the concept/pattern. GT is the study of a concept.

Theoretical Sensitivity

A significant process as for the reflective practitioner to bring to the stages of developing grounded theory essentially about theoretical sensitivity.

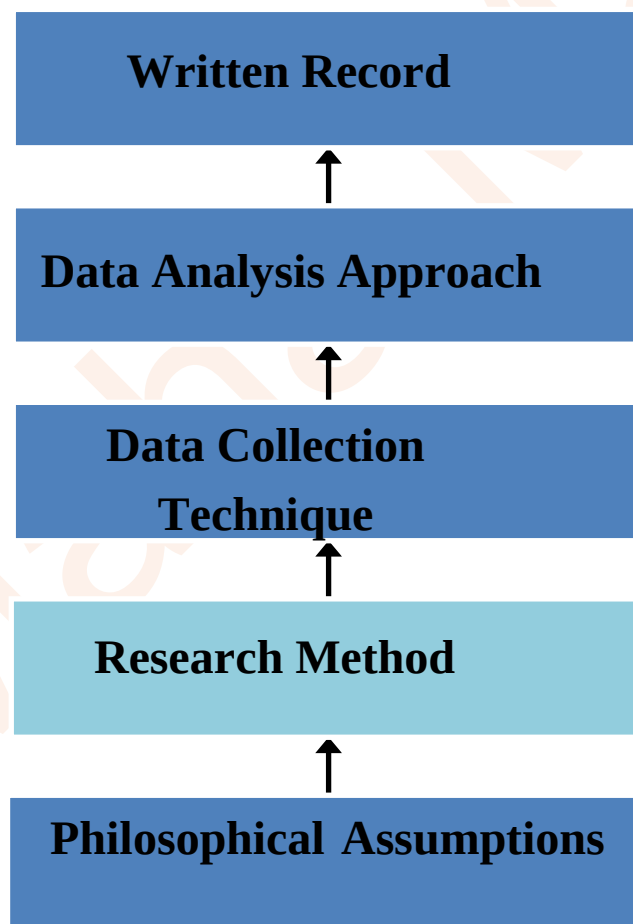
Sources of Theoretical Sensitivity:

- Professional experience
 - Personal experience of an event
-

- Analysis process itself: become sensitive to concepts, meanings and relationships
- Theoretical sensitivity is a good thing in that it helps us connect what we know to what we do not know about what we are studying (awareness of our biases)
- Forces us to step back and make sure what we are truly seeing is FOUND in the data. Think of it as a grounding tool for conducting grounded theory

Grounded Theory

- There is a continuous interplay between data collection and analysis



There are certain philosophical assumptions that we bring to the job of teaching as reflective practitioner our philosophical assumptions not only affect what we do but from the grounded theory perspective affect how we research and practice so our philosophical assumptions affect the research methods that we use and consequently the research methods that we are applying affect the collection of data the strategies that we used to collect data and also our approach to analysis of data and moreover once collection and analysis of data has occurred the way we write about our learning the way record information is affected by the research method and so we can see that our first philosophical assumptions are key in terms of developing grounded theory.

Some Approaches to Grounded theory

Two main variants:

1. Strauss and Corbin (1990) provide a one coding paradigm (context, conditions, interactions, conditions and consequences)
2. Glaser (1978) provides 18 coding _families 'giving many more options Strauss and

Corbin (1990) variant is the most widely used

- **Theoretical coding = The formulation of a theory**
- **The Process and Categories Within the Flow of Research in Grounded Theory for the Reflective Practitioner**

Types of Grounded Theory Design

There are certain types of grounded theory that we can consider as reflective practitioners:

- **Emergent Design:**

This is where grounded theory emerges from the data over time. Grounded theory exists at the most abstract conceptual level rather than the least abstract level as found in visual data presentations such as a coding paradigm. most importantly emergent design helps us understand that theory is strongly grounded in the data and it's not forced into pre-conceived categories the categories emerge from the

data and for emergent design we think of certain categories that may emerge for example the fit of data within the categories and the nature of the categories in relation to particular aspects of the work that we do. We are also considering about the relevance of the categories in relation to the experience and most importantly whether the categories can be modified refined over time.

- **Constructivist Design:**

This is a philosophical position between the positive perspective learning and post-modern ideas about research and in this sense it's very much focused on the feelings and explaining the feelings of individuals in relation to experiences and therefore putting all of this into a very focused context. Constructivist design then helps the reflective practitioner not only study experience but make connections between that experience and their beliefs and values as practitioners and also as researchers and therefore ignores any predetermined categories and develops categories as they become apparent in the data. Narrative of this whole process is more explanatory, discursive, and probing the assumptions and meanings for the individuals in the study.

Types of Grounded Theory Design: Criteria for Evaluating Grounded Theory

Types of grounded theory design mean that we need to consider the criteria for how we actually evaluate the grounded theory which emerges there are certain questions that we can ask for example:

- Is there an obvious connection between the categories and the raw data?
 - Is the theory useful as a conceptual explanation for the process being studied?
 - Does the theory provide a relevant explanation of actual problems and a basic process?
 - Can the theory be modified as conditions change or further data are gathered?
 - Is a theoretical model developed or generated that conceptualizes a process, action, or interaction?
-

This is a good way for us to test the robustness of the theory that's being developed. Another question we can ask:

- Is there a central phenomenon (or core category) specified at the heart of the model?
- Does the model emerge through phases of coding? (e.g., initial codes to more theoretically oriented codes or open coding to axial coding to selective coding)

If we can say that these this is the case then this is a good way for evaluating the quality of grounded theory, another aspect of this theory is:

- Does the teacher attempt to inter-relate categories?
- Does the study show how the teacher validated the evolving theory by comparing it to the data, examining how the theory supports or refutes existing theories in the literature, or checking theory with students' colleagues?

Building Theory From and For Practice: Framework for Organizing Teachers' Learning and Knowledge: Building Theory from and For Practice

We need to develop a framework for organizing our learning and knowledge in order to build practice and there's a simple model that we can use called the knowledge conversion model. The knowledge conversion model focuses primarily on knowledge creation and the transfer of knowledge between implicit and explicit forms and across individual and organizational levels.

There are four knowledge conversion stages (SECI):

- Socialization (S) between individuals using tacit knowledge.
 - Externalizing (E) it in individual reflections in an organizational context that builds collaborative knowledge
 - Combination (C) facilitates bringing tacit individual knowledge into explicit organizational knowledge
-

- Making knowledge reusable for individual learning in internalization (I)

Summary: Socialization is the process between individuals in terms of sharing and using that acquired knowledge from experience and the next phase of the model is the externalizing phase this is where individuals are taking that acquired knowledge that tacit knowledge and exposing it within the organization externalizing it to individuals to others so that we begin to build an organizational perspective on that knowledge and we can start to collaboratively construct and re-construct that knowledge and then this leads to another phase of the model which is known as the combination phase this is where the individual reflective practitioner is linking the acquired knowledge the tacit knowledge with the externalized knowledge the explicit organizational knowledge which is being created synthesizing understanding from those two perspectives and then that leads to the final stage of the model which is known as the internalization stage this is where the individual takes both the tacit and the explicit knowledge and through a process of reflection begins to build new understandings of process and practice in teaching.

Socialization Communication between the members of the community	Externalization Publishing reflections. Comments from community member to peers' reflection
Internalization Monitoring community members reflections and learning from them. Planning own competence development	Combination Collaborative creation of learning resources. Formulating community norms and visions

Building Theory from and For Practice: A Process for Personal Theory Building

Building theory from and for practice for the reflective practitioner is a process of personal theory building the aims of theory building are essentially about explanation and prediction.

The Aims of Theory Building

- Explanation: explaining things which are experienced which are occurring in the platform
-

Examples: what produces bad behavior?

- prediction: making predictions in relation to the teaching that were engaged with if “A” happens, “B” will follow

Theory Building

Theory building for the reflective practitioner is an attempt to construct and also evaluate statements explanatory statements about what we are experiencing and what is going on around us.

Some common-sense examples:

- punishment deters bad behavior
- improved teaching increases student achievement

Theory building operates both at the abstract level of concepts (ideas abstracted from an object) and at the empirical level (experience of reality). These two levels can interact with one another for the reflective practitioner as part of the theory building process.

Issues in developing your personal theory

- People are unlikely to change their beliefs unless they have opportunities to critically reflect upon them.
- The intention to learn springs from problems experienced in practice or the desire to maintain currency in knowledge
- The intention to learn is essential to the recursive process of personal theory building.

Building Theory from and For Practice: A Process for Personal Theory Building

What is Theory?

A coherent set of general propositions used as principles of explanation of the apparent relationships of certain observed phenomena.

What Makes a Good Theory?

- Validity

It fits to the facts.

- Generalization

Makes predictions about future or other events.

- Replication

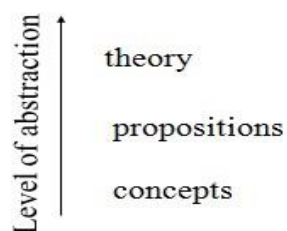
It can be repeated with similar findings

Constructs:

Teachers use concepts (or constructs) as variables. Examples: School leadership; honesty; efficiency.

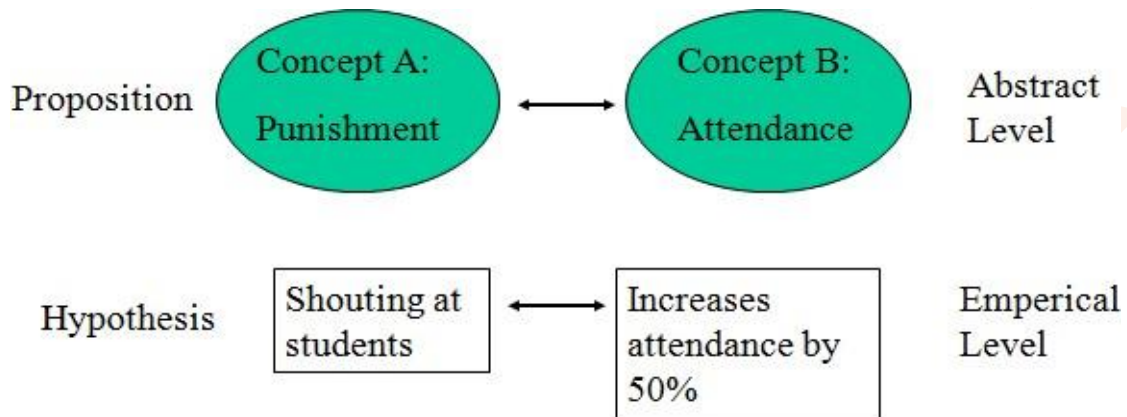
Propositions

- Concepts are the basic building blocks
- Propositions propose the linkages between these concepts



The process of personal theory building has certain propositions. There are concepts which for the individual for the reflective practitioner are the basic building blocks from which theory Springs and propositions are

used by the reflective practitioner to develop linkages between the concepts and ultimate theory and as we move from concept to theory the level of abstraction becomes greater or in other words the level of generalization becomes greater the more we move from concept to theory.



We can look at an example for example concept a punishment is related to attendance the proposition here is that punishment is going to have some impact on attendance and attendance is abstracted in this sense because the type of punishment could affect the amount of attendance and whether attendance is actually linked to punishment is contested and what this means for the reflective practitioner is that at an empirical level an observational experiential level we can look to see changes in attendance through certain forms of punishment so for example a hypothesis would be that by shouting at students we actually increase attendance by some amount. of course this is just an example we are not saying that these things are true but the idea is there that concept through proposition develops theory.

What Makes a Good Hypothesis

- precise
- specifies variables to measure
- specifies relationships between variables

The Double Movement of Reflexive Thought

- Induction occurs when we observe a fact and ask “why”.
- To answer this, we develop a tentative hypothesis as the explanation.
- Deduction is the process we use to test the hypothesis.

Building Theory from and For Practice: Using Evidence

In order to build theory from practice and to build theory for practice though reflective practitioner needs to use evidence and evidence comes in a variety of forms.

- Evidence from own students (formal and informal)
- Evidence of own practice (link to evidence from students)
- Evidence from others' research to inform practice

Using Evidence:

We use evidence for a number of reasons:

- Beliefs, Knowledge and Skills of Reflective Practitioners

- Inquiry habit of mind
- Evidence as a source of information for teaching and learning (not labels for students)
- Beliefs, Knowledge and Skills of Reflective Practitioners
- Knowledge and skills
- The meaning of the evidence for practice
- Sufficient understanding to make relevant adjustments to practice
- Evidence-informed Conversations



We can see this diagrammatically that evidence-informed conversations rely on three key components. One component is the inquiring habit of mind that an individual has that they bring to the process of reflection. This is in relation to the relevant evidence that is being used to inform practice. And also, the relationships between individuals in an organization that exists, relationships of respect and also challenge of thinking.

Building Theory from and For Practice: Using Evidence

Evidence from Students

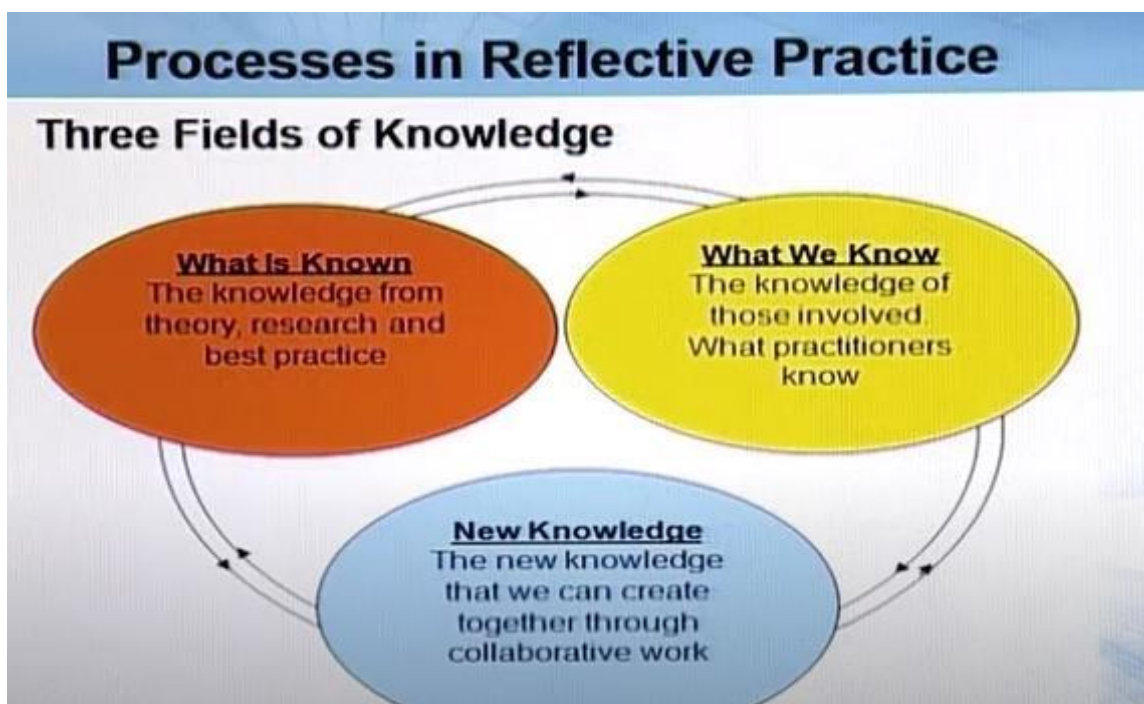
Building theory from practice and building theory for practice as we have seen uses evidence and evidence can come from students there are particular things that we can look for certain things that we can question as reflective practitioners for example:

- What do the students already know?
- How adequate are the sources of evidence we have used?
- What do the students need to learn and do?
- How do we build on what they know?

Evidence of Effectiveness of Own Knowledge and Practice

Evidence is also looked for in terms of our own growing knowledge and our own developing practice and so questions that we can ask in this area include things around knowledge and practice for ourselves for example:

- How we have contributed to existing student outcomes?
- What do we already know that we can use to promote improved outcomes for students?
- What do we need to learn and do to promote these outcomes?
- What sources of evidence / knowledge can we utilize?



The first field of knowledge is what is known the knowledge from theory, the knowledge from research and also the knowledge from best practice which is shared with us by colleagues what is known is a good starting point for the knowledge building process another component is what we know the knowledge of those involved in teaching, what the practitioners know not only the individual but the community of practitioners and another field of knowledge is new knowledge, the knowledge that we can create both individually and collaboratively from work and for work and we can see that these three fields of knowledge work together and inform one another what is known the knowledge from theory and research informs what we know about our practice and both of these can inform new knowledge and conversely the new knowledge that we can create individually or collaboratively with colleagues can inform what we know as practitioners, what we know about what works and what doesn't work and this can actually then be used to inform what is known generally about teaching and learning them and informs theory and research.

The use of evidence for professional learning cannot be a single event: Pervades all aspects of the cycle, it helps us to:

- Identifying what students know and need to learn
- Identifying what teachers know and need to learn
- Deciding what might be most effective
- Checking impact of changes to practice

The three-basic means of human knowledge acquisition are:

- To discover
- To study
- To communicate - obtaining knowledge from others

Significance of Knowledge Management

- Track, measure, share and make use of intangible assets in a school.
- To capture, share and use productive knowledge to enhance learning and improve performance

BASICS of knowledge:

Information; Values; Beliefs; Experiences; Rules and Procedures

Categories of Knowledge

We have to consider certain categories of knowledge,

- **Tacit Knowledge** - we have seen that for the reflective practitioner knowledge starts off as being quite tacit this therefore means the knowledge is very personal based on experiences that we are having in the classroom and therefore personal wisdom which is built up over time in this sense it's very context specific and therefore is very difficult to make explicit

and perhaps difficult to make it meaningful for a society as whole because it is so individual and because it is tacit it makes it difficult to actually quantify all to codify the components of that knowledge.

- Categories of knowledge can be transmitted through social interactions and socialization.
- **Explicit knowledge** - Explicit manner knowledge is what is what ends up being recorded tacit knowledge can be translated into explicit knowledge by being written down by being recorded in some way, easily identified and shared with others through different means.
- Cultural knowledge - Cultural Ethos specific to a school or region or language or religion or nation.

Three particular characteristics of action research are that it:

- Arises from practical questions
- Is participatory in nature
- Its validity is strengthened through peer examination and discussion.

The Objects of Enquiry Are:

- Observable social activities, patterns, structures;
- Intentions motivating those activities;
- Shared, available interpretations of these activities;
- Goal & interest to document, explicate, critique, transform.

Aspects of Action Research

Practical	Participatory
Studying practices involving individual or team-based enquiry	Studying social issues that constrain individual lives
Focusing on teacher development and student learning	Emphasizing equal collaboration
Implementing a plan of action leading to the teacher-as- researcher	Focusing on —lifeenhancing changes Resulting in the emancipated practitioner

The Reflective Practitioner looks at Action Research as a Cycle: Action Research is an “interacting cycle”

Look ↔ Think ↔ Act

Action research engages teachers in a cycle of:

- Experience
- Critical reflection
- Action
- It is a deliberate rather than a purely exploratory entry into a naturally-occurring educational setting. That is, it is a planned and self-consciously focused examination of changing practice.
- It is a solution-oriented investigation aimed explicitly at solving particular problems rather than simply documenting their instances, character or consequences

- It is group or personally owned and conducted. This emphasizes the importance of the practitioner's role as a determinant of the description of the problem, what counts as solutions, and what form the reporting will take.
- It takes the form of a series of iterations on and around the problem, its documentation and theorization, and the analyses that are used to display how it has been redefined and solved.
- These iterations are referred to as spirals but are more commonly known as the Action Research Cycle. This 'cyclic' feature of Action Research is taken to be central to its core emphasis on the documented improvement of practice
- The trying out of ideas is not undertaken solely for the purposes of re-theorizing practice, or adding to knowledge, but is also aimed at improving practice.

Ethical Practice

It is collaborative in nature:

- Provides opportunities for colleagues to share, discuss and debate aspects of their practice with the aim of improvements and development and involves responsible
- sense-making of data collected from within the field of teachers' own practice.

It is transformative in its intent and action:

Practitioner researchers engage in an enterprise which is about contributing to transformation of practice.

This provides opportunities for the reflective practitioner to work with colleagues, to share understandings with colleagues and this draws out the **participatory** nature of action research.

This **participatory** nature helps in meaning-making and sense-making from the information and from the actions which bring about the information that we are using.

Action Research: Reasons to do Action Research

- Help you build a reflective practice based on proven ideas or techniques.

- Allow you to try new ideas and reliably assess their effectiveness.
- It will help you build confidence in your teaching.
- It will contribute to the professional culture of teaching at your school.
- It can create meaningful and lasting change in order to improve student achievement.

Levels of Action Research

- The individual practitioner for classroom
- Group of teachers for department
- Teachers, administrators, and other stakeholders to affect change in the larger school community

How to Get Started – Identify a Problem, Issue or Question

- Decide on a FOCUS

Find your professional self...some guiding questions may be...

- What are your broad interests in teaching/ specific interests?
- What questions are manageable?
- What are you passionate about?
- Is there a problem or question that you would like an answer to as a teacher and how will an answer to your question or idea help you do a better job?

Sources of Information

Studies published in books, journals, periodicals, technical reports, and academic theses and dissertations available either in print or online.

Conducting Action Research Steps

- Identify the question, issue or problem
- Conduct a literature review
- Define a solution
- Apply the solution and collect the data
- Analyze your findings
- Report your findings
- Take ACTION

The Nature of Action Research

- Research is empirical.
- Research is systematic.
- Research should be valid.
- Research should be reliable.
- Research can take on a variety of forms.

Research is empirical

- Action research is characterized as empirical because empiricism is the concept that all knowledge is derived from sense experience.
- Information takes the form of data, which can include test scores, field notes, responses to questionnaire items, etc.

Research is systematic

Based on the scientific method, there are five steps:

1. Identify the problem
2. Review the information
3. Collect data
4. Analyze data
5. Draw conclusions

Research should be valid

- Validity is when something should be based on fact or evidence
- Internal validity is the extent to which results can be interpreted accurately.
- External validity is the extent to which results can be generalized.

Research should be reliable

Reliability of research concerns the replicability and consistency of the methods, conditions, and results.

Action Research can take on a number of forms:

- Basic and applied research
- The primary purpose of basic research is the extension of knowledge.
- The primary purpose of applied research is the solution of an immediate, practical problem.

Action Research for Continuous Improvement

For the reflective practitioner, the process of action research is a process of continuous improvement and there are certain aspects of this continuous improvement which we can look at diagrammatically.

- Successful schools will have learning rather than teaching as a focus. Teachers will work together to analyze student work and consider best practice.
- Instructional decisions will be based on data, emphasizing formative as well as summative data.

Professional Learning Communities

- Collaborative conversations will be a part of the regular school day for teachers.
- There will be a focus on results and staff will hold themselves accountable.
- Professional learning teams will function successfully on a variety of levels leading to improved student achievement
- SMART goals will be understood and used meaningfully.
- Action research or teacher enquiry will inform practice.
- Students who are experiencing difficulty will be supported
- Learning is for all students and teachers

Action Research: Gathering Data

- There are many options for gathering data.
- Action researchers must choose their data gathering methods.
- Choose the techniques that most effectively address the question.

Techniques for Gathering Data

- Reflective Journal
- Portfolio
- Observations
- Interviews
- Questionnaires
- Case studies
- Checklists

Guidelines for analyzing Data (Do)

- Design a system for analyzing your data.
- Look for themes or pattern to emerge from the data.
- Share your findings with a colleague.

Guidelines for analyzing Data (Don't)

- Let your previous assumptions guide your analysis.
- Censor the data you have recorded.
- Look through your data only once.

A Process for Analyzing Qualitative Data

- Write continuously...What are you seeing? What questions emerge? What are you

Learning?

- Look for themes, patterns, and big ideas
- Identify main points that appear most frequently and most powerfully
- Draw information together
- Include support for each of your themes

Action Research is: Process, Collaboration, Improvement, Practice, and Communication

Appreciative Inquiry: An Introduction

Appreciative Inquiry (AI) is the study and exploration of what gives life to human systems when they function at their best. This approach to personal change and organizational change is based on the assumption that questions and dialogue about strengths, successes, values, hopes, and dreams are themselves transformational.

AI simply put....

- If we continue to search for problems, we will continue to find problems.
- If we look for what is best and learn from it, we can magnify and multiply our success.

Problem-Solving vs. Appreciative Inquiry

Problem-solving is:

- Identify problems; Conduct Root Cause Analysis; Brainstorm Solutions & Analyze; Develop Action Plans
- Metaphor: Teaching involves problems to be solved.

And there's a particular metaphor, an expression that we can use to describe this notion of problem solving for the reflective practitioner and that's a **teaching** involves problems which must be solved.

Appreciative Inquiry is:

- Appreciate “What is”; Imagine “What might be”; Determine “What Should Be”; Create “What Will Be”.
- Metaphor: Teaching is a solution/ mystery to be embraced.

Focusing on Being Exceptional**Reflective Practice tends to focus on:**

- Making improvements
- Changing things for the better
- The language of ‘deficit’ - correcting what is wrong

For the Reflective Practitioner, using AI focuses on:

- Positive Attributes

Examples:

- Describe a peak experience or high point in your teaching.
- Identify a time in your experience when you felt most effective and engaged.
- What are three wishes you have to enhance the quality of your teaching?

Underlying Benefits**Appreciative Inquiry unleashes power by:**

- Building relationships.
- Creating opportunities for teachers to be heard.

- Generating opportunities for teachers to dream.
- Allowing teachers to choose how they will contribute.
- Giving teachers the support to act.
- Encouraging and enabling teachers to be positive and affirming.

Appreciative Inquiry as a Process for Reflective Practice

5 Principles of Appreciative Inquiry (AI)

- Constructionist: The way we know is fateful
- Simultaneity: Change begins the moment you ask the question
- Poetic: People in organizations are an open book
- Anticipatory: Deep change = change in active images of the future
- Positive: The more positive the question = greater/longer lasting change

Key Understandings of AI

- Appreciative/value the best of what is
- Envision what might be
- Engage in dialogue about what should be
- Innovate what will be
- A cooperative inquiry
- A collaborative process

- Generate new narratives/perspectives
- The 4D Cycle

The 4D Cycle Expanded

- Discover: The identification of organizational processes that work well
- Dream: The envisioning of processes that would work well in the future
- Design: Planning and prioritizing process that work well
- Destiny: The implementation of the proposed design

Critical Inquiry

- Systematic inquiry that seeks to reveal the operation of wider structures such as power;
- Claims to objective knowledge;
- Claims to ‘truth’ and understanding;
- Can be inclusive and dialogical;
- Includes a vision of a better world;
- It can be argued that it often results in a problem focus with an emphasis on solutions

Comparison of AI to Critical Inquiry and Similarities between AI and Critical Inquiry

	Critical inquiry	Appreciative inquiry
Driver	Logic of analysis Find what’s wrong, broken or missing/find solution to it.	Faith in strengths. Find what’s working, present or right/affirm and expand on it.
Knowledge focus	Negative experiences of power/conventions	Positive and unique experiences.
Epistemology	Social constructivist/realist. More stress on understanding.	Social constructivist. More stress on meaning significance.

Value	Rationality and reason, the intellect 'objective knowledge'	Innovation, affirmation, intuition, imagination, 'constructed knowledge'
Approach	Address the 'negative'; leverage the 'positive'	Amplify the 'positive'; reframe the 'negative'
Space-time dimension	Wide-continuous	Immediate (here/now)
Process	Does not have to be social. Dialogical or reflective criticism?	Narratives; Storytelling. Encouraging. Draws on principles of simultaneity.
Impact	Directive/intolerant resulting in defensiveness. Potentially paralyzing or deficit discourse – maintains status quo. Potentially emancipatory - shared meaning of Justice	Encouraging and inclusive resulting in sense of camaraderie; Potentially emancipatory - way forward. Spirals of 'dysfunctional' practices.

The **Driver** of critical enquiry is very much about the reflective practitioner engaging in a logical process of analyses; breaking components of their practice down in two aspects of the work that they can investigate and find solutions for, but the **Driver** of appreciative inquiry is about realizing the foundation of the strengths and find ways to either confirm those strengths or to build on those strengths.

Critical inquiry has a particular **knowledge focus** and which is based in problem solving, identifying negative experiences and attempting to change those. Whereas the **knowledge focus** for appreciative inquiry is always positive and unique to the individual in terms of their experiences. The **epistemological background** for critical enquiry is very much looking at things from individual perspective and how knowledge is created, constructed over time and there is a greater stress is on **understanding** compared to appreciative inquiry where the stresses is on **meaning** and on significance to the individual.

There is also a **value** component to critical inquiry which is the about **intellect and reason**. Whereas with appreciative inquiry it is very much about **intuition and imagination** and how the transformation **can occur** over time.

Another focus for critical inquiry is the **approach** that **is actually taken**. It is looking at negative aspects and trying to change them into positive components, whereas with appreciative inquiry the approach is actually to build on positive components and rather than looking at things negatively, looking at things from an opportunistic pointed of view.

For critical inquiry there is also a dimension around the context. The **space-time dimension** for critical enquiry is very wide and very long term but for appreciative inquiry, it is very much focused in the immediate work of the practitioner.

For critical enquiry the **process** can be dialogic or reflectively critical but **for** appreciative enquiry, it's very much about creating **a** narrative, telling a story and developing **that** story over time.

The **impact** of critical enquiry is very direct and does not bring itself to emancipation of freedom and developing choice for the reflective practitioner but appreciative inquiry is encouraging, stimulating, promoting freedom of choice and freedom of development for the individual.

Therefore, we can say that there are commonalities between critical inquiry and appreciative enquiry and also **difference** is in the process and the product. Appreciative inquiry is much more meaningful for the reflective practitioner from a personal change perspective and a professional change perspective. **Critical** inquiry is meaningful for an **organizational** change perspective and **also** the professional perspective.

Aspects of transformative inquiry (TI)

	Transformative inquiry
Driver	Social change – Logic and affirmation
Knowledge focus	Power in both negative and unique/shared experience
Epistemology	Postmodern stress on rational meaning. Retains a critical realist orientation.

Value	Reflectivity modulated inquiry. 'Not everything goes'
Space-time dimension	Natural, Modulated choices through social reflection.
Process	Wide-continuous through the immediate
Impact	Collaborative, situated judgments
Impact	Second order learning

Transformative Inquiry (TI)

- Claims to retain the mobilizing appeal of AI with the reason of CI.
- Claims to foster informed positions on change (whether transformative or conservative) conducive to action.
- Can result in the inclusion of new voices.
- Potentially expands the circle of influence.
- Recognizes the power/reality of the whole in the specific.
- More awareness of consequences (practical reason).
- Can result in second order learning.

The 5-I approach

- Provides a more detailed framework for AI
- Based on S.O.A.R
- Provides a strategic framework with an approach that focuses on strengths and seeks to understand the whole system (e.g., school) by including the voices of the relevant stakeholders.
- The '5-I approach' is: Initiate, Inquire, Imagine, Innovate & Inspire to Implement

based on a premise called SOAR. SOAR represents strengths, opportunities, aspirations, and results.

- **Strength:** We use SOAR so that we can identify the strengths of our practice. And from the appreciative inquiry point of view, this is identifying what is positive, and finding ways to build on that.
- **Opportunities:** And then in SOAR, we are looking at opportunities from the strengths, or in other words, what are people asking for, what do they want that we can use as a guiding light to help us develop our practice.
- **Aspirations:** The next component is aspirations, or in other words, as practitioners, as professionals, what do we care about, what do we want to achieve, and what are our goals.
- **Results:** And then, we have results, how do we know when we are actually succeeding in achieving higher levels of practice.

The 5-I approach provides a framework, a very strategic framework for the reflective practitioner to use which helps them understand the process of appreciative inquiry, and gives a strong structure to follow in terms of undertaking appreciative inquiry.

The 5-I approach uses a framework of key words to help us move through the process of inquiry. The first is **‘initiate’** how do we actually identify what we want to do, and then inquire the process of collecting data, and investigating. And then **‘imagine’**, imagine what can be, how we can change things or transform things for the future. And then **‘innovate’**, and **‘inspire’**, in other words, being creative, around what we're doing so build on that practice. And we can look at this diagrammatically.

The SOAR Process

- Can be done quickly or over an extended period of time.
 - Depends on purpose/goal
- This is best opportunity to involve various stakeholders

Reflection as Rationality

- John Dewey, saw reflection as a further dimension of thought, and as such in need of education; “while we cannot learn or be taught to think, we do have to learn to think well, especially acquire the general habit of reflection” (Dewey, 1933).

- For Dewey, reflection is a rational and purposeful act, an “active, persistent and careful consideration of any belief or supposed form of knowledge in the light of the grounds that support it, and further conclusions to which it leads...” (Dewey, 1933).
- Emphasized practical ideas in both his philosophical and educational theories, always striving to show how abstract concepts could work in everyday life.
- He emphasized "hands-on" learning, and opposed authoritarian methods in teaching.
- Recognizes the individual's own experience as the key element in learning – as both the ‘means’ and the ‘end product’.
- In its simplest form, reflection is comprised of direct experience, process and understanding of experience.
- To achieve understanding – to make meaning – the reflective practitioner MUST engage in his or her own inquiry cycle.

Dewey and Reflective Thinking**Reflection as a Meaning-Making Process:**

- Moves the learner from one experience to the next with deeper understanding of its relationships with, and connections to, other experiences and ideas.
- The thread that makes continuity of learning possible. It ensures the progress of the individual and, ultimately, society.
- It is a means to essentially moral ends.

Reflection as a Rigorous Way of Thinking:

- Systematic
- Rigorous
- Disciplined
- Roots in scientific inquiry
- Reflection needs to happen “in community”, in interaction with others.

Certainly, reflective practice needs to be:

- very systematic,
- very logical and

- analytical.

Reflection as a Set of Attitudes:

- Reflection requires attitudes that value the personal and intellectual growth of oneself and others.

Six Phases of Reflection:

- An experience
- Spontaneous interpretation of the experience
- Naming the problem(s) or the question(s) that arise out of the experience
- Generating possible explanations for the problem(s) or question(s)
- Constructing the explanations into full-blown hypotheses
- Experimenting or testing the selected hypotheses

Significant attitudes for reflection

- Empathy
- Open-mindedness
- Curiosity
- Self-awareness
- Inter-cultural communication skill
- Patience
- Ability to take risks/ act/ experiment
- Active seeking of feedback and alternative perspectives

Dewey's model of five stage cyclical process

Here we can see that Dewey's model is put into a five-stage cyclical process. Very simply, identifying a process where the reflective practitioner is inquiring into their practice, and developing meaning, and professionalism as they reflect.

Ask

The starting point is to ask a question about practice, asking what could be improved, asking what needs to change in order for things to improve.

Investigate

And through this then investigating, in some sort of detail, how to make those improvements and investigating or to understand the components of those change needs.

Create

From this, the reflective practitioner creates a new approach to teaching, a new technique to try out and puts this into practice in order to make the reflective process meaningful for themselves.

Discuss and reflect

And then to discuss with others, the outcomes of that action that they've engaged with helps the reflection process. Because it's from this deeper level of reflection, that the practitioner can ask further questions, and so engage in this continuous cycle.

Dewey: Pattern of reflective thinking

- Problem Definition/Identification
- Problem analysis
- Solutions
- Criteria
- Implementation

Problem Identification/Definition

- What is the specific question or problem?
- What are the key terms, and what do they mean?
- What are the group's limitations?

Problem Analysis

- What is the nature of the problem?
- What caused the problem?
- Why is it a problem?
- What are your goals?
- What are the obstacles preventing you from attaining your goals?
- How does it affect the persons involved?
- What are the short-range effects?
- What are the long-range effects?

Solutions – Selection

- What are the available solutions?
- What creative solutions can we generate?
- Brainstorming

Solutions - Criteria

- By what criteria will you judge a solution?
- How well does each solution meet these criteria?
- What additional effects will preferred solutions generate?
- What are possible negative side effects?
- What are possible bonus effects?
- What is the best solution or combination of solutions?

Dewey: Routine Action and Reflective Action

- Provides a useful distinction between routine action – in which the grounds for action have not been actively considered and where tradition, external authority and circumstance are guiding factors.
- Reflective action derives, in Dewey's view, from the need to solve a problem and involves 'the active, persistent and careful consideration of any belief or supposed form of knowledge in the light of the grounds that support it'.
- Teachers who are unreflective about their teaching tend to accept the everyday reality in their schools and concentrate their efforts on finding the most effective and efficient means to solve problems that have largely been defined for them.
- It's not that unreflective teachers aren't thinking – rather, their thinking does not allow the possibility of framing problems in more than one way.
- Main feature of Dewey's approach to reflection and reflective thinking are the generation of the process through 'perplexity', a sense of goal-directedness and the notion of testing or evaluation.

- This brings Dewey's approach into the realms of experiential learning. Dewey believes that the process can be improved by having an understanding of, and experimenting with, forms of thinking.

Dewey suggests aspects (phases) of reflective thought:

- Suggestions...where the mind leaps forward to a possible solution.
- Intellectualization...of the difficulty or perplexity that has been directly experienced into a problem to be solved.
- Hypothesis...the use of one idea after another as a leading idea to initiate and guide observation.
- Mental elaboration...which clarifies the idea by overt or imaginative action.
- Testing the hypothesis...Dewey did not believe that there was a strict order to these phases. He described them as being "fluid".

Dewey: Experience, Reflection and Learning

- As the father of the 20th Century progressive movement in education and an eminent philosopher, John Dewey's work is particularly helpful in defining and describing the relationships among experience, reflection, and learning.
- Dewey (1933) states that an experience is an interaction between the individual and the environment.
- An experience first includes more than participation in activities; experience could be reading a book, taking lecture notes, or talking with others.

Secondly, an experience contains what Dewey referred to as continuity, a continuous flow of knowledge from previous experiences.

- Learning, therefore, is a continuous and cumulative process.
- Prior learning becomes the fodder for further understanding and insight.

In his 1933 work, "How We Think", Dewey distinguishes between four different modes of thinking:

- Imagination
- Belief

- Stream of consciousness
- Reflection

Dewey acknowledges that imagination, belief, and stream of consciousness are certainly part of our thinking activities, yet they do not necessarily contribute to learning and even less to lifelong learning.

Reflection however plays a different role.

- Reflection is active. When we reflect, we examine prior beliefs and assumptions and their implications. Reflection is an intentional action.
- A “demand for a solution of a perplexity is the steady, guiding factor in the entire process of reflection” (Dewey, 1933, p. 14).
- A key point is that informed action follows this reflective thinking process and leads to more ideas and therefore generates more experience on which to reflect. “Reflective thinking impels to inquiry” (Dewey, 1933, p. 7)

Experience

Dewey states that experience is an interaction between the individual and the environment in which they're operating. And an experience includes participation by the individual and by others. And there are activities which the reflective practitioner can engage with in terms of that experience. What the extension to this model defines is that the development of that experience can first start with things like a literature review, or discussion with others or observing other people's practice and taking notes, rather than just the direct experience.

Reflection

- Consequently, learning from reflection is not only a continuous process, but it's also a cumulative process, building up an extending over time. And therefore, prior learning forms a significant part of this experiential process, because prior learning, provides the bedrock, the foundation for new learning.
- And what Dewey was always saying in terms of his model, because of this continuity aspect to it, is that there are certain **modes of thinking**, which are very important for the reflective practitioner. There are certain focus points that we need to think about.

- The reflective practitioner certainly needs to have a good deal of imagination around the teaching learning transaction, to be finding ways to experiment, to develop the skills and knowledge that they have. And along with this, a belief system that each individual has about teaching and learning, about what is good teaching and what is effective learning. And this in terms of continuity, comes together as a process of thinking over time, what do we called a stream of consciousness, where prior learning informs current experience and leads to new experience and a new learning, and this is all part of reflection and how experience, reflection and learning come together.
- Dewey recognizes and acknowledges that **imagination, belief, stream of consciousness** and **reflection** are important thinking activities for the individual teacher, but they don't necessarily contribute to learning as such. Certainly, he was saying that learning needs to be a thoughtful, active process by the individual in order to make reflection meaningful. Without learning there is no point in reflection.

Learning

Reflection plays a very clear role in terms of facilitating learning for the reflective practitioner. Reflection is that added component which makes the whole process of development work and makes the actual development meaningful.

Reflection needs to be active. It's not a passive process. When we reflect as teachers, we're examining our beliefs and our understandings, and we're looking at the implications of these. And the reflection then becomes an intentional process by the individual rather than a routine action.

Extension to Boud's Representation

- Returning to the experience and attending to feelings.
- The strength of this model is it addresses emotions.
- Returning to the experience and attending to feelings.
- The strength of this model is it addresses emotions.

Returning to the experience – recall and detail of the salient points.

Attending to – connecting with feelings – using helpful feelings and removing or containing obstructive ones.

Evaluating Experience – re-examining the experience in light of one's intent and existing knowledge.

- The Importance of Emotion
- Boud is concerned about the role emotion plays in blocking or facilitating reflective processes.
- In this way, reflection is essentially a private process where emotional influences - such as avoidance of an area of thought - can steer the process of reflection more strongly than any other influence.
- This requires the Reflective Practitioner to return to an event, incident or experience & record it.
- Then consider it in detail at an emotional and cognitive level.
- Then the Reflective Practitioner re-evaluates the event in the light of experience, knowledge & experimentation; seeking to understand the meaning of the experience.
- Then the reflective Practitioner plans for what might change.

And this is because Boud's triangle links together the experience that the reflective practitioner has in the classroom with the emotional response, the feelings that they have generated by that experience. This is very much seen as the strength of this particular model.

- The starting point for this deeper understanding of this model is to think about how to **return to the experience**. The reflective practitioner is using reflection as a tool to not only recall what happened, but also to consider in much greater detail the key points, the key parts of the experience from which to learn.
- And from this, the reflective practitioner is not only **attending to their responses**, but making connections between the experience and the response to the experience. And from a deeper perspective, we can also now consider that the emotional response to the experience can actually have barriers which need to be overcome.
- It is this whole process of attending to feelings and attending to the barriers to the feelings, which are added to the **evaluation of the experience**. And this is an important part of the re-

examining of the whole process as part of reflection. This is how the reflective practitioner is developing their own knowledge and also developing the knowledge of the profession.

- And so, what we can see is the most important, the most significant component of Boud's triangle is the **emotional response** of the practitioner. At this deeper understanding, what Boud is most concerned with as part of the reflection process are the actual barriers, the emotions which are barriers, and the emotions with which are facilitators of the reflection process.
- So, the **representation** in this model requires that the reflective practitioner in a way **revisits the experience**, is **looking at the incident in detail** and then **recording** it. Then considering it not only at a **cognitive level**, but also at an **emotional level**. It is these two things coming together, where the model becomes very powerful.
- Then the reflective practitioner **evaluates the experience**, and uses this to construct knowledge and then to add this to the experimentation, the practice in the classroom, in order to understand the experience in more detail and their process within the experience.
- This whole action then for the reflective practitioner leads to the **potential for change**. And it's this whole process, especially the emotional response, where change can actually occur.

Extension to Boud's Representation: Understanding the Three Stages of Boud's Model

Boud's Triangular Representation of Reflective Learning is one of the most popular and frequently cited models of RP.

Key components of Boud's Model of reflective practice

We also consider this as Boud's triangle. The three components are /There are three stages of reflection:

- 
- Experience
 - Reflection
 - Learning
- In the three-stage model, Boud recommends that the Reflective Practitioner first reflect on an experience by mentally replaying the experience and describing it in a descriptive, non-judgmental way.

- The second stage involves attending to feelings – both positive and negative – triggered by the experience, ‘discharging’ any negative feelings which may obstruct the reflection.

After attending to feelings, the teacher is then ready to re-evaluate the experience by progressing through four sub-stages:

- Association (relating new data to what is already known);
- Integration (seeking new relationships between the data);
- Validation (determining the authenticity of the new ideas and looking for inconsistencies or contradictions);
- Appropriation (making the new knowledge/attitudes one’s own).

Extension to Boud’s Representation-Issues

Issues in Boud’s Model

- Boud’s Triangular Representation of Reflective Learning is one of the most popular and frequently cited models of RP. One problem with this model is that it tends to confine reflection to a retrospective role: Reflection-on-action rather reflection-in-action.
- Also, the focus stays on individuals’ mental activity; practitioners are not encouraged to engage in reflective dialogue in a wider social arena.
- Boud has countered such criticism by highlighting the complexity of the reflective process given our understanding about the way emotions and cognition interact.
- For example, if Reflective Practitioners feel more positive about themselves, they will be more likely to persist with reflective activities.
- Ideas like these have since been applied in teachers’ professional education with the use of mentors and supervisors who understand the importance of giving teacher (Reflective Practitioners) external validation and positive feedback about their reflections.

Extension to Boud’s Representation-Naming Barriers to Reflection

- Boud’s model puts the process of reflection into context.
- When formulating the model in 1985 he was not aware of barriers which hindered the process of reflection.
- In his 1993 book he became more concerned with critical reflection and barriers.

- So finally, in 1993 we see the emergence of barriers as a separate entity within the process of experiential learning and reflection in the Boud model.
- This largely came about due to Boud's experience of being 'blocked'.
- Boud classifies barriers as internal and external based on their origins in relation to the practitioner;

Internal barriers come from within the learner, such as:

- Previous negative experiences
- Accepted presuppositions about what the learner can do
- What learning can take place

Internal barriers come from within the learner, such as:

- A lack of awareness of one's assumptions
- The emotional state of the teacher
- Established patterns of behavior

External barriers can come from:

- People
- The learning environment
- The larger personal situation and context of the teacher
- Social forces such as stereotyping, cultural expectations, classism and so on.

The internal barriers:

- Are the most significant because they come with the reflective practitioner,
- They are internal, they are part of the emotional response.
- can come from any area, any arena for the reflective practitioner

The external barriers include:

- the work environment,
- the learning environment,
- the wider community environment
- and how that impacts on the school and on the individual.

Extension to Boud's Representation-Naming Barriers to Reflection

Boud specifically focuses on emotions and feelings as a barrier to reflection.

- In Boud's model of reflection he talks about Stage 2 of the model - Attending to Feelings and stresses that in reflecting it is important to work with any feelings which we may have.
- However much valuable learning occurs in circumstances which we would never choose to experience, if we knew what the end point would be, and if we were aware of what we would need to experience so as to get to the end point!
- The significance of this is that the Reflective Practitioner may not want to engage fully with the affective elements of reflection which can lead to anxiety, pain and discomfort.

Personal awareness as a barrier to reflection

- This is the most important barrier in reflecting on experience. If we are not personally aware that a barrier exists, then how can we possibly seek to overcome it?
- The Reflective Practitioner's learning can be seriously impeded in that s/he does not see the benefit of formally reflecting:
- 'It wasn't in my nature' or 'I'm too practical for things like that' can be typical thoughts.
- Also coupled with this, not knowing 'how' to reflect comes the feelings of not being 'able' or 'good' enough to undertake critical reflection = 'impostership'.

The environment as a barrier to reflection.

Any environment can be disturbing to careful reflection if it is not comfortable for the Reflective Practitioner.

The keeping of a journal as a barrier to reflection

- A journal is a recognized method of evaluating personal experience by reflecting on it.
- It is not easy to pick up a pen and to start writing!
- Writing about personal experience can be profoundly difficult.
- Boud suggests a possible reason for this in that the culture of academic and professional writing has always devalued personal experience in the 'quest for objectivity and generality'.
- So, we can see two instant barriers that restrict personal writing, namely previous experience and the traditional nature of academic writing.
- Boud refers to the fact that the real battleground for working with barriers is the teacher.

Overcoming Barriers to Reflection: Overcoming Barriers by Oneself**Overcoming barriers by oneself**

- Boud identifies a process for overcoming barriers starting with acknowledging that they exist.
- An important landmark in overcoming barriers is called 'naming' - the more clearly, we can understand them, the more easily we can work with them.

Overcoming barriers with the help of a Group

- Reflective Practitioners are able to learn from a group, since their experiences of using reflection in practice are invaluable.
- Teachers working together can motivate each other and help sustain interest in the exercise.
- The Reflective Practitioner can commit to the group because the group becomes an invaluable part of the reflective process, because we need, as learners, appropriate support, trust and challenge from others.

Overcoming barriers with the help of a facilitator

- Boud refers to the importance of effective facilitation.
- But the facilitator should create conditions in which authentic dialogue and communicative discourse can occur.
- Boud reminds us that the facilitator should resist the temptation of running mindlessly giving answers and solving problems.

Extension to Gibb's model: Revisiting the six Phases of Reflection Using Gibb's Model

- **Gibbs' reflective cycle, as we saw previously, is a popular model for reflection.**
- **The model includes 6 stages of reflection:**
 - Description
 - Feelings
 - Evaluation
 - Analysis
 - Conclusion
 - Action Plan

- Gibbs' discussed the use of structured debriefing to facilitate the reflection involved in Kolb's 'experiential learning cycle'.
- **He presents the stages of a full structured debriefing as follows:**
 - Describe = Initial experience
 - Description = What happened? Don't make judgments at this point or try to draw conclusions, simply describe
 - Feelings = What were your reactions and feelings? Don't move too quickly to analysis.
 - Evaluation = What was good or bad about the experience? Make value judgments.
 - Analysis = What sense can you make of the situation? Bring in ideas from outside the experience to help you. What was really going on? Were different people's experience similar or different from yours in important ways?
 - Conclusions (general) = What can be concluded, in a general sense, from these experiences and the analysis you have undertaken?
 - Conclusions (Specific) = What can be concluded about your own specific, unique, personal situation or way of working?
 - Personal Action Plans = What are you going to do differently in this type of situation next time? What steps are you going to take on the basis of what you have learned?

Extension to Gibb's model: Describing the Phases of Reflection Using Gibb's Model

Stages of Gibb's Reflective Cycle

- Description
- Feelings
- Evaluation
- Analysis
- Conclusion
- Action Plan

Description

It is important to remember to keep the information provided relevant and to-the point. Don't focus on details that aren't required.

Feelings

- Discuss your feelings and thoughts about the experience.
- Consider questions such as: How did I feel at the time? What did I think at the time? What did I think about the incident afterwards?
- You can discuss your emotions honestly.

Evaluation

- For your evaluation, discuss how well you think things went. Perhaps think about: How did you react to the situation, and how did other people react?
- What was good and what was bad about the experience?
- If you are reflecting on a difficult incident, did you feel that the situation was resolved afterwards? Why/why not?
- This phase is a good place to include the theory – remember it is important to discuss the theory and not just describe it.

Extension to Gibb's Model: Describing the Phases of Reflection Using Gibb's Model

The **fourth and fifth stage of Gibbs reflective cycle** are the **analysis** and **conclusion** phase. And we can see that the analysis and conclusion phase, follow one after the evaluation phase. Just to remind ourselves the analysis phase is where the reflective practitioner is making sense of the experience in the classroom. And the conclusion phase is where the reflective practitioner is considering what else could have been done in relation to the experience.

Analysis

- In your analysis, consider what might have helped or hindered the event.
- You also have the opportunity here to compare your experience with the literature you have read.
- This phase is very important, particularly for higher level reflection.

- As a reflective practitioner you need to bring the theory and experience together.

Conclusion

- In your conclusion, it is important to acknowledge: Whether you could have done anything else; what you have learned from the experience; consider whether you could have responded in a different way.
- If you are talking about a positive experience discuss whether you would do the same again to ensure a positive outcome.
- Also consider if there is anything you could change to improve things even further.
- If the incident was negative, explore how you could have avoided it happening and also how you could make sure it doesn't happen again.

Extension to Gibb's Model: Action Plan

The sixth and final stage of Gibbs reflective cycle is the action planning stage. And we see the action planning stage in the cycle between the conclusion of a previous experience, and as part of leading to a new experience which needs to be described. And the action plan is where we consider the experience that we have had, and how we need to deal with it, what we would do in order to change it to improve it.

Action plan

- Action plans sum up anything you need to know and do to improve for next time.
- Perhaps you feel that you need to learn about something or attend some training. What can you do which means you will be better equipped to cope with a similar event?

Action plan- An Example

- Following is an example of an action plan.
- It demonstrates a number of options (aimed at support and activity) within the reflective thinking process for consideration as ways forward for the reflective practitioner.

Action Plan-An Example

- In future, I will ensure that I build up a relationship with colleagues. I am working alongside several different teachers and I intend to speak to each of them about my worries about students' behaviour.
- I have already had a useful conversation with one teacher and together we have developed a programme of team-teaching for the next few weeks so that I do not feel so pressurized.
- I plan to do this with the other class teachers, as it will help them to understand how I feel.
- I also need to speak to colleagues more often about how they feel, as I think I will be able to learn from them.
- In terms of training, I have booked onto a behaviour management workshop.

Extension to Gibb's model: Using Gibb's Reflective Cycle for Deeper Reflection

Gibb's Cycle for Deeper Reflection

Gibb's Model helps the Reflective Practitioner develop the type of thinking associated with deep thought; reflection which is aimed at achieving better understanding. It contains a mixture of elements:

1. Making sense of experience

- We don't always learn from experiences.
- Reflection is where we analyze experience, actively attempting to 'make sense' or find the meaning in it.

2. Standing back

It can be hard to reflect when we are caught up in an activity. 'Standing back' gives a better view or perspective on an experience, issue or action.

3. Repetition

Reflection involves 'going over' something, often several times, in order to get a broad view and check nothing is missed.

4. Deeper honesty

Reflection is associated with ‘striving after truth’. Through reflection, we can acknowledge things that we find difficult to admit in the normal course of events.

5. Weighing up

Reflection involves being even-handed, or balanced in judgement. This means taking everything into account, not just the most obvious.

6. Clarity

Reflection can bring greater clarity, like seeing events reflected in a mirror. This can help at any stage of planning, carrying out and reviewing activities.

7. Understanding

Reflection is about learning and understanding on a deeper level. This includes gaining valuable insights that cannot be just ‘taught’.

8. Making judgments

Reflection involves an element of drawing conclusions in order to move on, change or develop an approach, strategy or activity.

Extension to Gibbs’ model: Trans-disciplinary skills in Gibbs’ model

Trans-disciplinary skills in Gibbs’ model

Gibbs model helps the reflective practitioner develop a range of trans disciplinary skills.

- Social skills
- Communication skills
- Thinking skills
- Research skills
- Self-management skills

Social skills

Social skills include skills for:

- Accepting responsibility
- Respecting others
- Cooperating

- Group decision making
- Adopting a variety of roles

Communication skills**Communication skills include skills for:**

- Reading
- Writing
- Listening
- Speaking
- Non-verbal communication

Thinking skills**Thinking skills include skills for:**

- Acquisition of knowledge
- Comprehension
- Application
- Analysis
- Synthesis
- Evaluation
- Dialectical thought
- Metacognition

Research skills**Research skills include skills for:**

- Formulating questions
- Observing
- Planning collecting data
- Organizing data
- Interpreting data

Self-management skills**Self-management skills include skills for:**

- Time management
- Codes of behavior
- Informed choices
- Ethics

Extension to John's model of Structured Reflection: Describing the Stages of Structured Reflection

- John's model of structured reflection is an example of a 'staged model' of reflection.
- Another popular model used by reflective practitioners, it should be used to assist you in moving beyond a descriptive account of an experience to one that involves analysis, learning and plans of action.

Stage 1: Describe the event/experience

- What happened?
- Who was involved?
- What part did you/others play?
- What was the result?

Stage 2: Thinking and feeling

- What was significant about this experience to me?
- What was I thinking and feeling during the experience?
- What was I trying to achieve?
- How do I feel about the outcome of the event?

Stage 3: Evaluation

- What was good and bad about the experience?
- What were the consequences of my action/actions of others?

Stage 4: Analysis

- What sense can I make of the situation?

- What factors (e.g., values, assumptions, meaning perspective, experiences) influenced my feelings, thoughts, and actions?
- What sources of knowledge influenced or should have influenced my actions?
- How did others feel and how do I know?
- What could I have done differently?
- What would be the consequences of those other actions?

Stage 5: Conclusion and action plan

- How do I now feel about the experience?
- What have I learned about my practice/myself/my organizations?
- What would I do now in a similar situation?
- What factors might get in the way of me applying my learning from the experience?

Extension to John's Model of Structured Reflection: Reflecting on Cue Questions

1. Description questions

- ✚ Phenomenon questions
- ✚ Causal questions
- ✚ Context questions
- ✚ Clarifying questions

2. Reflection questions

3. Influencing factors

4. Could I have dealt with the situation better?

5. Learning

Extension to John's Model of Structured Reflection: Using John's Model

- This would be useful for more complex decision making and analysis.
- Johns supports the need for the reflective practitioner to work with a supervisor throughout their learning experiences.
- He refers to this as guided reflection, and recommends that teachers use a structured diary.
- Johns considered that through sharing reflections on learning experiences, greater understanding of those experiences could be achieved than by reflection as a lone exercise.

Carper's four patterns of knowing

John use additional frameworks, for example, Carper's (1978) four patterns of knowing in relation to each stage in the model.

- Aesthetics
- Personal
- Ethics
- Empirics
- Reflexivity

Epistemological base

- This is a strength of the model as it is one of the few models of reflection that refers to the development of an epistemological base to reflections.
- If you use this model for a situation that is ongoing, you could adapt the reflexive section using cues from another model or develop your own set of cues.

Using John's Model Looking in- Thoughts and Emotions**Looking inward**

- Find a space to focus on self

- Pay attention to your thoughts and emotions
- Write down those thoughts and emotions that seem significant in realizing desirable work.

Looking out

- Write a description of the situation surrounding your thoughts and feelings.
- What issues seem significant?

Aesthetic

In terms of looking out, we can apply those additional focus questions to the stage model, we can ask questions around the aesthetics of describing the experience, questions like,

- What was the point of what I did?
- What was I trying to achieve with it?
- Why did I respond in a particular way to that experience when I could have responded in a different way?
- What were the consequences of that response?
- What were the consequences to me as the teacher also to the students, or even to others in the teaching situation?
- How other people were affected?
- How, what were their feelings?
- What were their emotions, about the experience?

Personal

In terms of the personal attribute, questions,

- What the reflective practitioner is doing is asking questions about why they felt in a particular way,
- Why certain emotions were generated and trying to understand the reasons for that.

Ethics

And in terms of the ethical considerations, the questions that the reflective practitioner can use to look outward and to describe the situation, questions around understanding includes,

- Did I act for the best? (ethical mapping)
- What factors (either embodied with me or embedded within the environment) were influencing me?

In a sense, this is about creating a cognitive map of the ethics, which bound us as professionals. Some people call this ethical mapping. And we are looking at the sort of at the factors which are embodied both within us and within others, which influence everyone and influence the environment.

Empirics

In terms of the empirical questions that we can ask, we are looking at

- What particular knowledge do we have?
- Did we generate which informs our practice?
- What is the experience that we are using?
- What is the knowledge of the theories or the literature saying to us, which we can bring to the experience of reflection?

Reflexivity

- Can I support myself and others better as a consequence?
- How available am I to work with students/families and staff to help them meet their needs?

Considering the Cue:

- What internal factors were influencing me?
- Consider the following framework

Topic243

Extensions to Models		
Expectations from self: · obligation/duty · conscience · beliefs/values Normal practice - felt I had to conform to a certain action. Fear of sanction?	Negative attitude towards The student/family? What factors influenced my actions? Time/priorities?	Expectations from others: · in what way? Loyalty to staff versus loyalty to student/family? Anxious about ensuing conflict?

- Considering this cue helps the reflective practitioner to give equal weight to the internal perspective of reflection as much as
- Adding 'reflexivity' means that the danger of 'navel-gazing' (thinking with no point) will be avoided and along with looking in provides us the reflective practitioner with the opportunity for 'deep learning'.

Using John's Model: Writing Reflexively and Writing Reflectively/ John's Model provides a structured approach to reflexive writing and reflective writing.

- What is the difference?
- Develop style of writing that uses 'I' and personal experience

- Danger of being too personal
- A personal response to events/ experiences
- Immediate/ surface considerations
- Aiming at cataloguing – may be subjective – but this is OK!
- Provides foundations for the professional response
- Might feel strange at first. You may not be used to writing about your feelings or actions from an impersonal perspective
- Will take practice.
- Experiment with different models of reflection.
- It is still academic.
- It is not a diary entry, blog or email to a friend.
- It is not simply a description of events. Reflection is in the analysis of those events.
- It is the considered exploration of your own role in the experience.
- It should not be chatty (informal) in style.
- It should still contain a clear introduction, a main body, and a conclusion.
- It may even include evidence and references.
- It should be clearly linked to theory.
- It should show what you have learned from the process.
- It should consider other perspectives

John's Model of Structured Reflection is an example of a 'staged model' of Reflection

Stage 1: Describe the event/ experience

- What happened?
- Who was involved?
- What part did you/others play?
- What was the result?

Stage 2: Thinking and feeling

- What was significant about this experience to me?
- What was I thinking and feeling during the experience?
- What was I trying to achieve?
- How do I feel about the outcome of the event?

Stage 3: Evaluation

- What was good and bad about the experience?
- What were the consequences of my action/actions of others?

Stage 4: Analysis

- What sense can I make of the situation?
- What factors (e.g., values, assumptions, meaning, perspective, experiences) influenced my feelings, thought, and actions?
- What sources of knowledge influenced or should have influenced my actions?
- How did others feel and how do I know?
- What could I have done differently?
- What would be the consequences of those other actions?

Stage 5: Conclusion and Action Plan

- How do I now feel about the experience?
- What have I learned about my practice/self/my organizations?

Extension to John's Model of Structured Reflection: Reflecting on Cue Questions

- John's model for structured reflection can be used as a guide for analysis of a critical incident or general reflection on experience.
- This would be useful for more complex decision making and analysis.

- Johns supports the need for the reflective practitioner to work with a supervisor throughout their learning experience.
- He refers this as a guided reflection, and recommends that teachers use a structured diary.
- Johns considered that through sharing reflections on learning experiences, greater understanding of those experiences could be achieved than by reflection as a lone exercise.

Johns also uses Carper's (1978) four patterns of knowing:

- Aesthetic
- Personal
- Ethics
- Empirics, adding a fifth pattern, reflexivity.
- This is a strength of the model as it is one of the few models of reflection that refers to the development of an epistemological base to reflections.
- If you use this model for a situation that is ongoing, you could adapt the reflexive section using cues from another model or develop your own set of cues.

Extension to John's Model of Structured Reflection: Using John's Model

- Johns model for structured reflection can be used as a guide for analysis of a critical incident or general reflection on experience.
- This would be useful for more complex decision making and analysis.
- Johns supports the need for the reflective practitioner to work with a supervisor throughout their learning experience.
- He refers this as a guided reflection, and recommends that teachers use a structured diary.
- Johns considered that through sharing reflections on learning experiences, greater understanding of those experiences could be achieved than by reflection as a lone exercise.

Johns also uses Carper's (1978) four patterns of knowing:

- Aesthetic
 - Personal
 - Ethics
 - Empirics, adding a fifth pattern, reflexivity.
-
- This is a strength of the model as it is one of the few models of reflection that refers to the development of an epistemological base to reflections.
 - If you use this model for a situation that is ongoing, you could adapt the reflexive section using cues from another model or develop your own set of cues.

Using John's Model Looking in- Thoughts and Emotions

- Find a space to focus on self
- Pay attention to your thoughts and emotions
- Write down those thoughts and emotions that seem significant in realizing desirable work.
- Write a description of situation surrounding your thoughts and feelings.
- What issues seems significant?

Aesthetics

What was I trying to achieve?

Why did I respond as I did?

What were the consequences of that for the student/others/myself?

How were others feeling?

Personal

- Why did I feel the way I did within this situation?

Ethics

- Did I act for the best? (Ethical mapping)
- What factors (either embodied within me or embedded within the environment) were influencing me?

Empirics

- What knowledge did or could have informed me?
- Does this situation connect with previous experiences?
- How could I handle this situation better?

Reflexivity

- What would be the consequences of alternative actions for the student/others/myself?
- How do I feel about this experience?
- Can I support myself and others better as a consequence?
- How available am I to work with students/families and staff to help them meet their needs?

Using John's Model: Writing Reflexively and Writing Reflectively

- John's Model provides a structured approach to reflexive writing.
- What is the difference?

Reflexive writing

- Different to other forms of academic writing. Switch from 3rd to 1st person for example.

- Develop style of writing that uses 'I' and personal experience.
- Danger of being too personal.

Taking Reflective Practice Forward:

Reflection investigates pedagogy and the content of teaching, in a conscious on-going review and evaluation of theory, policy and practice.

Different levels have been identified, for example:

- Technical (focusing on teaching strategies only e.g., how to teach vocabulary)
- Contextual (relationships between problems and teacher actions)
- Critical (deep) thinking, seeking to get at the heart of practice while taking account of contextual and technical factors).
- The quality of reflection is also something to be borne in mind. Reflection has been seen as always beneficial among teachers.
- However, some reflection may just lead to stagnation, with some teachers just going round in circles
- In addition, power is a key issue; teachers without power cannot perhaps use their reflection to best advantage.
- In addition, power is a key issue: teachers without power cannot perhaps use their reflection to best advantage.

Taking Reflective Practice Forward: Professional Practice

These two domains are by no means separate, since the main objective of the former is to facilitate the later!

The reflective practitioner needs to embrace the diversity of available definition, including:

- Evaluating the effectiveness of one's teaching practice.
- Examining teaching from the perspective of the learner
- Conscious and self-aware

- Deliberation on professional practice (sometimes called *reflection on action*, after Schon)
- Intuitive and implicit application of professional knowledge to specific teaching and learning settings (called *reflection in action* after Schon)
- Deepening one's understanding of one's role as a professional teacher in the light of experience (again based on Kolb's cycle of experiential learning) basing professional decision upon feedback (e.g., from learners and/or colleagues).
- Theorizing from experience that is, constructing abstract models or analytical frames based on practical experience of teaching

Taking Reflective Practice Forward: A Clean Approach to Reflective Practitioner

A Clean Approach to Reflective Practitioner

A clean approach to RP aims to facilitate Reflective Practitioners to improve their practice by:

- Enhancing the effectiveness of their reflection
- Deepening their understanding of their process of reflection and its application to their practice (i.e. self-modelling).

The Lawley and Tompkins Model of Reflective Practice involves two parallel processes (each with two aspects):

- External behaviour (event)
- Verbal and non-verbal
- Internal behaviour (experience)
- State and strategy

The Lawley and Tompkins Model has 5 components:

1. Reflection
2. Desired Outcome

3. Plan

4. Practice

5. Feedback

Taking Reflective Practice Forward: Comparing Models

Comparing Models

The Disney Strategy for Reflective Practice is described in terms of roles the practitioner takes on:

- The Dreamer
- The Realist
- The Critic
- With all models it is possible to find points of comparison.

Look at the following, which compares three different models:

Lawley and Tompkins	Kolb Equivalent
1a What happened? (External) 1b What was experienced? (Internal)	Active Experimentation Concrete Experience
1c What meaning is given to 1a/ 1b? 1d How is that arrived at?	Reflective Observation
1e What is concluded/ learned? 1f How are 1a and 1b evaluated?	Abstract Conceptualisation



Lawley and Tompkins	Disney Strategy
2a What outcome is desired in the future?	
2b What evidence (feedback) will be used to monitor improvement?	Dreamer
3a what is the plan?	
3b how and when will it happen?	Realist
3c what might prevent the plan from working/	
3d what can be done about that in advance?	Critic

Taking Reflective Practice Forward: Self-Directed Learning -What is it?

- The process of learning in which the learner assumes primary responsibility for planning, implementing and evaluating learning.
- The learner chooses what to learn and how to learn, and also decides when to continue and when to end

Goals for Reflective Practice

- It enhances the ability of teachers to be more self-directed in learning (humanistic approach)
- Fosters transformational learning
- Promotes social action

Enhanced Abilities

- Reflective Practitioners can use SDL to develop both skills and attributes as learners which support lifelong development
- Better plan, carry out and evaluate their own learning
- The Reflective Practitioner who is a self-directed learner can participate fully and freely in the dialogue through which we test our interests and perspectives against those of others and accordingly modify them and our learning goals

Self-Directed Learning

- Is listed as a key component of 21st Century skills.
- It is intricately inked to lifelong learning
- Listed as a demand for modern societies (UNESCO, OECD)

For the Reflective Practitioner, self-directed learning is:

- A learning process
- A personality aspect
- Ownership of learning
- Self-management
- Self-monitoring
- Extension of learning

Taking Reflective Practice Forward: Self-Directed Learning 2**Ownership of Learning**

- The Reflective Practitioner 's characteristics or personal attributes are important to SDL
- Teachers who take personal responsibility in learning have ownership of their learning

Self Management

SDL, as an instructional process of the individual teacher assessing their needs, identifying learning resources, implementing learning activities and evaluating learning outcomes.

Self-Monitoring

This focuses the Reflective Practitioner on both cognitive and meta-cognitive aspects of learning, which are internal to the teacher (thinking, making meaning of information and integrating new knowledge into existing knowledge structures).

Extension of Learning

The Reflective Practitioner has total control about the choice of what to learn, how to learn and how to evaluate learning.

Some possible behavioural indicators

- Teacher identifies/ determines and articulates own learning goals
- Teacher identifies learning tasks to achieve the goals and charts learning progress
- Teacher challenges self for achievement of learning goal
- Teacher formulates a question and generates relevant inquiries
- Teacher explores a range of possibilities and makes decisions
- Teacher self-plans and self-manages time
- Teacher critically reflects on learning and initiates gathering of feedback
- Teacher applies learning in new contexts

Higher Order Thinking for a Unified Curriculum, Instruction and Learning:**Three Dimensions of Critical Thinking****Focus on the three dimensions of critical thinking**

- Analysis of thinking by focusing on the parts
- Analysis of thinking by focusing on teaching quality (Standards)
- Improvement of thinking by using what you learned

Education, as a concept, defines a set of higher order goals, but actual school learning typically culminates in a set of lower order results.

- How can we narrow the gap between goals and results?
- How can we make higher order goals a practical reality?
 - ...so that in math classes students learn to think mathematically,
 - ...so that in history classes students learn to think historically,
 - ...so that in science classes students learn to think scientifically,
- So that in general –Students begin to think critically in a disciplined, self-directed way
- The obstacle is that many teachers make assumptions about instruction, knowledge, and learning that are incompatible with HOT

Reflective Practice provides a new set of assumptions:

We need long-term commitment to Reflective Practice because of the deep-seated nature of the changes needed and the depth of resistance that can be expected.

Higher Order Thinking for a Unified Curriculum, Instruction and Learning
Conceptualizing Higher Order-Thinking

- The fundamental problems in schooling today at all levels are fragmentation and lower order learning.
- There is too little connection and depth.
- Fragmented lists dominate curricula
- Fragmented teaching dominates instruction
- Fragmented recall dominates learning
- What is missing is coherence, connection, and depth of understanding.
- The bottom line, as we all well know, is not what is taught but what is learned.
- Students are learning something very different from what we think we are teaching them.
 - “All too often we focus on a narrow collection of well-defined tasks and train students to execute those tasks in a routine, if not algorithmic fashion.”Alan Schoenfeld

“Then we test the students on tasks that are very close to the ones they have been taught. If they succeed on those problems, we and they congratulate each other on the fact that they have learned some powerful mathematical techniques.” Alan Schoenfeld

Reflective Practice and Higher-order Thinking

- Reflective practice is a cure for lower-order learning which is an increasingly common mode of learning in every subject area.
- In other words, though education by its very nature comprises a set of higher order goals, actual school learning, given established practice, culminates in a set of lower order results.

The issue that emerges from these harsh realities is unambiguous for the Reflective

- Reflective practitioners can work to improve student performance only by improving student thinking. We can improve student thinking only by creating opportunities and incentives for them to think.
- But we can only provide students with these opportunities and with these incentives if and only if we as teachers as reflective practitioners have time to think about what we’re teaching think about and thoughtfully redesign what we do and what we do is how we teach.
- We can give teachers time to thoughtfully redesign their instruction only if they do not feel compelled to cover huge amounts of subject matter.
- We can reduce the obsession to cover huge amounts of subject matter only if the curriculum is restructured to focus on basic concepts, understandings and abilities.
- We can restructure the curriculum to focus on basic concepts, understandings and abilities only if we understand why such a focus is essential to higher order learning.

Assumptions of Teaching

- The reflective practitioner understands why such a focus is essential to higher order learning only if teachers clearly understand the profound differences between didactic models of education.
- These confuse acquiring knowledge with memorization and the critical model of education which recognizes that acquiring knowledge intrinsically and necessarily depends higher order critical thoughts.

What lies behind uncritically held assumptions?

- That students learn how to think when they know what to think.
- That knowledge can be given directly to students without their having to think it through for themselves

21st Teaching Assumptions

- The Reflective Practitioner understands and values education focused on higher order learning as this holds a very different set of assumptions, namely:
- That students learn what to think only as they learn how to think
- That one gains knowledge only through thinking
- That the process of education is the process of each student gathering, analyzing, synthesizing, applying, and assessing information
- That classes with much student talk, focused on live issues, is a better sign of learning than quiet classes focused on a passive acceptance of what the teacher says

**Higher Order Thinking for a Unified Curriculum, Instruction and Learning:
The Paradigm Shift**

- The Reflective Practitioner will make a paradigm shift from a didactic to a critical model of education to make higher order thinking a classroom reality
- What are the basic changes that must be made to affect this shift?

What are these changes that we need to make in order to realize this paradigm shift in thinking from lower order to higher order?

- Re-conceive the curriculum
- Curricula play a significant role in school life. Instruction arises from goals and objectives stated in them.
- The redesign of instruction
- Teachers feel they have no time to focus on higher order learning and therefore the most basic ideas and issues within a content area can be neglected.
- The shift from a lecture-drill-recall paradigm to one focused upon engaged deep processing can only be achieved through long-term evolution.
- The Reflective Practitioner has the opportunity to place a major emphasis on a detailed formulation of philosophy, highlighting the essential role of thinking in the acquisition of knowledge, and contrasts lower order with higher order learning.

How does this translate for the Reflective Practitioner?

- Higher order learning multiplies comprehension and insight
- Lower order rote memorisation and performance multiply misunderstanding and prejudice.
- Higher order learning stimulates and empowers
- Lower order discourages and limits the learner.
- Good teaching focuses on high content, basic ideas and issues taught in ways which actively engage student reflection and thought.

The Future of Reflective Practice

- Reflective practice is something that was invented in the 60s, 70s, and 80s.
- It has been used worldwide for the past ten years. Many things have changed after 1994

We can begin to see some of these things:

- The concept of wikis, where many people collaborate to construct a knowledge base, an evolving encyclopedia.
- **Search engines are changing the way we access information. This will change our professional world.**
 - The Internet – the wisdom of crowds, combining people ‘s knowledge to make decisions.
 - The concept of collective intelligence, and similarly, communities of practice, which are small groups

We can begin to see some of these things:

- Constant renewal - Any profession will face the challenge of needing to adapt. The way I do my profession will need to be reinvented, and in some cases I will need to create new approaches, new beliefs and new understandings.
- My assumption is that many ideas of reflective practice are being used by authors today, and they are finding it very powerful. Its importance is growing in this Century.
- Are those authors citing Schon? If you analyse how they present their ideas, you can see similarities. The idea of Schon is basically how to improve the ways we think in action.

The Future of Reflective Practice: Perspective of RP in the New Professional Environment**Perspectives of RP in the New Professional Environment**

- Although a product mostly of the 60s, 70s and 80s, one name recurs through the years:
- Donald Schon, Displacement of Concepts, 1963
- Chris Argyris and Donald Schon Theory in Practice, 1974
- Donald Schon, The Reflective Practitioner, 1983
- Donald Schon, Educating the Reflective Practitioner, 1987
- Donald Schon and Martin Rein, Frame Reflection, 1994

- Many things have happened to quicken the pace of reflective practice, and these will continue to strongly influence the ways that Reflective Practitioners reflect on practice

The Web World Wide was invented and spread throughout the world.

- A new communicational platform was implanted, new collaborative software was developed and new business processes emerged.
- Cell phones have spread worldwide and internet communication (e.g. Skype and so on) are now spreading even faster.
- China and India have entered into the global market with highly dynamic economies.

The Future of Reflective Practice: Emerging New Ways for Collaborating in Professional Work

New Ways for Collaborating in Professional Work are emerging

- Wiki, or the gradual construction of a knowledge base from many people
- Search Engines, reaching out to the whole pool of knowledge worldwide.
- Collective Intelligence, combining intelligence of many people around shared interests.
- Communities of Practice, stable spaces for exploring and learning together.

Innovation is Being Distributed

- Democratising Innovation, teachers, researchers and educationists develop new ideas of teaching, learning and education.
- Social Innovation, many teachers working in diverse contexts can come together because of common interests and innovative initiatives.
- Research Networks, articulate capacities worldwide around a common question or inquiry

The Art of Conceptual Innovation is the Driver

- Innovation now is achieved in new concepts of the way professionals work.
- Artistry will be more important than before in the exercise of professions.
- The capacity of integrating technologies is key for creating new solutions and tap into the pool of knowledge of humanity
- When millions of people are innovating, the only way of keeping pace with change is through the creation of new concepts.
- Artistry will be more important than before in the exercise of professions.

Reflective Practice Will Evolve Continually: Traditional Teaching Skills

Reflective Practice Will Evolve Continually

- The diversity of situations and pace of change will require the capacity of:
- Updating
- Extending
- Adapting
- Reinventing the practice of professions, and
- Reinvention of the professions or the creation of new ones
- How can we fit the ideas of Reflective Practice into this emerging professional environment?
- These first 6 teaching skills are not new, but their importance has increased significantly for the modern teacher

Commitment:

It is essential that teachers are committed to their work and to the education of young people. The responsibility that lies with a teacher is huge, so a modern

Traditional Teaching Skills

Preparation

- There used to be a time when the right temperament enabled you to become a teacher. Nowadays it is impossible to find a teacher without formal academic training.
- This requirement is increasing as education levels improve in society.
- The better prepared you are as a teacher, the more effective you will be, so you should pursue your studies with this ethos in mind

Organization

Good organization and the planning of a course/ lesson in advance are key factors for success. It is very important that a teacher organizes the lesson properly and allocates the time to cover it in its entirety.

Tolerance

- In an increasingly diverse and multicultural society, it is necessary for teachers to manage any prejudices they may have and to treat all their students equally without showing favoritism.
- It is a very important teaching skill not to impose your world view on your students, instead you should openly discuss topics and let students decide for themselves

Story Telling

One of the best ways to teach and transfer ideas is through stories. The best teachers have used this method in their classes for centuries

Open to Questions

- Having discussions and collaborating in class are essential for encouraging students and implementing new teaching techniques
- Teachers must be open to answering their students' questions.

- Modern teachers truly listen to their students' questions and answer them honestly, not just with a cursory or textbook response.

Reflective Practice Will Evolve Continually: New Teaching Skills

New Teaching Skills

These new teaching skills complement the more traditional ones. These skills are associated with new technologies. Incorporating these into your teaching repertoire will ensure you become a modern teacher.

Innovative

- The modern teacher must be willing to innovate and try new things, both teaching skills and educational apps, ICT tools and electronic devices.
- The modern teacher must be an —early adopter

Tech Enthusiast

The modern teacher must not only be innovative but also be willing to explore new technologies. Whether it is iPads, apps or personal learning environments, modern teachers should be in constant search of new ICT solutions to implement.

Social

One of the traditional teaching skills was to be open to questions. The modern teacher should lead the conversation to social networks to explore possibilities outside of the class itself. Try using Twitter in the Classroom

Geek

- The internet is the greatest source of knowledge that humanity has ever known, so to be a modern teacher you must be a curious person and incorporate this resource at every available option. Your students are going to do it if you don't!
- You need to be someone who is always researching and looking for new information to challenge your students and engage them in a dialogue both in class and online.

Reflective Practice Will Evolve Continually: Better Teaching

Better Teaching

- The 21st Century reflective practitioner will favor student-centered methods—for example, problem-based learning and project-based learning
- Problem-based Learning and Project-based Learning allow students to collaborate, work on authentic problems, and engage with the learning community in school.
- Teachers know about them and believe they're effective. And yet, teachers don't use them. Recent data show that most instructional time is composed of whole class instruction led by the teacher.
- Even when class sizes are reduced, teachers do not change their teaching strategies or use these student-centered methods
- When students collaborate, one expects a certain amount of noise in the room, which could devolve into chaos in less-than-expert hands.
- These methods demand that teachers be knowledgeable about a broad range of topics and are prepared to make in-the-moment decisions as the lesson plan progresses
- Part of the 21st Century Reflective Practitioner skills relates to greater collaboration among teachers and learning from knowledge acquired from classroom experience